

# Supporting Information: Substituting Reference Solvent $\sigma$ -Profiles for Learned Ones Degrades COSMO-SAC Solubility Prediction

N. L. Polomoshnov<sup>\*,†,‡</sup> and A. V. Rudik<sup>‡</sup>

<sup>†</sup>*Faculty of Bioengineering and Bioinformatics, Lomonosov Moscow State University,  
Moscow, Russia*

<sup>‡</sup>*V. N. Orekhovich Institute of Biomedical Chemistry, Moscow, Russia*

E-mail: nikitapol@fbb.msu.ru

This document is the Supporting Information for the article named above. Its own sections, tables, figures, equations and lemmas carry an S—§S1, Table S4, Fig. S1, Eq. (S1), Lemma S1—and a reference without one, such as §3.5 or Table S30, is to the article, which defines the terminology used here—including IDAC, infinite-dilution activity coefficient—in the text and indexes it in Table S1. What this document adds to the article is evidence and not claim: the proof, the per-row table, the robustness grid, the control and the negative behind a statement the article has already made in its own text, with the set it is measured on, its uncertainty and its scope. That division is a rule the two documents are held to and not a property either can certify of itself. It was audited claim by claim against the compiled article, each failure resolved by moving the missing quantity into the article rather than by qualifying the sentence here; so a claim of the article whose set, uncertainty or scope prints only in this document is an error in this submission and not a division of labour between the two. One result here is not a claim of the article at all, and is declared rather than left implicit: the loss-simplification ablation of §S1, a pre-registered check that the paradox is not an artifact of the objective, whose numbers appear in this docu-

ment alone, which contradicts nothing the article states and which no article sentence rests on.

## Contents

|                                                                                                            |             |
|------------------------------------------------------------------------------------------------------------|-------------|
| <b>List of Tables</b>                                                                                      | <b>S-2</b>  |
| <b>List of Figures</b>                                                                                     | <b>S-3</b>  |
| <b>S1 Supplementary methods</b>                                                                            | <b>S-4</b>  |
| S1.1 The estimator rules in full:<br>cell, convention, stratification,<br>admissibility . . . . .          | S-8         |
| S1.2 Reading conventions:<br>what $P$ is, what each interval<br>resamples, and at what precision . . . . . | S-9         |
| <b>S2 Proofs</b>                                                                                           | <b>S-13</b> |
| <b>S3 The claims ledger, the scored<br/>row set, the decomposition and<br/>its robustness, and the map</b> | <b>S-16</b> |
| S3.1 The claims ledger . . . . .                                                                           | S-16        |
| S3.2 The scored row set, why it<br>is not arm-dependent, and<br>the per-arm table . . . . .                | S-17        |
| S3.3 The estimator grid, and<br>the analyst choices in it . .                                              | S-23        |

|           |                                                                                                   |             |                       |                                                                                                                                                               |             |
|-----------|---------------------------------------------------------------------------------------------------|-------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| S3.4      | The decomposition on both sets, the three kernel arms, and the bounds inventory . . . . .         | S-26        | S6.6                  | The 2002 $\rightarrow$ 2010 arms, the chemistry the closure error sits in, and the size of the profile-database exposure                                      | S-72        |
| S3.5      | The stratification rule, and what sets the level of the bound . . . . .                           | S-26        | S6.7                  | The deployed closure against a published evaluation of the same closure .                                                                                     | S-73        |
| S3.6      | Two nulls: one for the chemical partition, one for the fifty-nine-fold search .                   | S-30        | S6.8                  | Conventions, admissible cells, latent magnitudes, probe scarcity and the untested modern closure: limitations (ii), (iii), (v), (vii)–(xi) and (xiii) . . . . | S-74        |
| S3.7      | The out-of-sample test, declared before it was run . .                                            | S-31        | S6.9                  | Diagnostics that do not work, and why . . . . .                                                                                                               | S-76        |
| S3.8      | The map, stratum by stratum . . . . .                                                             | S-32        | S6.10                 | The second domain in full: construction, bounds, and the reversal . . . . .                                                                                   | S-77        |
| S3.9      | Concentration inside the map, and the cells that do and do not separate . . . .                   | S-42        | <b>S7</b>             | <b>The pKa closure: full construction, measured sweep, and trained comparison</b>                                                                             | <b>S-79</b> |
| S3.10     | Whether the closure defect is correctable, and the arithmetic behind the fidelity lever . . . . . | S-44        | <b>S8</b>             | <b>Data provenance and reproducibility</b>                                                                                                                    | <b>S-83</b> |
| S3.11     | Robustness of the closure/insufficiency separation                                                | S-47        | <b>S9</b>             | <b>Black-box probe programme: full tables</b>                                                                                                                 | <b>S-95</b> |
| S3.12     | A pre-declared attempt to tighten the insufficiency bound, and its failure . . .                  | S-51        | <b>S10</b>            | <b>Closure fidelity sets the reference-input penalty: a controlled synthetic family</b>                                                                       | <b>S-96</b> |
| <b>S4</b> | <b>External baselines and ablations</b>                                                           | <b>S-54</b> | <b>S11</b>            | <b>Positioning against adjacent literatures</b>                                                                                                               | <b>S-97</b> |
| <b>S5</b> | <b>Supplementary mechanism results</b>                                                            | <b>S-59</b> | S11.1                 | The nearest published construction, and the account from fluid dynamics . . . .                                                                               | S-98        |
| <b>S6</b> | <b>Supplementary diagnostics and broad negatives</b>                                              | <b>S-64</b> | <b>References</b>     |                                                                                                                                                               | <b>S-99</b> |
| S6.1      | Supporting diagnostics . .                                                                        | S-64        | <b>List of Tables</b> |                                                                                                                                                               |             |
| S6.2      | Identifiability and compensation . . . . .                                                        | S-68        | S1                    | Terminology and notation. Each term is defined in the text where it is first used; this table collects those definitions in one place. . . . .                | S-5         |
| S6.3      | Grounding the crystal factor: an arm underpowered by a pool the model already held . . . . .      | S-69        |                       |                                                                                                                                                               |             |
| S6.4      | Chemistry-resolved structure of the asymptotic physics penalty . . . . .                          | S-70        |                       |                                                                                                                                                               |             |
| S6.5      | The data-efficiency sweep, and what survives its confounds . . . . .                              | S-71        |                       |                                                                                                                                                               |             |

|     |                                                                                     |      |     |                                                                                                                                                                                                                                                                                                                                                                                                    |      |
|-----|-------------------------------------------------------------------------------------|------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| S2  | Loss registry: the thirteen terms with a non-zero default weight . . . . .          | S-14 | S23 | Fisher audit of the SLE design                                                                                                                                                                                                                                                                                                                                                                     | S-68 |
| S3  | The claims ledger: every load-bearing claim with its set, value and scope . . . . . | S-18 | S24 | The asymptotic physics penalty per solvent class . . .                                                                                                                                                                                                                                                                                                                                             | S-71 |
| S4  | Per-arm scaffold-split metrics                                                      | S-22 | S25 | The asymptotic physics penalty per solute class and novelty stratum . . . . .                                                                                                                                                                                                                                                                                                                      | S-71 |
| S5  | Separation margin over the estimator grid: the VT-2005-matched set . . . . .        | S-24 | S26 | The data-efficiency sweep . .                                                                                                                                                                                                                                                                                                                                                                      | S-72 |
| S6  | Separation margin over the estimator grid: the broad IDAC set . . . . .             | S-25 | S27 | The deployed closure against a published evaluation of the same closure, and this paper's own two sets through it. AAD is the mean absolute error in $\ln \gamma^\infty$ , $n$ the number of scored records. . . . .                                                                                                                                                                               | S-75 |
| S7  | Separation margin over the estimator grid: the glycol-ether stratum . . . . .       | S-25 | S28 | The fixed substituent list behind the Hammett constant .                                                                                                                                                                                                                                                                                                                                           | S-81 |
| S8  | The closure-insufficiency decomposition on both sets . . .                          | S-27 | S29 | Run families: which runs produced which numbers, and at how many seeds . . . . .                                                                                                                                                                                                                                                                                                                   | S-84 |
| S9  | The three COSMO-SAC arms, and the block split of the kernel contrast . . . . .      | S-28 | S30 | Solubility accuracy in two blocks that may not be read across: this work on the solute-scaffold split, external models re-run here on an easier by-solute split. Two sample sizes appear where a row's two metric columns are scored on different rows, one where they coincide. The grounded, ungrounded and $\sigma$ -oracle arms of the first block come from the five-seed leak-free re-run. . | S-87 |
| S10 | Upper bounds on the input-insufficiency term, both sets taken whole . . . . .       | S-29 | S31 | Data Set S2 in full: the VT-2005-matched set, row by row                                                                                                                                                                                                                                                                                                                                           | S-94 |
| S11 | The out-of-sample set, class by class . . . . .                                     | S-33 | S32 | Crystal-term decodability per seed, with its nulls . . . . .                                                                                                                                                                                                                                                                                                                                       | S-96 |
| S12 | The map, solvent axis, in full                                                      | S-35 | S33 | The model's temperature response against the empirical slope . . . . .                                                                                                                                                                                                                                                                                                                             | S-96 |
| S13 | The map, solvent axis at the fine granularity . . . . .                             | S-36 | S34 | Share of representation variance per organising factor . .                                                                                                                                                                                                                                                                                                                                         | S-96 |
| S14 | The map, solute axis . . . . .                                                      | S-37 |     |                                                                                                                                                                                                                                                                                                                                                                                                    |      |
| S15 | The map, the solvent-by-solute cross-tab and the set as a whole . . . . .           | S-38 |     |                                                                                                                                                                                                                                                                                                                                                                                                    |      |
| S16 | The map: every stratum the estimator can bound . . . . .                            | S-39 |     |                                                                                                                                                                                                                                                                                                                                                                                                    |      |
| S17 | The map, solute axis: the boundable strata . . . . .                                | S-40 |     |                                                                                                                                                                                                                                                                                                                                                                                                    |      |
| S18 | The map's error, bound and source curve, stratum by stratum . . . . .               | S-41 |     |                                                                                                                                                                                                                                                                                                                                                                                                    |      |
| S19 | The closure/insufficiency verdict across activity-coefficient strata . . . . .      | S-50 |     |                                                                                                                                                                                                                                                                                                                                                                                                    |      |
| S20 | Solubility accuracy, in three blocks . . . . .                                      | S-55 |     |                                                                                                                                                                                                                                                                                                                                                                                                    |      |
| S21 | Same-pair high-temperature extrapolation . . . . .                                  | S-64 |     |                                                                                                                                                                                                                                                                                                                                                                                                    |      |
| S22 | Solvent ranking under the two profiles . . . . .                                    | S-66 |     |                                                                                                                                                                                                                                                                                                                                                                                                    |      |

## List of Figures

|    |                                                             |      |
|----|-------------------------------------------------------------|------|
| S1 | The model, and where the oracle substitution enters . . . . | S-11 |
| S2 | The separation margin over the estimator grid . . . . .     | S-24 |

|     |                                                                                   |      |
|-----|-----------------------------------------------------------------------------------|------|
| S3  | Where the closure binds, and where the two terms do not separate . . . . .        | S-34 |
| S4  | Activity-coefficient parity on the sixty matched pairs . . . .                    | S-50 |
| S5  | The drift solubility fine-tuning imposes on the learned profile                   | S-59 |
| S6  | Explained-variance spectrum of the fine-tuning drift . . . .                      | S-61 |
| S7  | The data-efficiency curve . . .                                                   | S-72 |
| S8  | Per-molecule cosine between the measured drift and the first-order step . . . . . | S-76 |
| S9  | pKa through a Hammett LFER on a known-ground-truth construction . . . . .         | S-84 |
| S10 | The closure-fidelity sweep on synthetic teacher functions . .                     | S-97 |

## S1 Supplementary methods

**COSMO-SAC constants.** The differentiable COSMO-SAC layer uses the standard VT-2005 / Lin-Sandler constants: effective segment area  $a_{\text{eff}} = 7.5 \text{ \AA}^2$ , electrostatic-misfit constant  $\alpha' = 16466.72 \text{ kcal \AA}^4 \text{ mol}^{-1} \text{ e}^{-2}$ , hydrogen-bond coefficient  $c_{\text{hb}} = 85580$  with cutoff  $\sigma_{\text{hb}} = 0.0084 \text{ e \AA}^{-2}$ , coordination number  $z = 10$ , and Staverman-Guggenheim normalisers  $r_0 = 66.69 \text{ \AA}^3$ ,  $q_0 = 79.53 \text{ \AA}^2$ ; the  $\sigma$ -profile grid is 51 bins on  $[-0.025, 0.025] \text{ e \AA}^{-2}$ . The segment-activity fixed point is damped ( $\lambda = 0.7$ ) and run for 16 iterations at train time and 30 at evaluation.

**A usage pitfall in the reference implementation.** Supplying the 2002 kernel with a typed three-profile base produces a silently-wrong result that the library returns without warning. The reference implementation<sup>S1</sup> pairs its 2002 model with the single-profile Virginia Tech database; the demonstrated input is one untyped, Mullins-averaged profile. When that 2002 model is instead supplied with a typed three-profile Hsieh base — the configuration that arises when a 2002 and a 2010 arm share

one profile source — it silently reads the non-polar channel only, discards the donor and acceptor channels, keeps the total cavity area as prefactor, and returns a number without error. The prescribed way to recover a single 2002 profile from the typed set is instead their *sum*,  $p = p_{\text{nhb}} + p_{\text{OH}} + p_{\text{OT}}$  (Eq. 9 of Ref.<sup>S1</sup>); the nonpolar-only read is neither that sum nor a documented mode. This single-channel arm scores MSE 1.263 on the VT-2005-matched set, whereas the *2002, all channels* arm under the full convention (the same molecule supplied either to our differentiable layer or to cCOSMO with the full profile placed in the single nonpolar channel it actually reads) scores 1.757/1.758. This is an issue for the library maintainers (a silently-wrong return where an error would be safer), not a defect of any published benchmark. The kernel arms of §3.5 are deposited across two artifacts, and which file holds a value does not follow the set. The dispersion-*off* values (1.757 and 0.765 on the VT-2005-matched set, 1.911 and 1.978 on the broad IDAC set), all under the full convention in which the kernel comparison is run, are our differentiable layers on each kernel’s native profiles. Both dispersion values are the reference implementation’s own dispersion-on call: the VT-2005-matched set’s 0.785 from the kernel-flip artifact, where the reference returns no dispersion for the three bromine/iodine pairs and that arm therefore scores on  $n=57$ ; the broad IDAC set’s 1.855 from the dispersion-on field of the fidelity-lever artifact, whose VT-2005-matched counterpart in the same file is instead our own layer’s 7.43. That 7.43 is the measure of the gap: our transcribed dispersion term is not validated against the reference, which is why the dsp arm is deferred to that implementation throughout. The input-fixed control of §S3.10(a’) is a third deposited file: it runs the 2002 kernel on the summed typed profile and re-derives the 1.757 and 0.765 arms of the first file as its own gate.

**The differentiable COSMO-SAC-2010/dsp layer, and the scope of its dispersion caveat.** The 2010 layer implements the four mechanisms of:<sup>S2,S3</sup> the

Table S1: Terminology and notation. Each term is defined in the text where it is first used; this table collects those definitions in one place.

| Term                                       | Meaning in this manuscript                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| closure $g$                                | the fixed, non-fitted physical map from the intermediate to the observable: here COSMO-SAC, and a Hammett relation in the second chemical domain                                                                                                                                                                                            |
| kernel                                     | the segment-interaction energy inside COSMO-SAC; a closure variant is named for its kernel’s year ( <i>2002</i> , <i>2010</i> )                                                                                                                                                                                                             |
| arm                                        | one trained configuration, or one closure variant evaluated; <i>DirectGNN</i> is the closure-free control arm and <i>TGNNsolv</i> the closure-carrying one, abbreviated <i>TGNN</i> in Table S30                                                                                                                                            |
| $z^*$ , $\hat{z}$                          | the closure’s <i>entire</i> argument at its reference (tabulated) value, and the network’s own prediction of it; a $\sigma$ -oracle feeds $z^*$ where $\hat{z}$ would go                                                                                                                                                                    |
| grounding                                  | two opposite operations on one reference database: <i>supervising</i> $\hat{z}$ against it during training, and <i>substituting</i> $z^*$ for $\hat{z}$ at evaluation                                                                                                                                                                       |
| $B_{\text{closure}}$ , $B_{\text{insuff}}$ | the error of the closure as deployed—the fixed map read at the tabulated intermediate—and what its inputs cannot resolve about the target: the two terms of Eq. (2)                                                                                                                                                                         |
| margin                                     | $\text{MSE} - 2B_{\text{insuff}}^{\text{up}}$ , with the MSE of Eq. (2)—the closure read at $z^*$ against its own target $m$ —so in squared $m$ units, squared $\ln \gamma_2^\infty$ on the activity axis and not the solubility axis                                                                                                       |
| binning bound                              | the upper bound $B_{\text{insuff}}^{\text{up}}$ of Lemma 1(c): the variance of $m$ within bins of $g(z^*)$ , which fits nothing to the data                                                                                                                                                                                                 |
| estimator cell                             | one setting of the binning bound’s four analyst choices—bin count, bin edges, within-bin variance convention, combinatorial convention—together with the row or the pair as unit                                                                                                                                                            |
| $\mathcal{G}$                              | the grounding gap $R_{\text{orc}} - R_{\text{e2e}}$ ; distinct from $\Gamma_S$ , the segment activity coefficient inside COSMO-SAC                                                                                                                                                                                                          |
| stratum, map                               | a chemically defined row set of the broad IDAC set, and the 59 of them the estimator is run on                                                                                                                                                                                                                                              |
| boundable                                  | $n \geq 40$ , so the fixed cell’s eight bins hold five rows each                                                                                                                                                                                                                                                                            |
| admissible                                 | boundable <i>and</i> sign-stable under deletion of each contributing publication, in all four unit $\times$ convention cells                                                                                                                                                                                                                |
| full, residual-only                        | which part of COSMO-SAC is evaluated, a model-form choice and not an analyst convention: <i>full</i> carries the Staverman–Guggenheim size-and-shape term, <i>residual-only</i> runs the segment-activity part alone and is what this pipeline deploys, its $\sigma$ head predicting no cavity volume, so the term needs one from elsewhere |
| VT-2005, UD                                | the two $\sigma$ -profile databases, never interchangeable: VT-2005 the Virginia Tech 2005 tabulation the $\sigma$ head is supervised against and which is substituted for $\hat{z}$ at prediction time, UD the University of Delaware successor the activity-coefficient rows are matched to                                               |
| labelled test rows                         | the 5608 of the 8103 scaffold-split test rows carrying a solubility measurement; the other 2495 carry a melting point only and are scorable in no arm                                                                                                                                                                                       |
| broad IDAC set                             | 477 IDAC $\cap$ UD rows, 185 pairs, all temperatures, UD profiles                                                                                                                                                                                                                                                                           |
| VT-2005-matched set                        | 60 pairs at 298 K carrying a VT-2005 profile on both sides                                                                                                                                                                                                                                                                                  |
| $\sigma$                                   | alone, the screening-charge-density profile; $\sigma(\cdot)$ with an argument, the generated $\sigma$ -algebra; $\sigma_{\text{H}}$ , the Hammett substituent constant, $\sigma_{\text{H}}^*$ its tabulated value and $\hat{\sigma}_{\text{H}}$ the network’s; $\sigma_\eta$ , experimental label noise                                     |

donor/acceptor-typed three-profile representation, the temperature-dependent electrostatic misfit  $c_{\text{ES}}(T) = A_{\text{ES}} + B_{\text{ES}}/T^2$ , the typed hydrogen-bonding kernel, and an optional dispersion term in the tabulated parameter  $\epsilon/k_{\text{B}}$ . With dispersion *off* it reproduces the NIST COSMO-SAC-2010 reference to RMSE  $1.6 \times 10^{-4} \ln \gamma$  over 400 infinite-dilution pairs at 298.15 K on identical typed profiles, the largest single departure being  $1.7 \times 10^{-3}$  and the combinatorial term matching to zero, so

the whole of the discrepancy is in the residual term. The dispersion term is not validated to that standard, for two reasons we state rather than absorb. It carries the tabulated coefficient at a single sign, the class-dependent sign reversal of Ref.<sup>S3</sup> being unimplemented and unexercised on the matched set, which holds no water or carboxylic-acid pair; and no  $\epsilon$  is tabulated for the three matched pairs containing bromine or iodine, which our layer scores at  $\epsilon=0$ . That alone accounts for the MSE 7.43

our layer records over all 60 VT-2005-matched pairs; on the remaining 57 it returns the reference 0.785, showing no divergence where both are defined. Every dispersion-off contrast in the paper is therefore computed on our layers and cross-checked against the NIST `cCOSMO` reference, and the 2010, *dispersion on* runs are quoted from `cCOSMO` itself (§3.5).

**Solver, curriculum and loss.** The SLE fixed point is iterated as the damped map  $x_2^{(k+1)} = \lambda \exp(-\Phi(T) - \ln \gamma_2(x_2^{(k)}, T)) + (1 - \lambda)x_2^{(k)}$  with closure-specific kernels,  $\lambda = 0.7$ , for 5 iterations in training and 20 at evaluation. Those are a second pair of iteration counts, on the composition loop, distinct from the 16 and 30 of the segment loop above, and  $\lambda$  is one configuration field read by both, so the two dampings move together. Re-solving the NRTL arm’s deposited per-row kernels—which its deposit identifies exactly, the reconstruction returning that arm’s own  $\ln x_2$  column—at the training count instead of the evaluation count moves 2.5% of the labelled test rows by more than  $0.05 \ln x_2$  and its worst row by 9.20, and the seed-mean MAE by +0.0025. Carrying the same rows to the fixed point instead moves that mean a further +0.017, against this arm’s own +0.093 physics penalty over DirectGNN: the operator that arm is scored through is neither the one it was fitted against nor a converged one. The COSMO-SAC arms carry the same two counts and cannot be re-solved this way, their kernel being a  $\sigma$ -profile pair the deposit does not carry; nor can the weights be re-run, the tree retaining the three DirectGNN checkpoints of Fig. 2 and none of the COSMO-SAC ones. On tabulated profiles at 298 K the count is worth 0.014 in MSE over the low- $\gamma$   $n=60$  set, and that is the only bound available: the substituted arm produces 174 rows above  $\ln \gamma_2 = 10$  where the learned arm produces none, and no bound measured on a low- $\gamma$  set at one temperature carries to those rows. The convergence exposure on the substitution contrast is therefore unmeasured on these arms, and the two profiles it compares—a broad softmax against a tabulation with 57% exactly-zero bins—need not reach the fixed point at

the same rate. Where it can be measured it is large. The one trained COSMO-SAC head the tree retains—the retired 8-iteration run of §S5—scores 2.61 at 30 against 1.92 at 8 on the same labelled rows, and its substitution penalty over the 5571 rows whose solvent carries a reference profile is +4.32 at 8 against +6.43 at 30, a movement of half the penalty’s own size. That head’s penalty is fifteen times the one reported here, at the count that one is measured at, and its predictions move the other way under the substitution, so those figures size the lever on a head of this architecture and do not transfer to these arms. Gradients on the full NRTL kernel use the implicit-function theorem at the converged  $x_2^*$  (with  $\eta = \partial \ln \gamma_2 / \partial x_2$ ) rather than backpropagation through the iterations: exact at the fixed point, memory-constant in the iteration count, and well conditioned unless  $1 + x_2^* \eta \rightarrow 0$ . That branch is taken in training only, and only on the first of that arm’s two solves—the corrected re-solve is handed  $\tau$  directly, which turns it off—while the COSMO-SAC and single-parameter kernels return before it is reached. Those are back-propagated through the iterations at a memory cost linear in the count, and the pinned COSMO-SAC recipe’s implicit-differentiation setting has no effect on them. The segment counts follow a convergence audit of their own: a shorter  $n=8$  schedule fails a  $5.86 \ln x_2$  convergence test on strongly non-ideal pairs. The three-phase curriculum is (P1) auxiliary-property pretraining on crystal, Hansen and  $\ln \gamma_2^\infty$  targets with no SLE loss; (P2) full SLE training, with the adaptive correction unfrozen partway and an optional crystal→pair weight ramp; (P3) low-learning-rate stabilisation with monotonicity and van’t Hoff regularisers. Of the weighted loss, few terms are load-bearing for this paper: a bin-weighted Huber on  $\ln x_2$ ; auxiliary losses on the  $\sigma$ -profile (an earth-mover distance), on crystal  $T_m/\Delta H_{\text{fus}}$ , on  $\ln \gamma_2^\infty$  and on Hansen parameters; and structural regularisers for temperature monotonicity and the van’t Hoff slope. The remainder — channel orthogonality, contrastive, mixture-of-experts balance, priors — is auxiliary. The grounded arms begin with a warm-up on the  $\sigma$  pool of up to 40 epochs,

early-stopped on a held-out tenth, and the pinned recipe freezes the  $\sigma$  head for P2/P3, so that stream enters the warm-up and P1 only.

**The closure itself, and the remaining terms of the 2002 layer.** The COSMO-SAC layer of §2 maps a profile pair to  $\ln \gamma_2$  through a damped fixed point for the segment activity coefficient  $\Gamma_S$ ,

$$\Gamma_S(\sigma_m) = \left[ \sum_n p_{\text{norm}}(\sigma_n) \Gamma_S(\sigma_n) \exp(-\Delta W(\sigma_m, \sigma_n)/RT) \right]^{-1}, \quad (\text{S1})$$

with a precomputed exchange-energy matrix  $\Delta W$  (electrostatic misfit plus a hydrogen-bonding term), a restoring contribution assembled from those segment activities, and an optional Staverman–Guggenheim combinatorial term. The restoring contribution is  $\ln \gamma_2^{\text{res}} = (A_2/a_{\text{eff}}) \sum_m p_2^{\text{pure}}(\sigma_m) [\ln \Gamma_{S,\text{mix}}(\sigma_m) - \ln \Gamma_{S,2}^{\text{pure}}(\sigma_m)]$ , so the cavity area  $A_2$  sets the residual term’s prefactor as well as normalising the profile. The optional Staverman–Guggenheim combinatorial term needs a second size factor  $r_i = V_i/r_0$ , and where that volume comes from differs between the two places this closure is used, which matters for reading the oracle arms below. In the *decomposition*, where every input is a tabulated one,  $V_i$  is the reference COSMO cavity volume of the same calculation that produced  $A_i$ , so the two share a cavity basis. In the *trained model* the reference volume is never read: the volume channel is off in every arm this paper reports the paradox on (the volume channel is disabled, so no  $V_i$  reaches the closure and the combinatorial term is not evaluated at all), and in the one arm where it is on—the + Staverman–Guggenheim row of Table S4— $V_i$  is the auxiliary head’s predicted molar volume, detached. Small hydride solvents such as water carry explicit hydrogens so that the polar and hydrogen-bond channels activate. The pair representation is passed through an MLP and may be augmented with RDKit descriptors, Morgan fingerprints and an ionic-context feature; the crystal head additionally offers Walden-entropy and group-contribution  $T_m$  modes. The two NRTL control closures of §2 are the Non-Random Two-Liquid model,<sup>S4</sup> the most expres-

sive closure here, which fits interaction parameters per pair — predicting two interaction energies with a  $1/T$  law and a non-randomness factor, optionally regularised by a UNIFAC group-contribution prior<sup>S5</sup> — and its symmetric one-parameter collapse  $\ln \gamma_2(x_2, T) = \ln \gamma_2^\infty(T)(1 - x_2)^2$ , in which the identifiability analysis of §2.1 is cleanest. The crystal head’s bounded parametrisations are below.

**Crystal head.**  $T_m = 100 + 600 \text{ logistic}(\cdot) \in [100, 700]$  K,  $\Delta H_{\text{fus}} = S_H \text{ softplus}(\cdot)$ ; the group-contribution  $T_m$  prior is affine-calibrated on the train split before entering the head. Our *shared* message-passing encoder embeds solute and solvent, a bipartite cross-attention block lets them interact, and the resulting pair representation feeds every head. We implemented two further backbones, and both stay outside the numbers reported here. The physics-channelled one was run against this encoder and trailed it, an observation we recorded with no set, seed count or interval attached (§S1), so we claim no accuracy comparison between backbones. A fusion head predicts  $T_m$  and  $\Delta H_{\text{fus}}$ , giving the ideal term  $\Phi(T)$  of Eq. (1). The activity term  $\ln \gamma_2$  comes from one of three differentiable closures. Two are controls: the fitted NRTL (non-random two-liquid) excess-Gibbs-energy model, and its symmetric one-parameter collapse. The third, parameter-free, is the closure under study: a  $\sigma$ -profile head emitting a 51-bin screening-charge-density profile per molecule, and a COSMO-SAC layer mapping the profile pair to  $\ln \gamma_2$ . **That layer runs COSMO-SAC’s residual, segment-activity part alone.** The Staverman–Guggenheim combinatorial term (the contribution molecular size and shape make to the entropy of mixing, and the second factor of  $\gamma = \gamma^{\text{res}} \gamma^{\text{SG}}$  in Lin–Sandler) is off in every arm the substitution result is reported on. The  $\sigma$  head emits a profile shape and a cavity area while leaving the cavity volume unpredicted, so its own output falls short of evaluating the term. That is a model-form choice rather than an analyst convention: the kernel constants the layer uses are the 2002 values, regressed with the combinatorial term present, so the residual-only setting runs a fit-

ted kernel outside its parameterisation. Its size on a published evaluation is a mean shift of  $-0.74$  in  $\ln \gamma_2^\infty$  over 7889 records, from  $+0.03$  to  $-0.71$  in bias. The two settings are the *full* versus *residual-only* pair on which §2.5 and §3.5 turn. One arm of Fig. 2 restores the term by taking a molar volume from a separate auxiliary head and passing it detached, so the combinatorial term is active while standing beyond the reach of the solubility loss. Inside the layer, the segment activities are themselves a fixed point, truncated at a count that is a free hyperparameter and *differs* between training and scoring: the COSMO-SAC arms here train at 16 segment iterations and are scored at 30. We chose sixteen against a convergence regression test on hand-built Gaussian  $\sigma$ -profiles, where the 16-against-30 gap is  $6 \times 10^{-5}$ . Hand-built profiles are the whole of that test’s input, so agreement between the two counts on these arms stands unshown, and a checkpoint fitted at a different count is scored through an operator it was never fitted against.

On tabulated profiles that exposure is measurable without a checkpoint, and we measure it. Sweeping the count through the estimator itself on the broad IDAC set’s UD profiles, at the cell of §2.5, moves the set’s margin from  $+0.64$  at eight iterations to  $+0.57$  at sixteen,  $+0.52$  at thirty and  $+0.51$  at three hundred, and the glycol-ether stratum’s from  $+2.33$  to  $+2.13$ ,  $+2.05$  and  $+2.04$ . The sign never changes and the widest excursion over the whole range is  $0.29$ . Two things follow. The decomposition carries no exposure at all, because it is computed at three hundred iterations, which the sweep reproduces to the deposited value. And at the count the arms are *scored* at, thirty, the tabulated closure sits  $0.0024$  from converged in median  $|\Delta \ln \gamma_2|$  and  $0.039$  at its worst row, where eight sits at  $0.051$  and  $0.80$  – so the retired head’s movement with the count is a property of eight, not of thirty.

What stays unmeasured is the learned side. The  $\sigma$ -oracle arm’s closure input is the tabulation and is covered above; the learned arm’s profiles are a softmax against a table whose bins are 57% exactly zero, and no retained checkpoint carries them. The substitution contrast

on the solubility axis therefore keeps one unbounded component, and every margin this paper reports on that axis is conditional on the evaluation count it is read at. A second, independent truncation sits outside it: the SLE fixed point in  $x_2$ , run to a different count in training than in scoring. That second one is the one a deposit can re-solve, and the NRTL arm moves by  $+0.0025$  in seed-mean MAE at its training count and a further  $+0.017$  to convergence on the outer loop. That figure prices the outer loop rather than the segment one. Both truncations stay beyond re-resolution on the COSMO-SAC arms, their kernel being a  $\sigma$ -profile pair that the deposit and the retained checkpoints alike omit (§S1). The saturated mole fraction solves  $\ln x_2 = -\Phi(T) - \ln \gamma_2(x_2, T)$  as a damped fixed point in  $x_2$ , and a bounded, gated adaptive correction perturbs the physical parameters rather than the output, so the physical decomposition survives it.

## S1.1 The estimator rules in full: cell, convention, stratification, admissibility

**The estimator cell.** Eight equal-count bins of  $g(z^*)$ , the *unbiased* (Bessel-corrected) within-bin variance, and the row unit, applied identically to every set and every stratum whatever its size. The bin count is the choice with the most leverage (it takes the broad set’s aggregate margin from  $+0.01$  to  $+0.96$ ), so the cell never stands alone anywhere a finding rests on it: we run the same sweep for the strata the admissibility rule passes, and report it with them. Enumerating every equal-count contiguous partition, the bin-edge choice is immaterial on the broad IDAC set and material on the VT-2005-matched set, whose margin changes sign inside the envelope. We accordingly read that set’s margin as consistent with zero and carry the claim on the broad set in stratified form.

*The convention rule.* The margin depends on the combinatorial convention, so one rule governs both sets: every margin is read in the *residual-only* convention, pairing MSE and the bound computed under that same conven-



tion. Taking the residual-only convention on the smaller set and the full convention on the broad one would be the favourable choice on each set taken separately. One rule applied to both costs the broad set’s margin about half its value. The Jensen constant-offset bound is convention-specific: it separates under the full convention and inverts under the residual-only default. What we report therefore rests on the binning bound, which fits nothing and so leaves every held-out row to stand on its own instead of borrowing strength from a near-twin. Both bounds are valid on the one population quantity, by Lemma 1(d), and so is the smaller.

*The stratification rule.* The broad set is stratifiable where the VT-2005-matched set’s 60 pairs resist stratification. Its two axes are pure functions of molecular structure: an ordered nine-class cascade of substructure tests over the solvent, hydrogen-bonding role first and dominant functional group second, and the solute’s role as water or as an organic. Every class name is a substructure test rather than a chemical judgement: a *glycol ether* here is a solvent carrying a hydroxyl group and two or more oxygens, and an *aprotic* solvent one lacking an O–H or N–H hydrogen to donate. The classifier reads a SMILES string and stops there, leaving  $m$ ,  $g$ , the squared error, the source and the bootstrap unread, so every cell was fixed before its outcome, which is a property of the code rather than an assurance. A stratum is *boundable* at  $n \geq 40$ , where eight bins hold five rows. We report the smaller ones at that same cell, with the adaptive bin count  $\max(3, \lfloor n/10 \rfloor)$  in the rows  $n$  it is applied to carried beside them in the deposited table, nothing being dropped for being small.

*The admissibility rule.* A stratum is *admissible* only if it is (a) boundable at that fixed cell and (b) robust to leave-one-source-out (its margin keeps its sign when each source publication contributing to it is deleted in turn) in all four unit  $\times$  convention cells, since the map reports all four. That last clause is load-bearing and we keep it: drop it and the set taken whole passes too. A stratum drawing on one publication fails (b) by non-testability, there being no deletion to perform. A deletion leaving fewer

than 16 rows leaves the fixed cell undefined, and fails (b) as well. A third clause reduces the admissible *cells* to row sets, and it is stated here with the same precision rather than left to prose, because the one claim this paper carries from the map and the multiplicity null that tests that claim both turn on it. (c) Admissible cells holding identical rows count once. An admissible positive row set  $X$  is *demoted* to another,  $Y$ , when four things hold:  $Y$  is admissible and positive,  $Y$  holds more rows than  $X$ , more than half of  $X$ ’s rows are  $Y$ ’s, and the residual  $X \setminus Y$  falls short of admissible, so that  $X$ ’s margin stays inside  $Y$ . That residual decides it, not the overlap. A row set most of whose rows lie inside a larger one survives (c) whenever its own margin stands outside that larger one, so (c) tests margins and not containment, and overlap by itself demotes no cell (§S3.8). Clause (c) takes the six admissible cells of §3.5.1 to three distinct row sets. Positivity then removes one of the three and (c)’s residual test the second, leaving one. It is rebuilt on every draw of the multiplicity null, on the permuted labels, like the rest of the rule (§S3.6). Everything else is *reported in the map and not stated*, with its  $n$ , its sources, its margin, its interval and the reason. The rule governs the cells that favour us and those that count against us alike, and it tests whether a number is stable and not what it means. An admissible stratum whose margin sits at or below zero, and one that (c) demotes, both establish nothing.

## S1.2 Reading conventions: what $P$ is, what each interval resamples, and at what precision

A cluster-bootstrap frequency  $P$  is the frequency, over two-way (solute  $\times$  solvent) cluster resamples, with which the margin it is attached to stays positive. It is a resampling frequency rather than a  $p$ -value against any null. We quote such frequencies as bands (rounded to one decimal, with  $> 0.95$  and  $< 0.2$  closing the ends), because with 17 solvent and 41 solute clusters on the VT-2005-matched set they

carry at best one decimal of resolution. The deposited artifacts hold the two-decimal values. Our resampling intervals (the ranking and recalibration intervals of §3.2, the margins of §3.5 and the pKa intervals of §3.6) are *percentile* intervals: their endpoints are order statistics of a resampling distribution, uncorrected for bias and unstudentised, over the unit named beside each. §3.2 resamples the 107 solute clusters, 4000 draws, at the 95% level. §3.5 resamples solute and solvent clusters two-way, 3000 draws. In §3.6 the decomposition bounds resample molecules within each Hammett reaction series and the flip gap resamples the four series themselves, 4000 draws. Some  $\pm$  and bracketed spans below are of a different kind, two in particular: the kernel-transfer figures of §3.5.3 are a standard *error* over five fit seeds, and the leave-one-publication-out span of §3.5.1 is the range of that curve. Every claim below carries its set, its size and its scope in the sentence that makes it, and §S3.1 collects the same nineteen side by side as one table (Table S3).

**Architecture.** The headline COSMO-SAC model uses a message-passing encoder (a shared backbone with a 2-layer residual role-specific branch) of hidden width 64 over 3 message-passing layers, on 35-dimensional node and 8-dimensional edge features; a set2set readout (3 processing steps) pools each graph, solute and solvent interact through a 3-layer cross-attention block, and the 512-dimensional pair representation  $[g_s, g_v, g_s \odot g_v, |g_s - g_v|]$  (dropout 0.012) feeds the crystal head, the 51-bin  $\sigma$ -profile head, and the parameter-free COSMO-SAC closure. The full model has 5.40M trainable parameters. Whether that capacity is the binding constraint is not measured here: an in-sample linear probe does not bear on it (§S6.1, where isotropic noise features reach a *higher* train  $R^2$  than the trained encoder does), and the expectation that capacity is not the binding constraint is carried by the arm comparisons of §3.1 instead. The DirectGNN control shares this encoder, interaction, and readout verbatim, and adds a thermometer encoding of  $T$  in place of the temperature dependence the physics arm receives through  $\Phi(T)$  and the clo-

sure. Figure S1 draws the two arms and marks where each of the two error terms enters.

**What the oracle substitution replaces, and what it does not.** The deposited reference table carries 1432 rows, each a SMILES string, a cavity area in  $\text{\AA}^2$ , the 51 area-weighted profile bins in  $\text{\AA}^2$ , and a COSMO cavity volume in  $\text{\AA}^3$ . The loader canonicalises the SMILES with RDKit, giving 1424 distinct keys, and returns for each key exactly two objects: the 51-vector  $p^\star$  and the scalar  $A^\star$ , the tabulated cavity area. It does not return the cavity volume, and nothing in the model path reads it.

For each molecule the  $\sigma$ -head emits a shape  $\hat{p}^{\text{shape}}$  (a softmax over the 51 bins, so it sums to 1), a strictly positive area  $\hat{A}$  (a softplus scaled by 75 and offset by  $20 \text{\AA}^2$ ), and their product  $\hat{p} = \hat{A} \hat{p}^{\text{shape}}$ . The learned area is therefore the integral of the learned profile by construction, and the reference area is the integral of the reference profile as well: across the whole table  $|\sum_m p^\star(\sigma_m) - A^\star|/A^\star$  is at most  $7 \times 10^{-16}$ . The substituted object is thus one object and not two, an area-weighted profile, and the substitution is  $(\hat{p}, \hat{A}) \mapsto (p^\star, A^\star)$  on every row whose canonical SMILES is a key of the table, applied to solute and solvent *independently*, each on its own match. That single replacement is what carries the reference area into both places the closure reads an area: the residual prefactor  $A_2/a_{\text{eff}}$  and the normalisation  $p_2/A_2$ .

Nothing else is replaced. Not the size-factor volume—the combinatorial term is not evaluated at all in these arms, as above, so no volume of either provenance enters. Not the crystal head: it reads the solute-only pre-interaction embedding, which the substitution does not touch, so its  $T_m$  and  $\Delta H_{\text{fus}}$  are the same in both evaluations of a given checkpoint, and the separate crystal override of §2.4 is not applied in any oracle arm. Not the solver.

**Which channel of the role gap carries the identity violation.** The role-adapted readout feeds three numbers, and on the arms this paper reports one of the three carries the violation. The residual term is the bracket  $\ln \Gamma_{S,\text{mix}} - \ln \Gamma_{S,2}^{\text{pure}}$  scaled by  $A_2/a_{\text{eff}}$ , and when

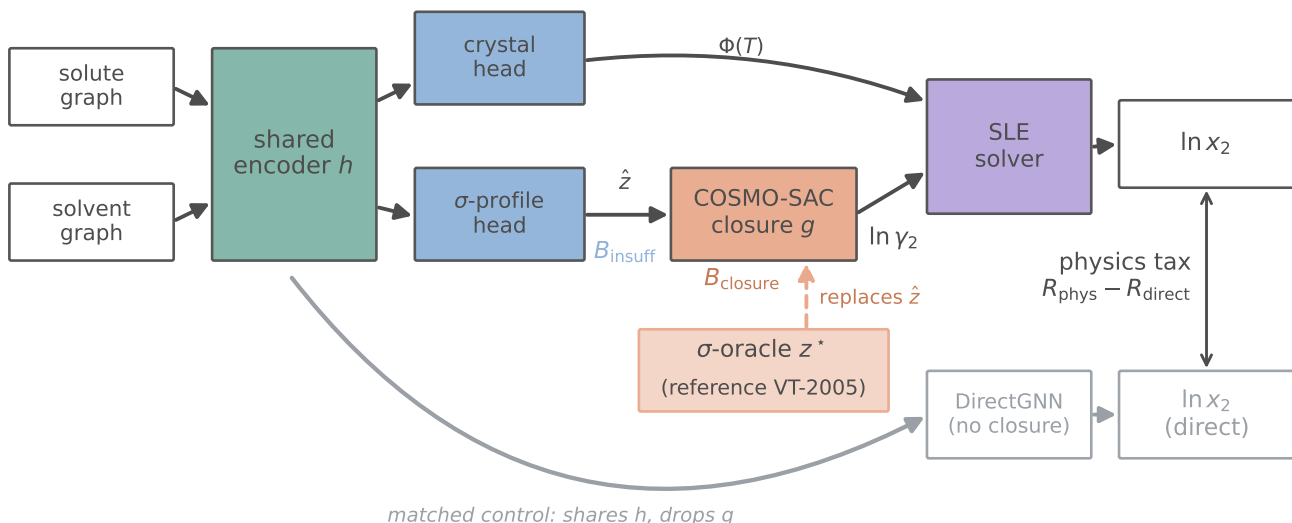


Figure S1: The model and the oracle substitution. Input insufficiency  $B_{\text{insuff}}$  enters at  $\hat{z}$  and closure misspecification  $B_{\text{closure}}$  where the fixed map  $g$  reads  $z^*$ , so  $B_{\text{closure}}$  is the error of map and profile calculation together; the  $\sigma$ -oracle (salmon) replaces  $\hat{z}$  with the reference VT-2005 profile  $z^*$ . DirectGNN (grey) is the capacity- and budget-matched control, separately tuned, and the double-headed arrow between the two  $\ln x_2$  outputs is the physics tax  $R_{\text{phys}} - R_{\text{direct}}$ .

the two roles agree on the shape that bracket is identically zero whatever the areas: giving them a 20% mismatch on the cavity area, or on the molar volume, moves  $\ln \gamma_2$  by exactly zero. The area stays live in general, multiplying the residual term through that prefactor, and on a self-pair it multiplies zero, so the profile disagreement is the whole of the identity violation there. On the one arm that restores the combinatorial term the same split also reaches  $\ln \gamma_2$  through the auxiliary head’s volume, and a mismatch on the volume alone gives a median  $|\ln \gamma_2|$  of 0.152 with the two profiles in perfect agreement. Tying the two roles is the change this calls for; where the size term is live it has to reach the auxiliary head as well, and either way it is a retraining.

The departure §3.3 reports between the learned profile and its reference,  $36 \pm 1\%$  and  $51 \pm 2\%$ , is a relative  $L_2$  distance between *shapes* with both sides normalised, so it prices the shape and leaves the area out. How far the deployed arms’ areas sit from the tabulated ones lies beyond the retained artifacts, which

keep none of their weights. On the one trained COSMO-SAC head the tree does keep, the learned cavity area is a median 1.01 times the tabulated one over the solutes it scores and 1.58 times it over the solvents; that head departs from the substitution penalty of §3.1 in both size and direction, so those two ratios price the quantity and not the arms.

One coupling is worth stating rather than glossing, because “only the profile changes” is not quite exact. The bounded correction reads a compact summary of the solver’s inputs which includes the closure’s own  $\tau$ , so a substituted profile perturbs the correction, and through it the corrected crystal parameters the solver receives. The perturbation is small, and at five seeds it is an order of magnitude smaller than it was at three. Over the five seeds, on the 8066 of the 8103 test rows where the substitution applied, it moves  $T_m$  by at most 0.007 K against a median of 450 K,  $\Delta H_{\text{fus}}$  by at most  $1.3 \text{ J mol}^{-1}$  against a median of  $3.5 \times 10^4$ , and  $\Phi$ —the quantity that actually enters the SLE—by at most  $3.2 \times 10^{-5}$  against a median mag-

nitude of 4.4. Each maximum is the largest of the five per-seed maxima—for  $\Phi$  those are 3.2, 1.2, 1.2, 1.1 and 1.1 in units of  $10^{-5}$ , the worst at seed 42—and each median is over those rows with the five seeds pooled; the per-seed medians of  $|\Phi|$  are 4.37, 4.53, 4.36, 4.24 and 4.53, so a single-seed reading of this quantity is not the five-seed one, and §2.4 prints the five-seed one. On the rows where the substitution did not apply the same quantities still move, by  $1.2 \times 10^{-4}$  K,  $0.018 \text{ J mol}^{-1}$  and  $4 \times 10^{-6}$ , which is the residue of forwarding one checkpoint twice on a GPU and sets the floor these figures should be read against. So  $\Phi$  is not bit-identical between the two evaluations, and the fourth decimal of a per-row  $\Phi$  comparison between them is not meaningful; nothing at the scale of the +0.41 turns on it. Restricted to the 5608 labelled test rows that every solubility figure in the article is scored on, the worst-seed maximum is the same  $3.2 \times 10^{-5}$ : that is the figure §3.2, Table S3 and §S6.1 print, and it equals the figure over the 8066 above because at these seeds the worst row of the larger set lies inside the smaller one, not because a different quantity is being measured. The separately trained arms differ over the same 5608 rows by 4.4 to 5.9 in  $\Phi$ , four orders of magnitude more.

All three oracle applications—evaluation-only, the training-time injection, and the channel swap—use the same table and the same both-sides rule; the two training-time arms differ from the evaluation-only one in *when* the replacement is made and in nothing about *what* is replaced.

**What the match rule reaches, and which side carries the effect.** On the 5608 labelled test rows the table matches the solvent in 5571 (99.3%; 64 of the 70 distinct solvents) and the solute in 297 (5.3%; 7 of the 147 distinct solutes). Every solute match is also a solvent match, so the rows split into 37 with neither molecule replaced, 5274 with the solvent only, and 297 with both; the substitution flag carried by the deposited per-row files is exactly the union, 5571 rows. That asymmetry is a property of the reference database, which covers common solvents and few drug-like solids, and

it is worth stating because it decides which side the reported gap is a statement about. Comparing the  $\sigma$ -oracle arm against the grounded arm it is evaluated from, row class by row class and averaged over the three seeds, the MAE difference is  $-0.000 \pm 0.000$  on the 37 neither-matched rows—the two are the same checkpoint forwarded twice, and agree to  $5 \times 10^{-6}$  there— $+0.431 \pm 0.036$  on the 5274 solvent-only rows, and  $+0.006 \pm 0.025$  on the 297 both-matched ones. The +0.41 of §3.1 is therefore carried by the solvent profile. On the rows where the reference solute area  $A_2$  is the one entering the prefactor, the two arms are not separated; with 297 rows over 7 solutes that is a failure to resolve a difference and not a demonstration that the substitution is harmless there, and we draw nothing from it beyond the scope it sets on the headline number.

**Encoder backbones.** The shared encoder has three interchangeable backbones: the headline message-passing network (MPNN); a hybrid local-message/global-attention graph transformer (GPS<sup>S6</sup>); and a physics-channelled variant, Thermodynamically-Informed Message Passing (TIMP), which splits messages into dispersive, polar, and hydrogen-bonding channels supervised toward the Hansen parameters to prevent channel collapse. TIMP does not outperform the plain MPNN across our runs. That is an observation and we do not explain it: the linear probe of §S6.1 has no untrained-encoder arm to read against and supports no explanation.

**Optimisation, and the two arms’ settings.** The physics arm uses Adam (weight decay  $5 \times 10^{-4}$ , gradient clipping at 2.0, batch 64) under the three-phase curriculum of §2: 30 epochs of auxiliary pretraining (P1), 70 of full SLE training (P2), and 10 of low-rate stabilisation (P3), at learning rates  $2.6 \times 10^{-4}$ ,  $8.5 \times 10^{-5}$ , and  $8.5 \times 10^{-6}$  respectively. The DirectGNN control was tuned separately and its settings differ: dropout 0.174 against 0.012, P2 learning rate  $8.6 \times 10^{-4}$  against  $8.5 \times 10^{-5}$ , weight decay  $4.2 \times 10^{-5}$  against  $5 \times 10^{-4}$ , gradient clipping 2.54 against 2.0, over 110 P2

epochs equal to the physics arm’s total budget at the same width (64) and depth (3). Width, depth, batch size and epoch budget are therefore matched and the optimisation is not, so  $\Delta_\infty$  and the physics-versus-control  $\Delta\text{MAE}$  are differences between two separately tuned pipelines and are not attributable to the thermodynamic bottleneck alone (§2.4). A hyperparameter-matched control was not run.

**Loss.** The objective is a weighted sum. Its registry holds 32 terms, of which 17 are hardwired to 0 and two more (bridge, solubility variance) default to 0 through the configuration—disabled experimental scaffolding, released with the code for completeness. Table S2 lists the 13 carrying a non-zero default, together with the solubility-variance term and the  $\sigma$ -profile earth-mover term, which is applied outside the registry when the grounding stream is on; only the top two groups are load-bearing for the claims, and the *Loss-simplification ablation* paragraph below tests which.

**Loss-simplification ablation.** For a diagnostic result the elaborate objective is a potential confound: the paradox could be an artifact of loss tuning. The test registered in advance of the run trains the end-to-end grounded arm with only the task ( $\ln x_2$  Huber), the crystal grounding ( $T_m, \Delta H_{\text{fus}}$ ), and the  $\sigma$ -profile warmup, *dropping* the activity-grounding auxiliaries ( $\ln \gamma_2^\infty$ ,  $\ln \gamma_2$ , Hansen) and every regulariser, and then reads the sign of the paradox (oracle vs. learned) and the surrogate drift. The registered prediction was that both persist, the mechanism being end-to-end training through a misspecified closure rather than the auxiliary terms, and both do at the level the ablation measures. The ablation re-measures the solubility *paradox* (reference minus learned  $\ln x_2$  MAE), the observable effect, not the profile drift directly: on the full test set it is +1.13 (95% CI [0.88, 1.38]) under the minimal objective, *larger* than the +0.30 ([0.20, 0.39]) under the full one. This is the *overall*, solvent-dominated paradox, and it strengthens once the activity-grounding auxiliaries—the only pressure toward physics—are removed; on the both-

reference subset ( $n=297$  rows over the  $\approx 7$  VT-covered solutes) the sign reverses (−0.23 minimal vs +0.22 full), so the ablation is evidence that the overall observable effect is loss-robust, not subset-level confirmation of the mechanism. It is *consistent with* the drift being intrinsic to end-to-end training through a misspecified closure rather than a regularisation artifact. It is an implication of the amplified paradox, not a direct re-measurement of  $\|\hat{\sigma} - \sigma\|$  under the minimal run (the minimal-loss checkpoint was not retained, and the larger reference-arm error is consistent with more profile drift but does not isolate it from closure-path recalibration). The minimal model trains normally (validation  $\ln x_2$  MAE 1.87, within 0.1 of the full-loss base), ruling out a degenerate-training reading (seed 42).

## S2 Proofs

*Proof of Lemma 1 (closure-insufficiency decomposition).* (a) *The identity, and the cross-term condition.*  $g(z^*)$  is  $\sigma(z^*)$ -measurable by construction, since  $z^*$  is the closure’s whole argument. Split the residual as  $m - g(z^*) = [m - \mathbb{E}(m \mid z^*)] + [\mathbb{E}(m \mid z^*) - g(z^*)]$ , square and take expectations; the squared terms are  $B_{\text{insuff}}$  and  $B_{\text{closure}}$ , and the cross term vanishes by the tower rule,  $\mathbb{E}[(m - \mathbb{E}(m \mid z^*))(\mathbb{E}(m \mid z^*) - g(z^*))] = \mathbb{E}[(\mathbb{E}(m \mid z^*) - g(z^*))\mathbb{E}(m - \mathbb{E}(m \mid z^*) \mid z^*)] = 0$ , because the second factor is  $\sigma(z^*)$ -measurable. **That measurability is the condition, and it fails if any argument of  $g$  is left out of the conditioning variable:** with  $g = g(z^*, T)$  and  $T$  excluded, the cross term need not vanish and the binning in (c) is no longer a coarsening of the conditioning  $\sigma$ -algebra, so neither the identity nor the upper bound survives. Where  $g$  reads a temperature—the broad IDAC set of §3.5.1,  $\dim = 103$ —we accordingly take  $z^* = (\text{profile pair}, T)$ ; on the VT-2005-matched set every row sits at 298 K and the two readings coincide. The same rule fixes  $z^*$  in the second domain: the Hammett closure of §3.6 reads a per-series constant pair as well as the substituent constant, so there  $z^* = (s, \sigma_{\text{H}}^*)$  and the conditioning is on both. (b) *Jensen lower*

Table S2: The 13 registry terms carrying a non-zero default weight, with the solubility-variance term (zero through the configuration) and the  $\sigma$ -profile earth-mover term, which is not a registry entry.<sup>a</sup> A further 17 registered terms are hard-wired to 0, and one more (bridge) is zero through the configuration.

| role                   | term                                          | default wt. |
|------------------------|-----------------------------------------------|-------------|
| task                   | solubility (bin-weighted Huber on $\ln x_2$ ) | 1.0         |
| crystal grounding      | melting point $T_m$                           | 0.3         |
|                        | fusion enthalpy $\Delta H_{\text{fus}}$       | 0.3         |
| $\sigma$ /act. ground. | infinite-dilution $\ln \gamma_2^\infty$       | 0.5         |
|                        | finite-composition $\ln \gamma_2$             | 0.5         |
|                        | Hansen parameters                             | 0.2         |
|                        | $\sigma$ -profile EMD (grounding stream on)   | †           |
| regulariser            | solubility monotonicity in $T$                | 0.1         |
|                        | physics preference                            | 0.1         |
|                        | adaptive-correction residual                  | 0.01        |
|                        | NRTL $\tau$                                   | 0.01        |
|                        | solubility variance                           | cfg         |
| control coupling       | DirectGNN NLL                                 | 0.2         |
|                        | DirectGNN regulariser                         | 0.05        |
| architecture           | mixture-of-experts balance                    | 0.02        |

<sup>a</sup> † marks the  $\sigma$ -profile earth-mover term, active only when the grounding stream is on. Per-phase weight overrides are pinned in each run’s resolved configuration.

*bound.* With  $\Delta(z^*) = \mathbb{E}(m \mid z^*) - g(z^*)$ ,  $B_{\text{closure}} = \mathbb{E}[\Delta^2] \geq (\mathbb{E}[\Delta])^2 = (\mathbb{E}[m] - \mathbb{E}[g])^2$ , using  $\mathbb{E}[\mathbb{E}(m \mid z^*)] = \mathbb{E}[m]$ . It needs *no* smoothness assumption; its one requirement is that  $m$  and  $g$  share the  $\ln \gamma_2^\infty$  reference state, which the closure validation of §2.3 establishes. (c) *The binning bound, a law-of-total-variance upper bound.* For any binning  $B = \text{bin}(g(z^*))$  one has  $\sigma(B) \subseteq \sigma(z^*)$ , and conditioning on a coarser  $\sigma$ -algebra cannot reduce expected conditional variance:  $\mathbb{E}[\text{Var}(m \mid B)] \geq \mathbb{E}[\text{Var}(m \mid z^*)] = B_{\text{insuff}}$ . (d) *The estimand is convention-free; the bound is not.*  $B_{\text{insuff}} = \mathbb{E}[\text{Var}(m \mid z^*)]$  is a functional of the law of  $(m, z^*)$  alone and does not involve  $g$ , so it is identical across closure conventions and any  $\text{MSE}(g_1) - \text{MSE}(g_2)$  is entirely  $B_{\text{closure}}(g_1) - B_{\text{closure}}(g_2)$ . The bound of (c) is not convention-independent, and no contradiction arises: two conventions coarsen the same  $\sigma(z^*)$  along two different statistics, giving two valid upper bounds on one quantity, of which the minimum is again valid but is biased low in finite samples; it is reported, and no margin below is read from it.  $\square$

**Lemma S1** (Structural non-identifiability of the SLE split). *Consider the infinite-dilution*

*SLE observable  $\mu(T) = -\Phi(T) - \ln \gamma_2^\infty(T)$  with  $\Delta C_p = 0$ , so that  $\Phi(T) = (\Delta H_{\text{fus}}/R)(1/T - 1/T_m)$  is affine in  $u = 1/T$ , and a symmetric two-suffix activity class ( $\ln \gamma_2 = \ln \gamma_2^\infty (1 - x_2)^2$ ) in which  $\ln \gamma_2^\infty(T)$  is affine in  $u$ . Then, for any design with at least two distinct temperatures,  $\mu$  depends on the parameters only through the two-dimensional statistic (its slope and intercept in  $u$ ), and the activity intercept and slope already span that plane; hence the crystal parameters  $(\Delta H_{\text{fus}}, T_m)$  are unidentified, the profiled Fisher information for  $\Delta H_{\text{fus}}$  is exactly zero, and the indeterminacy set is two-dimensional. The deficiency is closed only by a rank-completing external constraint (a direct crystal label, or a fixed activity slope). At finite dilution the composition factor  $(1 - x_2)^2$  breaks the exact affinity, leaving a non-zero but near-degenerate information of order  $x_2^2$ .*

*Proof of Lemma S1 (non-identifiability).* Write  $u = 1/T$  and  $h = \Delta H_{\text{fus}}/R$ ,  $u_m = 1/T_m$ . At infinite dilution and with  $\Delta C_p = 0$  the observable is

$$\begin{aligned}
 \mu(u) &= -\Phi(u) - \ln \gamma_2^\infty(u) \\
 &= -h(u - u_m) - (c + du) \\
 &= -(h + d)u + (hu_m - c),
 \end{aligned}$$

using  $\ln \gamma_2^\infty(u) = c + du$  (affine in  $u$  by hypothesis). Thus  $\mu$  depends on the parameters  $\theta = (\Delta H_{\text{fus}}, T_m, c, d)$  only through the pair  $(S, K) = (-(h + d), hu_m - c) \in \mathbb{R}^2$ . The activity block spans this plane,  $\partial(S, K)/\partial c = (0, -1)$  and  $\partial(S, K)/\partial d = (-1, 0)$ ; hence for *any* crystal value  $(h, u_m)$  and any target  $(S, K)$  there is an activity pair  $d = -S - h$ ,  $c = hu_m - K$  reproducing it. The likelihood level set therefore contains the full two-dimensional family  $\{(h, u_m)\}$ , so  $(\Delta H_{\text{fus}}, T_m)$  are unidentified. Equivalently, the crystal score  $\partial(S, K)/\partial h = (-1, u_m)$  lies in the span of the activity columns, so profiling out  $(c, d)$  annihilates it and  $\mathcal{I}_{\text{eff}}(\Delta H_{\text{fus}}) = 0$ . At finite dilution  $\ln \gamma_2(u) = \ln \gamma_2^\infty(u) (1 - x_2)^2$  adds  $\ln \gamma_2^\infty(u) (2x_2 - x_2^2) = O(x_2)$ , which is not affine in  $u$  because  $x_2 = x_2(u)$ ; the crystal score then has a component outside the activity span of order  $x_2$ , so  $\mathcal{I}_{\text{eff}}(\Delta H_{\text{fus}}) = O(x_2^2) > 0$ .  $\square$

**Lemma S2** (Class-dependent efficient information). *Let the activity nuisance lie in a class  $\mathcal{H}$  with tangent projection  $\Pi_{\mathcal{H}}$ , and let  $v = \partial\mu/\partial\Delta H_{\text{fus}}$  be the crystal score. The efficient information for  $\Delta H_{\text{fus}}$  is  $\mathcal{I}_{\text{eff}}(\Delta H_{\text{fus}}) = \sigma^{-2} \|v - \Pi_{\mathcal{H}}v\|^2$ . If  $\mathcal{H}$  contains the crystal score's span  $\{1, 1/T\}$  (a free intercept and slope) then  $\Pi_{\mathcal{H}}v = v$  and  $\mathcal{I}_{\text{eff}} = 0$ , recovering Lemma S1; if  $\mathcal{H}$  is restricted (the slope is known or constrained) then  $\mathcal{I}_{\text{eff}} > 0$  and the Cramér–Rao lower bound (CRLB) on  $\Delta H_{\text{fus}}$  is finite. (The melting point  $T_m$  is likewise an unknown crystal parameter, but is profiled jointly: its score  $\partial\mu/\partial T_m$  lies in  $\text{span}\{1\} \subset \text{span}\{1, 1/T\}$ , already inside the nuisance tangent whenever the slope is free, so the dichotomy is unaffected by treating  $T_m$  as free.)*

*Proof of Lemma S2 (class-dependent information).* This is the semiparametric efficient-score construction<sup>S7</sup> (here finite-dimensional and Gaussian, so equivalently the partitioned-information / Frisch–Waugh–Lovell identity). With Gaussian noise of variance  $\sigma^2$ , parameter-of-interest score  $v = \partial\mu/\partial\Delta H_{\text{fus}}$ , and nuisance tangent space  $\mathcal{T}_{\mathcal{H}}$  (the  $L^2$ -closure of the span of nuisance scores), the efficient score is the residual  $\tilde{v} = v - \Pi_{\mathcal{H}}v$  and the efficient information is  $\mathcal{I}_{\text{eff}} = \sigma^{-2} \|\tilde{v}\|^2$ . Here  $v = \partial\mu/\partial\Delta H_{\text{fus}} = -(u - u_m)/R \in \text{span}\{1, u\}$ . If

$\mathcal{T}_{\mathcal{H}} \supseteq \text{span}\{1, u\}$  then  $\Pi_{\mathcal{H}}v = v$  and  $\mathcal{I}_{\text{eff}} = 0$ , recovering Lemma S1; if  $\mathcal{T}_{\mathcal{H}}$  is a proper subspace not containing  $v$  (e.g. a fixed slope,  $\mathcal{T}_{\mathcal{H}} = \text{span}\{1\}$ ) then  $\Pi_{\mathcal{H}}v \neq v$  and  $\mathcal{I}_{\text{eff}} > 0$ , with a finite Cramér–Rao bound.  $\square$

Lemma 1, the identity this instrument rests on, is stated in the article at §2.1.1 and proved at the head of this section: the cross term vanishes because  $g(z^*)$  is  $\sigma(z^*)$ -measurable when  $z^*$  is the closure's whole argument, the Jensen bound needs no smoothness, the binning bound needs only that a binning of  $g(z^*)$  coarsens  $\sigma(z^*)$ , and  $B_{\text{insuff}}$  is a functional of the law of  $(m, z^*)$  alone and therefore the same quantity under both combinatorial conventions—which is why Table S8 carries two valid upper bounds per set. Two consequences of that one-sidedness reconcile the bounding apparatus with the injectivity of the design. First, because every reported  $B_{\text{insuff}}$  is an *upper* bound and every  $B_{\text{closure}}$  the matching *lower* one, the bounds are conservative in one direction only: were the superpopulation variance negligible, the finding would be the stronger  $B_{\text{closure}} = \text{MSE}$ . Nothing reported can therefore be an artifact of having over-estimated  $B_{\text{insuff}}$ , and no entry in any table is the share of the error the inputs contribute. Second, the resampling frequencies are statements about the *instrument* and not about the superpopulation: the clustered-bootstrap frequency  $P_{\text{boot}}$  is the fraction of cluster resamples in which the *measured bound* still falls below half the total error,  $\text{MSE}/2$  being the threshold that orders the two terms. It measures whether this design resolves the ordering, not the probability that an inequality between two population quantities holds, so a low  $P_{\text{boot}}$  is loss of resolution and never evidence for the inputs.

**Theorem S1** (Grounding-gap sandwich). *Assume all risks are finite ( $m, g(z^*) \in L^2$ ;  $g$  and each  $h \in \mathcal{E}$  measurable), (A1) the true encoder is representable,  $\phi^* \in \mathcal{E}$  with  $z^* = \phi^*(x)$ , and (A2) the intermediate is a lossless bottleneck,  $\mathbb{E}[m | x] = \mathbb{E}[m | z^*]$ . Then*

$$0 \leq \mathcal{G} \leq B_{\text{closure}}, \quad (\text{S2})$$

$$\mathcal{G} = B_{\text{closure}} - (R_{\text{e2e}} - R_{\text{free}}),$$

with  $\mathcal{G} = B_{\text{closure}}$  iff  $R_{\text{e2e}} = R_{\text{free}}$  (the composed class realizes the Bayes predictor by distorting the intermediate). In particular a well-specified closure ( $B_{\text{closure}} = 0$ ) gives  $\mathcal{G} = 0$ ; hence under A1–A2,  $\mathcal{G} > 0$  certifies  $B_{\text{closure}} > 0$ .

*Proof of Theorem S1 (grounding-gap sandwich).* Throughout assume  $m, g(z^*) \in L^2$  and  $g$ , every  $h \in \mathcal{E}$  measurable. (a)  $\mathcal{G} \geq 0$ . By A1,  $\phi^* \in \mathcal{E}$  is a feasible encoder with  $g(\phi^*(x)) = g(z^*)$ , so  $R_{\text{e2e}} = \inf_h \mathbb{E}[(m - g(h(x)))^2] \leq \mathbb{E}[(m - g(z^*))^2] = R_{\text{orc}}$ ; the infimum need not be attained. Hence  $\mathcal{G} = R_{\text{orc}} - R_{\text{e2e}} \geq 0$ . (b)  $R_{\text{e2e}} \geq R_{\text{free}}$ . Every  $g(h(x))$  is  $\sigma(x)$ -measurable, and  $R_{\text{free}} = \mathbb{E}[(m - \mathbb{E}(m | x))^2]$  is the infimum of  $\mathbb{E}[(m - p)^2]$  over all  $\sigma(x)$ -measurable  $p$ ; an infimum over the subclass  $\{g(h(\cdot)) : h \in \mathcal{E}\}$  is no smaller. (c)  $R_{\text{free}} = B_{\text{insuff}}$  under A2. From  $\text{Var}(m) = \mathbb{E}[\text{Var}(m | x)] + \text{Var}(\mathbb{E}(m | x)) = \mathbb{E}[\text{Var}(m | z^*)] + \text{Var}(\mathbb{E}(m | z^*))$  and A2 ( $\mathbb{E}(m | x) = \mathbb{E}(m | z^*)$ , so the two  $\text{Var}(\mathbb{E}(\cdot))$  terms agree), the two expected conditional variances agree:  $R_{\text{free}} = \mathbb{E}[\text{Var}(m | x)] = B_{\text{insuff}}$ . (d) *Sandwich and identity.* By Lemma 1,  $R_{\text{orc}} = B_{\text{insuff}} + B_{\text{closure}}$ . With (c),  $\mathcal{G} = R_{\text{orc}} - R_{\text{e2e}} = B_{\text{closure}} - (R_{\text{e2e}} - R_{\text{free}})$ , and (b) gives  $R_{\text{e2e}} - R_{\text{free}} \geq 0$ , so  $0 \leq \mathcal{G} \leq B_{\text{closure}}$ ; equality  $\mathcal{G} = B_{\text{closure}}$  holds iff  $R_{\text{e2e}} = R_{\text{free}}$ . Finally  $B_{\text{closure}} = 0$  forces  $0 \leq \mathcal{G} \leq 0$ , i.e.  $\mathcal{G} = 0$ ; contrapositively, under A1–A2,  $\mathcal{G} > 0 \Rightarrow B_{\text{closure}} > 0$ . (e) *A2 is necessary for the certificate.* If A2 fails, take  $x \in \{a, b\}$  equiprobable,  $z^* \equiv 0$  (so A1 holds),  $m(a) = +1, m(b) = -1$ , and  $g$  with  $g(0) = 0 = \mathbb{E}(m | z^*)$  so  $B_{\text{closure}} = 0$ ; an encoder  $h^\dagger(a) = 1, h^\dagger(b) = -1$  with  $g(\pm 1) = \pm 1$  gives  $R_{\text{e2e}} = 0$  while  $R_{\text{orc}} = 1$ , hence  $\mathcal{G} = 1 > 0 = B_{\text{closure}}$ . Thus  $\mathcal{G} > 0$  alone does not certify  $B_{\text{closure}} > 0$ ; the decomposition (Lemma 1) does.  $\square$

Remark 1 is stated in the article at §2.1.2, in the asymptote-plus-variance form  $\mathcal{T}(n) \approx \Delta_\infty + [V_{\text{phys}}(n) - V_{\text{direct}}(n)]$  with  $\Delta_\infty := R_{\text{e2e}} - R_{\text{direct}, \infty}$ , together with the scope the rule is read under. It is proved here, with the limiting refinement (iii) that the statement there defers to this section.

*Proof of Remark 1 (sign of the physics penalty).* Write  $D(n) := V_{\text{direct}}(n) - V_{\text{phys}}(n)$

and  $\Delta_\infty := R_{\text{e2e}} - R_{\text{direct}, \infty}$ , so that  $\mathcal{T}(n) = \Delta_\infty - D(n)$ . (i)  $\mathcal{T}(n) < 0$  exactly when  $D(n) > \Delta_\infty$ , immediately. (ii) If  $\Delta_\infty > 0$  and  $D(n) \rightarrow 0$ , choose  $n^*$  with  $|D(n)| < \Delta_\infty$  for all  $n > n^*$  (possible since  $D(n) \rightarrow 0$ ); then  $\mathcal{T}(n) = \Delta_\infty - D(n) > 0$ . Monotonicity of  $D$  is *not* required for existence of  $n^*$ ; it is needed only for the stronger claim “ $\mathcal{T}(n) > 0$  at every  $n$ ,” which we treat as an empirical observation. (iii) *A limiting refinement.* Since  $V_{\text{phys}}(n) \geq 0$ ,  $D(n) \leq V_{\text{direct}}(n)$ : the variance a prior can save is bounded by the control’s own estimation variance. As the black box approaches the aleatoric floor  $V_{\text{direct}}(n) \rightarrow 0$ , so  $\limsup_n D(n) \leq 0$ ; with both estimators consistent, so  $V_{\text{phys}}, V_{\text{direct}} \rightarrow 0$ , one has  $D(n) \rightarrow 0$  and  $\mathcal{T}(n) \rightarrow \Delta_\infty$ . That limit is  $\geq 0$  when the control is asymptotically Bayes, so  $R_{\text{direct}, \infty} = R_{\text{free}}$  (the risk of an unconstrained predictor), and not merely from  $V_{\text{direct}} \rightarrow 0$ ; under that condition and A2,  $\Delta_\infty = R_{\text{e2e}} - R_{\text{free}} = B_{\text{closure}} - \mathcal{G}$  by (S2).  $\square$

The variance term  $D(n)$  of Remark 1 is what carries the semiparametric phenomenon §2.1.2 appeals to, in which an estimated propensity outperforms the known-true one:<sup>S8–S10</sup> a fitted nuisance can be the more *efficient* estimator even when its bias is no smaller, so a true-valued input need not give the lower total risk at finite  $n$ .

## S3 The claims ledger, the scored row set, the decomposition and its robustness, and the map

### S3.1 The claims ledger

Table S3 collects the article’s nineteen load-bearing claims in one place: what is claimed, the data set it was measured on and that set’s size, and the value with its uncertainty and its scope. It is an index and not the place those scopes live. Every qualification printed in a value cell below—an  $n$ , a per-seed triple, “one seed”, “one checkpoint”, an interval that spans zero, a named exception, the one-sidedness of a



margin—is also carried by the article sentence that makes the claim; nothing in the article depends on this table being open beside it. What the table adds is that the nineteen can be read against one another. Two of the three columns do that at a glance, and the second is where a shared data set shows: the same 5608 labelled test rows carry the first of the six blocks and the last. The confounds and the estimator conventions do not read at a glance, and the table is not offered as showing them that way—each is a sentence, and the cells that carry them run to a few hundred characters, the two longest to five and seven hundred, so the third column is read a row at a time and not scanned.

### S3.2 The scored row set, why it is not arm-dependent, and the per-arm table

Every accuracy number of §3.1 is on 5608 rows, and the scaffold partition gives 8103 test rows. The 2495-row difference decomposes with no remainder, identically at every seed of both families: 0 rows are lost to duplicate (solute, solvent,  $T$ ) keys—there are none—and 2495 carry no solubility flag, a melting point and no solubility measurement, with the  $\ln x_2$  field a placeholder zero. All 2495 carry a melting point; none is a self-solvation row; they are unscorable for solubility in every arm, the control included.

The clause that could have made this arm-dependent removes nothing. The code that builds the set intersects, over arms, the rows that are both labelled and finite in that arm’s  $\ln x_2$  prediction, and the second half of that conjunction is outcome-dependent by construction. Audited over all thirty-seven per-row prediction files in the deposit—twenty-two of the published family, being the five intersecting arms and the  $\sigma$ -oracle at each of its three seeds plus the two co-adaptation controls and the two output-residual arms at seed 42, and fifteen of the leak-free re-run, being its three arms at each of five—the count of non-finite  $\ln x_2$  predictions is 0 in every file, over every row that file carries and not merely the labelled ones. All thirty-seven are full length, covering all 8103 test rows

and each carrying exactly 5608 labelled ones. (Twenty-one of the published family’s twenty-two were full length as first deposited; the seed-44  $\sigma$ -oracle file was repaired, and the repair is audited below.) The five intersecting arms’ labelled row sets are moreover identical as sets, not merely equal in size. Dropping the finiteness clause outright therefore returns the same 5608 rows, so no arm’s failures are removed from any arm’s score and no number in this paper moves. The set is fixed by which rows carry a label.

Non-finiteness in  $\ln x_2$  is not the same event as the saturation that Table S20’s clip note reports. That note counts rows whose predicted  $x_2$  approaches 1, where the *unit conversion* to  $\log_{10} S$  diverges while  $\ln x_2$  itself stays finite; those rows are inside the 5608 and are scored, under the clip that note discloses.

**What the scored set holds that the ideal term is not written for.** 796 of the 5608 rows—14.2%, carried by 26 of the 147 test solutes—have a solute of more than one component: chloride, sodium, lithium, tosylate, malate and phosphate salts, two hexafluorophosphate ionic liquids and four hydrates. For none of those is the solid in equilibrium with the saturated solution the one-component crystal that  $\Phi = (\Delta H_{\text{fus}}/R)(1/T - 1/T_m)$  is written for, and not one of the 26 carries a measured melting point—against 31 of the 121 single-component solutes—so on each of those rows the crystal head supplies a  $T_m$  that no label anchors. Whether the arms pay for that there is unresolved, and they do not agree on the sign. Three are worse on the stratum than off it in the seed mean—by  $+0.27 \ln x_2$  for DirectGNN,  $+0.18$  for the grounded learned- $\sigma$  arm and  $+0.04$  for NRTL—while the ungrounded arm is *better* on it, by  $-0.16$ . The effective sample size is 26 solutes against 121, only DirectGNN holds its sign at all three seeds, and all twelve solute-clustered 95% intervals, the ungrounded arm’s included, cross zero. What is resolved is the arithmetic of removing it, which is why it stays: dropping the stratum would move the DirectGNN mean by  $-0.038$ , the grounded learned- $\sigma$  mean by  $-0.026$  and

Table S3: One row per load-bearing claim: what is claimed, the data set and its size, and the value measured with its uncertainty and its scope. The grounded, ungrounded and  $\sigma$ -oracle arms come from the five-seed leak-free re-run. Notes *a–h*, at the foot of the last part, qualify a row or a column.

| Claim                                                                                  | Set ( <i>n</i> ) <sup>a</sup>                                                        | Value, with its uncertainty and scope <sup>b</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>The two senses of grounding</i> (§3.1)                                              |                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Reference profiles hurt solubility at evaluation (MAE)                                 | labelled test rows (5608 of 8103), 5 seeds                                           | 1.93 $\rightarrow$ 2.34; the per-seed values do not overlap; the gap carries the test-time input swap with it                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Training-time $\sigma$ supervision has <i>no established sign</i> (MAE)                | labelled test rows, 5 seeds                                                          | 2.04 $\rightarrow$ 1.93 in the seed mean, but the per-seed values overlap and one of the five gains reverses: 0.149/0.122/0.175/ $-$ 0.017/0.122. They straddle the co-adapted arm’s +0.09, so the two operations are not ordered by magnitude either. This is the contrast the pre-committed leak-free re-run was run to settle (§2.2), and it settles it as unresolved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| The evaluation substitution degrades the <i>choice</i> of solvent, not its level alone | labelled test rows grouped by solute and <i>T</i> (823 groups, 107 solutes), 5 seeds | per-solute Spearman $-0.185$ $[-0.269, -0.107]$ pooled, all five per-seed intervals excluding zero; nDCG@3 $-0.071$ $[-0.114, -0.029]$ ; top-1 $-0.088$ $[-0.172, -0.009]$ pooled only, three of five per-seed intervals spanning zero. Within-checkpoint—one set of weights, $\max  \Delta\Phi  = 3.2 \times 10^{-5}$ over the labelled rows of this row’s own set at the worst seed, the same quantity over the full test set, labelled rows and unlabelled together, being larger (§S1)—so free of the separate-tuning confound of §3.7, though not of the evaluation-time swap the first row of this block carries; a per-group affine recalibration fitted in sample removes 91% of the <i>MAE</i> gap and refitted out of sample returns an interval that holds both all of it and none, and leaves this one untouched either way where the fitted map increases, as it does in 485 of the 823 groups in both arms at all five seeds. The substitution falls on the solvent channel |

Table S3: *Continued.* The three closure blocks of the ledger: two on the activity axis, one on a second closure in another domain. Notes *a–h* are at the foot of the third part.

| Claim                                                                                                                                               | Set ( <i>n</i> ) <sup>a</sup>                                                                                                          | Value, with its uncertainty and scope <sup>b</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Closure against inputs on the activity axis</i> (§3.5); margins in squared $\ln \gamma_2^\infty$ units, not the $\ln x_2$ MAE of the first block |                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| $B_{\text{closure}} > B_{\text{insuff}}$ on glycol-ether solvents: the instrument’s output, not a claim of this work <sup>d,g</sup>                 | broad IDAC, 182, 3 sources                                                                                                             | the margin, a lower bound on $B_{\text{closure}} - B_{\text{insuff}}$ , runs +2.04 [+1.25, +2.87] to +2.94 over the four cells, $P > 0.95$ in all of them; worst deletion +1.60—and the survivor of a 59-fold search, at the ninetieth percentile of a chemistry-blind null on all 477 rows that certifies some row set in 58% of draws, and not testable out of sample, where no rows of that cell exist; the pre-declared cross-fitted $B_{\text{insuff}}^{\text{up}}$ , which came back looser and is not the estimator of record, puts the same cell at +1.76 [+1.02, +2.49]—one replicate of the map’s own bootstrap, its second decimal seed and not estimator (§S3.12)                                                                                                                                                                                                                                                     |
| ... on every other stratum of the map                                                                                                               | broad IDAC, 59 strata                                                                                                                  | §S3.8, Tables S12–S18, each with its margin, interval and the test it fails; none is stated as a finding <sup>h</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| ... for either set <i>as a whole</i>                                                                                                                | broad IDAC (477); VT-2005-matched (60)                                                                                                 | +0.51 (full +0.95); –0.38 compounding two cuts (+0.01 full); +0.18 less one publication; +0.05 on the smaller set: an average weighted by composition <sup>f</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| $B_{\text{insuff}}^{\text{up}}$ near stratum-independent, MSE not                                                                                   | broad IDAC, 3 boundable classes; 6 with an estimate                                                                                    | $\times 2.0$ against $\times 25$ ; $\times 6.8$ against $\times 51$ ; $p=0.02$ against 2000 null partitions of 9 classes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| $B_{\text{closure}} > B_{\text{insuff}}$ , 2010 on its own typed latent <sup>f</sup>                                                                | broad IDAC (477); VT-2005-matched (60)                                                                                                 | +1.29, $P > 0.95/\approx 0.9/\approx 0.9^e$ ; –0.77 at $P \approx 0.3$ on the smaller set, where the ordering is lost: loss of resolution, not a reversal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <i>Whether a better closure moves the ceiling</i> (§3.5.3)<br>The 2002 → 2010 step lowers the error                                                 | broad IDAC (477)                                                                                                                       | –0.99 at $P \approx 0.2$ : no improvement, and not a measured equality either; under the full convention –0.07, $P \approx 0.6$ , and +0.06, $P \approx 0.6$ with dispersion on                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| ...but does on the VT-2005 set’s chemistry                                                                                                          | 57 shared pairs, UD                                                                                                                    | 1.78 → 0.77, $P > 0.95$ , at fixed profile database and temperature, full convention; those 57 pairs are also broad-set rows, so not an independent sample                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| A fitted kernel residual transfers across solvents                                                                                                  | VT-2005-matched (60), grouped folds                                                                                                    | –5 ± 16%: spans zero; the ± prices optimiser noise only, and the fold spread that dominates it is not propagated, so no bound follows                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <i>The same split on a second closure</i> (§3.6); MAE in $\text{p}K_a$ , not the squared $\ln \gamma_2^\infty$ of the two blocks above              |                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| A Hammett $\text{p}K_a$ closure reverses the substitution’s sign between its two poles                                                              | OPERA $\text{p}K_a$ , clean pole (158) and degraded (846); $K=20$ within-scaffold splits, $G=4$ series; certified in distribution only | MAE gap –0.165 [–0.28, –0.04] on the clean pole, the reference better in 19 of 20 splits and in three of the four series, against +0.79 [+0.57, +0.95] on the degraded, worse in 20 of 20 and in all four; 90% intervals over the four series, which as a sign test read $p \approx 0.31$ on the clean pole and $p=1/16$ on the degraded—so that test carries the degraded half and not the positive control. Neither pole is shown well specified: the clean pole’s Jensen bound is $B_{\text{closure}} \geq 0.083$ in squared $\text{p}K_a$ , 27% of its 0.303, and its binning-free floor (–0.2 filtered, +0.55 on the 163 rows before the filter) is negative for any closure whatever once $B_{\text{insuff}}^{\text{up}}$ exceeds the MSE. On a Bemis–Murcko split the degraded pole’s gap turns negative with its cluster interval spanning zero, so neither that half nor the reversal is established out of sample (§S7) |

Table S3: *Continued.* The mechanism and accuracy blocks, and notes *a–h*, which qualify rows and columns of all three parts. The grounded, ungrounded and  $\sigma$ -oracle arms come from the five-seed leak-free re-run.

| Claim                                                                      | Set ( $n$ ) <sup>a</sup>         | Value, with its uncertainty and scope <sup>b</sup>                                                                                                                                                                                        |
|----------------------------------------------------------------------------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Mechanism</i> (§3.3)                                                    |                                  |                                                                                                                                                                                                                                           |
| The latent departs from its reference along a shared direction             | matched molecules (44), 3 seeds  | $51 \pm 2\%$ ; the magnitude is not stable across analysis configurations, and these are a separate three-seed set of runs, not the arms of Fig. 2, whose latent departure is not measured                                                |
| ...and that departure’s top-two mode share exceeds a phase-randomised null | one grounded checkpoint, same 44 | $0.40$ against $0.22 \pm 0.01$ , the $\pm$ being the null’s spread; one checkpoint, and a wrong-molecule null is not cleared                                                                                                              |
| ...and that direction <i>transfers</i> to molecules outside the fit        | matched molecules (44), 3 seeds  | $4.1 \pm 0.5\times$ a zero-drift null, with no constant-offset control on these seeds; the $\pm$ spans seeds at one partition; and the held-out half is homologue-saturated, so “outside the fit” is largely within-cluster interpolation |
| That departure <i>cancels</i> closure error                                | not measured                     | the one direct check is under-powered and runs against it                                                                                                                                                                                 |

<sup>a</sup> The sets are defined in Table S1, except the two  $pK_a$  poles of the second-closure block, which §3.6 defines and Table S1 does not.

<sup>b</sup>  $P$  is a two-way (solute  $\times$  solvent) cluster-bootstrap frequency that the margin stays positive, reported as a band by the rounding rule at the head of this section.

<sup>c</sup> The same 5608 rows: the seed-42 control arms carry the identical labelled set, so their cross-arm intersection with the arms of Table S4 is that set exactly.

<sup>d</sup> One estimator cell (§2.5) for both sets, both kernels and every stratum.

<sup>e</sup> Bootstrap band over pair / two-way / source clusters.

<sup>f</sup> An aggregate margin is a composition-weighted average, and the deletions and cuts recorded here are failures of the headline cell rather than robustness passes. These rows record what the aggregate was and how it fails; the rows above replace it.

<sup>g</sup> The comparison is one-sided in both directions it can be read: a positive margin establishes  $B_{\text{closure}} > B_{\text{insuff}}$ , a non-positive one is failure to separate and licenses nothing. The margin’s size is a lower bound on  $B_{\text{closure}} - B_{\text{insuff}}$  and not an estimate of it, so an interval on a margin bounds that bound and not the difference (Lemma 1, §2.1.1).  $B_{\text{closure}}$  is the error of the fixed map *read at a computed profile*, so what a positive margin establishes is a property of that composite and not of the kernel alone (§3.5.3).

<sup>h</sup> Of fifty-nine strata, fifteen are boundable and six admissible; clause (c) of §2.5 makes those six three distinct row sets, of which one is also positive and not demoted by its residual test—the glycol ethers, the one row set the rule leaves standing.

the NRTL mean by  $-0.006$ , widening the NRTL physics penalty from  $+0.0928$  to  $+0.1252$  and the learned- $\sigma$  penalty from the  $+0.1435$  read off the table below to  $+0.1560$ . A filter applied at this stage would widen the very gap this paper reports, on the row set every number in both documents is keyed to.

Table S4 gives the exact per-arm metrics behind the six three-seed bars of Fig. 2: three seeds on the labelled test rows ( $n=5608$  of 8103, those carrying a solubility measurement); the two single-seed co-adaptation bars are valued in the paragraph below. Per-seed values are printed to three decimals; every mean, difference and  $\pm$  quoted anywhere is computed on the unrounded values, so the printed triples reproduce them only to that precision. Every  $\pm$  that spans *seeds* in this paper is a population standard deviation over them; a sample standard deviation would be larger by  $\sqrt{3/2}$ ,

and the error bars of Fig. 2 follow the same convention. Three kinds of  $\pm$  in this paper are *not* that: the kernel-residual figures of the correctability test ( $46 \pm 5\%$ ,  $-5 \pm 16\%$ ,  $+2 \pm 15\%$ ) carry a standard *error* over five fit seeds; a  $\pm$  on a permutation or phase-randomised null (Table S3; §3.3) is that null’s own spread over draws, not a spread over seeds at all; and the  $pK_a$  arm errors of §S7 span the 20 within-scaffold splits. Reading the table off the rounded means mis-states differences by up to 0.01: the learned- $\sigma$  and DirectGNN arms differ by 0.2253 from their unrounded means, which we quote as  $+0.23$ , while their rounded values 1.93 and 1.70 differ by 0.23. Both closure families—NRTL and COSMO-SAC over a learned  $\sigma$ -profile—trail the DirectGNN control in the seed mean (the physics penalty), though the arms are separately tuned and that gap is not attributable to the bottleneck (§3.7). Only

the COSMO-SAC family trails it at every seed: the per-seed column below puts the NRTL arm at 1.734 against the control’s 1.749 at seed 42, so the two arms’ per-seed values overlap and that ordering is a property of the seed-mean column and not of the arms. The  $\sigma$ -oracle (reference VT-2005 profile) is the worst arm. Restoring the Staverman–Guggenheim combinatorial term makes the *grounded learned- $\sigma$*  arm slightly worse ( $1.8457 \rightarrow 1.8788$ )—the signature of a misspecified map (§3.5). Both sides of that pair are the published family at three seeds, which is what makes it a contrast: the +SG arm was not re-run, so against the re-run’s grounded mean of 1.9275 the ordering reverses, and that reversal is the baseline moving between families rather than the term acting. No oracle-plus-combinatorial arm was run either, so the combinatorial contrast is against the grounded learned- $\sigma$  row and not against the oracle row. To four decimals the means are 1.7022, 1.7950, 1.8788, 1.9275, 2.0374 and 2.3356, with population standard deviations 0.033, 0.071, 0.091, 0.094, 0.064 and 0.108. The NRTL row is where the two roundings part: averaging its printed per-seed triple gives 1.7953, whose two-decimal form is 1.80, but the arm’s mean is 1.794975, so the entry is 1.79. The two senses of grounding are the fourth and the last of these: training-time supervision moves the arm from 2.0374 to 1.9275, and evaluation-time substitution from 1.9275 to 2.3356 (hurts, +0.41). The first of those two is  $-0.11$  as a mean and is not a sign: its five per-seed gains are +0.149, +0.122, +0.175,  $-0.017$  and +0.122, one of them against the other four.

**$\sigma$ -head schedule of these arms.** The four COSMO-SAC rows come from three training runs under one pinned configuration — 30 P1, 70 P2 and 10 P3 epochs with the  $\sigma$ -head freeze on, so that head’s weights are held from P2 onward. The two grounded rows precede that with a  $\sigma$  warm-up on the grounding pool (up to 40 epochs, early-stopped on a held-out tenth of the pool) and interleave the pool through P1; under the freeze the stream is inactive in P2/P3. The ungrounded row sets warm-up and stream budget to zero, so its  $\sigma$  head is shaped

in P1 by the corpus-internal  $\ln \gamma_2^\infty$  auxiliary loss alone, never by an external  $\sigma$  label, and is frozen from P2 in the same way. The  $\sigma$ -oracle row is the grounded checkpoint re-evaluated with  $z^*$  injected, so it shares the grounded schedule exactly. Because the freeze holds the head and not the encoder that feeds it, the predicted profile is not pinned at its warm-up value in these arms. We have not run the surrogate isolation (§3.3) on these checkpoints, so how far their latent drifted is not measured. The supervision contrast quoted there ( $+0.40 \rightarrow +0.14$ ) is a separate matched pair of runs — one warm-up-plus-short-P1 base resumed twice for SLE, with the freeze off and on, on  $n=5571$  matched rows, so a single pair carrying no seed replicate and no interval — and not these arms.

**Oracle-application controls: two of the three are single-seed, and are reported as an illustration rather than as a result.** All three oracle applications (evaluation-only, training-time injection, channel-swap; §2) are computed on the same 5608 labelled test rows as Table S4, and their per-arm predictions are released with the code. Only the evaluation-only substitution is a result. **The training-time injection and the channel swap are stated here and nowhere else in the submission:** each was run at one seed, against a comparator that its own two run families give two values for, and neither is a matched contrast. They are not claimed in the article, they carry no row in Table S3, and no quantity reported anywhere rests on them. What follows prices the illustration, not the paradox. Training-time injection (MAE 1.98) and the channel swap (MAE 2.08) were run at seed 42 alone, so their matched comparator is the seed-42 learned- $\sigma$  value 1.803 on the same  $n=5608$  lock, not a seed mean. That comparator is 1.803 in their own family and 1.887 in the re-run, so the same two measured arms admit two readings and neither is a matched five-seed contrast. Both readings are stated here and nowhere else, §3.1 having stopped reporting these arms. Both exceed the worst learned- $\sigma$  seed of either family, so the direction of both survives a worst-case seed com-

Table S4: Per-arm scaffold-split metrics (the  $n=5608$  labelled test rows), with per-seed MAE to three decimals. Ordered by MAE. The ungrounded, grounded and  $\sigma$ -oracle arms come from the five-seed leak-free re-run and carry 5 seeds; the DirectGNN, NRTL and +SG rows are the published runs at their own three. The  $\pm$  is a population standard deviation over the five seeds where a row has five and over the three where it has three. Mean,  $\pm$  and per-seed column are each rounded once from the unrounded values, the column independently of the mean beside it.

| Arm                                               | MAE             | $R^2$            | per-seed MAE                  |
|---------------------------------------------------|-----------------|------------------|-------------------------------|
| DirectGNN (no closure)                            | $1.70 \pm 0.03$ | $+0.42 \pm 0.03$ | 1.749/1.674/1.684             |
| NRTL closure                                      | $1.79 \pm 0.07$ | $+0.34 \pm 0.03$ | 1.734/1.758/1.894             |
| COSMO-SAC, learned $\sigma$ (grounded)            | $1.93 \pm 0.09$ | $+0.29 \pm 0.07$ | 1.887/1.818/1.856/2.056/2.020 |
| + Staverman–Guggenheim combinatorial              | $1.88 \pm 0.09$ | $+0.32 \pm 0.05$ | 1.805/2.007/1.824             |
| COSMO-SAC, ungrounded $\sigma$                    | $2.04 \pm 0.06$ | $+0.21 \pm 0.06$ | 2.036/1.940/2.031/2.039/2.142 |
| COSMO-SAC, reference $\sigma$ ( $\sigma$ -oracle) | $2.34 \pm 0.11$ | $-0.07 \pm 0.08$ | 2.145/2.342/2.418/2.456/2.317 |

parison. The channel-swap margin ( $+0.27$ ) is 2.3 times the learned arm’s full between-seed range; the training-time margin ( $+0.18$ ) is one and a half times it, so that control fixes a sign rather than an effect size. This matters for how the paradox’s magnitude should be read. The evaluation-only substitution ( $+0.41$  against the three-seed mean) mixes closure misspecification with the ordinary degradation any pipeline suffers when an input distribution is swapped at test time; the training-time injection, in which the model co-adapts to the reference profile, is the arm that removes that component and it costs  $+0.18$ . The co-adapted arm is therefore free of that confound and the evaluation-only  $+0.41$  is not. It is not free of a second: neither this control nor the channel swap is certified to have trained on the same  $\sigma$ -stream build as the seed-42 comparator both are read against, and the file times exclude no history in either direction (§S8). Both are penalties of uncertified stream provenance. How large a share of the gap co-adaptation carries off can be read only on the seed the two arms share: at seed 42 the injection costs  $1.981 - 1.887 = 0.094$  where the evaluation-only substitution costs  $2.145 - 1.887 = 0.258$ , so co-adaptation carries off  $0.258 - 0.094 = 0.164$  of that gap, a share of 0.64. We report that share as a description of seed 42 and not as a magnitude, because it is not resolved—and at five seeds it is not a magnitude at all. The injected arm is the quantity that was measured once, so its

own between-seed spread is unmeasured, and a spread the size of the learned arm’s 0.238 would move the share by about  $\pm 0.92$ , which is larger than the share. Holding that arm at its single value 1.981 and recomputing at the other four seeds, where the learned baseline and the substituted arm both move, gives 0.69, 0.78, 1.19 and 1.13: the last two exceed one, which no share can. They exceed it because at seeds 45 and 46 the learned baseline is worse than the once-measured injected arm, so the injection gap changes sign and the numerator overruns its own denominator. It is that one arm’s MAE that is held fixed across the readings and not the numerator, which is 0.164, 0.362, 0.437, 0.475 and 0.337 over denominators 0.258, 0.524, 0.562, 0.400 and 0.297 (from the unrounded arm values, the injected arm being 1.9806). What survives that latitude is the sign—at the matched seed part of the evaluation-only gap does go with co-adaptation, which is what makes the co-adapted arm the one free of the test-time swap—and the sign, not the share, is what the claim rests on. The ratio  $(0.406 - 0.177)/0.406 = 0.56$  sets the one-seed injection against the three-seed mean gap; it is a share of nothing and is reported nowhere. Neither  $+0.18$  nor  $+0.27$  can be ordered against the  $-0.20$  supervision gain either, whose own per-seed values are 0.237/0.075/0.281 and straddle  $+0.18$ .

### S3.3 The estimator grid, and the analyst choices in it

Every MSE, bound and margin on the activity axis is in squared  $\ln \gamma_2^\infty$  units, and not on the  $\ln x_2$  axis the per-arm MAEs above are measured on. Three analyst choices set the separation margin on the VT-2005-matched set, and Table S5 reports every combination of them rather than one setting of each. The margin is positive in 66 of the 72 cells. The six exceptions are the coarsest conditionings—three bins under either combinatorial convention and four bins residual-only—where the binning-bound bound is valid but weakest. The grid is not a set of interchangeable settings, because two opposing biases run across it. Coarser bins condition on less, so the bound is weaker; finer bins condition on more but leave few points per bin, and the within-bin variance is then downward-biased in finite samples, which inflates the margin favourably. Neither end is unbiased. Every margin is read at the eight-bin cell under the *unbiased* within-bin variance, on an occupancy rule fixed in advance of the grid—the coarsest binning that still leaves fewer than eight rows per bin, here 7.5 at  $n=60$ —and we carry the same *bin count* to the broad IDAC set, so the two are conditioned at the same granularity. On 477 rows that is not the same occupancy rule (which would call for about 60 bins) and is deliberately conservative. The broad set’s margin grows with bin count over the grid as a whole, though not cell by cell: in the residual-only convention under the unbiased variance it falls from +0.10 at three bins to +0.01 at four, from +0.57 at seven to +0.51 at eight, and from +0.83 at twenty to +0.82 at twenty-four. The endpoints are what the choice turns on: +0.51 at eight bins against +0.96 at forty-eight. Eight bins therefore gives a looser bound and a smaller margin there than the occupancy rule would. It is *not* the least favourable cell of the middle range: seven bins residual-only under the unbiased variance gives +0.01 against the eight-bin +0.05, and twenty bins gives +0.08. Nothing turns on the choice—the whole middle range of the residual-only convention lies between +0.01 and +0.47, all positive—but the

reason for it is occupancy and not conservatism, and taking the least favourable middle cell instead gives +0.01 and the same verdict of an unresolved margin. The per-cell  $P$  holds that cell’s bin count fixed across resamples. The main text instead quotes the deposited adaptive bin count  $\max(3, \lfloor n/10 \rfloor)$ , here in the rows  $n$  of the resample, which conditions on less on small resamples and is therefore more conservative: it gives  $\approx 0.78$  (maximum-likelihood) and  $\approx 0.72$  (unbiased) residual-only, against  $\approx 0.89$  and  $\approx 0.86$  full. The two schemes measure different things and we report both rather than the higher.

**A fourth analyst choice: what “equal-count” means when  $n$  is not divisible by the bin count.** The rule this paper deploys, and which §2.5 names as the fourth analyst choice without restating, is linearly interpolated quantile edges of  $g$  at  $j/8$ ,  $j = 1, \dots, 7$ , with a value falling exactly on an edge assigned to the upper bin. Neither  $n$  divides by eight ( $60/8 = 7.5$ ,  $477/8 = 59.625$ ), and that alone is what makes standard implementations of the same phrase disagree; there are no exact ties in  $g$  on either set under either convention, so each of them is a genuine coarsening of  $\sigma(g)$  and the bound is valid under all of them. On the VT-2005-matched set at eight bins, the residual-only convention and the unbiased variance,  $B_{\text{insuff}}^{\text{up}}$  is 0.713 under our rule, under the same edges with ties sent to the lower bin, and under the standard quantile-cut implementation alike; 0.624 under an equal-length split of the  $g$ -sorted order (margin +0.225); 0.669 under  $\lfloor k \text{rank}/n \rfloor$  (+0.134); and 0.736 under order-statistic quantile edges (+0.001). Exact enumeration by dynamic programming over *every* partition of the  $g$ -ordered rows into eight contiguous bins of sizes in  $\{7, 8\}$  puts the VT-2005-matched bound in  $[0.620, 0.742]$  and its margin in  $[-0.012, +0.232]$ ; at seven bins the margin envelope is  $[-0.134, +0.157]$ , and one such rule realises  $-0.089$  there. The sign of the VT-2005-matched margin is therefore set by the binning rule and not by the data, which is a further reason to read its +0.05 as consistent with zero. The broad IDAC set absorbs the same

choice: the six strictly equal-count implementations give 0.694 to 0.700, a spread of 0.006, and the exact envelope over all sizes- $\{59, 60\}$  layouts is  $[0.687, 0.701]$ , margin  $[+0.499, +0.528]$  (at seven bins,  $[+0.564, +0.570]$ ). Our 0.697 sits mid-range in both enumerations. Nothing load-bearing moves, since the VT-2005-matched set’s margin is unresolved either way and the claim is carried on the broad set; we name the rule because the phrase “equal-count” does not by itself fix it.

Table S5: Separation margin  $\text{MSE} - 2B_{\text{insuff}}^{\text{up}}$  on the VT-2005-matched set ( $n=60$ ), over bin count, within-bin variance (ML, Be) and combinatorial convention.<sup>a</sup> Negative margins in bold.

| bins | full/ML            | full/Be            | res/ML             | res/Be             |
|------|--------------------|--------------------|--------------------|--------------------|
| 3    | <b>-0.06</b> (.64) | <b>-0.16</b> (.59) | <b>-0.43</b> (.43) | <b>-0.53</b> (.38) |
| 4    | +0.18 (.82)        | +0.07 (.78)        | <b>-0.14</b> (.63) | <b>-0.26</b> (.57) |
| 5    | +0.50 (.90)        | +0.38 (.87)        | +0.21 (.77)        | +0.10 (.70)        |
| 6    | +0.66 (.94)        | +0.54 (.92)        | +0.35 (.84)        | +0.22 (.77)        |
| 7    | +0.76 (.96)        | +0.62 (.94)        | +0.18 (.89)        | +0.01 (.83)        |
| 8    | +0.74 (.97)        | +0.58 (.96)        | +0.24 (.91)        | +0.05 (.86)        |
| 9    | +0.90 (.98)        | +0.75 (.97)        | +0.42 (.93)        | +0.24 (.88)        |
| 10   | +1.12 (.98)        | +0.99 (.97)        | +0.42 (.94)        | +0.21 (.89)        |
| 11   | +1.11 (.99)        | +0.95 (.98)        | +0.38 (.96)        | +0.14 (.91)        |
| 12   | +0.96 (.99)        | +0.75 (.98)        | +0.49 (.96)        | +0.24 (.93)        |
| 13   | +1.01 (.99)        | +0.80 (.98)        | +0.59 (.97)        | +0.34 (.94)        |
| 14   | +1.05 (.99)        | +0.81 (.98)        | +0.50 (.97)        | +0.21 (.94)        |
| 15   | +1.00 (.99)        | +0.73 (.99)        | +0.52 (.98)        | +0.20 (.95)        |
| 16   | +1.03 (1.0)        | +0.77 (.99)        | +0.56 (.98)        | +0.22 (.96)        |
| 17   | +1.25 (1.0)        | +1.05 (.99)        | +0.56 (.99)        | +0.20 (.96)        |
| 18   | +1.08 (1.0)        | +0.76 (.99)        | +0.69 (.99)        | +0.39 (.97)        |
| 19   | +1.11 (1.0)        | +0.77 (1.0)        | +0.79 (.99)        | +0.47 (.97)        |
| 20   | +1.23 (1.0)        | +0.94 (.99)        | +0.55 (.99)        | +0.08 (.97)        |

<sup>a</sup> ML, maximum likelihood; Be, unbiased. In parentheses, the fixed-bin two-way cluster-bootstrap frequency that the margin is positive.

The reported cell is not a favourable one. Figure S2 runs the same estimator over the whole grid of bin counts and conventions, where the aggregate spans  $-0.53$  to  $+1.12$  and changes sign. Of the 34 cells returning a positive margin, the reported one is second from the bottom.

The same grid on the broad IDAC set is Table S6, now with its own per-cell bootstrap frequency rather than a bare range of point margins. None of its 56 cells is negative. Within the *residual-only* convention the margin runs from  $+0.01$  (four bins) and  $+0.10$  (three bins) up to  $+0.96$ , with the eight-bin unbiased headline at

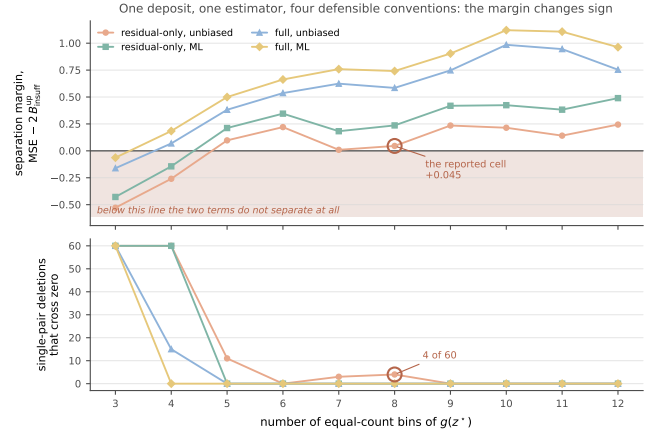


Figure S2: **The same rows, the same estimator, and four defensible conventions.** The VT-2005 matched set ( $n = 60$ ) at every bin count from three to twelve, under both within-bin variance conventions (unbiased, maximum-likelihood) and both combinatorial ones (residual-only, full). (a) the separation margin  $\text{MSE} - 2B_{\text{insuff}}^{\text{up}}$ ; below zero Lemma 1 separates nothing. (b) how many of the 60 single-pair deletions send that margin below zero. The ringed point is the cell §2.5 fixes in advance and this section reports. The aggregate spans  $-0.53$  to  $+1.12$  across cells that are each individually defensible, which is why the choices are fixed before the data are seen rather than justified after.

$+0.51$ ; the frequencies at the two coarsest cells, 0.65 and 0.42, say that those conditionings do not resolve the ordering, and they are why the broad set’s aggregate margin is not resolved at every conditioning. Under the full convention the same grid runs from  $+0.23$  to  $+1.25$  and reaches  $P > 0.95$  from five bins upward.

**The same grid on the stratum the paper states.** The two grids above sweep the bin count for the sets taken *whole*, and they are what establishes that the bin count is the analyst choice with the largest leverage here—the broad set’s residual-only margin runs from  $+0.01$  to  $+0.96$  across Table S6. Every one of the 59 strata of §S3.8 is then computed at the single fixed eight-bin cell, so the one row set the rule leaves standing was reported at one cell of a grid that moves. Table S7 runs the grid of Ta-



Table S6: Separation margin  $\text{MSE} - 2B_{\text{insuff}}^{\text{up}}$  on the broad IDAC set ( $n=477$ ), over bin count, within-bin variance (ML, Be) and combinatorial convention.<sup>a</sup> No cell is negative; the headline is the 8/res/Be cell.

| bins | full/ML     | full/Be     | res/ML      | res/Be      |
|------|-------------|-------------|-------------|-------------|
| 3    | +0.24 (.60) | +0.23 (.59) | +0.11 (.42) | +0.10 (.42) |
| 4    | +0.50 (.93) | +0.49 (.93) | +0.02 (.67) | +0.01 (.65) |
| 5    | +0.84 (.98) | +0.82 (.98) | +0.50 (.77) | +0.48 (.76) |
| 6    | +0.84 (.99) | +0.82 (.99) | +0.51 (.86) | +0.50 (.85) |
| 7    | +0.94 (.99) | +0.93 (.99) | +0.59 (.89) | +0.57 (.88) |
| 8    | +0.96 (1.0) | +0.95 (1.0) | +0.53 (.93) | +0.51 (.92) |
| 10   | +1.06 (1.0) | +1.04 (1.0) | +0.67 (.96) | +0.64 (.94) |
| 12   | +1.05 (1.0) | +1.03 (1.0) | +0.74 (.97) | +0.71 (.97) |
| 16   | +1.11 (1.0) | +1.09 (1.0) | +0.76 (.99) | +0.72 (.99) |
| 20   | +1.17 (1.0) | +1.14 (1.0) | +0.88 (1.0) | +0.83 (.99) |
| 24   | +1.19 (1.0) | +1.15 (1.0) | +0.87 (1.0) | +0.82 (1.0) |
| 32   | +1.21 (1.0) | +1.16 (1.0) | +0.93 (1.0) | +0.86 (1.0) |
| 40   | +1.30 (1.0) | +1.25 (1.0) | +1.03 (1.0) | +0.95 (1.0) |
| 48   | +1.29 (1.0) | +1.22 (1.0) | +1.06 (1.0) | +0.96 (1.0) |

<sup>a</sup> ML, maximum likelihood; Be, unbiased. In parentheses, the fixed-bin two-way cluster-bootstrap frequency that the margin is positive. Regenerates from the deposited convention audit.

ble S6 on that row set. At  $n=182$  the eight-bin cell holds 22.75 rows per bin, where the occupancy rule that fixed the cell on the  $n=60$  set would ask for 23 bins: the fixed cell is *coarser* than the rule that chose it and therefore conditions on less, which weakens the bound and lowers the margin.

The glycol-ether margin is positive in all 56 cells of that grid at  $P > 0.95$  throughout, the residual-only unbiased column running from +1.06 at three bins to +2.19 at forty-eight, with the fixed cell’s +2.04 sitting below the midpoint of that column and the occupancy rule’s own 23 bins giving +2.17 residual-only and +2.36 full. The pair unit,  $n=43$ , supports ten of the fourteen bin counts and is positive in all 40 of the cells it defines, [+1.41, +2.83] residual-only and [+1.69, +3.10] full. Across the 36 cells of the sweep where the leave-one-publication-out test can be run at all, the sign survives every deletion in all 36, the worst leaving +1.08. Neither the sign nor the admissibility of that stratum is a property of the bin count. The same stratum under solvent deletion. Leaving one glycol out at a time the margin reads +1.72, +1.94, +2.05 and +2.29; taken one at a time it reads

+3.17, +2.43, +1.71 and +1.21 on 41, 36, 56 and 49 rows, monotone in chain length. Diethylene glycol’s 36 rows sit below the  $n \geq 40$  boundability threshold, so that single-solvent cell is reported and not bounded. The spread across these four, and not the bootstrap interval at any one of them, is the honest width on the solvent axis. The other two admissible row sets behave differently and are reported here rather than left out: the acceptor-only aprotics are non-positive in every cell of the same grid, [−0.55, −0.03] over the 44 its  $n=57$  defines, which licenses nothing by one-sidedness; and the alkane solutes are positive in all 56 row-unit cells but lose the sign in two of the twenty pair-unit residual-only cells and fail the deletion test at eight of the forty cells where it runs, so that set is admissible at the fixed cell and not across the sweep—one more reason the finding is stated on the glycol ethers and not on it.

Table S7: Separation margin  $\text{MSE} - 2B_{\text{insuff}}^{\text{up}}$  on the *glycol-ether stratum* of the broad IDAC set ( $n=182$ , row unit), over the same bin-count  $\times$  variance  $\times$  convention grid the aggregate is swept on.<sup>a</sup> No cell is negative; the fixed cell of the map is 8/res/Be.

| bins | full/ML     | full/Be     | res/ML      | res/Be      |
|------|-------------|-------------|-------------|-------------|
| 3    | +1.29 (1.0) | +1.27 (1.0) | +1.08 (.99) | +1.06 (.99) |
| 4    | +1.92 (1.0) | +1.91 (1.0) | +1.72 (1.0) | +1.71 (1.0) |
| 5    | +1.96 (1.0) | +1.95 (1.0) | +1.77 (1.0) | +1.76 (1.0) |
| 6    | +2.02 (1.0) | +2.01 (1.0) | +1.82 (1.0) | +1.80 (1.0) |
| 7    | +2.12 (1.0) | +2.11 (1.0) | +1.93 (1.0) | +1.91 (1.0) |
| 8    | +2.24 (1.0) | +2.23 (1.0) | +2.05 (1.0) | +2.04 (1.0) |
| 10   | +2.24 (1.0) | +2.22 (1.0) | +2.05 (1.0) | +2.04 (1.0) |
| 12   | +2.34 (1.0) | +2.33 (1.0) | +2.13 (1.0) | +2.13 (1.0) |
| 16   | +2.36 (1.0) | +2.35 (1.0) | +2.16 (1.0) | +2.15 (1.0) |
| 20   | +2.37 (1.0) | +2.36 (1.0) | +2.18 (1.0) | +2.17 (1.0) |
| 24   | +2.39 (1.0) | +2.38 (1.0) | +2.18 (1.0) | +2.17 (1.0) |
| 32   | +2.39 (1.0) | +2.37 (1.0) | +2.19 (1.0) | +2.18 (1.0) |
| 40   | +2.40 (1.0) | +2.38 (1.0) | +2.20 (1.0) | +2.19 (1.0) |
| 48   | +2.41 (1.0) | +2.39 (1.0) | +2.20 (1.0) | +2.19 (1.0) |

<sup>a</sup> ML, maximum likelihood; Be, unbiased. In parentheses, the fixed-bin two-way cluster-bootstrap frequency that the margin is positive. The pair unit, the other five admissible cells, and the leave-one-publication-out curve at every bin count are in the deposited stratum bin sweep.

### S3.4 The decomposition on both sets, the three kernel arms, and the bounds inventory

Table S8 is the closure-insufficiency decomposition of §3.5.1–§3.5.2 on both sets under both conventions, with the maximum-likelihood variance variant, the constant-offset (Jensen) row and the bootstrap bands. Its headline columns are the residual-only ones on *both* sets, one rule applied throughout; the full-convention columns beside them are the comparison and not a second headline. Table S9 is the three COSMO-SAC arms of §3.5.3 on both sets under both conventions, with the bootstrap frequencies, the 90% margins and the block split of the 2002 → 2010 contrast into the 57 pairs shared with the VT-2005-matched set, the complementary 420 and the 80 of those at 298 K. Every claim either table carries has a row in Table S3 with its set, value and its uncertainty.

**The bounds inventory, named and valued.** Table S10 lists every bound either set supports at the aggregate level, together with the corroborating estimators Table S8 points here for instead of listing them row by row, and one estimator the VT-2005-matched set cannot support. The per-stratum bounds that replace the aggregate margin are the map of §S3.8. The inclusion rule is note *a* of the table, and it is a rule for that table and not a claim of completeness over the paper: the binning-bound half is a selection from two bin grids far too large to reproduce, and the  $\ln \gamma_2^\infty$ -stratified bounds of Table S19 are computed on subsets of the VT-2005-matched set rather than on either whole set and are not in it. The deposited figure script regenerates it, printing the inventory as it goes, so the table can be re-checked against the artifacts. The regeneration’s guard against an artifact changing under the inventory is *not* the same on the two sets, and we state which is which rather than claim one rule. On the VT-2005-matched set it stops both ways: on an estimator the artifact records and the inventory does not name, and on a named one the artifact no longer records. It additionally checks

the entries that set’s two audit artifacts share against each other. On the broad set it stops only the first way, so a named estimator absent from that set’s audit artifact would be dropped rather than raised.  $B_{\text{insuff}}^{\text{up}}$  and the  $B_{\text{closure}}$  lower bound it implies are contiguous shares of one total, so what separates the terms is the margin  $\text{MSE} - 2B_{\text{insuff}}^{\text{up}}$  and not a gap between the two. Two rows of the table carry the whole risk on each set. On the broad set the coarsest and the least favourable cell differ—three bins gives 0.901, four bins 0.948 against a threshold of 0.951, a margin of +0.007 at bootstrap frequency 0.65 and the tightest anywhere on that set. On the VT-2005-matched set the two coincide at three bins (1.000), and both it and the four-bin cell (0.866) sit *above* that set’s threshold of 0.736: the negative-margin cells Table S5 sets in bold, and both are listed here.

### S3.5 The stratification rule, and what sets the level of the bound

§3.5.1 reports the decomposition per stratum rather than per set, and the rule is set out here in full because it is the object that has to be checkable against the charge of having been fitted to its own outcome.

*Axis 1, solvent class.* An ordered cascade over functional groups; the first class a solvent matches is the class it takes, and the order is hydrogen-bonding role first—the axis COSMO-SAC’s H-bond term acts on—then dominant functional group. (1) *water* (111 rows / 15 pairs), the only liquid whose entire surface is H-bond active, with  $\sigma$ -profile mass at both extremes and the smallest cavity of any of them, so the Staverman-Guggenheim term is at its most extreme and the two combinatorial conventions disagree most. (2) *glycol ether*, at least one O–H and at least two oxygens: the ethylene-glycol oligomers (182 / 43), whose rows are 108 alkane, 48 aromatic hydrocarbon, 24 alkene and 2 halogenated aromatic, 175 of the 182 measured chromatographically; donor and acceptor on one flexible chain, so intramolecular H-bonds form and the profile is

Table S8: Closure–insufficiency decomposition on the reference-input activity path, as bounds rather than a point split. Each set is given under both combinatorial conventions, the residual-only column of each pair in bold.

| Quantity                                                                | broad IDAC (477) <sup>a</sup>               |                    | VT-2005-matched (60) <sup>a</sup> |                   |
|-------------------------------------------------------------------------|---------------------------------------------|--------------------|-----------------------------------|-------------------|
|                                                                         | res <sup>b</sup>                            | full <sup>b</sup>  | res <sup>b</sup>                  | full <sup>b</sup> |
| MSE <sub>total</sub> = $B_{\text{closure}} + B_{\text{insuff}}$ (exact) | 1.90                                        | 1.91               | 1.47                              | 1.80              |
| $B_{\text{insuff}}$ upper bound <sup>c</sup>                            | 0.70                                        | 0.48               | 0.71                              | 0.61              |
| $B_{\text{closure}}$ lower bound <sup>d</sup>                           | 1.21                                        | 1.43               | 0.76                              | 1.19              |
| $B_{\text{closure}}$ lower, constant-offset (Jensen) <sup>e</sup>       | 0.18                                        | 0.71               | 0.43                              | 0.72              |
| separation margin <sup>f</sup>                                          | + <b>0.51</b>                               | +0.95              | + <b>0.05</b>                     | +0.58             |
| ... bootstrap band <sup>g</sup>                                         | > 0.95/≈0.9/≈0.9                            | > 0.95 (all three) | ≈0.7                              | ≈0.9              |
| ... maximum-likelihood variance <sup>h</sup>                            | 0.68/+0.53                                  | 0.47/+0.96         | 0.62/+0.24                        | 0.53/+0.74        |
| ... tighter of the two conventions <sup>i</sup>                         | 0.48/+0.94                                  |                    | 0.61/+0.26                        |                   |
| Further $B_{\text{insuff}}$ upper bounds <sup>j</sup>                   | Table S10                                   |                    |                                   |                   |
| Label noise $\sigma_{\eta}^2$ (IDAC)                                    | already inside every row above <sup>k</sup> |                    |                                   |                   |

<sup>a</sup> Broad IDAC,  $n=477$  on UD profiles; VT-2005-matched,  $n=60$  on VT-2005 profiles throughout (§3.5.3 scores the same pairs on UD). <sup>b</sup> *full*, with the Staverman–Guggenheim combinatorial term; *res*, residual-only. <sup>c</sup> Law of total variance, at eight equal-count bins of  $g(z^*, T)$  under the unbiased within-bin variance. It fits no model to the data.

<sup>d</sup>  $\text{MSE} - B_{\text{insuff}}^{\text{up}}$ ; this row decides the ordering. <sup>e</sup>  $(\mathbb{E}[m] - \mathbb{E}[g])^2$ . <sup>f</sup>  $\text{MSE} - 2B_{\text{insuff}}^{\text{up}}$ , the quantity that orders the two terms. <sup>g</sup> Pair / two-way / source clusters. The VT-2005-matched set supports the two-way clustering only, so it carries one entry per column. <sup>h</sup> Same rows and bins, maximum-likelihood within-bin variance, as  $B_{\text{insuff}}$  upper / margin. <sup>i</sup> Charged against the residual-only MSE, as  $B_{\text{insuff}}$  upper / margin. Not the reported pairing.

<sup>j</sup> The entries of Table S10: out-of-fold RF and ridge (random and blocked folds),  $k$ NN-in- $z^*$ , and the binning-free  $\mathbb{E}[\text{Var}(m \mid \text{pair})]$ . Not a paper-wide inventory: the bin grids behind the binning-bound selection are Tables S5 and S6, and the  $\ln \gamma_2^\infty$ -stratified bounds, computed on VT-2005-matched subsets, are Table S19. <sup>k</sup> A *component* of  $B_{\text{insuff}}$ , not a term beside it (§2.1.1). <sup>l</sup> The two sets are not independent samples: 57 of the VT-2005-matched set’s 60 pairs also appear in the broad set at 298 K. Agreement between the columns is a consistency check and not corroboration, and the two are never combined.

conformer-dependent—the documented worst case for a model that scores one gas-phase conformer as a bag of independent surface segments with no memory of connectivity. (3) *mono-alcohol*, exactly one O–H, exactly one O, no N, no halogen (35 / 17), the single-donor self-associating chemistry the 2002 H-bond term was fitted on. (4) *N–H protic* (33 / 33): formamide, *N*-methylformamide, primary and secondary amines, piperidine—nitrogen donors, weaker and differently shaped than O–H and lumped with them under one shared  $\sigma_{\text{hb}}$  threshold. (5) *aryl ether* (13 / 2), anisole and phenetole, tested *before* the general acceptor class so as not to be swallowed by it: lone pairs delocalised into the ring put the acceptor strength far below a dialkyl ether’s, where a single cut-off is at its crudest. (6) *aprotic acceptor* (57 / 57): carbonyl, nitrile, azine nitrogen, ether oxygen, nitro or tertiary-amine nitrogen, with no donor—pentyl acetate, pyridine and four methylpyridines, acetone, acetonitrile.

(7) *halogenated* (30 / 7), any C–F/Cl/Br/I and no donor: large soft polarisable surfaces against a kernel with no dispersion term and one effective contact area. (8) *aromatic hydrocarbon* (14 / 9), a quadrupolar  $\pi$  face with  $\sigma$  near zero over the ring, and (9) *aliphatic hydrocarbon* (2 / 2), pure dispersion,  $\sigma \approx 0$  everywhere, so the residual term nearly vanishes and the prediction is almost entirely combinatorial. The nine are exhaustive and disjoint over the set’s 36 solvents. Ether, nitro and tertiary-amine tests sit in the cascade because the same cascade is applied to the *solutes*, where those groups occur; no solvent reaches them.

*Two further granularities, so that the granularity is not itself a choice.* Coarse: donor-and-acceptor oxygen (classes 1–3,  $n=328$ ), donor nitrogen (4, 33), acceptor-only (5–6, 70), halogenated non-H-bonding (7, 30), hydrocarbon non-H-bonding (8–9, 16). Fine: amide N–H 25 against amine N–H 8; ester 30, azine 23, ketone or aldehyde 2, nitrile 2; halogenated

Table S9: Activity error (MSE, squared  $\ln \gamma_2^\infty$  units) of the three COSMO-SAC arms, and the block split that reads the 2002  $\rightarrow$  2010 reversal against profile database and temperature held fixed.

| (a) Arm <sup>a</sup>                            | VT-2005-matched (60), UD <sup>b</sup>   |                  | broad IDAC (477), UD <sup>b</sup> |                         |
|-------------------------------------------------|-----------------------------------------|------------------|-----------------------------------|-------------------------|
|                                                 | full <sup>b</sup>                       | res <sup>b</sup> | full <sup>b</sup>                 | res <sup>b</sup>        |
| 2002, all channels                              | 1.757                                   | 1.436            | 1.911                             | 1.902                   |
| 2010, dispersion off                            | 0.765                                   | 0.831            | 1.978                             | 2.893                   |
| 2010, dispersion on                             | 0.785 <sup>e</sup>                      | — <sup>d</sup>   | 1.855                             | —                       |
| $P$ (2010 better) <sup>c</sup> , dispersion off | >0.95                                   | —                | $\approx 0.6$                     | $\approx 0.2$           |
| 90% margin <sup>c</sup>                         | [+0.06, +2.18]                          | —                | [−1.95, +0.93]                    | [−3.06, +0.70]          |
| $P$ (2010 better) <sup>c</sup> , dispersion on  | —                                       | —                | $\approx 0.6$                     | —                       |
| 90% margin <sup>c</sup>                         | —                                       | —                | [−2.24, +1.31]                    | —                       |
| (b) Block (UD profiles, full convention)        | 2002 $\rightarrow$ 2010, dispersion off |                  | $P^c$                             | 90% margin <sup>c</sup> |
| shared with the VT-2005-matched set (57)        | 1.783 $\rightarrow$ 0.767               |                  | >0.95                             | [+0.06, +2.36]          |
| complementary (420)                             | 1.928 $\rightarrow$ 2.143               |                  | $\approx 0.5$                     | [−2.41, +0.87]          |
| ... of these, at 298 K (80)                     | 0.796 $\rightarrow$ 1.284               |                  | $\approx 0.3$                     | —                       |

<sup>a</sup> The three arms, named here and in the text: *2002, all channels*, COSMO-SAC-2002 in our NIST-validated layer summing the segment contributions of all three profile channels; *2010, dispersion off* and *2010, dispersion on*, COSMO-SAC-2010 without and with the dispersion term (§3.5.3). <sup>b</sup> *full* / *res* as in Table S8. *Both* sets are scored on UD profiles here, the VT-2005-matched set included—Table S8 scores the VT-2005-matched set on VT-2005 instead, and at 30 segment iterations against the 300 behind the 1.757 here, so its VT-2005/full 1.80 and that 1.757 part on two axes. At 30 iterations on both databases the two part by 0.029 on that set’s full-convention MSE and the count carries a further 0.014. The two tabulations are not one calculation—no molecule of the 44 agrees to  $10^{-3} \text{ Å}^2$  per bin—but only two are a different calculation outright, naphthalene and butyl acetate at 12.4 and  $2.7 \text{ Å}^2$  of worst-bin deviation against 0.058 for the other 42. The 0.029 therefore sizes a re-tabulation and not a displaced reference (§3.5.3). Within a set, each arm uses the  $\sigma$ -profile averaging its own model prescribes (Mullins for 2002, Hsieh for 2010). <sup>c</sup>  $P$  is the two-way cluster-bootstrap frequency that the 2010 arm is the better one, with the 90% margin on the paired squared-error difference beneath it. <sup>d</sup> A dash marks a cell the deposited artifacts do not carry. <sup>e</sup> This one cell is  $n=57$ , not 60: cCOSMO tabulates no dispersion  $\epsilon$  for the three bromine- or iodine-containing VT-2005-matched pairs. On those common 57 the *dispersion off* arm scores 0.767, so the 0.765-versus-0.785 ordering is not an artifact of the differing  $n$ , and no claim rests on that ordering (§3.5.3).

aliphatic 18 against halogenated aromatic 12. All three levels are in the deposited table, and two of their cells are worth naming. The coarse *acceptor-only* class, which merges the aryl ethers into the aprotic acceptors, carries a *positive* margin (+1.43 residual-only at  $P \approx 0.7$ ) where its dominant child is negative—the composition effect of §3.5.1 reappearing one level up, on 13 rows. At the fine level the azine sub-family changes sign between conventions at  $n=23$ . Neither overturns a headline-level verdict, and both are why the granularity is reported rather than chosen.

*Axis 2, solute role.* Water as the dilute solute (37 / 6) against organic solutes (440 / 179), reported alone and crossed with axis 1. Water as the solute is a different physical situation from water as the solvent: an isolated  $\text{H}_2\text{O}$  in an organic continuum has all four H-bond

sites unsatisfied by like molecules, a state the segment-activity expression is not fitted near and the one COSMO-SAC-2002’s documented weakness is specific to. The same cascade applied to the solute gives a secondary solute-family axis, also deposited, because a practitioner deciding where to ground needs both halves of a pair.

*Two mechanical points.* The solvent classifier takes one argument, a SMILES string, and the whole cascade is a sequence of SMARTS tests: it never reads  $m$ ,  $g$ , the squared error, MSE,  $B_{\text{insuff}}$ , the margin, the source DOI or the bootstrap. That is a property of the code and not an assurance. Second, the University of Delaware compound list stores formamide and *N*-methylformamide as imidic-acid tautomers ( $\text{N}=\text{CO}$ ,  $\text{CN}=\text{CO}$ ); RDKit’s canonical tautomer restores the amide so that the N–H donor defin-

Table S10: Upper bounds on the input-insufficiency term  $B_{\text{insuff}}$ , on both sets taken whole. Columns: the value, what each estimator fits, and the note qualifying it. Which estimators are listed and which are not: note *a*.

| Estimator <sup>a</sup>                                                                                            | $B_{\text{insuff}}^{\text{up}}$ | fits <sup>b</sup> | note <sup>c</sup>           |
|-------------------------------------------------------------------------------------------------------------------|---------------------------------|-------------------|-----------------------------|
| <i>Broad IDAC set</i> ( $n=477$ , UD profiles). MSE = 1.902; threshold MSE/2 = 0.951; headline margin +0.51.      |                                 |                   |                             |
| Binning bound, 8 bins, unbiased variance <b>[headline]</b>                                                        | 0.697                           | no model (○)      |                             |
| Binning bound, 8 bins, maximum-likelihood variance                                                                | 0.685                           | no model (○)      |                             |
| Binning bound, 3 bins (coarsest)                                                                                  | 0.901                           | no model (○)      |                             |
| Binning bound, 4 bins (least favourable)                                                                          | 0.948                           | no model (○)      | tightest margin on this set |
| Binning bound, 8 bins, full convention (also valid) <sup>d</sup>                                                  | 0.482                           | no model (○)      |                             |
| RF out-of-fold, folds blocked on the molecule pair                                                                | 0.882                           | out-of-fold (◇)   |                             |
| ridge out-of-fold, folds blocked on the molecule pair                                                             | 1.467                           | out-of-fold (◇)   | above thr.                  |
| RF out-of-fold, folds blocked on the pair, conditioning on $(z^*, T)$                                             | 0.823                           | out-of-fold (◇)   |                             |
| RF out-of-fold, folds blocked on the solute                                                                       | 2.043                           | out-of-fold (◇)   | vacuous                     |
| RF out-of-fold, random folds                                                                                      | 0.146                           | out-of-fold (◇)   | leaky                       |
| ridge out-of-fold, random folds                                                                                   | 0.801                           | out-of-fold (◇)   | leaky                       |
| $\mathbb{E}[\text{Var}(m \mid \text{pair})]$ , binning-free, on the 359 rows that support it                      | 0.046                           | no model (○)      | single-source               |
| <i>VT-2005-matched</i> ( $n=60$ , VT-2005 profiles). MSE = 1.472; threshold MSE/2 = 0.736; headline margin +0.05. |                                 |                   |                             |
| Binning bound, 8 bins, unbiased variance <b>[headline]</b>                                                        | 0.713                           | no model (○)      |                             |
| Binning bound, 8 bins, maximum-likelihood variance                                                                | 0.618                           | no model (○)      |                             |
| Binning bound, 3 bins (coarsest, least favourable)                                                                | 1.000                           | no model (○)      | above thr.                  |
| Binning bound, 4 bins                                                                                             | 0.866                           | no model (○)      | above thr.                  |
| Binning bound, 8 bins, full convention (also valid) <sup>d</sup>                                                  | 0.608                           | no model (○)      |                             |
| $k$ NN-in- $z^*$ (convention-independent; least conservative)                                                     | 0.400                           | out-of-fold (◇)   | deflated                    |
| $k$ NN, neighbour from a distinct solute                                                                          | 0.448                           | out-of-fold (◇)   |                             |
| $k$ NN, neighbour from a distinct solvent                                                                         | 1.458                           | out-of-fold (◇)   | above thr.                  |
| $k$ NN, neighbour from a distinct pair                                                                            | 1.831                           | out-of-fold (◇)   | vacuous                     |
| RF out-of-fold, random folds                                                                                      | 0.621                           | out-of-fold (◇)   | deflated                    |
| ridge out-of-fold, random folds                                                                                   | 0.625                           | out-of-fold (◇)   |                             |
| RF out-of-fold, folds blocked on both molecules                                                                   | 3.653                           | out-of-fold (◇)   | vacuous                     |
| ridge out-of-fold, folds blocked on both molecules                                                                | divergent                       | out-of-fold (◇)   |                             |
| ridge out-of-fold, deliberately aggressive                                                                        | 0.877                           | out-of-fold (◇)   | above thr.                  |
| $\mathbb{E}[\text{Var}(m \mid \text{pair})]$ <sup>e</sup>                                                         | —                               | —                 | no repeated pairs           |

<sup>a</sup> One inclusion rule governs both sets, in two halves that differ in kind. The binning-bound half is a *selection* from bin grids too large to draw (56 cells on the broad set, 72 on the VT-2005-matched set; Tables S6 and S5): *within the residual-only convention and under the unbiased within-bin variance*, the headline cell, the *coarsest*, the *least favourable* and every such cell above the threshold, with the headline cell’s maximum-likelihood variant and its full-convention counterpart listed beside it. Cells above the threshold under the other variance or the other convention are not listed; they are in those two grids, where Table S5 sets the negative-margin ones in bold. The estimator half is every out-of-fold estimator that set’s audit artifact records, in every fold design it records, vacuous designs included—the per-set guards on that are stated with the regeneration command above. <sup>b</sup> What the estimator fits to the data. *no model* (○) for the binning-bound cells and for  $\mathbb{E}[\text{Var}(m \mid \text{pair})]$ , *out-of-fold* (◇) for a fitted estimator. The binning bound, fitting nothing, is unaffected by any fold design. <sup>c</sup> *above thr.*, the bound exceeds that set’s separation threshold MSE/2, so that cell does not separate; *vacuous*, it exceeds the total MSE, which two of the VT-2005-matched set’s estimators and one of the broad set’s do; *leaky*, random folds leave a held-out row’s own molecule pair in the training fold at a neighbouring temperature; *deflated*, biased low by the crossed design, in which two pairs sharing a molecule agree on 51 of the 102  $z^*$  coordinates; *single-source*, every multi-row pair supporting the bound comes from one publication, so it is valid on those rows but samples no between-laboratory component and is not promoted. <sup>d</sup> Computed under the other combinatorial rule and reported against the residual-only-convention threshold beside it. It is valid there, and it is the only entry on either set that is not residual-only. <sup>e</sup> The estimator that set cannot support.

ing their class is visible, and that normalisation is applied to every molecule alike rather than carved for two of them.

### What sets the level of the bound, and why it barely moves between strata.

From above,  $B_{\text{insuff}}^{\text{up}}$  is capped by the within-stratum label variance: the law of total variance on a single bin is  $\text{Var}(m)$ , and refining the bins can only lower it, so under the maximum-likelihood within-bin variance  $B_{\text{insuff}}^{\text{up}} \leq \text{Var}(m)$  identically—which the deposited maximum-likelihood column satisfies in every stratum (0.12 against 0.80 on the aprotic acceptors, 0.10 against 3.07 on the glycol ethers, 0.20 against 1.42 on water, 0.22 against 5.61 on the N–H protics, 0.44 against 3.66 on the halogenated, 0.58 against 0.60 on the mono-alcohols). The unbiased-variance estimate the manuscript reports can exceed  $\text{Var}(m)$  once bins hold only a few rows, and does so exactly once: the mono-alcohols, 0.733 against 0.604 at  $n=35$  and four rows a bin, where the artifact raises its own exceedance flag. That flag fires only on a stratum already declared not boundable, which is the boundability criterion doing its job rather than an unexplained anomaly—and that same value is what sets the  $\times 6.8$  span quoted in §3.5.1.

From below the received story needs a correction.  $z^*$  is a deterministic function of the pair, so  $\mathbb{E}[\text{Var}(m \mid \text{pair}, T)]$  is  $B_{\text{insuff}}$  on rows that repeat; but no two of the 477 share a pair *and* a temperature, so pure replicate disagreement is not measurable on this set. The coarser  $\mathbb{E}[\text{Var}(m \mid \text{pair})]$ , which additionally absorbs the whole within-pair temperature spread, is 0.0003 on the mono-alcohols, 0.037 on water, 0.044 on the glycol ethers, 0.075 on the halogenated and 0.046 over the 359 rows that support it on the whole set—one to two orders of magnitude below the binning bound of 0.11–0.73. The level of  $B_{\text{insuff}}^{\text{up}}$  here is therefore not set by label noise but by the residual spread of  $m$  inside a quantile bin of a scalar, which is a property of the estimator rather than of the chemistry. That is the sharper form of the structural claim, and the reason the bound is so nearly stratum-independent while MSE is free to move by a factor of 25 to 51.

## S3.6 Two nulls: one for the chemical partition, one for the fifty-nine-fold search

*The structural claim as a test.* 2000 random partitions of the 477 rows into blocks of exactly the observed class sizes, at the same estimator cell and under the same boundability filter. Deployed convention: observed MSE range 2.163 against a null median of 0.405, no draw reaching it; observed  $B_{\text{insuff}}^{\text{up}}$  range 0.107 against 0.165,  $p = 0.26$ ; observed ratio 20.2 against 2.46,  $p = 0.02$ . Full convention: 2.353 against 0.337, again no draw reaching it; 0.058 against 0.133,  $p = 0.12$ ; ratio 40.5 against 2.48,  $p = 0.003$ . The chemical partition separates the error far beyond chance and separates the bound no more than chance, which is what makes an aggregate margin a composition statistic.

*A multiplicity null for the search itself.* That test asks whether the chemical partition separates error; it does not ask how often a 59-fold search certifies a stratum by chance, and the admissibility rule is a stability filter rather than a multiplicity correction. The missing null is this one. Each of the 36 solvents keeps its rows and receives the label *triple* (nine-class, coarse, fine family) of a uniformly drawn other solvent, and each of the 78 solutes another solute’s (family, role) pair, the triples moving together so that the nesting between granularities is preserved exactly; the stratum family is rebuilt from the permuted labels by the same code and put through the same rule—boundability, leave-one-publication-out in all four unit  $\times$  convention cells, positivity in all four, and clause (c) of §2.5, the reduction to distinct row sets together with its asymmetric residual test. Preserved: the number of strata, the number of molecules per stratum, the containment lattice, and every row’s  $m$ ,  $g$ , source and pair. Destroyed: the association between chemistry and error. Every frequency here is that null on all 477 rows under the binning bound, the row set and the estimator the map itself deploys; §S3.12 re-runs it on 473 and its figures differ for that reason alone. Over 2000 draws the chemical map’s six admissible strata

are above chance ( $p = 0.01$ ) and its two admissible positive row sets are not ( $p = 0.286$ ): a chemically blind relabelling certifies at least one row set in 58% of draws and two or more in 28.6%. What the search returns is therefore not rare; its *size* is closer to being so. The largest certified margin reaches the observed +2.04 or more in 10.1% of all draws (17% of the draws that certify anything), against a median of +1.57 among those, a 95th percentile of +2.39 and a maximum of +3.14. The consequences for what may be claimed are in §3.5.1 and limitation (vi); in one line, the finding sits at about the ninetieth percentile of what this search returns under a chemistry-blind null, which is not multiplicity control at a conventional level.

*Two biases in the null, and what each does to a frequency.* Both are measured rather than argued, and neither is what one would guess. (1) *Provenance survives the permutation.* Permuting the chemistry labels does not leave the *provenance* structure alone—publications measure related sets of solvents—so we recorded what it does: a label-permuted map has *more* boundable cells than the chemical one (62 against 50, median over draws) and each passes the deletion test *less* often (0.56 against 0.70). Those two halves run in opposite directions and are not netted here, so every frequency in this paragraph is unsigned: it is neither an upper nor a lower bound on a chance rate, and the 10.1% in particular cannot be read as conservative. (2) *A positive aggregate seeds every draw.* Because the set taken whole has a positive margin at this cell, a randomly assembled stratum inherits a positive margin on average; a null frequency near one half for “certifies something” is expected and is not evidence that the rule is loose. This one runs in a single direction, and it inflates the 58% rather than the 10.1%. The exactly matched null would permute chemistry and provenance jointly, which this design cannot do because the two are confounded in these data. The draw-by-draw record is deposited.

### S3.7 The out-of-sample test, declared before it was run

A selection objection is settled by rows the search never saw. The only such rows this work holds are the PGL 6th-edition IDAC database scored by the deployed closure on VT-2005 profiles (Table S27): 7889 records over 2279 pairs, another group’s compilation, sixteen times the broad set’s rows on a different profile database. Two other candidates were considered and rejected as not out of sample: the 14900-row pool the broad set was drawn from contains 184 neutral glycol-ether-solvent rows of which 182 are already in the broad set, its remaining glycol-ether-classified rows being hydroxy-functionalised ionic liquids with no reference profile and no scoreable closure; and the VT-2005-matched set is inside the search. Before any margin was computed on the PGL rows we fixed the row set (the same solvent-classifier cascade, unmodified, minus any pair shared with the broad set), the estimator cell (the headline one: row unit, residual-only convention, eight equal-count bins, unbiased within-bin variance), the admissibility rule (the manuscript’s, with the source axis declared to be the record file refined by the compilation label, which is the finest provenance these files carry and is coarser than the broad set’s DOI), and the three verdicts available—confirmed, not confirmed, and not testable, the last counting as not confirmed for what the title may say. The verdict was **not testable, hence not confirmed**. The cell the finding is about does not exist there: no glycol-ether solvent carries an organic solute in that database, its non-aqueous half listing 88 solvent molecules and no glycol ether. That emptiness needed no margin to see. The geometry of the candidate row sets—how many rows exist, which solvents and solutes they carry, which files they come from—was established before the declaration was written, because a test cannot be designed without knowing whether its data exist, and the declaration records the zero-row count for the exact analogue cell along with the two mismatches of the nearest one. What the pre-registration delivered on this cell is therefore the unavailabil-

ity of confirming data and not a replication that ran and failed; what it protected is the rule and the row set that would have decided the verdict had rows existed, fixed and staged before anything was scored. The nearest cell is 52 rows of water in seven glycol-ether and polyol solvents over seven pairs, drawn 50–2 from the two compilations of one record file, so the one deletion that matters leaves two rows, the fixed cell undefined and the sign untested—the same reason the rule refuses water-as-solute in sample. Its margin, which licenses nothing under the rule declared in advance, is +0.35 [+0.28, +0.58], against +2.04 in sample. Of the four solvent molecules in the in-sample cell only ethylene glycol appears; the three oligomers carrying 141 of its 182 rows do not. Item 7 of the declaration fixed the same cell over that database’s ten solvent classes as description and *not* as a test, so no verdict is read from Table S11 and no row of it is stated as a finding. It is printed whole, nothing dropped for being small or for being negative, because a ranking quoted in part is a different statement: it runs from –2.53 (water, 3074 rows, the largest stratum there) to +1.73 (aliphatic hydrocarbons, 875 rows). The same admissibility rule admits two of the ten—the mono-alcohols, whose +0.09 carries an interval covering zero, and water at –2.53—so no out-of-sample class both passes the rule and separates the two terms, and the classes with the larger positive margins are ones the rule refuses. The glycol ethers are not where this closure fails hardest out of sample, and the instrument behaves out of sample as §3.5.1 says it behaves in sample—a map, not a verdict. The deposited artifact carries the pre-declaration in full.

*The leave-one-out curves.* At the headline cell under the residual-only convention, baseline +0.509: leaving out 10.1021/acs.jced.7b00114 gives +0.184 and leaving out 10.1021/je020196d (105 rows) +0.419, while the remaining thirteen deletions run from +0.503 to +0.875. Under the full convention the same curve is flat, [+0.821, +1.073] against a baseline of +0.947, so the single-publication dependence is specific to the residual-only convention. All 185 leave-one-pair-out margins lie in [+0.422, +0.721]

with median +0.507 and none non-positive, the seven most damaging deletions being *five* of the six water-as-solute pairs interleaved with two glycol-ether ones; the sixth water pair ranks 178th of 185, so those six rows do not move the aggregate in one direction. Leave-one-solvent-class-out spans [+0.352, +1.156]—removing water raises it most, by +0.647, and removing the halogenated solvents lowers it most, by –0.157—so exactly one class of the nine moves the aggregate by more than the aggregate’s own distance from zero, and six of the other eight move it by less than 0.38. The span is wide because one class is; it is not evidence that the aggregate is broadly composition-sensitive, and the composition argument of §3.5.1 rests on the compounded cut and the null-partition test rather than on this curve. Per-stratum leave-one-source-out, and the VT-2005-matched set’s own curves, are in the artifact and summarised in §3.5.1–§3.5.2. Figure S3 draws twelve of those strata at the granularity the article reports, each with its bound, its total error, its margin and its share of the set’s squared error.

### S3.8 The map, stratum by stratum

The map runs over 59 strata: the solvent axis at three granularities and the solute axis at two, plus their cross-tab and the set as a whole, over the broad IDAC set’s 477 rows, 185 pairs, 78 solutes, 36 solvents and 15 source publications. Tables S12–S15 carry all 59 between them, so nothing is dropped for being small, and Table S18 carries on the same 59 the columns those four do not. Rows at different granularities, and rows on the two axes, can be the same measurements: a finer solvent class is a subset of a coarser one, so the fine table re-partitions measurements rather than adding new ones, 362 of the 1711 stratum pairs share rows, and every row that keeps its sign carries its overlap with the others. Ten of the map’s 43 row sets are row-identical under more than one name. A bootstrap seeded from the name would draw an independent replicate for each of them, and the resulting intervals would sit



Table S11: **The out-of-sample set, class by class:** every solvent class of the PGL 6th-edition set at the map’s own headline cell—eight equal-count bins of  $g(z^*)$ , unbiased within-bin variance, row unit, residual-only convention. Rows in descending margin; the ten classes carry all 7889 records. The last column gives, for each class, either what it shows or the clause of the admissibility rule it fails.

| Solvent class         | $n$  | pairs | sources <sup>a</sup> | margin | 90% margin <sup>b</sup> | boundable <sup>c</sup> | deletions <sup>d</sup> | admissible | what it shows, or the test it fails                                                                                                                                                                             |
|-----------------------|------|-------|----------------------|--------|-------------------------|------------------------|------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| aliphatic hydrocarbon | 875  | 358   | 3                    | +1.73  | [−0.18, +5.26]          | yes                    | 3/3                    | no         | the margin changes sign when one source is deleted: WinOrg:J takes +1.726 → −0.151                                                                                                                              |
| halogenated           | 428  | 211   | 3                    | +0.67  | [−0.07, +2.23]          | yes                    | 3/3                    | no         | the margin changes sign when one source is deleted: WinOrg:J takes +0.670 → −0.275                                                                                                                              |
| N–H protic            | 368  | 119   | 3                    | +0.60  | [+0.22, +1.50]          | yes                    | 2/3                    | no         | 1 of 3 deletions leave fewer than 16 rows and the fixed cell is then undefined (Lazzaroni), so the sign is not verifiable                                                                                       |
| glycol ether          | 52   | 7     | 2                    | +0.35  | [+0.29, +0.57]          | yes                    | 1/2                    | no         | 1 of 2 deletions leave fewer than 16 rows and the fixed cell is then undefined (WinOrg:J), so the sign is not verifiable                                                                                        |
| unclassified          | 143  | 53    | 2                    | +0.10  | [−0.13, +1.15]          | yes                    | 1/2                    | no         | 1 of 2 deletions leave fewer than 16 rows and the fixed cell is then undefined (Lazzaroni), so the sign is not verifiable                                                                                       |
| mono-alcohol          | 1027 | 344   | 3                    | +0.09  | [−0.06, +0.32]          | yes                    | 3/3                    | yes        | boundable, and the sign holds when any one source is deleted, so the rule admits it; the margin is positive but its interval covers zero, so nothing is separated at this cell                                  |
| aprotic acceptor      | 1577 | 714   | 3                    | +0.02  | [−0.19, +0.31]          | yes                    | 3/3                    | no         | the margin changes sign when one source is deleted: WinOrg:J takes +0.019 → −0.005; WinOrg:T takes +0.019 → −0.114                                                                                              |
| aromatic hydrocarbon  | 344  | 133   | 3                    | −0.13  | [−0.28, +2.00]          | yes                    | 3/3                    | no         | the margin changes sign when one source is deleted: Lazzaroni takes −0.130 → +10.249                                                                                                                            |
| water                 | 3074 | 339   | 3                    | −2.53  | [−3.09, −1.77]          | yes                    | 3/3                    | yes        | boundable, and the sign holds when any one source is deleted, so the rule admits it; the margin is not positive, so the two terms are not separated here—a bound from above cannot show the inputs are at fault |
| aryl ether            | 1    | 1     | 1                    | —      | —                       | no                     | 0/1                    | no         | the fixed cell is undefined at $n = 1$ : no estimate                                                                                                                                                            |

<sup>a</sup> Source labels under the provenance axis declared before the run: the record file, refined by the compilation label inside the two Jaubert/TDE files (J Jaubert, T NIST TDE, Y Yaws). *WinOrg* is the water-in-organic file, *OrgInW* the organic-in-water one, *Lazzaroni* the third file, which carries no label and counts as one source. Six labels in all, against the broad set’s fifteen source DOIs: this axis is coarser than the in-sample one and the test it supports is therefore weaker, which the declaration states in advance.

<sup>b</sup> Two-way solute × solvent cluster bootstrap, 3000 draws, drawn separately at each class: the glycol-ether row is the same 52 rows as the test above and repeats its margin exactly, while its endpoints differ from that row’s in the last digits because the draw is a different one. This table keeps both printings and both second decimals, and at 3000 draws neither second decimal is determined—the per-cell precision rule of §S3.8 was applied to the map and not here, since no margin in this table is stated as a finding. <sup>c</sup>  $n \geq 40$ , so eight equal-count bins hold at least five rows. <sup>d</sup> How many of the class’s leave-one-source-out deletions can be run: a deletion leaving fewer than 16 rows leaves the fixed cell undefined, and a sign that cannot be checked reads as untested rather than as a pass.

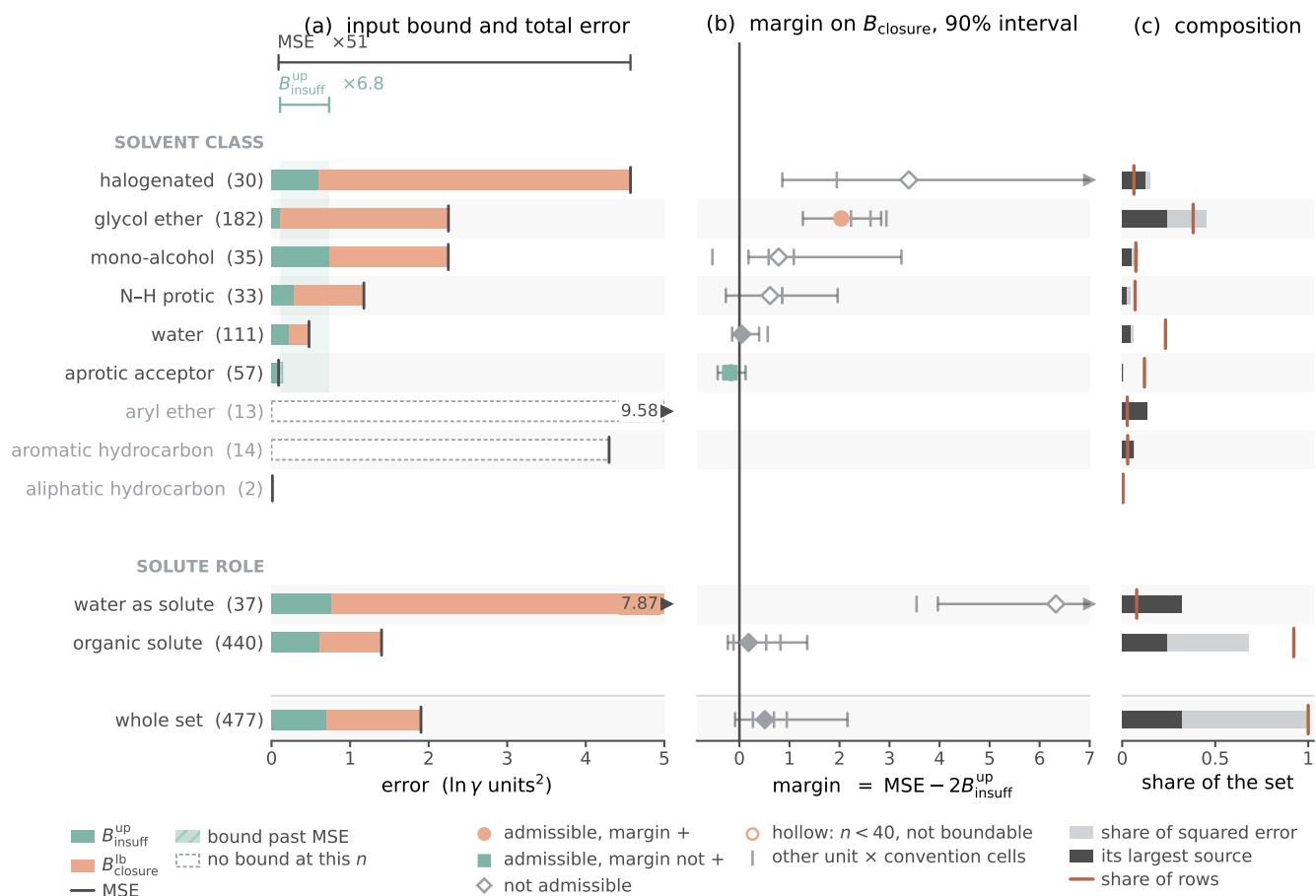


Figure S3: **Where the closure binds, and where the two terms do not separate**, by solvent class and by solute role, on the broad IDAC set (477 rows, 185 pairs, 78 solutes, 36 solvents, 15 source publications). Bars, points and shares are at the row unit under the residual-only convention, at one estimator cell for every stratum—eight equal-count bins of  $g(z^*)$  with the unbiased within-bin variance. **(a)** each stratum’s input-insufficiency bound and its total error; **(b)** the margin between them with its 90% two-way cluster-bootstrap interval, the other three unit  $\times$  convention cells as grey ticks; **(c)** each stratum’s share of the set’s squared error, the part carried by its largest source publication drawn dark. Twelve cells at the granularity reported here, drawn from the 59-stratum map of Tables S12–S15. Strata on the two axes overlap, so panel (c)’s shares do not sum down the column.

side by side in these tables as though they were separate measurements; the seed is a function of the row set, so one draw serves all of a row set’s names and each printed row carries the row-set identity the deposited record keys on. The  $\pm 0.02$  of Monte-Carlo spread at 3000 draws quoted for this cell belongs to this cell and is not a property of the map. The map’s own bootstrap draws 3000 resamples per row set, which does not determine a second decimal: over 200 seeds the Monte-Carlo spread of an endpoint any seed moves runs 0.0009 to 0.223, and 42 of the 46 endpoints drawn at that count stand

inside four Monte-Carlo standard errors of the rounding boundary their second decimal turns on—the closest reaching 0.8 and the median 0.1 of the four required. The interval column of Tables S12 to S15 is therefore not that output: every row set was re-drawn at a count chosen from its own Monte-Carlo error— $1.8 \times 10^5$  to  $1.2 \times 10^8$  draws—and prints at the precision that count supports, cell by cell. Note  $b$  of Table S12 gives the rule and the count at each precision. The whole-set row of Table S15 is the aggregate this work declines to state, printed with the rest.

Table S12: **The map, solvent axis, in full:** every solvent stratum of the broad IDAC set, including the ones the estimator cannot bound, at the pre-declared cell—eight equal-count bins of  $g(z^*)$ , unbiased within-bin variance, row unit, residual-only convention—applied identically to all of them. *full* is the same cell under the full combinatorial convention. Every margin here and in Tables S13–S15, and the MSE and bound behind it, is in squared  $\ln \gamma_2^\infty$  units. The last column states either what the row shows or the test it fails.

| Stratum                                                       | $n$ | pairs | sources <sup>a</sup> | margin | 90% margin <sup>b</sup>       | full  | cells <sup>c</sup> | what it shows, or the test it fails <sup>d</sup>                                                                                                                                                                                                   |
|---------------------------------------------------------------|-----|-------|----------------------|--------|-------------------------------|-------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Solvent class</i><br>glycol ether                          | 182 | 43    | 3 (P8, 0.53)         | +2.04  | [+1.25, +2.87]                | +2.23 | 4                  | the closure’s own error exceeds the bound on its inputs, and the sign holds when any one publication is deleted; the same rows as the two other glycol-ether cells, 59% shared with the alkanes                                                    |
| water                                                         | 111 | 15    | 3 (P14, 0.75)        | +0.05  | [−0.15, +0.39] <sup>0.7</sup> | +0.57 | 1                  | the margin changes sign when one publication is deleted: P2 takes +0.046 → −0.075; P14 takes +0.046 → −0.194                                                                                                                                       |
| aprotic acceptor                                              | 57  | 57    | 4 (P7, 0.32)         | −0.19  | [−0.4, +0.1]                  | −0.15 | 4                  | the sign holds under every deletion, but the margin is not positive, so the two terms are not separated here—a bound from above cannot show the inputs are at fault; the same rows as the other aprotic-acceptor cell, 23% shared with the alkanes |
| mono-alcohol                                                  | 35  | 17    | 2 (P3, 0.55)         | +0.78  | [+0.28, +3.32] <sup>11</sup>  | +1.09 | 0                  | $n < 40$ : too few rows for eight bins of five; too few rows remain after deleting P3 (1 of 2 deletions), so the sign is untested there                                                                                                            |
| N–H protic                                                    | 33  | 33    | 3 (P11, 0.62)        | +0.61  | [−0.3, +2.0]                  | +0.86 | 0                  | $n < 40$ : too few rows for eight bins of five; too few rows remain after deleting P12 (1 of 3 deletions), so the sign is untested there                                                                                                           |
| halogenated                                                   | 30  | 7     | 2 (P13, 0.83)        | +3.39  | [+1.02, +8.43] <sup>22</sup>  | +1.95 | 0                  | $n < 40$ : too few rows for eight bins of five; too few rows remain after deleting P13 (1 of 2 deletions), so the sign is untested there                                                                                                           |
| aromatic hydrocarbon                                          | 14  | 9     | 2 (P13, 0.88)        | —      | —                             | —     | 0                  | $n < 40$ : too few rows for eight bins of five; no estimate at this $n$                                                                                                                                                                            |
| aryl ether                                                    | 13  | 2     | 1 (P13, 1.00)        | —      | —                             | —     | 0                  | $n < 40$ : too few rows for eight bins of five; no estimate at this $n$                                                                                                                                                                            |
| aliphatic hydrocarbon                                         | 2   | 2     | 2 (P10, 0.86)        | —      | —                             | —     | 0                  | $n < 40$ : too few rows for eight bins of five; no estimate at this $n$                                                                                                                                                                            |
| <i>Solvent class, coarse</i><br>H-bond donor and acceptor (O) | 328 | 75    | 8 (P8, 0.40)         | +0.53  | [−0.2, +2.2]                  | +1.34 | 3                  | the margin changes sign when one publication is deleted: P15 takes +0.527 → −0.417                                                                                                                                                                 |
| H-bond acceptor only                                          | 70  | 59    | 5 (P13, 0.96)        | +1.43  | [−0.30, +6.21] <sup>15</sup>  | +0.77 | 0                  | the margin changes sign when one publication is deleted: P13 takes +1.434 → −0.192                                                                                                                                                                 |
| H-bond donor (N)                                              | 33  | 33    | 3 (P11, 0.62)        | +0.61  | [−0.3, +2.0]                  | +0.86 | 0                  | $n < 40$ : too few rows for eight bins of five; too few rows remain after deleting P12 (1 of 3 deletions), so the sign is untested there                                                                                                           |
| halogenated, non-H-bonding                                    | 30  | 7     | 2 (P13, 0.83)        | +3.39  | [+1.02, +8.43] <sup>22</sup>  | +1.95 | 0                  | $n < 40$ : too few rows for eight bins of five; too few rows remain after deleting P13 (1 of 2 deletions), so the sign is untested there                                                                                                           |
| hydrocarbon, non-H-bonding                                    | 16  | 11    | 3 (P13, 0.88)        | +2.99  | [−0.07, +8.78] <sup>13</sup>  | +1.84 | 0                  | $n < 40$ : too few rows for eight bins of five; too few rows remain after deleting P10, P11, P13 (3 of 3 deletions), so the sign is untested there                                                                                                 |

<sup>a</sup> Number of source publications, then the largest one and its share of that stratum’s squared error. The keys, used in all five map tables: P1 10.1016/j.fluid.2013.06.028, P2 10.1016/j.fluid.2015.01.010, P3 10.1016/j.fluid.2015.11.037, P4 10.1016/j.fluid.2016.08.030, P5 10.1016/j.fluid.2016.10.009, P6 10.1016/j.jct.2013.05.006, P7 10.1016/j.jct.2015.12.034, P8 10.1016/j.jct.2016.10.013, P9 10.1016/j.jct.2018.05.003, P10 10.1016/j.jct.2019.05.011, P11 10.1016/j.tca.2012.02.005, P12 10.1016/j.tca.2015.11.010, P13 10.1021/acs.jced.7b00114, P14 10.1021/acs.jced.9b00726, P15 10.1021/je020196d. <sup>b</sup> Two-way solute × solvent cluster bootstrap, 5th and 95th percentiles. Row-identical strata are one row set: drawn once and printed once, under the row set’s first name above and restated under the rest (10 of 43 row sets, 16 rows). Each prints at the precision its own draw count supports, decided cell by cell over 200 seeds and verified on the run that bought it. 2 carry two decimals (organic solute; glycol ether, the latter at  $1.2 \times 10^8$  draws) and 9 carry one, both endpoints of each standing four Monte-Carlo standard errors clear of the nearest rounding boundary at that precision. For 11 no precision this table can print is determined: those carry two decimals with a superscript giving their endpoints’ Monte-Carlo standard deviation in units of that last digit, and the superscript is the warning rather than a refinement. A further 2 (aldehyde; halogenated | water as solute) no seed moves at all—each has one cluster on one of its two axes, so the resample is one-way over three clusters and about 10 distinct compositions exist: a bootstrap with nothing left to resample, not a converged one. The per-cell verdict, its draw count and its cost at either precision are on the deposited record (§S3.8). <sup>c</sup> In how many of the four unit × convention cells the stratum is both boundable ( $n \geq 40$ , so eight bins hold five rows) and keeps its margin’s sign under deletion of each contributing publication; the rule of §2.5 requires all four, which six of the 59 strata—three distinct row sets—satisfy. <sup>d</sup> The condition the row is in, not a label for it. A one-source cell cannot be tested and a deletion leaving fewer than 16 rows leaves the cell undefined, so both read as untested rather than as passes. A margin is one-sided: positive puts  $B_{\text{closure}} > B_{\text{insuff}}$  on those rows at that cell, non-positive is failure to separate and licenses nothing (Table S3, note *g*). Every row here is a cell of a 59-fold search, and the row that passes is not separated from it: a chemistry-blind relabelling of the same molecules, on all 477 rows, certifies at least one row set in 58% of 2000 draws and a margin of +2.04 or more in 10.1% (§S3.6, where the null’s own two biases are given).

Table S13: **The map, solvent axis at the fine granularity**—the third solvent granularity, at the same estimator cell and under the same rule as Table S12, whose notes apply here unchanged.

| Stratum                                     | $n$ | pairs | sources <sup>a</sup> | margin | 90% margin <sup>b</sup>       | full  | cells <sup>c</sup> | what it shows, or the test it fails <sup>d</sup>                                                                                                                                                |
|---------------------------------------------|-----|-------|----------------------|--------|-------------------------------|-------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Solvent family, fine</i><br>glycol ether | 182 | 43    | 3 (P8, 0.53)         | +2.04  | [+1.25, +2.87]                | +2.23 | 4                  | the closure’s own error exceeds the bound on its inputs, and the sign holds when any one publication is deleted; the same rows as the two other glycol-ether cells, 59% shared with the alkanes |
| water                                       | 111 | 15    | 3 (P14, 0.75)        | +0.05  | [−0.15, +0.39] <sup>0.7</sup> | +0.57 | 1                  | the margin changes sign when one publication is deleted: P2 takes +0.046 → −0.075; P14 takes +0.046 → −0.194                                                                                    |
| mono-alcohol                                | 35  | 17    | 2 (P3, 0.55)         | +0.78  | [+0.28, +3.32] <sup>11</sup>  | +1.09 | 0                  | $n < 40$ : too few rows for eight bins of five; too few rows remain after deleting P3 (1 of 2 deletions), so the sign is untested there                                                         |
| ester                                       | 30  | 30    | 1 (P9, 1.00)         | −0.06  | [−0.1, +0.0]                  | −0.05 | 0                  | $n < 40$ : too few rows for eight bins of five; one source (P9), so there is no deletion to run and the sign was never put at risk                                                              |
| amide N–H                                   | 25  | 25    | 2 (P12, 0.96)        | −0.24  | [−0.3, +0.8]                  | −0.02 | 0                  | $n < 40$ : too few rows for eight bins of five; too few rows remain after deleting P12 (1 of 2 deletions), so the sign is untested there                                                        |
| azine                                       | 23  | 23    | 2 (P11, 0.58)        | −0.05  | [−0.15, +0.14] <sup>0.9</sup> | +0.02 | 0                  | $n < 40$ : too few rows for eight bins of five; too few rows remain after deleting P5 (1 of 2 deletions), so the sign is untested there                                                         |
| halogenated aliphatic                       | 18  | 5     | 2 (P3, 0.87)         | +0.11  | [−0.2, +2.3]                  | −0.02 | 0                  | $n < 40$ : too few rows for eight bins of five; too few rows remain after deleting P3, P13 (2 of 2 deletions), so the sign is untested there                                                    |
| aromatic hydrocarbon                        | 14  | 9     | 2 (P13, 0.88)        | —      | —                             | —     | 0                  | $n < 40$ : too few rows for eight bins of five; no estimate at this $n$                                                                                                                         |
| aryl ether                                  | 13  | 2     | 1 (P13, 1.00)        | —      | —                             | —     | 0                  | $n < 40$ : too few rows for eight bins of five; no estimate at this $n$                                                                                                                         |
| halogenated aromatic                        | 12  | 2     | 1 (P13, 1.00)        | —      | —                             | —     | 0                  | $n < 40$ : too few rows for eight bins of five; no estimate at this $n$                                                                                                                         |
| amine N–H                                   | 8   | 8     | 1 (P11, 1.00)        | —      | —                             | —     | 0                  | $n < 40$ : too few rows for eight bins of five; no estimate at this $n$                                                                                                                         |
| aliphatic hydrocarbon                       | 2   | 2     | 2 (P10, 0.86)        | —      | —                             | —     | 0                  | $n < 40$ : too few rows for eight bins of five; no estimate at this $n$                                                                                                                         |
| ketone/aldehyde                             | 2   | 2     | 1 (P7, 1.00)         | —      | —                             | —     | 0                  | $n < 40$ : too few rows for eight bins of five; no estimate at this $n$                                                                                                                         |
| nitrile                                     | 2   | 2     | 1 (P7, 1.00)         | —      | —                             | —     | 0                  | $n < 40$ : too few rows for eight bins of five; no estimate at this $n$                                                                                                                         |

### What is measured and not established.

Water carried as the dilute solute is 37 rows and six pairs from 10.1021/acs.jced.7b00114 alone, which supplies 100% of its rows and 100% of its squared error: at  $n=37$  it is not boundable, and with one publication there is no deletion to run, so its +6.33 was never put at risk and is not evidence about water solutes. The halogenated solvents (30 rows, +3.39) are not boundable either, and 83% of that cell’s squared error is the same publication; deleting it leaves 12 rows, fewer than the fixed cell needs, so the sign cannot be verified. Neither may be used to explain a reversal. The coarse acceptor-only class, whose +1.43 this table names, is 96% carried by that publication’s 13 aryl-ether rows and goes to −0.19 when it is removed. Water as a *solvent* (+0.05 at  $P \approx 0.8$  residual-only, +0.57 full) flips under two of its three deletions; the coarse donor-and-acceptor-

oxygen class—“the hydroxylic solvents”—flips to −0.42 when 10.1021/je020196d is removed, so it is the glycol-ether sub-class and not the hydroxylic solvents as a class that carries the finding; the organic-solute subset flips to −0.09 under the same deletion; the mono-alcohols (35 rows, +0.78) and the N–H protics (33 rows, +0.61) are unboundable and each has a deletion that cannot be run; the aryl ethers are 13 rows from that same publication and carry no estimate at all, as do the aromatic hydrocarbons (14 rows) and the aliphatic hydrocarbons (2); and the ester (30 rows, one publication, −0.06) and azine (23 rows, one deletion unverifiable) sub-families, which the acceptor-only reading would lean on, are inadmissible on the same grounds. Of the 59 strata only 15 are boundable at all, and 24 carry no estimate at any  $n$  this set supplies. The whole-set row is admissible at the headline cell and

Table S14: **The map, solute axis**—the solute half of the map, at the same estimator cell and under the same rule as Tables S12 and S13, whose notes apply here unchanged. Every row that keeps its sign carries its overlap with the others.

| Stratum               | <i>n</i> | pairs | sources <sup>a</sup> | margin | 90% margin <sup>b</sup>       | full  | cells <sup>c</sup> | what it shows, or the test it fails <sup>d</sup>                                                                                                                                                                                               |
|-----------------------|----------|-------|----------------------|--------|-------------------------------|-------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Solute role</i>    |          |       |                      |        |                               |       |                    |                                                                                                                                                                                                                                                |
| organic solute        | 440      | 179   | 14 (P8, 0.35)        | +0.18  | [−0.23, +1.35]                | +0.82 | 2                  | the margin changes sign when one publication is deleted: P15 takes +0.184 → −0.094                                                                                                                                                             |
| water as solute       | 37       | 6     | 1 (P13, 1.00)        | +6.33  | [+3.84, +9.50] <sup>18</sup>  | +3.54 | 0                  | <i>n</i> < 40: too few rows for eight bins of five; one source (P13), so there is no deletion to run and the sign was never put at risk                                                                                                        |
| <i>Solute family</i>  |          |       |                      |        |                               |       |                    |                                                                                                                                                                                                                                                |
| alkane                | 129      | 45    | 7 (P15, 0.53)        | +1.38  | [+0.8, +2.3]                  | +1.72 | 4                  | the sign holds under every deletion, but 84% of these rows are the glycol ethers' and the 21 outside give −0.22, so this margin does not stand outside the glycol ethers: demoted to them by clause (c); 10% shared with the aprotic acceptors |
| aromatic hydrocarbon  | 71       | 32    | 7 (P15, 0.53)        | −0.19  | [−1.0, +0.7]                  | −0.07 | 0                  | the margin changes sign when one publication is deleted: P6 takes −0.188 → +0.081                                                                                                                                                              |
| ketone                | 63       | 11    | 3 (P14, 1.00)        | −0.53  | [−0.85, +0.63] <sup>1.8</sup> | −0.33 | 0                  | too few rows remain after deleting P14 (1 of 3 deletions), so the sign is untested there                                                                                                                                                       |
| mono-alcohol          | 39       | 21    | 5 (P11, 0.46)        | +0.27  | [+0.09, +2.15] <sup>3.1</sup> | +0.44 | 0                  | <i>n</i> < 40: too few rows for eight bins of five; the sign holds, but all 5 deletions leave <i>n</i> < 40                                                                                                                                    |
| water                 | 37       | 6     | 1 (P13, 1.00)        | +6.33  | [+3.84, +9.50] <sup>18</sup>  | +3.54 | 0                  | <i>n</i> < 40: too few rows for eight bins of five; one source (P13), so there is no deletion to run and the sign was never put at risk                                                                                                        |
| alkene                | 32       | 16    | 4 (P8, 0.98)         | +2.29  | [+0.15, +3.40] <sup>3.1</sup> | +2.76 | 0                  | <i>n</i> < 40: too few rows for eight bins of five; too few rows remain after deleting P8 (1 of 4 deletions), so the sign is untested there                                                                                                    |
| aldehyde              | 29       | 3     | 1 (P2, 1.00)         | −0.20  | [−0.51, +0.13]                | +0.68 | 0                  | <i>n</i> < 40: too few rows for eight bins of five; one source (P2), so there is no deletion to run and the sign was never put at risk                                                                                                         |
| halogenated aliphatic | 27       | 9     | 3 (P3, 0.99)         | +1.06  | [+0.15, +2.00] <sup>2.4</sup> | +1.36 | 0                  | <i>n</i> < 40: too few rows for eight bins of five; too few rows remain after deleting P3 (1 of 3 deletions), so the sign is untested there                                                                                                    |
| amine N–H             | 12       | 12    | 1 (P11, 1.00)        | —      | —                             | —     | 0                  | <i>n</i> < 40: too few rows for eight bins of five; no estimate at this <i>n</i>                                                                                                                                                               |
| ether                 | 12       | 4     | 3 (P2, 0.98)         | —      | —                             | —     | 0                  | <i>n</i> < 40: too few rows for eight bins of five; no estimate at this <i>n</i>                                                                                                                                                               |
| halogenated aromatic  | 8        | 8     | 4 (P1, 0.75)         | —      | —                             | —     | 0                  | <i>n</i> < 40: too few rows for eight bins of five; no estimate at this <i>n</i>                                                                                                                                                               |
| azine                 | 6        | 6     | 1 (P11, 1.00)        | —      | —                             | —     | 0                  | <i>n</i> < 40: too few rows for eight bins of five; no estimate at this <i>n</i>                                                                                                                                                               |
| tertiary amine        | 4        | 4     | 1 (P11, 1.00)        | —      | —                             | —     | 0                  | <i>n</i> < 40: too few rows for eight bins of five; no estimate at this <i>n</i>                                                                                                                                                               |
| ester                 | 3        | 3     | 2 (P5, 0.56)         | —      | —                             | —     | 0                  | <i>n</i> < 40: too few rows for eight bins of five; no estimate at this <i>n</i>                                                                                                                                                               |
| alkyne                | 2        | 2     | 2 (P5, 0.86)         | —      | —                             | —     | 0                  | <i>n</i> < 40: too few rows for eight bins of five; no estimate at this <i>n</i>                                                                                                                                                               |
| nitrile               | 2        | 2     | 2 (P9, 0.98)         | —      | —                             | —     | 0                  | <i>n</i> < 40: too few rows for eight bins of five; no estimate at this <i>n</i>                                                                                                                                                               |
| nitro                 | 1        | 1     | 1 (P9, 1.00)         | —      | —                             | —     | 0                  | <i>n</i> < 40: too few rows for eight bins of five; no estimate at this <i>n</i>                                                                                                                                                               |

Table S15: **The map, the solvent  $\times$  solute cross-tab and the set as a whole**—the last two granularities, at the same estimator cell and under the same rule as Tables S12–S14, whose notes apply here unchanged. The last row is the set taken whole.

| Stratum                                              | $n$ | pairs | sources <sup>a</sup> | margin | 90% margin <sup>b</sup>       | full  | cells <sup>c</sup> | what it shows, or the test it fails <sup>d</sup>                                                                                                                                                                                                   |
|------------------------------------------------------|-----|-------|----------------------|--------|-------------------------------|-------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Solvent class <math>\times</math> solute role</i> |     |       |                      |        |                               |       |                    |                                                                                                                                                                                                                                                    |
| glycol ether   organic solute                        | 182 | 43    | 3 (P8, 0.53)         | +2.04  | [+1.25, +2.87]                | +2.23 | 4                  | the closure's own error exceeds the bound on its inputs, and the sign holds when any one publication is deleted; the same rows as the two other glycol-ether cells, 59% shared with the alkanes                                                    |
| water   organic solute                               | 111 | 15    | 3 (P14, 0.75)        | +0.05  | [−0.15, +0.39] <sup>0.7</sup> | +0.57 | 1                  | the margin changes sign when one publication is deleted: P2 takes +0.046 $\rightarrow$ −0.075; P14 takes +0.046 $\rightarrow$ −0.194                                                                                                               |
| aprotic acceptor   organic solute                    | 57  | 57    | 4 (P7, 0.32)         | −0.19  | [−0.4, +0.1]                  | −0.15 | 4                  | the sign holds under every deletion, but the margin is not positive, so the two terms are not separated here—a bound from above cannot show the inputs are at fault; the same rows as the other aprotic-acceptor cell, 23% shared with the alkanes |
| mono-alcohol   organic solute                        | 35  | 17    | 2 (P3, 0.55)         | +0.78  | [+0.28, +3.32] <sup>11</sup>  | +1.09 | 0                  | $n < 40$ : too few rows for eight bins of five; too few rows remain after deleting P3 (1 of 2 deletions), so the sign is untested there                                                                                                            |
| N–H protic   organic solute                          | 33  | 33    | 3 (P11, 0.62)        | +0.61  | [−0.3, +2.0]                  | +0.86 | 0                  | $n < 40$ : too few rows for eight bins of five; too few rows remain after deleting P12 (1 of 3 deletions), so the sign is untested there                                                                                                           |
| halogenated   water as solute                        | 18  | 3     | 1 (P13, 1.00)        | +5.84  | [+2.70, +9.43]                | +3.39 | 0                  | $n < 40$ : too few rows for eight bins of five; one source (P13), so there is no deletion to run and the sign was never put at risk                                                                                                                |
| aryl ether   water as solute                         | 13  | 2     | 1 (P13, 1.00)        | —      | —                             | —     | 0                  | $n < 40$ : too few rows for eight bins of five; no estimate at this $n$                                                                                                                                                                            |
| halogenated   organic solute                         | 12  | 4     | 1 (P3, 1.00)         | —      | —                             | —     | 0                  | $n < 40$ : too few rows for eight bins of five; no estimate at this $n$                                                                                                                                                                            |
| aromatic hydrocarbon   organic solute                | 8   | 8     | 1 (P11, 1.00)        | —      | —                             | —     | 0                  | $n < 40$ : too few rows for eight bins of five; no estimate at this $n$                                                                                                                                                                            |
| aromatic hydrocarbon   organic solute                | 6   | 1     | 1 (P13, 1.00)        | —      | —                             | —     | 0                  | $n < 40$ : too few rows for eight bins of five; no estimate at this $n$                                                                                                                                                                            |
| aliphatic hydrocarbon   organic solute               | 2   | 2     | 2 (P10, 0.86)        | —      | —                             | —     | 0                  | $n < 40$ : too few rows for eight bins of five; no estimate at this $n$                                                                                                                                                                            |
| <i>The set as a whole</i>                            |     |       |                      |        |                               |       |                    |                                                                                                                                                                                                                                                    |
| whole set                                            | 477 | 185   | 15 (P13, 0.32)       | +0.51  | [−0.1, +2.2]                  | +0.95 | 3                  | passes here and fails at the pair unit (residual-only convention), where deleting P13 takes +0.270 $\rightarrow$ −0.119                                                                                                                            |

Table S16: **The map: every stratum the fixed estimator can bound:** the 15 of the 59 with  $n \geq 40$ , at the pre-declared cell (eight equal-count bins of  $g(z^*)$ , unbiased within-bin variance, row unit, residual-only convention), applied identically to all of them. *full* is the same cell under the full combinatorial convention. Every margin, and the MSE and bound behind it, is in squared  $\ln \gamma_2^\infty$ . The last column states what the row shows or the test it fails. The other 44 strata, the 24 of them carrying no estimate at all included, are in §S3.8 at this same cell.

| Stratum                       | $n$ | pairs | sources <sup>a</sup> | margin | 90% margin <sup>b</sup>       | full  | cells <sup>c</sup> | what it shows, or the test it fails <sup>d</sup>                                                                                                                                                                                                   |
|-------------------------------|-----|-------|----------------------|--------|-------------------------------|-------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Solvent class</i>          |     |       |                      |        |                               |       |                    |                                                                                                                                                                                                                                                    |
| glycol ether                  | 182 | 43    | 3 (P8, 0.53)         | +2.04  | [+1.25, +2.87]                | +2.23 | 4                  | the closure’s own error exceeds the bound on its inputs, and the sign holds when any one publication is deleted; the same rows as the two other glycol-ether cells, 59% shared with the alkanes                                                    |
| water                         | 111 | 15    | 3 (P14, 0.75)        | +0.05  | [−0.15, +0.39] <sup>0.7</sup> | +0.57 | 1                  | the margin changes sign when one publication is deleted: P2 takes +0.046 → −0.075; P14 takes +0.046 → −0.194                                                                                                                                       |
| aprotic acceptor              | 57  | 57    | 4 (P7, 0.32)         | −0.19  | [−0.4, +0.1]                  | −0.15 | 4                  | the sign holds under every deletion, but the margin is not positive, so the two terms are not separated here—a bound from above cannot show the inputs are at fault; the same rows as the other aprotic-acceptor cell, 23% shared with the alkanes |
| <i>Solvent class, coarse</i>  |     |       |                      |        |                               |       |                    |                                                                                                                                                                                                                                                    |
| H-bond donor and acceptor (O) | 328 | 75    | 8 (P8, 0.40)         | +0.53  | [−0.2, +2.2]                  | +1.34 | 3                  | the margin changes sign when one publication is deleted: P15 takes +0.527 → −0.417                                                                                                                                                                 |
| H-bond acceptor only          | 70  | 59    | 5 (P13, 0.96)        | +1.43  | [−0.30, +6.21] <sup>15</sup>  | +0.77 | 0                  | the margin changes sign when one publication is deleted: P13 takes +1.434 → −0.192                                                                                                                                                                 |
| <i>Solvent family, fine</i>   |     |       |                      |        |                               |       |                    |                                                                                                                                                                                                                                                    |
| glycol ether                  | 182 | 43    | 3 (P8, 0.53)         | +2.04  | [+1.25, +2.87]                | +2.23 | 4                  | the closure’s own error exceeds the bound on its inputs, and the sign holds when any one publication is deleted; the same rows as the two other glycol-ether cells, 59% shared with the alkanes                                                    |
| water                         | 111 | 15    | 3 (P14, 0.75)        | +0.05  | [−0.15, +0.39] <sup>0.7</sup> | +0.57 | 1                  | the margin changes sign when one publication is deleted: P2 takes +0.046 → −0.075; P14 takes +0.046 → −0.194                                                                                                                                       |

<sup>a</sup> Number of source publications, then the largest one and its share of that stratum’s squared error. The keys:

P1 10.1016/j.fluid.2013.06.028, P2 10.1016/j.fluid.2015.01.010, P3 10.1016/j.fluid.2015.11.037, P4 10.1016/j.fluid.2016.08.030, P5 10.1016/j.fluid.2016.10.009, P6 10.1016/j.jct.2013.05.006, P7 10.1016/j.jct.2015.12.034, P8 10.1016/j.jct.2016.10.013, P9 10.1016/j.jct.2018.05.003, P10 10.1016/j.jct.2019.05.011, P11 10.1016/j.tca.2012.02.005, P12 10.1016/j.tca.2015.11.010, P13 10.1021/acs.jced.7b00114, P14 10.1021/acs.jced.9b00726, P15 10.1021/je020196d. <sup>b</sup> Two-way solute × solvent cluster bootstrap, 5th and 95th percentiles. Row-identical strata are one row set:

drawn once and printed once, under the row set’s first name above and restated under the rest (10 of 43 row sets, 16 rows). Each prints at the precision its own draw count supports, decided cell by cell over 200 seeds and verified on the run that bought it. 2 carry two decimals (organic solute; glycol ether, the latter at  $1.2 \times 10^8$  draws) and 9 carry one, both endpoints of each standing four Monte-Carlo standard errors clear of the nearest rounding boundary at that precision. For 11 no precision this table can print is determined: those carry two decimals with a superscript giving their endpoints’ Monte-Carlo standard deviation in units of that last digit, and the superscript is the warning rather than a refinement. A further 2 (aldehyde; halogenated | water as solute) no seed moves at all—each has one cluster on one of its two axes, so the resample is one-way over three clusters and about 10 distinct compositions exist: a bootstrap with nothing left to resample, not a converged one. The per-cell verdict, its draw count and its cost at either precision are on the deposited record (§S3.8). <sup>c</sup> In how many of the four unit × convention cells the stratum is both boundable and sign-stable under deletion of each contributing publication; the rule of §2.5 requires all four, which six of the 59 satisfy, in three distinct row sets. <sup>d</sup> The condition the row is in, rather than a label for it. A one-source cell is unstable, and a deletion leaving fewer than 16 rows leaves the cell undefined, so both read as untested rather than as passes. A margin is one-sided: positive puts  $B_{\text{closure}} > B_{\text{insuff}}$  on those rows at that cell, while a margin at or below zero is a failure to separate and licenses nothing. <sup>e</sup> The search behind every row, in the two nulls of the paragraph below. Under the binning bound of record, a chemistry-blind relabelling of the same molecules certifies at least one row set in 58% of 2000 draws, reaches +2.04 or more in 10.1% of all draws and in 17% of the draws that certify anything, and separates the six admissible strata from chance ( $p = 0.01$ ) while leaving the two admissible positive row sets at chance ( $p = 0.286$ ). Under the pre-declared cross-fitted estimator, on the 473 rows it scores, the same relabelling reaches the observed maximum in 13.7% of all draws and in 54% of the certifying ones, against 9.4% and 16% for the binning bound on those same rows.

Table S17: **The map, solute axis:** the rest of the fifteen boundable strata, at the same estimator cell and under the same rule as Table S16, whose notes apply here unchanged. The alkane row is the second admissible positive row set, and clause (c) demotes it to the glycol ethers.

| Stratum                                                                    | <i>n</i> pairs sources <sup>a</sup> |     |                | margin 90% margin <sup>b</sup>      | full cells <sup>c</sup> | what it shows, or the test it fails <sup>d</sup>                                                                                                                                                                                                     |
|----------------------------------------------------------------------------|-------------------------------------|-----|----------------|-------------------------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Solute role</i><br>organic solute                                       | 440                                 | 179 | 14 (P8, 0.35)  | +0.18 [−0.23, +1.35]                | +0.82                   | 2 the margin changes sign when one publication is deleted: P15 takes +0.184 → −0.094                                                                                                                                                                 |
| <i>Solute family</i><br>alkane                                             | 129                                 | 45  | 7 (P15, 0.53)  | +1.38 [+0.8, +2.3]                  | +1.72                   | 4 the sign holds under every deletion, but 84% of these rows are the glycol ethers' and the 21 outside give −0.22, so this margin does not stand outside the glycol ethers: demoted to them by clause (c); 10% shared with the aprotic acceptors     |
| aromatic hydrocarbon                                                       | 71                                  | 32  | 7 (P15, 0.53)  | −0.19 [−1.0, +0.7]                  | −0.07                   | 0 the margin changes sign when one publication is deleted: P6 takes −0.188 → +0.081                                                                                                                                                                  |
| ketone                                                                     | 63                                  | 11  | 3 (P14, 1.00)  | −0.53 [−0.85, +0.63] <sup>1.8</sup> | −0.33                   | 0 too few rows remain after deleting P14 (1 of 3 deletions), so the sign is untested there                                                                                                                                                           |
| <i>Solvent class</i> × <i>solute role</i><br>glycol ether   organic solute | 182                                 | 43  | 3 (P8, 0.53)   | +2.04 [+1.25, +2.87]                | +2.23                   | 4 the closure's own error exceeds the bound on its inputs, and the sign holds when any one publication is deleted; the same rows as the two other glycol-ether cells, 59% shared with the alkanes                                                    |
| water   organic solute                                                     | 111                                 | 15  | 3 (P14, 0.75)  | +0.05 [−0.15, +0.39] <sup>0.7</sup> | +0.57                   | 1 the margin changes sign when one publication is deleted: P2 takes +0.046 → −0.075; P14 takes +0.046 → −0.194                                                                                                                                       |
| aprotic acceptor   organic solute                                          | 57                                  | 57  | 4 (P7, 0.32)   | −0.19 [−0.4, +0.1]                  | −0.15                   | 4 the sign holds under every deletion, but the margin is not positive, so the two terms are not separated here—a bound from above cannot show the inputs are at fault; the same rows as the other aprotic-acceptor cell, 23% shared with the alkanes |
| <i>The set as a whole</i><br>whole set                                     | 477                                 | 185 | 15 (P13, 0.32) | +0.51 [−0.1, +2.2]                  | +0.95                   | 3 passes here and fails at the pair unit (residual-only convention), where deleting P13 takes +0.270 → −0.119                                                                                                                                        |

fails once the pair unit is required, where deleting 10.1021/acs.jced.7b00114 takes +0.270 to −0.119; the rule therefore does not by itself retire the aggregate at the headline cell, and its retirement rests on the composition argument of §3.5.1 rather than on this rule.

**The two axes cross, and the map measures the crossing.** 362 of the 1711 stratum pairs share rows, so cells drawn from the two axes are not independent evidence. All 37 water-as-solute rows and all six of its pairs sit inside three solvent classes and all come from one publication: 18 rows in the halogenated class (60% of it), 13 in the aryl ethers (all of it) and 6 in the aromatic hydrocarbons. Naming water-as-solute and the halogenated solvents as two findings would count those 18 rows twice, and adding the aryl ethers or the coarse acceptor-only class a further 13. The two positive admissible cells overlap the same way: alkane ∩ glycol-ether is 108 rows over 24 pairs

and carries the whole effect (+2.51 on those rows), while the glycol ethers minus the alkanes remain admissible—74 rows at +1.40 residual-only and +1.69 full, under all three deletions—and the alkanes minus the glycol ethers, 21 rows, do not, being below the boundability threshold at the fixed cell and returning −0.22 there (−0.01 full). That is clause (c) of §2.5 run on these two cells: the glycol ethers hold the more rows and hold 84% of the alkanes', and the alkane residual is inadmissible, so the alkane solutes are demoted to the glycol ethers and not the reverse. The overlap by itself decides nothing—had the 21 rows carried the margin, the alkanes would have stood. Those rows are counted once.



Table S18: **The map’s error, bound and source curve**, stratum by stratum, at the same headline cell as Tables S12–S15—eight equal-count bins of  $g(z^*)$ , unbiased within-bin variance, row unit, residual-only convention. These are the columns those tables do not carry, on the same 59 strata, whose margins, intervals and admissibility are printed there.

| Stratum                                | $n$ | MSE   | $B_{\text{insuff}}^{\text{up}}$ <sup>a</sup> | $P^b$         | share <sup>c</sup> | leave-one-source-out <sup>d</sup> |
|----------------------------------------|-----|-------|----------------------------------------------|---------------|--------------------|-----------------------------------|
| <i>Solvent class</i>                   |     |       |                                              |               |                    |                                   |
| glycol ether                           | 182 | 2.252 | 0.108                                        | > 0.95        | 45.2%              | [+1.60, +2.87] (3)                |
| water                                  | 111 | 0.477 | 0.215                                        | $\approx 0.8$ | 5.8%               | [−0.19, +0.25] (3)                |
| aprotic acceptor                       | 57  | 0.089 | 0.140                                        | < 0.2         | 0.6%               | [−0.41, −0.13] (4)                |
| mono-alcohol                           | 35  | 2.250 | 0.733                                        | > 0.95        | 8.7%               | +1.95 (one deletion)              |
| N–H protic                             | 33  | 1.175 | 0.281                                        | $\approx 0.9$ | 4.3%               | [−0.24, +0.87] (2)                |
| halogenated                            | 30  | 4.569 | 0.592                                        | > 0.95        | 15.1%              | +5.84 (one deletion)              |
| aromatic hydrocarbon                   | 14  | 4.297 | —                                            | —             | 6.6%               | none possible                     |
| aryl ether                             | 13  | 9.582 | —                                            | —             | 13.7%              | none possible                     |
| aliphatic hydrocarbon                  | 2   | 0.011 | —                                            | —             | 0.0%               | none possible                     |
| <i>Solvent class, coarse</i>           |     |       |                                              |               |                    |                                   |
| H-bond donor and acceptor (O)          | 328 | 1.651 | 0.562                                        | $\approx 0.9$ | 59.7%              | [−0.42, +0.97] (8)                |
| H-bond acceptor only                   | 70  | 1.852 | 0.209                                        | $\approx 0.7$ | 14.3%              | [−0.19, +2.94] (5)                |
| H-bond donor (N)                       | 33  | 1.175 | 0.281                                        | $\approx 0.9$ | 4.3%               | [−0.24, +0.87] (2)                |
| halogenated, non-H-bonding             | 30  | 4.569 | 0.592                                        | > 0.95        | 15.1%              | +5.84 (one deletion)              |
| hydrocarbon, non-H-bonding             | 16  | 3.761 | 0.385                                        | $\approx 0.9$ | 6.6%               | none possible                     |
| <i>Solvent family, fine</i>            |     |       |                                              |               |                    |                                   |
| glycol ether                           | 182 | 2.252 | 0.108                                        | > 0.95        | 45.2%              | [+1.60, +2.87] (3)                |
| water                                  | 111 | 0.477 | 0.215                                        | $\approx 0.8$ | 5.8%               | [−0.19, +0.25] (3)                |
| mono-alcohol                           | 35  | 2.250 | 0.733                                        | > 0.95        | 8.7%               | +1.95 (one deletion)              |
| ester                                  | 30  | 0.043 | 0.049                                        | $\approx 0.2$ | 0.1%               | none possible                     |
| amide N–H                              | 25  | 0.591 | 0.414                                        | $\approx 0.8$ | 1.6%               | +0.47 (one deletion)              |
| azine                                  | 23  | 0.096 | 0.072                                        | $\approx 0.5$ | 0.2%               | −0.00 (one deletion)              |
| halogenated aliphatic                  | 18  | 1.526 | 0.709                                        | $\approx 0.9$ | 3.0%               | none possible                     |
| aromatic hydrocarbon                   | 14  | 4.297 | —                                            | —             | 6.6%               | none possible                     |
| aryl ether                             | 13  | 9.582 | —                                            | —             | 13.7%              | none possible                     |
| halogenated aromatic                   | 12  | 9.135 | —                                            | —             | 12.1%              | none possible                     |
| amine N–H                              | 8   | 2.999 | —                                            | —             | 2.6%               | none possible                     |
| aliphatic hydrocarbon                  | 2   | 0.011 | —                                            | —             | 0.0%               | none possible                     |
| ketone/aldehyde                        | 2   | 0.404 | —                                            | —             | 0.1%               | none possible                     |
| nitrile                                | 2   | 0.397 | —                                            | —             | 0.1%               | none possible                     |
| <i>Solute role</i>                     |     |       |                                              |               |                    |                                   |
| organic solute                         | 440 | 1.400 | 0.608                                        | $\approx 0.8$ | 67.9%              | [−0.09, +0.41] (14)               |
| water as solute                        | 37  | 7.866 | 0.770                                        | > 0.95        | 32.1%              | none possible                     |
| <i>Solute family</i>                   |     |       |                                              |               |                    |                                   |
| alkane                                 | 129 | 2.315 | 0.467                                        | > 0.95        | 32.9%              | [+1.39, +1.67] (7)                |
| aromatic hydrocarbon                   | 71  | 0.534 | 0.361                                        | $\approx 0.5$ | 4.2%               | [−0.86, +0.08] (7)                |
| ketone                                 | 63  | 0.628 | 0.577                                        | $\approx 0.5$ | 4.4%               | [−0.14, −0.14] (2)                |
| mono-alcohol                           | 39  | 1.394 | 0.563                                        | > 0.95        | 6.0%               | [+0.17, +1.37] (5)                |
| water                                  | 37  | 7.866 | 0.770                                        | > 0.95        | 32.1%              | none possible                     |
| alkene                                 | 32  | 2.661 | 0.184                                        | > 0.95        | 9.4%               | [+2.00, +2.68] (3)                |
| aldehyde                               | 29  | 0.091 | 0.144                                        | $\approx 0.4$ | 0.3%               | none possible                     |
| halogenated aliphatic                  | 27  | 1.631 | 0.286                                        | > 0.95        | 4.9%               | [+1.23, +1.71] (2)                |
| amine N–H                              | 12  | 3.362 | —                                            | —             | 4.5%               | none possible                     |
| ether                                  | 12  | 0.469 | —                                            | —             | 0.6%               | none possible                     |
| halogenated aromatic                   | 8   | 0.619 | —                                            | —             | 0.5%               | none possible                     |
| azine                                  | 6   | 0.135 | —                                            | —             | 0.1%               | none possible                     |
| tertiary amine                         | 4   | 0.278 | —                                            | —             | 0.1%               | none possible                     |
| ester                                  | 3   | 0.013 | —                                            | —             | 0.0%               | none possible                     |
| alkyne                                 | 2   | 0.350 | —                                            | —             | 0.1%               | none possible                     |
| nitrile                                | 2   | 0.124 | —                                            | —             | 0.0%               | none possible                     |
| nitro                                  | 1   | 0.068 | —                                            | —             | 0.0%               | none possible                     |
| <i>Solvent class × solute role</i>     |     |       |                                              |               |                    |                                   |
| glycol ether   organic solute          | 182 | 2.252 | 0.108                                        | > 0.95        | 45.2%              | [+1.60, +2.87] (3)                |
| water   organic solute                 | 111 | 0.477 | 0.215                                        | $\approx 0.8$ | 5.8%               | [−0.19, +0.25] (3)                |
| aprotic acceptor   organic solute      | 57  | 0.089 | 0.140                                        | < 0.2         | 0.6%               | [−0.41, −0.13] (4)                |
| mono-alcohol   organic solute          | 35  | 2.250 | 0.733                                        | > 0.95        | 8.7%               | +1.95 (one deletion)              |
| N–H protic   organic solute            | 33  | 1.175 | 0.281                                        | $\approx 0.9$ | 4.3%               | [−0.24, +0.87] (2)                |
| halogenated   water as solute          | 18  | 6.294 | 0.228                                        | > 0.95        | 12.5%              | none possible                     |
| aryl ether   water as solute           | 13  | 9.582 | —                                            | —             | 13.7%              | none possible                     |
| halogenated   organic solute           | 12  | 1.982 | —                                            | —             | 2.6%               | none possible                     |
| aromatic hydrocarbon   organic solute  | 8   | 0.873 | —                                            | —             | 0.8%               | none possible                     |
| aromatic hydrocarbon   water as solute | 6   | 8.862 | —                                            | —             | 5.9%               | none possible                     |
| aliphatic hydrocarbon   organic solute | 2   | 0.011 | —                                            | —             | 0.0%               | none possible                     |
| <i>The set as a whole</i>              |     |       |                                              |               |                    |                                   |
| whole set                              | 477 | 1.902 | 0.697                                        | $\approx 0.9$ | 100.0%             | [+0.18, +0.87] (15)               |

<sup>a</sup> A dash is a stratum at which the fixed cell is undefined. Strata with  $n \geq 40$  are *boundable*: eight bins hold at least five rows each. The others are reported anyway, and additionally carry the adaptive bin count  $\max(3, \lfloor n/10 \rfloor)$ , here in the stratum’s own rows  $n$ , in the deposited table. Every stratum here is at the one fixed bin count; the sweep over bin count for the strata the rule passes is Table S7. On the mono-alcohols  $B_{\text{insuff}}^{\text{up}} = 0.733$  exceeds that stratum’s own  $\text{Var}(m) = 0.604$ , which the artifact flags; it is the value that sets the  $\times 6.8$  span of §3.5.1, and it occurs only on a stratum already not boundable. <sup>b</sup> Two-way solute  $\times$  solvent cluster bootstrap, 3000 draws, quoted as a band by the rounding rule of §3. <sup>c</sup> That stratum’s share of the set’s squared error. Shares do not sum to 100% down the column: the granularities and the two axes overlap, which §3.5.1 measures. <sup>d</sup> Range of the margin over the deletions that can be run, with how many that is in parentheses; a deletion leaving fewer than 16 rows is not runnable, and a stratum with one source has none. Whether the sign survives—which is what admissibility turns on—is in Table S12.

### S3.9 Concentration inside the map, and the cells that do and do not separate

*Concentration, inside the map as well as under it.* Three solvent classes holding 47% of the rows carry 74% of the squared error, the 12% of rows that are acceptor-only aprotics carry 0.6%, and three of fifteen publications carry 76%. A map cell can be as source-carried as the aggregate was, so each is put through the same deletion (Tables S12–S15). Eleven multi-source cells have every deletion runnable in all four cells and six keep the sign throughout—the admissible ones. What singles the glycol ethers out among them is positivity and clause (c), not a deletion nobody else survives. On the halogenated solvents and the mono-alcohols only one of the two deletions can be run at all—in both, the one that cannot is the one removing the dominant source—and it leaves +5.84 and +1.95, so those two cells are unverified rather than shown to fail. At the level of the individual pair the broad set is stable, all 185 leave-one-pair-out margins falling in  $[+0.42, +0.72]$ , so what the aggregate is fragile to is the source and the stratum, which is what a composition effect looks like. The chromatographic-only cut that *inverts* the margin on the enlarged 298 K VT-2005-matched set (§3.7, limitation (i)) does not invert here under either convention.

**The broad set: which chemistry does not separate, cell by cell.** Three of the six admissible cells are that same row set at three granularities. The remaining three are two row sets, and neither establishes anything. The alkane solutes (129 rows, 45 pairs, +1.38,  $[+0.8, +2.3]$ —one decimal, which is what that cell’s draw count determines, note *b* of Table S12) are admissible and positive, and clause (c) demotes them to the glycol ethers, which hold more rows and hold 108 of these 129: the alkane residual outside them is 21 rows, too few to bound at the fixed cell, and returns  $-0.22$  there, so the alkanes’ margin does not stand outside the glycol ethers—where the glycol ethers’ does stand outside the alkanes, on the 74 rows at the +1.40 above. What de-

motes the cell is that residual, not the 84% overlap and not the  $-0.22$ , which is itself sub-threshold. The acceptor-only aprotics (57 rows,  $-0.19$ , negative in all four cells,  $P < 0.2$  at both units under the residual-only convention and  $P \approx 0.3$  under the full one, at an MSE of 0.089 that is at the scale of the measurements) pass the rule and license nothing, the instrument being one-sided; their deletion test is additionally *anti-conservative*, two of the four deletions leaving  $n < 40$ , where the unbiased within-bin variance inflates—0.140 to 0.258 on the  $n=38$  remainder—and drives a negative margin further negative. That is the whole of the uniqueness claimed for the glycol ethers: the only row set admissible, positive and left standing by clause (c)—*not* the only admissible stratum and not the only one sign-stable under every deletion. Of the rest, water carried as the dilute solute (37 rows, +6.33) is too small to bound and draws on one publication, so its deletion cannot be run and that margin was never put at risk; the halogenated solvents (30 rows, +3.39) are too small to bound and only one of their two deletions runs, leaving +5.84; water as a solvent, the coarse hydroxylic class and the organic-solute subset each lose their sign under a deletion, which is why the finding belongs to the glycol-ether sub-class and not to the hydroxylic solvents as a class; the mono-alcohols, the N–H protics and the ester and azine sub-families, which an acceptor-only reading would lean on, are unboundable. None may be used to explain a reversal. The whole-set row passes at the headline cell and fails once the pair unit is required, so the rule does not by itself dispose of the aggregate there; the composition argument above does.

**The VT-2005-matched set, stratum by stratum and deletion by deletion.** §3.5.2 states what this set is and that its aggregate does not stand; here is the arithmetic. The two worst-fit pairs each rest on a single record and carry 34% of the squared error, a concentration we do not read as established chemistry; the separation does not survive their deletion at the headline cell (margin  $-0.002$ ), a margin of +0.05 being unable to absorb 34% of

the squared error leaving the sample, and that deletion is a failure of the headline cell rather than a robustness pass. Four neighbouring cells do survive it, and the full composition audit is above. These are not 60 exchangeable observations, and the folds hinge on the same molecules. The stratification rule of §2.5 carries the composition half of §3.5.1 here and not the bound half: the 17 solvents span six of the nine classes—no water, no halogenated solvent, no glycol ether beyond ethylene glycol—and the largest reaches 24, so the set’s composition is missing two of the three classes that carry most of the broad set’s error, and of its 38 strata not one is admissible in every unit  $\times$  convention cell. Within it, scored on UD profiles so that the two sets are commensurable, the ordering of the error is the broad set’s—2.71 on the monoalcohols, 3.00 on the N–H protics and 2.53 on the glycol ethers against 0.092 on the acceptor-only aprotics, that last cell again returning a negative margin,  $-0.05$  at  $P \approx 0.5$ —but 57 of the 60 pairs are the broad set’s own rows, so that agreement is a consistency check and not independent corroboration. What is this set’s own is the granularity of its fragility: deleting the publication supplying 33 of the pairs takes the margin to  $-0.23$ . Scored against VT-2005 profiles under the residual-only convention and the *unbiased* within-bin variance, the binning bound gives  $B_{\text{insuff}} \lesssim 0.713$  and the separation margin of  $+0.05$ ; the maximum-likelihood variant biases the within-bin variance downward and so the margin upward to  $+0.24$ . That number stands with the broad set’s aggregate, and on weaker evidence, four analyst choices setting it (§2.5).

**Which defect the broad set points at, and why it states neither.** §3.5.3 places the closure’s failure on the VT-2005-matched set in a *chemistry*—90% of that set’s squared error falls on strong hydrogen-bond-donor solvents, the chemistry the single-parameter H-bond kernel governs—and not in one component of the composite. On the broad set that reading does not reproduce, and neither does its replacement. The donor classes carry 64% of the squared error on 76% of the rows, which

is no enrichment, while the halogenated and aryl-ether solvents—which the 2002 kernel has no dispersion term for at all—carry 29% on 9%, which is. But 31 of those 43 rows are 10.1021/acs.jced.7b00114, and they are the same water-as-solute rows the map already declines to state; they carry 91% of that 29%. Put those two shares through the deletion the map applies to every cell and both invert: without that publication the halogenated and aryl-ether rows fall to 2.7% of the rows and 3.9% of the error, and the donor classes rise to 82% and 94%, which *is* the VT-2005-matched set’s localisation. So one publication decides the reading in either direction, which is why the article states neither.

**The constant-offset (Jensen) bound on both sets.** §2.5 promises these here. On the broad IDAC set  $\mathbb{E}[m] = 3.424$ ,  $\mathbb{E}[g_{\text{res}}] = 2.998$  and  $\mathbb{E}[g_{\text{full}}] = 2.579$ , so the constant-offset bound is  $B_{\text{closure}} \geq 0.182$  under the residual-only convention and  $B_{\text{closure}} \geq 0.715$  under the full one. Against the binning bound on  $B_{\text{insuff}}$  (0.697 residual-only, 0.482 full) the assumption-free bound therefore separates under the full convention and inverts under the residual-only one, exactly as it does on the VT-2005-matched set (0.426 against 0.713) and by a wider margin—the residual-only ratio is 0.26 here against 0.60 there. On this set it is in addition strictly slacker than the bound the decomposition already supplies,  $B_{\text{closure}} \geq \text{MSE} - B_{\text{insuff}}^{\text{up}}$  (0.182 against 1.205 residual-only, 0.715 against 1.429 full), so it adds nothing to the ordering under either convention; the residual-only-convention separation rests on the binning bound alone, which is what §2.5 says.

**The broad set’s matching rule, and two bounds not carried forward.** Three details of §3.5.1 follow. (a) *The matching asymmetry.* The VT-2005-matched set is matched to VT-2005 by canonical SMILES, a rule we defend there on the ground that a profile imported across a tautomer or protonation variant is the wrong  $z^*$ . The broad IDAC set is matched to UD by InChIKey with a first-block

fallback, which is looser. A displaced  $z^*$  inflates MSE and so  $B_{\text{closure}}$ , but it also inflates the conditional variance the binning bound estimates, and under a Berkson-type displacement it loads onto  $B_{\text{closure}}$  through the curvature of  $g$  as well (§S8); its direction on the margin is unsigned and we do not read it as conservative. The exposure is small and we measure it rather than argue it: of the 90 distinct molecules there, 86 resolve by full InChIKey and four only through the first block, touching 12 of the 477 rows. On the 465 rows matched exactly the verdict is unchanged—residual-only convention MSE 1.930,  $B_{\text{insuff}} \lesssim 0.70$ , margin +0.53 at  $P \approx 0.9$ ; full convention 1.926, 0.49, +0.94 at  $P > 0.95$ —so the looser rule is not carrying the result. (b) *The tighter of the two conventions’ bounds.*  $B_{\text{insuff}}$  is the same population quantity under both conventions (step (d) of the proof of Lemma 1, §S2); what differs is the statistic the binning bound bins on,  $g_{\text{full}}(z^*, T)$  or  $g_{\text{res}}(z^*, T)$ , each a deterministic function of the same argument and each therefore giving a separately valid upper bound. The smaller of the two is consequently also a valid upper bound in population, and charging it against the residual-only error gives 1.902 against  $2 \times 0.48$ , a margin of +0.94 (on the VT-2005-matched set, 1.472 against  $2 \times 0.61$ , +0.26). It is reported and no margin is read from it: a minimum over two finite-sample estimates is anti-conservative even when each is valid in population, and the same-convention pairing is the conservative reading. (c) *The binning-free pair bound.*  $\mathbb{E}[\text{Var}(m \mid \text{pair})] = 0.046$  is thirteen times tighter than the binning bound on the same 359 rows (0.61), and would put the margin *there* at +2.16 against the +1.02 the binning bound gives on those rows. It is not promoted because all 67 multi-row pairs draw their rows from a single publication, so it samples no between-laboratory spread (§3.5.1).

### S3.10 Whether the closure defect is correctable, and the arithmetic behind the fidelity lever

*Within a solvent, not across one.* Where  $B_{\text{closure}}$  dominates—on this set, the one row set of §3.5.1 the rule leaves standing—the ceiling is in the map *as deployed*, kernel and computed profile together (§2.1.1); that does not say the composite is *fixably* wrong. We tested this with a small structure-preserving residual on the fixed COSMO-SAC exchange-energy kernel ( $\Delta = B \text{diag}(a) B^\top$  over the  $\sigma$ -grid, symmetric, zero-initialised), fit on the *reference* VT-2005 profiles and cross-validated five-fold over five fit seeds. The answer depends on the fold design, and only one design is free of leakage. With random folds over the 60 matched pairs, a rank-one residual (52 parameters) removes  $46 \pm 5\%$  of the reference-input error out of sample (activity MSE  $1.47 \rightarrow 0.79$ ). The same solvent, however, then appears in training and test, and the closure error is predominantly a between-solvent systematic. Under folds grouped so that no solvent is shared, the reduction is  $-5 \pm 16\%$  at rank one and  $+2 \pm 15\%$  at rank two. Each  $\pm$  is a standard error over the five fit seeds on one fixed 60-pair set and prices optimiser noise only; the held-out spread across folds is an order of magnitude larger (sd 1.02 in MSE under the grouped design, 0.54 under random folds) and is not propagated into it. That spread is itself about 70% of the reference-input MSE the reduction is measured against, so the grouped design yields a null and no upper bound: an interval wide enough to price it would exclude nothing about a cross-solvent reduction, and the narrow one quoted prices the wrong thing. What the two designs do support is the contrast between them—the 46% appears only when the same solvent may sit on both sides of the split, so it is a within-solvent effect.

That fold spread also exceeds the gaps between ranks ( $\leq 0.06$ ) by an order of magnitude, so these data do not resolve a rank ordering; and the 60 pairs form a single connected component under shared molecule iden-

tity, so a molecule-disjoint holdout does not exist at any fold design.  $B_{\text{closure}}$  is therefore an error that a handful of kernel parameters can absorb *within* a solvent, and that we have not shown to be correctable across solvents— weaker than a general repair, and a statement about what a correction of that shape absorbs rather than about which component of the composite the error came from. The fitted correction also gains nothing over a free one: the parameter-free 2010 kernel reaches 0.765 on the same molecule pairs (§3.5.3) against the 52-parameter residual’s random-fold 0.79. The residual’s leakage-free number, however, is the grouped-fold one, a  $-5 \pm 16\%$  reduction, i.e. no reduction at all, so the comparison that holds is a free 2002  $\rightarrow$  2010 step against a fitted correction that does not transfer. The two also run on different profile sources and conventions (residual-only on VT-2005 here, full on the University of Delaware (UD) profiles there<sup>S11</sup>), so this is indicative rather than a head-to-head; but each stays above its own convention’s insufficiency ceiling (0.62–0.71 residual-only, 0.53–0.61 full), so neither erases  $B_{\text{closure}}$ .

*Where the residual error sits chemically.* The composition audit behind §3.5: 90% of the VT-2005-matched set’s squared error falls on the strong hydrogen-bond-donor solvents, 84% once the two doubted amine/methanol pairs are deleted, and the acceptor-only solvents are near-exact. That localisation places  $B_{\text{closure}}$  on the chemistry COSMO-SAC-2002’s single-parameter hydrogen-bonding term governs. It does not place it in the kernel rather than in the profile: those solvents’ reference profiles are single gas-phase conformers as well, and the glycol ethers—the one stratum the rule leaves standing, and the class the cascade above names conformer-dependent—are where that displacement is expected to be largest, so a displaced reference would load onto the rows that carry the error. Neither set separates the two (§S8).

On the conformational part of that displacement the stratum is no longer hedged but bounded, and the bound is computed rather than argued. Its four solvent references—ethane-1,2-diol and the di-, tri- and tetraethylene glycols—are single conformers, and we re-

placed each with a Boltzmann ensemble computed for this purpose: ETKDgV3 embedding, MMFF pre-screening, geometry optimisation at BP86/def2-SVP with CPCM( $\epsilon=\infty$ ), single points at BP86/def2-TZVP with Klamt radii, and Mullins averaging onto the same 51-bin grid, weighted at 298.15 K. The ensembles displace their references by 0.038, 0.038 and 0.074 of the profile norm for the tri-, tetra- and diethylene glycols; ethane-1,2-diol collapses to one representative under the fixed clustering tolerance and is carried separately at 0.134. Substituting all four moves the margin from +2.036 to +2.270, 90% interval  $[+1.36, +3.29]$  over the same two-way solute  $\times$  solvent resample—away from zero, not toward it—and an equal-norm random displacement of the same four profiles leaves +1.678  $[+1.520, +1.807]$ , so the measured direction sits outside the random family and moves the margin the other way. Two scopes travel with that. It bounds conformational uncertainty in the four solvent references and nothing wider: displacing the 19 solutes by the same norm moves the margin as much or more (1.09 and 2.67 times, against a pre-declared bar of 0.5), so reference uncertainty at large is not bounded here. And the ensemble direction is nearly orthogonal to the smooth broadening a displaced reference was expected to produce (cosine 0.04 to 0.30), so the two cannot stand in for one another. What that bound does not reach is the reference calculation itself, and the size of what it does not reach is measured on the same four molecules. Against a BP86/def2-TZVP CPCM( $\epsilon \rightarrow \infty$ ) recomputation, rigid single-conformer molecules put the cross-code floor—two defensible  $\sigma$ -profile calculations of the same molecule—at a median relative- $L_2$  of 0.282, range 0.190 to 0.403, four to eight times the conformational displacement bounded above. Swapping the recomputed lowest conformer for the tabulated profile moves the glycol margin from +2.036 to +7.466, against +0.234 for the conformer ensemble, so 95.7% of that movement is the code-and-level difference. Those raw swaps mix two profile conventions and are not read against the decision rule; what they establish is that the clo-

sure’s response to the choice of reference calculation dwarfs the effect this stratum reports, and that every substitution result in this paper is scoped to the VT-2005 tabulation’s level of theory.

A counter-test on the same closure points the same way, and is the one §3.5 cites. The Staverman–Guggenheim combinatorial term is the physically *more* complete treatment and acts only where the two molecules differ in size; restoring it *increases* the VT-2005-matched set’s reference-input error,  $\text{MSE} = 1.47$  under the residual-only convention against 1.80 with the term in place (Table S8), the gap between the two being carried by the size-asymmetric pairs. A term that adds physics and costs accuracy is what a misspecified map does and not what missing information about the inputs would do. It is equally what compensating errors do, and this design cannot tell the two apart. The residual part alone is not a model anyone fitted: its constants were regressed with the combinatorial term present (§2.3), so deleting that term removes a one-signed, size-dependent contribution from a kernel carrying its own size-dependent misfit, and the two can cancel. Improved agreement under the deletion is then a statement about the cancellation and not about which term is misspecified. The size-asymmetric pairs—where  $\gamma^{\text{SG}}$  is largest—are where the gap sits, which is what that reading predicts and not what distinguishes it. The same signature appears on the solubility axis, where restoring the term makes the grounded learned- $\sigma$  arm slightly worse (Table S4).

**The closure-fidelity lever: the arithmetic behind §3.5.3.** (*a'*) *The input-fixed kernel control.* The prescribed reduction of the typed Hsieh profile to a single 2002 profile is its sum  $p = p_{\text{nhb}} + p_{\text{OH}} + p_{\text{OT}}$  (Eq. 9 of Ref.<sup>S1</sup>), which holds database, cavity area, volume and averaging fixed and varies only the kernel. On the VT-2005-matched set that contrast is  $2.047 \rightarrow 0.765$ , two-way cluster bootstrap  $P > 0.95$  with 90% margin  $[+0.05, +2.90]$ ; the averaging step taken alone moves 2002 by  $+0.29$  in the opposite direction and is not resolved ( $P <$

$0.2$ ,  $[-0.79, +0.11]$ ). That averaging term is why the control’s 2002 endpoint is 2.047 and not the 1.757 of the *2002, all channels* arm, and why the input-fixed step (1.28) is *larger* than the native-input one (0.99): the control is favourable to the conclusion it supports. The control cannot separate the donor/acceptor typing, which the 2002 kernel has no channel to consume and which is therefore charged to the kernel by construction. The *2002, all channels* arm sits at 1.757 rather than lower because the NIST reference implementation, handed a typed three-profile database, reads the nonpolar channel alone, an arm scoring 1.26 on the VT-2005-matched set (§S1 gives it unrounded), which is not the reference closure, so the arm is taken from our own NIST-validated layer instead. (*a*) *The 2010, dispersion on arm’s coverage.* cCOSMO returns no dispersion value for the three bromine- or iodine-containing VT-2005-matched pairs—bromobenzene and iodobenzene in 1,2-ethanediol, and 2-bromopropane in pyridine—so that arm’s VT-2005-matched run scores on  $n=57$ . On those common 57 the *2010, dispersion off* arm scores 0.767, so the 0.765-versus-0.785 ordering is not an artifact of the differing  $n$ . They are the *same* three pairs the broad set’s construction excludes, its ladder also requiring a tabulated  $\epsilon$ , so the dispersion-defined 57 and the 57 shared between the two sets are one set, not two. (*b*) *A rounding coincidence.* The 1.783 quoted for the shared 57 and the 1.783 quoted for the 334 chromatographic rows are distinct numbers that round alike (unrounded 1.78257 against 1.78245), the smaller set lying wholly inside the larger. (*c*) *The 2010 kernel on its own typed latent.* That latent is three 51-bin profiles per molecule, 153 in all, so the conditioning variable is 306-dimensional and the resolution ratios are  $307/477 \approx 0.64$  per row and  $307/185 \approx 1.66$  per distinct pair on the broad IDAC set, and  $306/60 \approx 5.1$  on the VT-2005-matched set, where all rows sit at 298 K and  $T$  adds no dimension. Under the residual-only convention the VT-2005-matched set gives 0.831 against  $B_{\text{insuff}} \lesssim 0.80$ , hence  $B_{\text{closure}} \gtrsim 0.03$  and a margin of  $-0.77$  at  $P \approx 0.3$ ; the broad IDAC set gives 2.893 against  $B_{\text{insuff}} \lesssim 0.80$ ,

hence  $B_{\text{closure}} \gtrsim 2.09$  and a margin of  $+1.29$  at  $P \approx 0.9$ . Under the literature-standard full convention the same rows give 1.978 against  $B_{\text{insuff}} \lesssim 0.60$ , margin  $+0.78$  at  $P \approx 0.8$ , and the VT-2005-matched set 0.765 against 0.69, margin  $-0.61$ . The switch to the residual-only convention raises the 2010 kernel’s broad-set error from 1.978 to 2.893 and its floor bound from 0.60 to 0.80, so the wider margin comes from a worse closure and not from a tighter floor, and the reading does not move. On the broad set the binding clustering is the source-DOI bootstrap on 15 clusters, whose 90% interval  $[-0.41, +4.53]$  still admits zero, as does the two-way one  $[-0.24, +4.71]$ ; only the pair clustering is strictly positive, and it was so before the convention switch as well. *(d) The published comparison.* The comprehensive assessment<sup>S11</sup> covers 29,173  $\gamma^\infty$  points for the 2010 and dsp kernels only. Its largest documented infinite-dilution step—adding the whole dispersion term—yields a  $\sim 10$ -point mean-absolute-deviation gain ( $95\% \rightarrow 86\%$ ) with a *reversal* on aqueous systems (203% vs. 194%); 2010’s gains are otherwise concentrated in liquid-liquid equilibria and in the temperature dependence of the electrostatic term, where an infinite-dilution design has little leverage. *The scope of the search behind the null of §3.5.3:* the papers introducing each kernel,<sup>S2,S3,S12</sup> the benchmark implementation,<sup>S1</sup> this comprehensive assessment<sup>S11</sup> and the works citing them. It was not exhaustive, which is why the main text reads the result as an absence of evidence rather than as evidence of absence. *(e) The floor checks.* Each is same-database, same-latent and same-convention, stated under the residual-only rule with the full-convention reading beside it. *(1) Broad IDAC set:* the 2002, all channels error 1.902 against  $2B_{\text{insuff}} \lesssim 1.39$  on the same UD profiles, margin  $+0.51$  at  $P \approx 0.9$  (1.911 against 0.96,  $+0.95$ ,  $P > 0.95$  full). *(2) VT-2005-matched, VT-2005 against VT-2005:* 1.472 against  $2B_{\text{insuff}} \lesssim 1.427$ , so  $B_{\text{closure}} \gtrsim 0.759$  against  $B_{\text{insuff}} \lesssim 0.713$  and a margin of  $+0.05$  (1.80 against 1.22,  $+0.58$  full; Table S8). The 1.757 of (a’) is those same 60 pairs scored on UD profiles, and two axes separate it from the 1.80: the profile database and

the segment-iteration count, which the two artifacts do not share. At a common count the databases part by  $+0.029$  of the full-convention  $+0.043$  and the count carries  $+0.014$  (note  $b$  of Table S8). Charged against a UD-native floor recomputed for those pairs, the residual-only convention gives 1.436 against  $B_{\text{insuff}} \lesssim 0.71$ , a margin of  $+0.01$  at  $P \approx 0.8$ , against 1.757 versus a UD-native  $B_{\text{insuff}} \lesssim 0.54$  and a margin of  $+0.69$  full—so on UD profiles the VT-2005-matched set’s own floor check is, under the rule we deploy, at zero. That last reading is the one the two axes reach: the residual-only 1.472-against-1.436 gap is  $+0.027$  database and  $+0.009$  count, and the count term is the size of the  $+0.01$  margin it is read against, so the residual-only floor check on UD profiles is at zero on either attribution and is not resolved by this set.

### S3.11 Robustness of the closure/insufficiency separation

We stress-test the residual-only-convention verdict  $B_{\text{closure}} > B_{\text{insuff}}$  (§3.5)—equivalently  $B_{\text{insuff}} < \text{MSE}/2 = 0.74$  at  $\text{MSE}_{\text{res}} = 1.47$ —four ways on the  $n=60$  set.

*(i) Estimator sensitivity.* The four random-fold  $B_{\text{insuff}}$  upper bounds of Table S10 (the binning-bound one is also the headline row of Table S8)—0.40 ( $k$ -NN), 0.71 (the binning bound at the headline unbiased variance; 0.62 at the maximum-likelihood one), 0.62 (random forest), 0.62 (ridge out-of-fold)—all fall below the 0.74 threshold, the binning bound only just. The forest is the one estimator our two VT-2005-matched audit artifacts disagree on: the standalone decomposition run’s own forest gives 0.565 on these rows against the audit’s 0.621, a spread of the size a forest’s own randomisation produces. The audit value is the one Table S10 carries, because the audit artifact is what the inclusion rule names; both are below the threshold and neither is the bound relied on. The  $k$ -NN value is the smallest and least conservative input to  $B_{\text{closure}} \geq \text{MSE} - B_{\text{insuff}}^{\text{up}}$ , so the leakage-immune binning bound is the

one relied on instead. The binning bound bins  $g(z^*)$  and is therefore finite-sample downward-biased when points per bin are few, which would *inflate*  $B_{\text{closure}}$  favourably. The two non-binning bounds (RF 0.62, ridge 0.62) are immune to that particular per-bin sparsity, but they do not provide independent corroboration: under random folds every held-out row shares its solvent or its solute with a training row, so both can borrow strength from a near-twin, anti-conservatively and in the same direction as the binning (§3.5). Blocking folds on both molecules leaves no informative estimator on this set. The broad IDAC set behaves the same way once its own leak—a held-out row’s molecule pair recurring at a neighbouring temperature—is blocked: the random forest goes from 0.15 to 0.88 and the ridge from 0.80 to 1.47, past its threshold. The reported separation should therefore be read as resting on the leakage-immune binning bound alone, on either set, not on agreement between three estimators. The sole non-certifying configuration is a deliberately aggressive ridge, which lifts  $B_{\text{insuff}}$  to 0.88 (then  $B_{\text{closure}} \geq 0.59 < 0.88$ ): a sensitivity edge, not a refutation.

(0) *The VT-2005-matched set’s design.* The 60 pairs span 41 solutes and 17 solvents; two formamide pairs are excluded for a canonical-SMILES tautomer mismatch. Methanol occupies 25 of the 60 rows (12 as solute, 13 as solvent) and pyridine is the largest solvent stratum (20 rows);  $\text{Var}(m) = 2.47$  across the set. The two worst-fit pairs, 1-hexanamine and cyclohexylamine in methanol, carry 34% of the squared error; both rest on a single record, and 1-hexanamine sits far off its own homologous series ( $\ln \gamma_2^\infty = 2.62$  against  $-0.78$  for 1-butanamine in the same solvent), which is why we do not read that concentration as established chemistry. Deleting both leaves MSE 1.01; charged against this the headline cell of the residual-only convention—eight equal-count bins under the *unbiased* within-bin variance—gives  $B_{\text{insuff}}^{\text{up}}$  0.5056 and margin  $-0.0015$ , so the separation does *not* survive the deletion at the cell the paper reports. The four neighbouring cells do: 0.4370 (margin  $+0.14$ ) at eight bins under the maximum-likelihood

variance, 0.4717 ( $+0.07$ ) at six bins unbiased, 0.4241 ( $+0.16$ ) at six bins maximum-likelihood, and, under the full convention on the same 58 rows, MSE 1.20 against 0.3309 ( $+0.53$ ) at eight bins unbiased. The reading we take from the cell that fails is that the VT-2005-matched set’s  $+0.05$  headline margin has no room to absorb the loss of 34% of the squared error, which is the conclusion the leverage folds below reach by a different route.

(i bis) *Leverage folds on the VT-2005-matched set.* The separation survives all 17 leave-one-solvent and all 60 leave-one-pair folds and deletion of the eight highest-error pairs, but only 40 of 41 leave-one-solute folds (six equal-count bins): the exception drops methanol, the maximum-leverage solute (12 pairs), and loses the separation ( $-0.26$ ), as does dropping methanol as solute with pyridine as solvent ( $n=28$ ;  $-0.36$ ). Deleting methanol from both roles ( $n=35$ ) holds only marginally—the margin falls from 0.35 to 0.11 at six bins and from 0.24 to 0.003 at eight, and inverts under the unbiased within-bin variance ( $-0.03$ ,  $-0.21$ ). Those fold counts and the bootstraps use the maximum-likelihood variance at six bins. Repeated at the headline cell—eight bins, unbiased variance—the leave-one-solvent count is unchanged at 17/17 (minimum margin  $+0.03$ ), the leave-one-pair count falls to 56/60 (minimum  $-0.03$ ) and the leave-one-solute count to 36/41 (minimum  $-0.22$ ); deletion of the eight highest-error pairs still holds there ( $+0.04$ ).

(ii) *Leave-one-solvent-out.* Dropping each of the 17 solvents in turn, the VT-2005-matched set’s aggregate margin stays positive in all 17 folds under the binning bound—a fold curve on a quantity §3.5.1 retires, so it records what that aggregate did and not a robustness pass. Under the random-forest estimator it holds in only 2/17, because an RF over-fits conditional variance in 102 dimensions at  $n=60$  (its out-of-fold MSE approaches  $\text{Var}(m)$ ), so its leave-one-out estimate is unstable—an estimator artifact, not a sign flip.

(iii) *Label noise.* Injecting zero-mean Gaussian noise  $\sigma_\eta \in \{0.2, \dots, 0.8\}$  into  $m$  leaves the  $B_{\text{closure}}$  lower bound stable ( $0.84 \rightarrow 0.70$ ), as expected since  $B_{\text{closure}} = \text{MSE} - \mathbb{E}[\text{Var}(m \mid z^*)]$



is invariant to zero-mean noise.

(iv) *Clustered-bootstrap interval.* A solvent-clustered bootstrap (3000 resamples of the 17 solvents, binning bound) makes the confidence explicit: the  $B_{\text{closure}}$  lower bound has median 0.93 (CI<sub>90</sub> [0.40, 1.62]), and the margin  $B_{\text{closure}} - B_{\text{insuff}}$  has median 0.36 (CI<sub>90</sub> [-0.22, +0.97]) with  $P(B_{\text{closure}} > B_{\text{insuff}}) \approx 0.89$  (single-axis; the conservative two-way solute $\times$ solvent bootstrap, which also charges for repeated solutes, gives  $\approx 0.78$ , the value reported in the main text). With  $G = 17$  solvent and 41 solute clusters this small- $G$  regime has optimistic interval coverage, and the finite-cluster correction acts in the direction of less confidence and non-certification. We therefore treat the  $P$  values as a coarse band, not two-decimal quantities, and the main text quotes them as bands (rounded to one decimal, with  $> 0.95$  and  $< 0.2$  closing the ends) throughout; the two-decimal values are in the deposited artifacts and in the tables here, where the estimator that produced each is named. The separation on the VT-2005-matched set does not clear the 90% level under either clustering, which is why the article treats it as supporting rather than decisive. On the broad IDAC set, with 78 solute and 36 solvent clusters and a third clustering over 15 source publications, the bands mean something; but they do not all clear. Under the residual-only convention the margin bootstraps to  $P > 0.95$  over the 185 molecule pairs (90% margin [+0.23, +1.16]) and only  $\approx 0.9$  two-way over the  $78 \times 36$  solute/solvent clusters ([-0.08, +2.16]) and over the 15 sources ([-0.07, +1.74]), with two of the three 90% intervals extending just below zero, so that set is not resolved either. Read as a ratio, the broad set’s +0.51 is 1.1 times its own 90% bootstrap half-width under pair clustering and 0.46 times it under the two-way one, which is the sense in which §3.5 calls that margin unresolved. It is only under the full convention that all three clear ( $P > 0.95$ ), in the same order: [+0.71, +1.27] over pairs, [+0.48, +1.85] two-way, [+0.50, +1.57] over sources.

(v) *A label systematic tracking  $z^*$ , and what the VT-2005-matched set’s fold designs leave.* Zero-mean noise cannot manufacture the sep-

aration: it raises MSE and  $B_{\text{insuff}}^{\text{up}}$  alike, leaves the  $B_{\text{closure}}$  lower bound unchanged, and shrinks the margin by  $\sigma_\eta^2$ , so it can understate the separation but not create it. What the margin does bound is a *systematic* label error correlated with  $z^*$ , which enters  $B_{\text{closure}}$  rather than  $B_{\text{insuff}}$ : on the VT-2005-matched set it would close the separation at root-mean-square  $0.49 \ln \gamma$  under the maximum-likelihood binning but at only  $\approx 0.21$  under the unbiased one. The matched IDAC is essentially single-method (158/165 gas chromatography), which makes a within-method systematic of 0.5 implausible; one of 0.21 is not, so on the corrected binning that threat is *not* excluded. The VT-2005-matched set’s out-of-fold estimators cannot police it either: under random folds every held-out row shares its solvent ( $47/60 = 78\%$  of rows) or its solute ( $27/60 = 45\%$ ) with a training row, and under folds blocked on both molecules three of the four estimates (3.65, 1.83, and the divergent ridge) exceed the total MSE = 1.47, beyond which an upper bound on  $B_{\text{insuff}}$  constrains  $B_{\text{closure}}$  vacuously.

**In-regime ( $\gamma$ -stratified) separation.** The  $n=60$  measurement is matched at low  $\gamma$ , while the paradox spans all  $\gamma$ . Stratifying the  $n=60$  set by  $\ln \gamma_2^\infty$  tests whether closure-dominance is a regime artifact. It is not: the ordering  $B_{\text{closure}} > B_{\text{insuff}}$  (leakage-immune binning bound) holds in every  $\gamma$  stratum and *strengthens* toward higher  $\gamma$  (Table S19)—in the highest stratum ( $\ln \gamma_2^\infty = 2.94$ , approaching the non-ideal regime)  $B_{\text{closure}} \geq 2.68$  exceeds  $B_{\text{insuff}} \leq 0.67$  several-fold. The high- $\gamma$  strata are small ( $n = 17\text{--}30$  in 102 dimensions): the binning bound can be pushed toward zero by too-few-points-per-bin, which mechanically widens the margin exactly where data is thinnest, so it is the several-fold separations rather than the precise values that carry the qualitative reading. The bound here bins each stratum into three equal-count bins (against eight in the main text) and uses the maximum-likelihood within-bin variance, as in the headline row of Table S8; recomputed with the unbiased one it rises to 0.08/0.95/0.81 across the three strata, so each still separates, but the mid-

dle stratum only marginally (0.95 against its own  $\text{MSE}/2 = 0.96$ ). The  $m > \text{med.}$  stratum separates through its upper (high- $\gamma$ ) half, and the *isolated* moderate band ( $\text{med.} < m \leq 1.5$ ,  $n=13$ ) does *not* separate. This is not a counter-example. The total error there is only  $\approx 0.06$ , so  $B_{\text{closure}} \leq \text{MSE} \approx 0.06$  assumption-free (since  $B_{\text{insuff}} \geq 0$ ), and there is nothing to attribute—near ideality the target is small and the split is uninformative. We read no closure-quality claim from it in either direction. This supports—but does not prove—the conjecture that the same closure misspecification drives the all- $\gamma$  paradox.

Table S19: Closure/insufficiency verdict across  $\ln \gamma_2^\infty$  strata of the matched set (residual-only convention;  $B_{\text{insuff}}^{\text{up}}$  is the leakage-immune binning bound throughout, at the maximum-likelihood within-bin variance).

| Stratum                                 | $n$ | $\overline{\ln \gamma_2^\infty}$ | $B_{\text{insuff}} \leq$ | $B_{\text{closure}} \geq$ |
|-----------------------------------------|-----|----------------------------------|--------------------------|---------------------------|
| lower $\gamma$ ( $m \leq \text{med.}$ ) | 30  | -0.40                            | 0.07                     | 0.95                      |
| higher $\gamma$ ( $m > \text{med.}$ )   | 30  | +1.90                            | 0.85                     | 1.07                      |
| high $\gamma$ ( $m > 1.5$ )             | 17  | +2.94                            | 0.67                     | 2.68                      |

The closure/insufficiency decomposition (full and residual-only conventions; binning-bound, random-forest (RF), ridge, and kNN estimators; solvent-clustered bootstrap; the pyridine stratum) is tabulated in §3.5. Figure S4 shows the shape of that error on the VT-2005-matched set: the reference- $\sigma$  closure’s parity plot against the measured  $\ln \gamma_2^\infty$ .

**Four qualifications on the decomposition (full).** Four qualifications bear on the decomposition of §3.5. None of them inverts the sign of the VT-2005-matched set’s aggregate margin; each sharpens its scope. That aggregate is retired on other grounds (§3.5.1–§3.5.2), so these are recorded as what the four choices do and not as robustness passes for a live verdict.

1. The combinatorial default is residual-only. Under it the *constant-offset* (Jensen) lower bound inverts ( $B_{\text{closure}} \geq 0.43 < B_{\text{insuff}} \leq 0.71$ ; it separates only under the full convention,  $B_{\text{closure}} \geq 0.72$  against  $B_{\text{insuff}} \leq 0.61$ ),

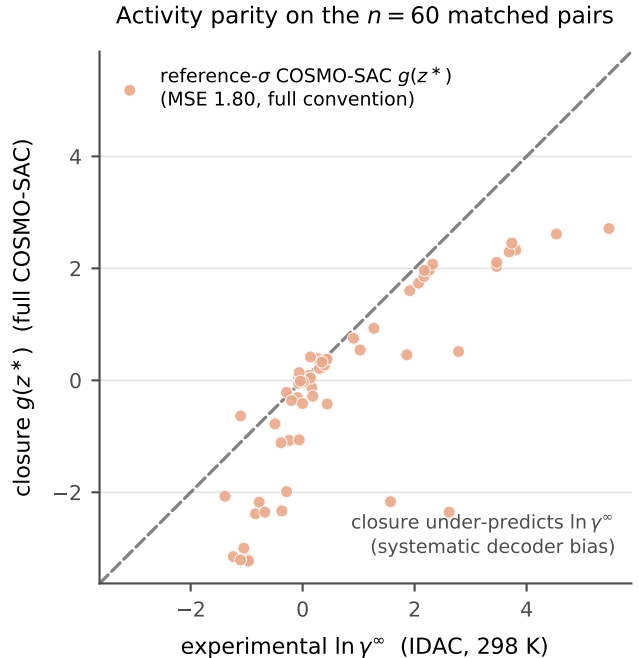


Figure S4: Activity-coefficient parity on the  $n=60$  matched pairs. The reference- $\sigma$  COSMO-SAC closure  $g(z^*)$  (salmon) falls below the diagonal at larger  $\ln \gamma_2^\infty$ , under-predicting the activity coefficient (MSE 1.80, full convention).

so that assumption-free bound does not separate under deployment; Table S8 carries that row for both sets, and the full-convention margin (the sharpened  $\text{MSE} - B_{\text{insuff}}$  bound, 1.19 vs 0.61) is a full-COSMO-SAC-only result. What separates under the residual-only convention is the leakage-immune bound  $B_{\text{closure}} \geq \text{MSE}_{\text{res}} - B_{\text{insuff}}^{\text{up}} \approx 0.76 > 0.71$ , which carries the headline claim, together with the combinatorial argument above. Every  $B_{\text{insuff}}$  ceiling quoted in this item is the unbiased within-bin variance, the headline setting of Table S8; the maximum-likelihood pair (0.85 against 0.62) is the demoted variant and is not quoted here.

2. The closure error is predominantly a *between-solvent* systematic bias ( $\sim 64\%$  of total error); within the largest single-solvent stratum (pyridine,  $n=20$ ) the closure is nearly correct (stratum MSE 0.07). Because solvent  $\sigma$  is part of  $z^*$ , these offsets are still decoder error and

give an independent solvent-conditional lower bound  $B_{\text{closure}} \geq 0.89$  (full)/0.56 (residual-only).

3. Confidence on the full-convention constant-offset separation is the solvent-clustered bootstrap  $P(B_{\text{closure}} > B_{\text{insuff}}) \approx 0.85$  (0.83 vs the random-forest upper bound, 0.89 vs the binning bound); we report the solvent-clustered scheme rather than the less conservative i.i.d.-pair one because solvent dependence dominates. The leakage-immune margin (0.76 vs 0.71) is a point estimate. The decomposition is checkpoint/seed-independent (the closure has zero learnable parameters).
4. The 60 matchable pairs are the low-to-moderate  $\gamma$  regime (matched mean  $\ln \gamma_2^\infty = 0.75$  vs 3.56 for VT-2005-unmatchable pairs,  $t = -7.44$ ; the strongly non-ideal amide-solvent regime is untested because VT-2005 lacks those profiles). This narrows scope but is *conservative*, since the untested regime is a harder closure test. The target is  $\ln \gamma_2^\infty$  at 298 K, so the numeric ceiling does not necessarily transfer unchanged to the finite-composition solubility regime.

### S3.12 A pre-declared attempt to tighten the insufficiency bound, and its failure

*What was declared, and when.* The cross-fitting below is the sample-splitting construction of double/debiased machine learning.<sup>S10</sup> In the solubility domain  $B_{\text{insuff}}^{\text{up}}$  is the binning bound throughout: the within-bin variance of  $m$  over eight equal-count bins of the scalar  $g(z^*)$ , whose slack is the information the binning destroys and is not estimable from these data (§2.5). The other domain does not coarsen—in the  $pK_a$  closure the same quantity is the out-of-fold residual mean square of a cross-fitted forest of  $m$  on  $z^*$  (§S7)—and because every margin in the map is

$\text{MSE} - 2B_{\text{insuff}}^{\text{up}}$ , a regressor that predicts  $m$  from  $z^*$  better than an eight-bin histogram of a scalar summary of  $z^*$  does would raise all of them at once. The experiment that asks whether it does was pre-declared, in the header of the estimator itself (sha256 `d158989fe9cfdc79...`, the full digest deposited beside it), written and staged in version control before any cross-fitted margin existed. It fixed in advance: the feature set ( $z^*$  and nothing else—the two  $\sigma$ -profiles, their areas and  $T$ , 105 dimensions—with  $v_{\text{cosmo}}$  excluded because it is an argument of the full convention only, so conditioning on it would not bound the residual-only convention’s  $B_{\text{insuff}}$ ); the regressor, the  $pK_a$  domain’s own, untuned, with a ridge sensitivity reported beside it and never adopted; the folds, five-fold grouped on the molecule pair, with a source-grouped variant reported beside every cell; the row set, 473 of the 477, the four rows whose solutes carry no VT-2005 profile dropped; an outcome-blind validity gate; a ban on forming the per-cell minimum of the two bounds; the rule fixing which estimator the paper would adopt; and three verdicts. Nothing above the declaration was edited afterwards, and both bounds are deposited for every one of the 1164 cells.

*It did not tighten.* On the same 473 rows, broad set, row unit, primary folds, the cross-fitted bound is  $B_{\text{insuff}}^{\text{cf}} = 0.856$  against the binning bound’s 0.704 (residual-only convention) and 0.489 (full)—looser by 22% and by 75%. It is looser having *passed* its own gate: no pair straddles a fold, the global out-of-fold  $R^2$  is +0.823, and the value sits well above the deposited replicate floor  $\mathbb{E}[\text{Var}(m \mid \text{pair})] = 0.046$ . This is the estimator working and losing, not failing. The arithmetic of the loss is one comparison: with  $\text{Var}(m) = 4.85$  on these rows, a fitted bound beats the binning bound only above  $R^2 = 1 - B_{\text{insuff}}^{\text{bin}}/\text{Var}(m)$ , which is 0.855 under the residual-only convention and 0.899 under the full one. The forest reached 0.823.

Every consequence runs against this work, and the numbers are printed here for that reason. The glycol-ether margin—the paper’s one surviving stratified finding—falls from +2.036 to +1.757 at the fixed esti-

mator cell, its 90% two-way cluster interval from  $[+1.25, +2.87]$  to  $[+1.02, +2.49]$  and its leave-one-publication-out envelope from  $[+1.597, +2.866]$  to  $[+1.342, +2.339]$ , on the same 182 rows either side—no dropped row carries a glycol-ether solvent. The envelopes are exact deletion extrema and print to three decimals. The two intervals are not on the same footing:  $[+1.25, +2.87]$  is the two-way bootstrap run to convergence at  $1.2 \times 10^8$  draws, while  $[+1.02, +2.49]$  is one 3000-draw replicate of this scoring pass, whose bootstrap is seeded per stratum name rather than per row set. That row set is drawn here under three names and prints  $[+1.02, +2.49]$ ,  $[+1.05, +2.47]$  and  $[+1.02, +2.49]$ , so the second decimal of the cross-fitted endpoint is seed and not estimator. The fall it is quoted for is larger than that spread by six times on the lower endpoint and twenty-four on the upper, which is why the comparison survives its own precision. The map is poorer in every count that reads a margin: admissible cells  $35 \rightarrow 32$ , strata admissible in all four unit  $\times$  convention cells  $6 \rightarrow 5$ , and distinct admissible positive row sets after the maximality reduction  $2 \rightarrow 1$ , the alkanes being the row set that drops. Under the source-grouped fold variant—the conservative scheme the declaration required beside every cell, and the one the closure error’s stated character as a between-publication systematic argues for—that count is 0. No admissible positive row set survives it, the glycol-ether stratum included: at source-grouped folds its cross-fitted bound is 4.92 against an MSE of 2.25, so its margin is  $-7.58$ . Fifteen source groups over five folds leave each fold’s regressor extrapolating across publications, which is the design the scheme is chosen for and also what makes its bound this loose. The stratum’s own rows separate them. No pair in it is measured by more than one publication—none of the 43, and none of the 182 pair-by-temperature cells—and the two publications carrying 177 of the rows supply disjoint glycols, the ethylene and diethylene oligomers against the tri- and tetraethylene ones. The source partition therefore almost coincides with the solvent partition, and where it does not the exception is small:

ethylene glycol alone is carried by two publications, 36 rows against 5. A source-grouped fold accordingly holds out a whole solvent in three of the four cases and all but five of its rows in the fourth, so it prices chain-length extrapolation and samples almost no between-laboratory component, which is what the pair bound is refused for above. Its bound exceeds the stratum’s own total error, which  $B_{\text{insuff}}$  itself cannot do, an upper bound on it being free to be arbitrarily loose, and its regressor scores  $R^2 = -0.60$  out of fold on these rows: at this fold count the estimator failed, and  $-7.58$  is uninformative about this closure rather than a reversal of it. The count of *boundable* strata does not move, 15 of 59 under both estimators, because boundability is the  $n \geq 40$  test and the declaration froze it. The retired whole-set aggregate does not return: its margin falls from  $+0.944$  to  $+0.211$  (full) and from  $+0.506$  to  $+0.203$  (residual-only), and it now fails the deletion test outright, two publications flipping its sign. Nor does the chemistry-blind multiplicity null improve where it matters. *Both* arms of that null are re-run on the 473 rows, so the comparison is like for like and its binning-bound column is not the one the rest of the submission prints:  $p(C)$  moves from 0.297 to 0.255, but  $p(D)$ , the null  $p$  of the largest certified margin, moves the wrong way,  $0.094 \rightarrow 0.137$ ; conditioned on drawing a map that certifies anything at all, a chemistry-blind relabelling reaches the observed maximum in 54% of draws against 16% under the binning bound. The declared verdict is therefore LOSS on branch L1, with L2 firing independently. Restricting  $477 \rightarrow 473$  rows leaves the map itself untouched—the four dropped rows leave the 477-row baseline’s 35/6/2 and its  $+2.036$  exactly where they were—but it does move the null: on all 477 the same binning bound gives  $p(C) = 0.286$  and  $p(D) = 0.1005$ , that is 17% of certifying draws. Every statement of this null outside this subsection is that 477-row one.

*A deviation from the declaration, named as one.* The declaration’s Sec. 3 fixed an unconditional rule—“THE RULE. The cross-fitted estimator headlines UNCONDITIONALLY ... provided it passes the validity gate of Sec. 3.1, which

never reads a margin, an MSE, a  $g$  or a stratum label”—and spelled out what it committed to: “If the cross-fit is LOOSER than the binning bound, the headline margins SHRINK, cells that are certified today lose their certification, and the map gets poorer. That outcome is reported as the result, in the paper, with the same prominence a gain would have had.” **We did not follow that rule.** The binning bound remains the estimator of record, and every margin printed in this paper is the binning bound’s. The reason is that the rule was written against one hazard—choosing the smaller bound cell by cell after seeing the cells, which is biased low and which the paper disowns elsewhere—and it did not anticipate the other, that one valid bound would be uniformly *looser* than the other. Where that happens, obeying the rule prints a bound that is valid and strictly less informative: every claim weakens and none becomes more correct. What we did instead is what the declaration’s Sec. 2 already required independently—both bounds, both margins and their difference are printed for every cell of the deposited map, the cells that favour this paper and the cells that do not, so that the poorer map is legible in the deposit and every number in the paragraph above re-derivable from it. Two things about this deviation should be said rather than absorbed. It runs in this paper’s favour, which is exactly why it is named here instead of being folded into a methods sentence. And it is a choice made *after* seeing a number, which is what the declaration existed to prevent; what limits it is that the number is one global comparison of two bounds on one row set, not a per-cell selection, and that the alternative is on the page.

*Why the binning bound wins here, and where it would not.* The mechanism is measurable, and it is not the obvious one. It was established on synthetic data matched to the real design—477 rows in 185 pairs from 15 sources, the real rows-per-pair multiplicities, real  $\sigma$ -profiles drawn from the deposited 1432-row artifact so the regressor faces the real collinear geometry, and  $B_{\text{insuff}}$  known in closed form by construction. The binning bound is handed  $g$ : a physics-built statistic that was designed to predict  $m$  and is close to sufficient for  $\mathbb{E}[m \mid z^*]$ , so bin-

ning it starts from the answer. The forest is handed  $z^*$  and must rediscover  $g$  from about 145 of the 181 pair groups in each training fold. *It is not a dimensionality effect:* coarsening the regressor’s features to the leading unsupervised principal components of  $z^*$ —a legitimate operation on the bound, since a smaller  $\sigma$ -field can only raise  $\mathbb{E}[\text{Var}(m \mid \cdot)]$ —never helps. On the synthetic design the raw 105 dimensions give the tightest cross-fitted bound of the six feature sets tried (median 3.39, against 3.71 at eight components, 4.04 at sixteen, 4.59 at four, 5.00 at two and 6.10 at one), and all six sit above the eight-bin binning bound’s 2.3–2.6 on the same draws. What separates the two estimators is sample size alone: the ratio of the two bounds falls from 3.2 at  $n=40$  to 1.24 at  $n=960$ , and a log–log extrapolation of the two curves crosses near  $n \approx 2 \times 10^3$  rows, three to four times the 473 actually scored ( $1.7 \times 10^3$  fitting all eight sizes,  $2.0 \times 10^3$  omitting the two smallest; it is an extrapolation and is quoted as one). The same account explains why the cross-fitted estimator is the right one in the  $pK_a$  domain and is not an inconsistency between the two: there  $z^*$  is four series labels and a scalar,  $\mathbb{E}[m \mid z^*]$  is estimable directly within a scaffold at the sample sizes available, and the closure confers no head start because there is no high-dimensional statistic left for it to summarise (§S7). One further asymmetry is worth recording because a maximum-over-strata statistic inherits it: across the 15 boundable strata the binning bound has mean 0.315 and standard deviation 0.213, the cross-fitted one 0.658 and 0.519, and the same ordering holds under a chemistry-blind relabelling (0.241 against 0.553, median over draws), so the extra spread is a property of the estimator and not of the chemistry—which is why  $p(D)$  worsened while the bound merely loosened.

*The design rule this yields.* For anyone reusing this instrument on another closure, the choice of  $B_{\text{insuff}}^{\text{up}}$  is not a choice between an old estimator and a modern one. Both are valid upper bounds; the choice is about power, and it turns on three things that are known before any margin is formed. (i) If the closure’s own output  $g(z^*)$  is available as a feature and is

near-sufficient for  $\mathbb{E}[m \mid z^*]$ —the normal case when  $g$  is a physical model built to predict  $m$ —coarsening  $g$  begins from a statistic the regressor would have to learn, and it wins at every sample size that does not let the regressor learn it. (ii) If  $\dim(z^*)$  is small enough that  $\mathbb{E}[m \mid z^*]$  is directly estimable—a handful of labels and a scalar, as in the  $pK_a$  domain—condition directly and do not coarsen, since the coarsening would then discard information for no return. (iii) Between those poles the crossing is a sample-size question with a computable answer: fit both and prefer the regressor only if its out-of-fold  $R^2$  exceeds  $1 - B_{\text{insuff}}^{\text{bin}}/\text{Var}(m)$ , the requirement being a function of  $(m, z^*, g)$  that reads no MSE and no margin. Here that requirement was 0.855 and the forest returned 0.823, which is the whole of the result. The rule must be declared before the margins are read; choosing per cell between two valid bounds after seeing them is biased low, and choosing once and globally after seeing them, as we did above, is a disclosed deviation and not a method.

*The fold warning, which is the part that can invalidate a bound rather than merely loosen it.* The cross-fitted quantity is an upper bound only if the regressor is independent of the held-out row’s label, and on clustered data that is a property of the fold scheme, not of the estimator. This was measured before the real rows were touched, against known truth. With pair-grouped or source-grouped folds the estimator is a genuine upper bound: 0 violations in 636 cells over the declared evaluation scope—fit once globally, evaluate on label-blind sub-strata, both grouped schemes, random and chemically coherent sub-strata alike—including all 100 cells at  $n=40$ , the map’s own boundability threshold. With row-level folds it is not a bound at all: 9.4% of cells (18 of 192) fall below truth in the realistic regime,  $B_{\text{insuff}}/\text{Var}(m) \leq 0.30$ , and 71.9% (46 of 64) at 0.60. In the null design, where  $m$  is independent of  $z^*$  and the truth is  $\text{Var}(m) = 4.85$ , row-level folds return 1.73 under a pair shock and 1.42 under a source shock—64% and 71% *below* truth. The numerical gates do not catch this. The leaky version returns out-of-fold  $R^2 = +0.61$  and  $+0.64$  in exactly those two cells and clears the replicate-

floor check by a factor of thirty; the only gate that fires is the assertion that no group straddles a fold, which is a statement about the folds and not a number. **One deposited number is compromised by this and must not be cited as a bound:** the deposited decomposition artifact records an out-of-fold forest upper bound of 0.5646, fitted on row-level folds, with no grouping. Nothing in this paper rests on it: Table S10 carries the audit artifact’s 0.621 instead, and the disagreement between the two forests is reported above. The deposit now carries the warning in the file itself, beside the number. The scheme’s exposure differs by set and both readings reach the same verdict: on the VT-2005-matched set where that number lives no two rows share a pair, but 60 rows carry 41 solutes and 17 solvents, so every held-out row has a near-twin in training; on the broad set the pair channel is open outright, 359 of the 477 rows sharing a molecule pair with another row (§S8), and blocking it moves the same forest from 0.15 to 0.88. The cross-fitted map, its per-cell table, its multiplicity null, its within-stratum sensitivity, its verdict and its synthetic validation are all deposited, together with the declaration above, the scoring pass and the validation run that produced them.

## S4 External baselines and ablations

**What the three blocks may and may not be read across.** The three blocks of Table S20 may not be read across, and Table S30 prints the first two of them, without the random-pair rows, in the article. §3.7 states the three caveats under which this table may be read—two splits and two corpora, none of them like-for-like—and the one unfavourable sentence that survives them. Two of the three caveats take a further clause here. Scoring released FastSolv weights on our test rows is invalid specifically because FastSolv is trained on BigSolDB, which is this corpus. And the third block’s unit matches only through the conversion that table’s note defines, which puts our  $S$  per litre of *solvent* where the quoted block’s

Table S20: Solubility accuracy in three blocks: this work on the solute-scaffold split, external models re-run here on a by-solute and a random-pair split, and published figures on their own corpora and splits. A row carries two sample sizes where its two metric columns are not scored on the same rows, one where they coincide, and a dash where the value is quoted rather than computed here. The grounded, ungrounded and  $\sigma$ -oracle arms of the first block come from the five-seed leak-free re-run.

| Model                                                                                                                                 | Test set                                                           | $n$ ( $\ln x_2 / \log_{10} S$ ) | MAE $\ln x_2$   | RMSE $\log_{10} S$ | $R^2 \ln x_2$    |
|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|---------------------------------|-----------------|--------------------|------------------|
| <i>This work, solute-scaffold split; mean <math>\pm</math> s.d. over seeds: five for the two COSMO-SAC rows, three for the others</i> |                                                                    |                                 |                 |                    |                  |
| DirectGNN, same encoder, no closure<br>(control)                                                                                      | solute scaffold, this corpus                                       | 5608 / 5440                     | $1.70 \pm 0.03$ | $1.00 \pm 0.02$    | $0.42 \pm 0.03$  |
| TGNN, NRTL closure (physics)                                                                                                          | solute scaffold, this corpus                                       | 5608 / 5440                     | $1.79 \pm 0.07$ | $1.54 \pm 0.07$    | $0.34 \pm 0.03$  |
| TGNN, COSMO-SAC closure,<br>supervised $\hat{\sigma}$ (physics)                                                                       | solute scaffold, this corpus                                       | 5608 / 5440                     | $1.93 \pm 0.09$ | $1.44 \pm 0.11$    | $0.29 \pm 0.07$  |
| TGNN, COSMO-SAC, reference $\sigma^*$<br>substituted at evaluation                                                                    | solute scaffold, this corpus                                       | 5608 / 5440                     | $2.34 \pm 0.11$ | $1.64 \pm 0.09$    | $-0.07 \pm 0.08$ |
| <i>Scale reference on the same rows</i>                                                                                               |                                                                    |                                 |                 |                    |                  |
| predict the test-set mean of $\ln x_2$                                                                                                | solute scaffold, this corpus                                       | 5608 / 5440                     | 2.32            | 1.33               | 0.00             |
| <i>External models re-run on this corpus, one run each—a different split, so not a ranking against the block above</i>                |                                                                    |                                 |                 |                    |                  |
| FastSolv, retrained on this corpus <sup>a</sup>                                                                                       | by-solute, this corpus                                             | 10287                           | 1.44            | 0.84               | 0.55             |
| SolProp cycle, released model +<br>3-parameter recalibration                                                                          | by-solute, this corpus                                             | 10287                           | 2.34            | 1.34               | -0.18            |
| SolProp cycle, released model,<br>zero-shot                                                                                           | by-solute, this corpus                                             | 10287                           | 2.36            | 1.87               | -0.47            |
| SolProp encoder retrained on $\ln x_2$ ,<br>cycle discarded                                                                           | by-solute, this corpus                                             | 10570 / 10287                   | 1.62            | 1.07               | 0.39             |
| FastSolv, retrained on this corpus <sup>a</sup>                                                                                       | random pair, this corpus                                           | 10696                           | 1.02            | 0.61               | 0.76             |
| SolProp encoder retrained on $\ln x_2$ ,<br>cycle discarded                                                                           | random pair, this corpus                                           | 10887 / 10696                   | 0.64            | 0.45               | 0.88             |
| <i>As published, on their own corpora and splits; <math>\log_{10} S</math> in <math>\text{mol L}^{-1}</math></i>                      |                                                                    |                                 |                 |                    |                  |
| fastsolv (solution-FASTPROP<br>ensemble) <sup>S13</sup>                                                                               | SolProp test set (solutes<br>disjoint from the training<br>corpus) | —                               | —               | 0.83               | —                |
| solution-CHEMPROP<br>(companion) <sup>S13</sup>                                                                                       | SolProp test set                                                   | —                               | —               | 0.83               | —                |
| SolProp cycle, <sup>S14</sup> scored in <sup>S13</sup>                                                                                | SolProp test set                                                   | —                               | —               | 1.43               | —                |
| fastsolv (solution-FASTPROP<br>ensemble) <sup>S13</sup>                                                                               | Leeds set (Boobier et al.), near<br>room temperature               | —                               | —               | 0.95               | —                |
| solution-CHEMPROP<br>(companion) <sup>S13</sup>                                                                                       | Leeds set                                                          | —                               | —               | 0.99               | —                |
| SolProp cycle, <sup>S14</sup> scored in <sup>S13</sup>                                                                                | Leeds set                                                          | —                               | —               | 2.16               | —                |

*What the block-1 cells are.* Every one is the mean over seeds of that seed’s own metric, with the population standard deviation over seeds beside it; no cell pools the seeds’ rows. *The unit bridge.*  $\log_{10} S$  is obtained from  $\ln x_2$  through the solvent’s molarity, by one converter run on every row of the first two blocks and set out below. It carries *no* temperature dependence and it needs a molarity, so the 168 of the 5608 labelled test rows whose solvent has none carry no true  $\log_{10} S$  and leave that column for *every* arm: hence the two  $n$ . Block 1’s 5608 is itself the labelled subset of the 8103 test rows (Table S1); both exclusions are read off the data alone, neither off a prediction. The scale-reference row runs its one constant  $\ln x_2$  prediction through the same bridge on the same 5440 rows, so it is a reference in this column too. *The clip, and what the column owes it.* Predictions are clipped at  $x_2 = 0.999999$  before conversion, so an arm that saturates keeps its row with a large finite error instead of converting to  $+\infty$  and being dropped. Without that clip the exclusions would differ *by arm*—12 to 195 rows per seed for the three physics arms, none for the control, which never saturates—and would read the  $\sigma$ -oracle arm 0.26 lower, the COSMO-SAC-grounded arm 0.31 and the NRTL arm 0.48, the control unchanged. The clip is a chosen constant and this column moves with it: swept from  $x_2 = 1 - 10^{-3}$  to  $1 - 10^{-10}$  it leaves the control at 1.00 throughout and carries every physics arm across a range wider than the gaps between them. Two readings hold at every clip in that span, and separately at each of the three seeds: the control is below every physics arm, and every physics arm is above every published leader. Four do not, and each of the four fails on the seed axis as well as the clip one—the printed order of the NRTL and  $\sigma$ -oracle rows; whether a physics arm beats the scale reference; whether they sit below the better end of the published physics-informed cycle; and whether all three trail their control by more in this column than in MAE. The sweep and every cell of the four are below, and §3.7 takes only the readings that survive them. <sup>a</sup> *Re-scored:* the same checkpoint over the same rows with the single scaling it was trained under, the doubly standardising prediction path diagnosed below withdrawn rather than printed. This section gives the diagnosis, the withdrawn values and the control that reproduces them. The SolProp rows were always scored through a different path and are unaffected. The third block is quoted as published, in  $\text{mol L}^{-1}$ , on other corpora and splits: no conversion of ours stands behind those six values. MAE is not available for that block, which reports RMSE and  $\% \log S \pm 1$ , and we do not convert between them.

basis is unstated: the dimension matches and the task does not. The six figures of the third block are all  $\log_{10} S$  RMSE in  $\text{mol L}^{-1}$  and come from one evaluation, Attia et al.,<sup>S13</sup> on two corpora: the SolProp test set, whose solutes are disjoint from its training corpus, and the Leeds set of Boobier et al.<sup>S15</sup> near room temperature. The *leaders* are that work’s two direct models, the solution-FASTPROP ensemble and its solution-CHEMPROP companion, at 0.83 and 0.83 on the first corpus and 0.95 and 0.99 on the second; the *physics-informed cycle* is SolProp,<sup>S14</sup> scored in the same evaluation at 1.43 and 2.16 on those same two corpora. Neither range is a single number with a spread: each is two point values on two different corpora, and no uncertainty is published for any of the six. Against the leaders’ 0.83 to 0.99 and the cycle’s 1.43 to 2.16, the control’s seed mean of 1.00 sits just past the worst leader—at one of its three seeds (0.97) it is under it. The NRTL and grounded arms trail their control by more here—an RMSE deficit—than in MAE, at every clip of the note’s sweep and at every seed; the  $\sigma$ -oracle does not, and the sweep below says where. Where they sit *within* the cycle’s range is not a reading this column supports: it moves with the saturation clip the table’s note defines—the control’s figure does not move with that clip and all three of theirs do—and at the tight end of that sweep the NRTL and grounded arms fall below the cycle’s better end at every seed, the  $\sigma$ -oracle in its seed mean and at two of its three seeds but not at seed 43. They are also worse, in this column, than predicting one number for the entire test set: the scale reference reads 1.33 on the same rows, below all three printed figures. That is a reading of the seed-mean column at the deposited clip and not of the models—at seed 42 the grounded arm is below the reference at every clip, 1.13 against 1.33 at the deposited one, and seed-averaged the grounded and NRTL arms fall under it at the tight end as well; the table’s note counts the cells. What may be read from the three physics arms’ figures in this column is what survives the sweep—all three above every published leader and above their own control—and not the two decimals. The 0.75 aleatoric estimate<sup>S13</sup>—itself

a different functional on a different base from this paper’s own floor (§2.2)—sits below every row of the table but two, the closest published figure being the leaders’ 0.83; both exceptions, at 0.61 and 0.45, are on the random-pair split, where the same solute appears in training and the task is not extrapolation. The two arms run on both splits move the way the split predicts: the FastSolv retrainings score MAE 1.44 in  $\ln x_2$  by solute against 1.02 on random pairs, the retrained SolProp encoder 1.62 against 0.64. That FastSolv pair is re-scored, and it replaces one that read the other way round. Both runs had been scored through a prediction path of ours that standardises the model’s inputs twice, which saturates the network’s input activation on temperature and leaves the predictions exactly constant in  $T$ ; the fitted checkpoints were unaffected, so the repair is an inference pass on those same checkpoints and not a retraining (diagnosis and withdrawn values below). Re-scored, the by-solute row reads  $\log_{10} S$  RMSE 0.84, inside the published 0.83 to 0.95 but on an easier split than this work’s arms are held to.

We evaluated the leading published direct model (FastSolv<sup>S13</sup>) and the leading published physics-informed model (SolProp<sup>S14</sup>) on a by-solute test split ( $n \approx 10\text{--}12\text{k}$ ). The two arms are not obtained the same way, and the difference matters below: FastSolv is retrained on our corpus, whereas SolProp is the released pre-trained thermodynamic cycle—the arm that instantiates the physics—mapped onto our target by a three-parameter linear recalibration fitted on the training rows, not by retraining. That cycle is run at 298 K on every row, so the temperature of a measurement reaches the SolProp arm only through the calibrator’s temperature term. In  $\ln x_2$  they reach MAE 1.44 (FastSolv,  $R^2$  0.55) and 2.34 (SolProp, calibrated,  $R^2$  -0.18); in  $\log_{10} S$ , MAE 0.65 and 1.04, the same rows reading RMSE 0.84 and 1.34 in Table S30’s column. These are on a *by-solute* split that is *not comparable* to our scaffold numbers (Table S4): they bound the task’s difficulty, not a ranking of the models against one another, so this does *not* establish that our scaffold-split DirectGNN (1.70) outperforms them. A like-



for-like scaffold evaluation of the external models was not run, the deposited external runs being by-solute and pair-random only. The internal ordering—the physics-informed SolProp trailing the direct FastSolv, the same ordering we find between our physics closures and DirectGNN—is therefore a comparison of the *cycle* against a direct model. Discarding the cycle and retraining SolProp’s encoder directly on  $\ln x_2$  closes most of that gap without closing it: that arm reaches MAE 1.62 ( $R^2$  0.39) in  $\ln x_2$  on the  $n=10,570$  rows it predicts, and MAE 0.76 in  $\log_{10} S$  on the 10,287 of those the conversion is finite on, against FastSolv’s MAE 1.44 and 0.65 on 10,287 rows in both units, and both direct arms are well ahead of the cycle’s 2.34. The two arms’  $\ln x_2$  figures are therefore not on quite the same rows; Table S20 prints both sample sizes for the two rows of that block whose two metric columns disagree, and one figure for the four where they coincide. We quote the cycle as the physics-informed arm because only it assembles the separately learned components; the retrained arm measures what discarding the closure achieves on this corpus, not a better physics-informed model. The regime-level reading is that on this corpus no black box, ours or external, escapes the accuracy ceiling—consistent with the structural account of §3.7. Per-arm ablations (NRTL vs COSMO-SAC, combinatorial on/off) are in Table S4.

**The two FastSolv rows are re-scored, and what was wrong with the first pair.** The bundles first written for these two runs were read through a prediction path that standardises the model’s inputs *twice*. Our own inference wrapper scales solute descriptors, solvent descriptors and temperature with the checkpoint’s own buffers and then hands them to the released package’s prediction step, which scales the same tensors again; the training and validation steps do not re-scale, so the fitted models were never what was wrong. The network’s sigmoid input activation then annihilates the twice-scaled temperature, which reaches it at about  $-19.3$  rather than at the order  $-1$  a single scaling gives, a sigmoid of  $4 \times 10^{-9}$ . The

signature was unmistakable: those predictions were *exactly* constant in temperature—span 0 to machine precision—for all 1102 by-solute and 1149 random-pair (solute, solvent) pairs that span more than 5 K, where the measurements move a median 0.62 in  $\log_{10} S$  over a median 40 K. What survived the double scaling was a systematic offset, and it was the whole of the anomaly: the by-solute run’s mean  $\log_{10} S$  error was  $-0.36$  and the random-pair run’s  $-1.12$ , and once each was centred the two runs’ RMSEs were 1.04 and 1.07—the same dispersion, on both splits. The offset was present in validation too ( $-1.10$ ), so checkpoint selection did not catch it.

Because the fault was in the scoring and not in the fit, the repair is one inference pass per saved checkpoint. A re-scoring pass forwards the same two checkpoints over the same rows with the single scaling the models were trained under, and re-runs the doubly scaled path beside it as a control. That control reproduces the withdrawn bundles to every digit they printed—by-solute MAE 1.940994 in  $\ln x_2$  and RMSE 1.096639 in  $\log_{10} S$  on 10,287 rows, random-pair 2.910154 and 1.544042 on 10,696—which is what establishes that the scaling is the only thing that changed. Re-scored, the by-solute run reads MAE 1.44 in  $\ln x_2$  ( $R^2$  0.55) where the withdrawn bundle read 1.94, and the random-pair run 1.02 where it read 2.91; the temperature response returns, the predictions spanning a median 0.500 (by solute) and 0.631 (random pair) in  $\log_{10} S$  over those same pairs against a measured median 0.623, and the offset falls to  $-0.048$  and  $-0.024$ . Both re-scores are on the deposited runs’ own row sets, which the block’s other rows use; the current data preparation resolves a solvent molarity for 100 by-solute rows more, and scoring those as well moves that row to 1.43 and 0.84. Nothing is retrained: the checkpoints, the descriptors and the row sets are those of the original runs, and the withdrawn bundle is kept beside the re-scored one. The SolProp arms go through a different path and always carried a temperature response: on the by-solute run their predictions span a median 1.31 in  $\ln x_2$  across the temperatures of a pair, against a measured 1.35.

**What the  $\log_{10} S$  column owes to the saturation clip.** Converting  $\ln x_2$  to  $\log_{10} S$  sends a prediction at  $x_2 = 1$  to  $+\infty$ , so predictions are clipped at  $x_2 = 0.999999$  first, the standing prediction-side constant of the released code. The constant is a choice, and three of the four block-1 cells of Table S30 are a function of it, because a saturating row’s error is whatever the clip makes it. Sweeping  $x_2$  over  $1 - 10^{-3}$ ,  $1 - 10^{-4}$ ,  $1 - 10^{-5}$ ,  $1 - 10^{-6}$  (used),  $1 - 10^{-7}$ ,  $1 - 10^{-8}$  and  $1 - 10^{-10}$  gives, in RMSE  $\log_{10} S$ : DirectGNN 0.995 at every one of the seven; NRTL 1.233, 1.324, 1.429, 1.544, 1.667, 1.797, 2.072; COSMO-SAC-grounded 1.183, 1.242, 1.311, 1.387, 1.470, 1.558, 1.746;  $\sigma$ -oracle 1.421, 1.471, 1.531, 1.598, 1.672, 1.752, 1.925. Two readings hold across the whole span, and separately at each of the three seeds: the control is below every physics arm, and every physics arm is above every published leader (0.83 to 0.99). Four do not, and each of the four fails on the seed axis as well as the clip one. The NRTL and  $\sigma$ -oracle rows exchange places between  $1 - 10^{-7}$  and  $1 - 10^{-8}$  in the seed-averaged sweep, between  $1 - 10^{-4}$  and  $1 - 10^{-5}$  at seed 42, and not at all at seeds 43 and 44, where NRTL is the lower at all seven; their printed order is not a fact about the models. The scale reference predicts one number for the whole test set, does not saturate, and reads 1.326 at every clip and at every seed, the locked row set being the same 5440 rows at all three; the physics arms are under it in 15 of the 63 arm–seed–clip cells. Seed-averaged, the grounded arm is below it at the three tightest clips (1.183, 1.242, 1.311) and the NRTL arm at the two tightest (1.233, 1.324); the  $\sigma$ -oracle is above it at all seven. Per seed, the grounded arm is below it at all seven clips at seed 42 (at the deposited clip 1.125 against 1.326, the one cell of the printed column that beats the reference), at the tightest clip at seed 43 and at the two tightest at seed 44; the NRTL arm at the tightest clip at seed 42 and at the two tightest at seeds 43 and 44; the  $\sigma$ -oracle at no clip and no seed. So whether a physics arm beats that reference depends on the constant *and* on the seed: at the value we deposit the seed-mean column does not beat it, and one arm at one seed does. In MAE  $\ln x_2$ , by con-

trast, every arm including the  $\sigma$ -oracle is below the same reference (2.325) at every seed, so the reading is confined to this column. “Near the cycle’s better end” (1.43) fails at the tight end, where the NRTL and grounded arms sit below it at every seed and the  $\sigma$ -oracle’s seed mean (1.421, printed as 1.42) does too, though not its seed-43 value (1.479, printed as 1.48); §3.7 does not place the physics arms within that range. And the physics arms do not *uniformly* trail their control by more in this column than in MAE (a comparison of an RMSE deficit against an MAE one). The NRTL and COSMO-SAC-grounded arms do, at all seven clips and separately at each of the three seeds: the smallest such excess anywhere is +0.011, the grounded arm at seed 42 and the tightest clip. The  $\sigma$ -oracle does not. Its seed-averaged  $\log_{10} S$  deficit over the control runs 0.425, 0.476, 0.536, 0.603, 0.677, 0.757, 0.930 across the seven clips against an MAE deficit of 0.549, so the reading fails at the three tightest clips and holds at the deposited one by 0.054; per seed it fails at all seven clips at seed 42 and at the two tightest at seeds 43 and 44. §3.7 therefore states that reading of the other two arms only. A fifth reading fails on the seed axis alone, and neither document carries it: the grounded arm is the lowest of the three physics arms at every clip in the seed-averaged sweep above, but not at every seed. At the deposited clip the per-seed values are, at seeds 42, 43 and 44, grounded 1.125, 1.558, 1.478 against NRTL 1.643, 1.472, 1.515; at seed 43 the NRTL arm is lower at all seven clip values. It is a property of the seed-mean column and not of the models. What may be carried away from that column is what survives the sweep, and that is the two readings above and not the printed decimals.

**The unit bridge behind the  $\log_{10} S$  column.**  $\log_{10} S$  is obtained from  $\ln x_2$  as  $S = x_2 c_{\text{solv}} / (1 - x_2)$ , the solvent’s molarity  $c_{\text{solv}}$  in  $\text{mol L}^{-1}$  being  $1000\rho/M$  from a tabulated *mass* density  $\rho$  ( $\text{g cm}^{-3}$ ) and the molar mass  $M$  ( $\text{g mol}^{-1}$ ) where the solvent has one, and  $1000/V_m$  from the molar volume  $V_m$  ( $\text{cm}^3 \text{mol}^{-1}$ ) of one generated conformer where it has none (this  $\rho$  is unrelated to the Hammett

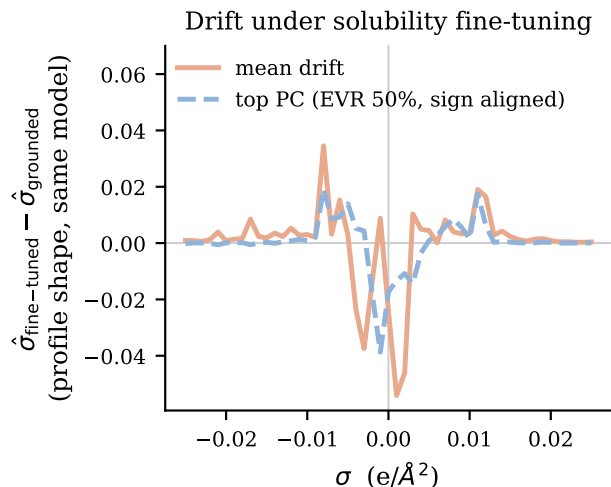


Figure S5: The drift solubility fine-tuning imposes on the learned  $\sigma$ -profile ( $n=44$  VT-2005-matched molecules):  $\hat{\sigma}$  after fine-tuning minus the *same model’s grounded profile*, not minus the external reference. Shown are the mean drift and its dominant PCA mode, defined only up to sign and drawn in the mean drift’s direction; the legend’s EVR is that mode’s explained variance ratio.

reaction constant of §3.6). Since  $x_2/(1 - x_2)$  is a mole ratio,  $S$  is moles of solute per litre of *solvent* rather than of solution, and the two conventions diverge as  $x_2 \rightarrow 1$ . The same converter runs on every row of the first two blocks of Table S20. It carries *no* temperature dependence—one ambient  $c_{\text{solv}}$  at every  $T$ —and it needs a molarity, so the 168 of the 5608 labelled test rows whose solvent has none carry no true  $\log_{10} S$  and leave that column for *every* arm, which is why that table’s rows carry two sample sizes. Predicting the mean of the true  $\log_{10} S$ , rather than running the constant- $\ln x_2$  scale reference through this bridge, would read 1.28.

## S5 Supplementary mechanism results

**What the drift runs are, and what they are not.** §3.3 states the departure, the three nulls, the transfer ratio and the direct cancellation check, each with the caveat that scopes

it. This section is the working behind them, and three things belong with the working rather than there. The design holds architecture and initialisation fixed and varies only the objective: one grounded model checkpointed before and after solubility fine-tuning with the  $\sigma$  head unfrozen, at three seeds, reported mean $\pm$ sd. Those runs are not the four COSMO-SAC arms of §3.1, which hold the head from P2 onward (Table S4); and that freeze holds the head and not the encoder feeding it, so those arms’ profiles are not pinned at their warm-up value either, while the deposited per-arm predictions carry no  $\sigma$ -profile and so cannot be made to yield one. Their drift is unmeasured in both directions, which is why what follows is not established of the checkpoints whose paradox it is offered to explain. And the margin by which the top-two share clears the one null it clears is small: 0.399 against 0.221 is roughly  $1.8\times$ .

**The dominant mode, and what the two cancellation checks share.** The dominant drift mode (Fig. S5) transfers area *out of* the nonpolar bins ( $\sigma \approx 0$ ) and *into* the polar wings; the mode is defined only up to sign and drawn there in the mean drift’s direction, and the plotted drift is taken against the same model’s grounded profile rather than against the external reference. Whether that shift *cancels* part of  $B_{\text{closure}}$ —the input compensating the map, as the non-identifiability of the split allows (Lemma S1)—is the comparison  $\text{MSE}(m, g(\hat{\sigma}))$  against  $\text{MSE}(m, g(z^*))$  on the same pairs, and §3.3 reports that it argues against the cancellation reading and names the three limits on it. Both that check and the closure’s own-axis sweep below sit on the infinite-dilution axis, so neither refutes cancellation in the finite-composition regime that generates the drift: they remove the one place it could have been measured, and the cancellation reading remains inferred from the paradox itself. The two paragraphs that follow give the power calculation that limits what the first can settle, and the sweep that is the second.

**The activity map’s Fisher geometry, and why  $\sigma$  is under-determined.** At infinite di-

lution the aggregate activity Fisher  $J^\top J$  ( $J = \partial \ln \gamma_2^\infty / \partial \sigma$ ) is rank-one (participation ratio 1.0 by autograd; finite differences decorrelate the redundant flat-profile coordinates and inflate it, §S6.9, so we read PR against a permuted control and not absolutely). The per-molecule Fisher that governs the drift has participation ratio  $\approx 1.4$  of 51 even at full solvent coverage. Those two figures are the measurement behind the rank-one statement of §3.3. The resolved direction has a chemical locus as well as a rank: the dominant eigenvector concentrates on the extreme- $|\sigma|$  bins the closure’s hydrogen-bonding term acts on—the same axis the solvent cascade of §S3.8 orders its classes by—so the coordinates the map leaves free are the nonpolar and weakly polar interior of the profile. The participation ratio is a concentration statistic and not an exact count. It is measured at infinite dilution, so its transfer to the finite-composition regime that generates the drift is not established, the same scope gap that qualifies  $B_{\text{closure}}$  (§3.7). The external reference is itself *computed* (VT-2005 quantum chemistry) rather than experimental, so none of this establishes that recovery of the physical profile is impossible in principle.

**Why the top-two share is read against a phase-randomised null, and how unstable the magnitudes are.** A top-two explained-variance share is not interpretable against an isotropic null, because smoothness alone concentrates a spectrum: smoothing isotropic noise reaches a top-two share of 0.25 to 0.83 as the correlation length grows from one bin to eight, spanning the observed value without any between-molecule alignment at all. The *phase-randomised* null of §3.3 preserves each deviation’s amplitude spectrum exactly—and so its smoothness—and destroys only the alignment across molecules that a shared compensating direction asserts; the observed 0.399 against its  $0.221 \pm 0.011$  ( $p < 0.001$ ) is therefore a statement about alignment and not about smoothness. The wrong-molecule null ( $0.772 \pm 0.023$ ,  $p = 1.00$ ), which the observed share does *not* clear, is a difference between two points on the reference manifold and inherits that

manifold’s own concentration, so it demands more alignment than a shared direction implies. That is the argument for preferring the phase-randomised one, and §3.3 states that if it is not accepted the spectrum is uninformative rather than supporting. In each case the  $\pm$  is the null’s spread over resamples, not an uncertainty on the observed value. The three magnitudes §3.3 reports but does not carry move in both directions across analysis configurations: a shortened run gives only  $\sim 16\%$  departure and a transfer of  $1.7\times$ , below what a constant offset alone reaches on the one run where that control can be computed, while uncontrolled two-model comparisons over-state both the departure ( $\sim 82\%$ ) and the transfer ( $16\text{--}26\times$ ). With  $n=3$  the  $\pm$  on the three-seed figures is indicative and not a precision. Nor does mean-centred low-rank structure exclude a shrinkage of each profile toward the corpus-mean shape: the reference profiles themselves sit at 0.77 mean-centred, so any drift linear in the profile inherits that concentration.

### The drift spectrum, and the null it lacks.

Figure S6 is the spectrum of the drift whose mean and dominant mode are Fig. S5. Its top two modes carry 72.9% in the companion run plotted there and  $71.8 \pm 1.3\%$  over the three seeds (quoted as  $72 \pm 1\%$  in §3.3): the same statistic on different runs. The companion run’s spectrum and the three-seed aggregate are both deposited; the aggregate’s top-two share has mean 0.71819 and standard deviation 0.01265 over seeds 0–2. Both this figure and Fig. S5 are drawn by one deposited script. That share has no matched null, so the panel is titled by what it plots and not by an inference. The phase-randomised and wrong-molecule nulls of §3.3 are run on the  $\hat{\sigma}$ -versus-reference deviation instead, a different quantity; the isolation drift has not been re-tested that way, because the two locally available checkpoints share a  $\sigma$  head and so have identically zero drift between them.

### The grounded warm-up checkpoint’s 0.73, and the power of the two-MSE check.

The direct cancellation check of §3.3 has almost no power in the direction that matters. Holding

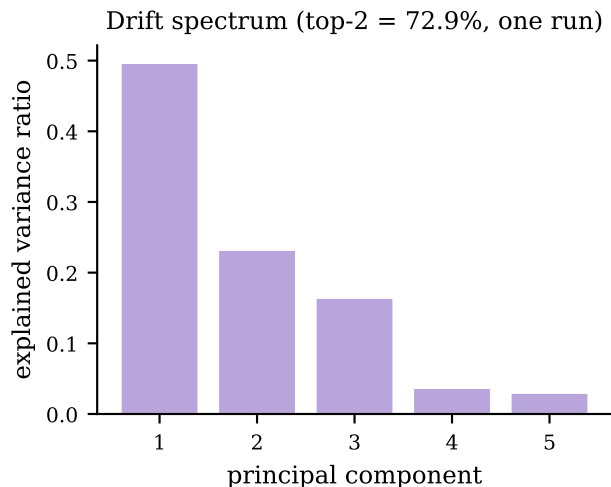


Figure S6: Explained-variance spectrum of the fine-tuning drift on the 44 matched molecules (companion run).

the cavity area at its reference value, the sole retained end-to-end SLE-trained COSMO-SAC checkpoint scores 18.75 against the reference input’s 1.47 on the  $n=60$  VT-2005-matched pairs: on the activity axis the drifted profile is an order of magnitude *worse* through the same closure, which is the check the article reads and the direction it runs. On the same set, 18.75 against  $\text{Var}(m) = 2.47$  is  $7.6\times$  worse than predicting the mean, the signature of a broken profile rather than of a partly compensating one, so the comparison separates “no cancellation” from “large cancellation” and not from “some cancellation”. The same comparison run on a grounded warm-up checkpoint, whose profile sits 28% from the reference, scores 0.73 where the end-to-end SLE-trained checkpoint scores 18.75. That is a statement about the closure’s sensitivity to profile roughness and about an in-sample fit to  $z^*$  on molecules inside the grounding pool, not evidence that the encoder carries information the reference lacks, and we do not read it as a both-channel reproduction of the paradox.

#### The closure’s own-axis check on polarity.

The second check of §3.3: on the  $n=60$  VT-2005-matched set the reference-input closure *under*-predicts, its mean output being +0.10 under the residual-only convention and −0.10

under the full one against a mean target  $m = +0.75$ . Sweeping its hydrogen-bond coefficient at zero, one and two times the standard value gives mean squared errors 2.42/2.83/10.15 on the strong-donor pairs, 0.090/0.092/0.918 on the acceptor-only pairs and 1.205/0.867/0.578 on the inert ones. All three classes are printed because they do not agree: the strong donors worsen with the coefficient, the acceptor-only pairs are flat until the doubling and then worsen, and the inert pairs improve monotonically. What the three together are consistent with is  $B_{\text{closure}}$  sitting in the hydrogen-bond kernel; they do not localise it there, since the classes are assigned by solvent alone and five of the eight inert-class pairs carry a solute with donor-window mass of its own—five primary or secondary amines, carrying 1.2 to  $3.8 \text{ \AA}^2$  of it against benzene—whose segment activity the coefficient scales at infinite dilution; for the other three—3-methylpyridine once and tripropylamine twice, all at  $0.000 \text{ \AA}^2$  in that window—it has no route at all, the exchange energy being a product of a donor and an acceptor segment, and those three are unaccounted. Nor is any of it a statement about profile polarity: the sweep varies a kernel constant, and no step connects a constant to a profile. The strong-donor column carries one further result, printed because it is not favourable to the closure this paper deploys: at  $c_{\text{hb}} = 0$  the error on its 28 pairs is 2.42 against 2.83 at the standard value, so on that class the 2002 kernel’s single-parameter hydrogen-bonding term costs more than it buys. Those pairs are concentrated rather than spread: 13 of the 28 carry one solvent and 20 carry two. No interval is quoted because none exists to quote: the sweep re-evaluates one fixed closure on one fixed set of pairs at three coefficient values, so its only sampling variation is the set’s, and what it establishes is the sign of a derivative and not a magnitude. It is independent of the two-MSE check above in what it varies—the closure’s own parameter rather than the profile handed to it—and dependent on it in the set it varies them over. What it is not is a second check against cancellation, and §3.3 does not read it as one.

**Whether a downstream correction repairs it.** If the closure binding of §3.3 were merely an inaccurate point estimate, a learned output-space correction on top of the physics prediction should recover accuracy. It fails to. An additive bounded residual with a learned confidence gate leaves the gate effectively shut. Forcing it open removes the systematic *bias* ( $+0.45 \rightarrow +0.06$ ) while recovering little *accuracy* ( $R^2$  0.236  $\rightarrow$  0.274 on the same rows), as expected if the closure and not the estimate is the binding constraint. Both arms run at seed 42 alone, and the statement is about the one correction tested: a gated additive bounded residual on the output, on this closure and this corpus.

**The reference ladder.** A relative deviation carries no absolute scale, which is what the ladder supplies. Its rungs are built from the reference profiles the grounded arm was supervised on, so it calibrates that deviation rather than competing with it. The rungs calibrating the 51% departure of §3.3, all on the same 44 molecules and the same relative- $L_2$  metric: a leave-one-out corpus-mean profile (one fixed shape reused for every molecule, carrying no molecular information) 0.493; each molecule assigned another molecule’s reference profile, 0.651–0.684 over five derangements; a uniform profile 0.889. Against these, the grounded base sits at 0.364 and the fine-tuned latent at 0.507. The grounded base is the  $\sigma$ -supervised recovery §3.3 reports as reproducing the reference profile far more closely: it is  $36 \pm 1\%$  over the three seeds in the norm ratio  $\|\hat{\sigma}_{\text{grounded}} - \sigma\|/\|\sigma\|$ , and it is an *in-sample* fit—all 44 probe molecules lie in the  $\sigma$ -grounding pool and the warm-up early-stops on it—so no out-of-sample recovery estimate exists on this set.

**The distortion operator  $D$ .** The transfer test of §3.3 fits a dense affine map on the 51-bin grid: a  $51 \times 51$  linear part plus a 51-bin offset, 2652 parameters, each output bin a ridge regression of 52 coefficients ( $\ell_2$  penalty  $10^{-3}$ ) on 22 of the 44 molecules. The constant-offset control replaces  $D$  by a single molecule-independent shift of the whole profile, fitted on

the same 22 molecules and applied unchanged to the held-out 22. It can be computed on one companion run only—the two-model comparison whose two checkpoints are both still on disk—and there it is  $2.1\times$  better than the identity null on both of that run’s arms. That run is the uncontrolled one, whose own transfer of 16–26 $\times$  §3.3 reports as over-stated, so its residual factor is not read across. The three seed checkpoints behind the  $4.1 \pm 0.5\times$  were not retained, and the remaining companion run records only an ephemeral checkpoint path, so no offset control exists for either and the  $4.1\times$  is a ratio against the identity null with no constant-offset control of its own. The homologue saturation that makes “held out” largely within-cluster interpolation is six  $n$ -alkanes, four halobenzenes, five pyridines and four butyl and hexyl amines among the 44; the median held-out molecule sits 0.048 in  $L_2$  from its nearest training-half neighbour, about 16% of its own profile norm.

**Non-identifiability, empirically (the corrupted twin).** Training a twin model on data whose crystal labels are corrupted (every  $T_m$  shifted by +273 K) leaves the *solubility* fit essentially unchanged— $\ln x_2$  MAE 1.835 versus 1.812 on the corrected data—while the recovered crystal grounding differs by hundreds of kelvin ( $T_m$  MAE 258 K versus 45 K). Two very different crystal/activity splits produce the same solubility, which is Lemma S1 made concrete.

**Crystal grounding on physical labels.** When the crystal head is supervised on measured melting points it predicts  $T_m$  to 45 K MAE, about  $4\times$  better than a Joback group-contribution reference—so the unresolved downstream result of §S6.3 is not an inability to learn crystal properties. That experiment’s external pool was 97.76% melting points the model was already trained on.

**The  $\Delta C_p = 0$  approximation.** The ideal term omits the constant-heat-capacity correction; a group-contribution audit finds a nonzero  $\Delta C_p$  for 1441 of 1525 solutes (108,287 rows), so

this is a model-form omission and not a rounding. It cannot produce either headline *value*, and the arithmetic that says so is about the evaluation and not about the training. The closure/insufficiency decomposition is measured on the activity target ( $\ln \gamma_2^\infty$ , at 298 K on the VT-2005-matched set and at each row’s own temperature over 273–403 K on the broad IDAC set), where  $\Delta C_p$  does not enter at all on either set: the ideal term is not evaluated in the decomposition, so no temperature range of it is at issue. The grounding paradox is an arm-to-arm difference (learned- $\sigma$  vs. oracle- $\sigma$ ) whose two arms share the *identical* crystal term ( $\max |\Delta \Phi|$  of  $3.2 \times 10^{-5}$  over the scored rows), so any  $\Delta C_p$  error cancels exactly in that comparison. What it is not orthogonal to is the *mechanism* of §3.3. Setting  $\Delta C_p = 0$  leaves  $\Phi$  too large by a median 0.32 and a mean 0.40 in  $\ln x_2$  on the 31 solutes carrying the 1318 labelled test rows with a measured  $T_m$  (at  $60 \text{ J mol}^{-1} \text{ K}^{-1}$ ), positive on every one of them, and the crystal head is externally supervised toward measured  $T_m$  and  $\Delta H_{\text{fus}}$ , so it is not free to absorb the bias; what absorbs it is  $\ln \gamma_2$ , which is the  $\sigma$  head. A drift toward the polar wings is what compensating a one-signed excess in  $\Phi$  would look like. The latent drift of §3.3 therefore has a live crystal-side explanation alongside the activity-kernel one, and nothing here separates them. Settling it means retraining the grounded and  $\sigma$ -oracle arms at nonzero  $\Delta C_p$ ; the configuration field exists and the run was not made.

**Fusion-supervision scarcity.** Direct  $\Delta H_{\text{fus}}$  labels are extremely sparse: 1261 training rows across 30 solutes, none at all in validation, and 18 rows on a single solute in test. This is the quantitative basis of the identifiability argument—there is almost no signal with which to pin the crystal term from data.

**The NRTL branch trains (a trainability check).** The asymptotic physics penalty is not an under-training artifact of the activity branch. In the headline NRTL arm the head produces solvent-varying interaction parameters— $\tau_{12}$  over the  $n=8103$  test pairs has mean 0.97 and std 2.5 (range  $[-2.6, 22]$ ),

far from collapsed—yet the arm still trails the closure-free DirectGNN in the seed mean (1.79 vs 1.70; Table S4), at two of the three seeds and not at seed 42, where it is ahead (1.734 against 1.749). A trained, non-degenerate activity branch that the control outperforms at two seeds of three is evidence that the penalty is structural rather than a failure to train the branch, and weaker evidence than a clean sweep of the seeds would be. (The core closure-misspecification result—the  $\sigma$ -oracle and the decomposition—is in any case training-*independent*: it feeds the reference VT-2005 profiles through the fixed closure with no encoder optimisation, so it cannot be a training pathology.)

### Structured temperature extrapolation.

Withholding *temperature* rather than structure (train at  $T \leq 309.75 \text{ K}$ , test at higher  $T$ ;  $n = 3343$ ), the affine-in- $1/T$  hypothesis classes extrapolate far better than an unstructured descriptor regressor (Table S21); the van’t Hoff class also predicts the *direction* of the high- $T$  shift correctly 99% of the time versus 40% for the random forest. This is a property of the structured hypothesis class and not an accuracy claim for our model.

It is also not a claim in the other direction, and the reason is a defect in the evidence rather than a result. The one TGNN checkpoint ever scored on this split predicts, statistically, a constant: prediction standard deviation 0.58 against a measured 2.56, and an MAE of 1.95 against 2.00 for the best global constant and 1.46 for the per-pair constant already in Table S21. Its activity and correction outputs never left initialisation— $\ln \gamma_2$ ,  $\tau_{12}$ ,  $\alpha$ , the gate and the correction magnitude have standard deviations between  $1 \times 10^{-6}$  and  $6 \times 10^{-4}$  over the test rows—and it was trained at a one-epoch first phase, on the pre-repair crystal labels of the corrupted-twin paragraph above. What it measures is the trainability of the bottleneck at that budget, not the temperature behaviour of a trained physics path, so no trained arm appears in Table S21 and none should be inferred from its absence. Settling the question needs the closure-carrying and closure-free arms re-



trained on this split at the budget used everywhere else here, read on the *bias* of the fitted per-pair slope as well as its dispersion; that is a GPU run we have not made.

Table S21: Same-pair high-temperature extrapolation ( $n = 3343$ , train  $T \leq 309.75$  K). “dir.” is the fraction of pairs whose high- $T$  shift direction is predicted correctly.

| Model                         | MAE  | $R^2$ | dir. |
|-------------------------------|------|-------|------|
| per-pair van’t Hoff ( $1/T$ ) | 0.37 | 0.89  | 0.99 |
| per-pair linear in $T$        | 0.41 | 0.85  | 0.99 |
| per-pair last-low- $T$ hold   | 1.20 | 0.69  | 0.01 |
| RF (Morgan + $T$ )            | 1.29 | 0.66  | 0.40 |
| per-pair mean                 | 1.46 | 0.56  | 0.01 |

## S6 Supplementary diagnostics and broad negatives

### S6.1 Supporting diagnostics

The interpretive claims of the preceding sections are based on a set of controlled diagnostics, each run under a named protocol. We report them here with their exact numbers, including the results that are unfavourable to the physics-informed hypothesis.

**Aleatoric noise floor.** The derivation, the two aggregating functionals and their values are in the article at §2.2. One property of the per-group distribution is not: it is strongly right-skewed, which is why the mean and the median differ by a factor of five. On the groups backed by two sources the 90th percentile of the per-group disagreement is  $0.42 \ln x_2$  units and the 99th is 1.41; on those backed by three, 0.83 and 3.55. The floor’s mean is therefore set by a minority of badly disagreeing systems rather than by a broad band, and on either reading it lies far below the scaffold-split error.

**A linear probe on the encoder, and what it does not establish.** To separate what the frozen graph embedding can represent from what it transfers out of distribution, we fit a

ridge probe from the encoder representation onto fifteen RDKit descriptors and compare in-distribution and held-out fit. The median train  $R^2$  is 0.758 against a test  $R^2$  of 0.448 ( $n = 700$  solutes in each partition), a gap of 0.310. Individual descriptors split sharply: FractionCSP3 transfers well (test  $R^2 = 0.96$ ) while HeavyAtomCount does not (0.23), so the train–test gap partly reflects covariate shift in the descriptor *targets* across scaffolds rather than an encoder property alone, and the probe measures descriptor-representability and not task-relevant expressivity.

**That gap does not establish that the encoder is transfer-limited rather than expressivity-limited.** The probe carries no control that would separate the two, and it does not survive one. An untrained, randomly initialised encoder of the same architecture scores 0.707/0.415/0.291 on the same three quantities and is awarded the *identical* verdict by the same decision rule; trained-minus-random bootstrap intervals cross zero on all three; a second random initialisation moves the gap to 0.194 and flips the rule’s verdict string, so the 0.25 threshold that rule turns on sits inside its own null’s spread. Nor is a high train  $R^2$  evidence that the embedding is expressive: isotropic Gaussian features at the same  $d = 768$  and  $n = 700$  reach a train  $R^2$  of 0.979, above the trained encoder’s, which is ridge geometry at  $d \approx n$  and not chemistry. What the probe establishes, once the random-encoder arm is subtracted, is only that training adds about 0.11 in  $R^2$  of linear descriptor decodability on train solutes and  $0.00 \pm 0.07$  on test. Whether a new graph encoder would move out-of-distribution accuracy is therefore *unmeasured* here rather than settled; the expectation that pretraining and coverage are the more likely levers is cited to the arm comparisons and not to this probe.

**Temperature extrapolation.** A structured van’t Hoff activity class (log-solubility affine in  $1/T$ ) encodes the correct temperature dependence of  $\ln \gamma_2$  by construction, and this is apparent when the split withholds temperature rather than structure. Training on measurements at  $T \leq 309.75$  K and testing at  $T \geq$



330 K ( $n_{\text{test}} = 3343$ ), the structured class attains a high-temperature MAE of 0.37 against 1.29 for a random forest over Morgan fingerprints plus temperature (in  $\ln x_2$  units), consistent with the temperature-aware  $\ln \gamma_2^\infty$  modelling of.<sup>S16</sup> This is a property of the structured hypothesis class, not a performance claim for the trained model: the functional form carries temperature extrapolation that a descriptor-plus-temperature regressor does not. Neither is it a claim that our trained arms fail to carry it. The one checkpoint scored on this split is a near-constant predictor whose activity branch never left initialisation, so it separates nothing (§S5); what would separate it is a retraining of both arms on this split at the budget used elsewhere here.

**Ranking and calibration.** Absolute error and decision quality diverge, and within one checkpoint the divergence has a direction. Table S22 ranks the solvents of each solute-temperature group of the labelled test rows under the two  $\sigma$ -profiles of §3.2: the same weights, the same rows, and only the profile fed to the closure differing ( $\max |\Delta\Phi| = 3.2 \times 10^{-5}$  over these 5608 labelled rows at the worst of the five seeds, against 4.4 to 5.9 between separately trained arms). With its own learned profile the model orders solvents above two solute-blind references at every one of the forty margins that comparison has—four metrics, five seeds, two floors—and 27 of the forty intervals exclude zero. With the reference profile substituted, 36 of its own forty margins do not separate it from those references, and the four that do are the shuffled-profile floor at seeds 43 and 44; both counts are broken out below. Both arms remain far above the within-group permutation floor, so the substitution does not reduce the ordering to chance; what it leaves cannot be told apart from what a ranker never shown the solute produces on most of the design, which is a failure to separate the two and not a demonstration that they are the same. Of the paired reference-minus-learned differences, all five per-seed intervals exclude zero for  $\rho$ , four of the five for  $\tau$  and nDCG@3, and two of the five for top-1, which is why that row is quoted pooled. A

floor built from the other arm is not a comparison, so each is rebuilt from the predictions of the arm it is set against. Groups are formed on solute and temperature rounded to 1 K, kept when they hold at least three distinct solvents, with duplicate measurements averaged per solvent. The effective sample is the 107 solute clusters, not the 823 groups, and every interval below is a 95% bootstrap over those clusters.

The shuffled-profile floor gives each group another randomly drawn solute’s predictions over the same solvents, 200 to 300 draws; the solute-blind label floor ranks the group’s solvents by their leave-one-solute-out mean measured solubility, which is label-derived and therefore an optimistic reference. Both are paired per group against the arm they are built from. The reading is of the intervals and not of the point estimates: the reference arm’s forty margins are all small and 36 of them positive (the four exceptions are top-1 at seeds 42, 45 and 46 against its own shuffled-profile floor,  $-0.009$ ,  $-0.016$  and  $-0.028$ , and Kendall at seed 45 against the label floor,  $-0.010$ ), and what the intervals say is that 36 of the forty do not separate that arm from its floor. The four that do are one arm of the design rather than a scatter: the shuffled-profile floor at seeds 43 and 44, on  $\rho$ , Kendall and nDCG@3 at seed 43 and on nDCG@3 alone at seed 44, the largest of them  $+0.101$  [ $+0.006$ ,  $+0.202$ ]. So *indistinguishable from its floor* is what the design supports at three of its five seeds and not at the other two, and *below* it is supported nowhere. The learned arm loses its own unanimity in the same pass and in the same place: all forty of its margins are positive and 27 of the forty intervals exclude zero, the thirteen that do not being seven of seed 42’s eight and four of seed 45’s. The contrast survives excluding the 16 groups where some solvent has no reference profile and the arm therefore runs on a mixture (807 groups:  $-0.097/-0.195/-0.215/-0.158/-0.177$ ), survives a leave-one-solute-out jackknife over the 107 clusters (every seed’s range lying wholly below zero and none of the five wider than 0.036: at seed 42, the narrowest gap of the five, it runs  $[-0.132, -0.096]$ ), and exceeds at four of the five seeds the largest difference be-

Table S22: Solvent ranking under the two  $\sigma$ -profiles, on the labelled test rows grouped by solute and temperature (823 groups, 107 solute clusters,  $\geq 3$  solvents per group). Both columns are one checkpoint per seed; only the profile injected into the closure differs, and per-seed values are 42/43/44/45/46 from the five-seed leak-free re-run. The last column is the paired per-group difference averaged over the five seeds, with a 95% cluster bootstrap over solutes. The lower block gives each arm’s margin over its *own* solute-blind floors, each rebuilt from the predictions of the arm it is set against, then counts over four metrics  $\times$  five seeds  $\times$  two floors how many of those margins have an interval excluding zero. The last row is the within-group permutation floor.

|                                                                   | learned $\sigma$                   | reference $\sigma$                 | reference – learned     |
|-------------------------------------------------------------------|------------------------------------|------------------------------------|-------------------------|
| Spearman $\rho$                                                   | 0.474/0.612/0.596/0.515/0.556      | 0.361/0.399/0.365/0.343/0.362      | −0.185 [−0.269, −0.107] |
| Kendall $\tau$                                                    | 0.388/0.536/0.522/0.457/0.490      | 0.304/0.338/0.301/0.281/0.300      | −0.174 [−0.251, −0.103] |
| nDCG@3                                                            | 0.711/0.786/0.793/0.740/0.771      | 0.684/0.705/0.691/0.675/0.690      | −0.071 [−0.114, −0.029] |
| Top-1 accuracy                                                    | 0.471/0.565/0.586/0.502/0.520      | 0.422/0.453/0.469/0.435/0.424      | −0.088 [−0.172, −0.009] |
| <i>Margin over its own solute-blind floor (<math>\rho</math>)</i> |                                    |                                    |                         |
| shuffled-profile floor                                            | +0.050/+0.223/+0.222/+0.099/+0.176 | +0.024/+0.100/+0.053/+0.001/+0.040 |                         |
| solute-blind label floor                                          | +0.110/+0.262/+0.268/+0.158/+0.203 | +0.018/+0.070/+0.030/+0.003/+0.017 |                         |
| of 40, excluding zero                                             | 27                                 | 4                                  |                         |
| <i>Groups whose measured best solvent the arm picks, by seed</i>  |                                    |                                    |                         |
| picks the best solvent                                            | 18.2/22.1/24.2/21.4/19.6%          | 13.2/10.9/12.5/14.7/10.0%          |                         |
| permutation floor, $\rho$                                         | 0.00 $\pm$ 0.02, both arms         |                                    |                         |

tween seed replicates of one arm; that benchmark is now  $|\Delta\rho| \leq 0.132$  over twenty seed pairs, four of which exclude zero, all four of them the learned arm’s and all four involving seed 42 or seed 45. Two further readings are not available from it. The floors are built at the prediction level, not by a forward pass on a permuted profile, so they are faithful to *which* profile the closure sees and not to how its non-linearity would answer a profile no molecule has; and the closure-carrying arm cannot be ranked against DirectGNN here at all on this tree, the re-run training no DirectGNN arm. In the published three-seed family, where both arms exist, that difference—DirectGNN minus the learned- $\sigma$  arm—is  $-0.056/+0.053/-0.040$  by seed, sign-flipping, each interval spanning zero and each smaller than that family’s seed-replicate spread.

**Per-group recalibration of the MAE gap, in sample and out.** The reading on which the substitution merely miscalibrates is granted in §3.2 at three strengths, each fitted inside each group, on exactly the row set the ranking uses: the 823 groups, 4912 solute–solvent–temperature cells, 107 solute clusters. No recalibration; an offset only, each group’s own mean subtracted from prediction and from

truth; and an affine map, slope and intercept fitted per group by ordinary least squares of measured on predicted  $\ln x_2$ . What is reported is the mean of the per-group MAEs and the paired reference-minus-learned difference, bootstrapped over solute clusters as everywhere here. Every interval in this ladder is printed to two decimals: the convergence-and-printing rule the glycol-ether interval obeys (§S3.8) was not run for these, so their third decimal is not certified against Monte-Carlo error. In sample, pooled over the five seeds, the ladder is  $+0.426$  [ $+0.30, +0.55$ ] raw,  $+0.370$  [ $+0.25, +0.49$ ] offset-only and  $+0.040$  [ $-0.00, +0.08$ ] affine; per seed the affine residual is  $+0.003/+0.063/+0.082/+0.039/+0.011$ , three of the five spanning zero, and at five seeds the pooled value spans zero as well. What the affine map leaves is therefore not distinguishable from nothing, where at three seeds a residual of  $+0.060$  [ $+0.02, +0.10$ ] stood clear of it.

Those three maps are fitted on the rows they are then scored on, and the groups are small: sizes run from 3 to 18 solvents with a median of 5 and a mean of 6.0, 103 groups holding exactly three and 273 four or fewer, so in one group in eight two parameters are being fitted to three points. The out-of-sample variant

refits the same map leaving out the row it scores—for each solvent of a group the map is fitted on that group’s other solvents and the held-out solvent scored under it. Leave-one-out rather than a held-out half, because a half-split is undefined at three solvents, where a two-parameter fit consumes two points and leaves one; and rather than fitting on one seed and applying to another, which reuses the same rows and would test seed stability instead of overfitting. Out of sample neither map absorbs the gap. The offset-only map, one parameter, transfers as a level correction—it takes the learned- $\sigma$  arm’s mean per-group MAE from 1.94 to 1.07 and the reference- $\sigma$  arm’s from 2.36 to 1.55—but not on the quantity printed here, which is the gap between them: that stands at +0.481 [+0.336, +0.640] pooled, no smaller than the +0.426 raw gap and larger than the +0.370 in sample. The affine map does not absorb it either, and at five seeds it does not open it: the gap becomes +0.321 [−0.615, +1.202] pooled, an interval that holds the whole gap and none of it alike. Per seed it is +0.380/−0.861/+1.220/+0.634/+0.232, one of the five negative and only seed 45’s excluding zero, the leave-one-out affine statistic being heavy-tailed on the three-solvent groups where two parameters are fitted to two points. The three-seed version of this paragraph read the same statistic as +1.09 [+0.3, +2.2] and called it larger than the raw gap; at five seeds that ordering is gone and what is left is an estimator too unstable to order against anything. The direction is what the two arms do separately, and it has changed with the seeds: the refit now *worsens* both, the learned- $\sigma$  arm 1.94 to 2.15 pooled and the reference- $\sigma$  arm 2.36 to 2.47, where at three seeds it improved the first and worsened the second. Both worse and by nearly the same amount is why the gap no longer opens. That instability is carried by the small groups. On the 550 groups with five solvents or more the out-of-sample affine gap is +0.166 [−0.03, +0.36], +0.144 [−0.08, +0.36], +0.283 [+0.11, +0.47], +0.225 [+0.01, +0.43] and +0.187 [−0.01, +0.39] by seed, against in-sample values of +0.035/+0.085/+0.098/+0.028/+0.048

there—the same small residual, measured too imprecisely at three of five seeds to be distinguished from zero. The in-sample 91% is therefore a bound on how much of the MAE gap a per-group affine map *can* absorb when it is fitted on the answer, not an estimate of how much a recalibration a user could apply would absorb, and §3.2 states it in that direction. Both fits, the ranking, the floors and every per-group fitted slope are deposited.

**The fitted slopes, and where the rank invariance holds.** Rank statistics are invariant under a per-group affine map only when that map is *increasing*; a fitted slope that comes out negative reverses the group’s order, and the invariance argument of §3.2 does not cover those groups. Of the 823 groups the fitted slope is negative in 124/83/83/125/118 of the learned- $\sigma$  arm’s (15.1/10.1/10.1/15.2/14.3% by seed) and in 166/127/147/148/171 of the reference- $\sigma$  arm’s (20.2/15.4/17.9/18.0/20.8%); the median slopes are 1.12/1.18/1.14/1.08/1.17 and 0.57/0.50/0.46/0.50/0.50, the reference arm’s the smaller because its within-group spread is the wider—median  $\text{sd}(p)/\text{sd}(t)$  of 1.03 to 1.34 against 0.58 to 0.69—while its alignment with the measurement is the weaker. Both arms’ maps increase in 621/639/649/626/598 groups by seed and in 485 at all five seeds together, 59% of the groups, a set that shrinks with every seed added. Restricted to those 485, the reference-minus-learned ranking gap is  $\rho$  −0.107 [−0.179, −0.031] pooled; per seed it is −0.069/−0.166/−0.141/−0.165/−0.131, and those five are on each seed’s *own* both-increasing set—the 621/639/649/626/598 above, a different row set at each seed—not on the 485 the pooled value is on. Also on the 485: Kendall −0.105 [−0.177, −0.028],  $\text{nDCG@3}$  −0.047 [−0.093, +0.003] and top-1 −0.085 [−0.194, +0.027]: about a third smaller than on the full set and of the same sign, with two of the four metrics spanning zero there where at three seeds only top-1 did. Groups whose predictions are constant have no fitted slope—there are 6/15/4/8/16 of them—so the 485 excludes them by construction. They leave two of the four metrics and not the

other two: Spearman and Kendall are undefined where every prediction in the group is tied, which is why those two contrasts run on 817/808/819/815/807 groups by seed rather than 823, while top-1 and nDCG@3 are defined there and score all 823. That seed-46 count of 807 is a coincidence of arithmetic and not the 807 of the mixed-coverage check above, which is a different subset.

Uncertainty, by contrast, is badly miscalibrated in the native model—a nominal-90% interval covers only  $\text{PICP@90} = 0.13$  of held-out points—but split-conformal wrapping repairs it to 0.92 empirical coverage at the same nominal level. Those two figures come from an earlier proxy checkpoint on a superseded split and are a property of an uncalibrated regressor of this family rather than of any arm of Fig. 2; the repair is the generic split-conformal guarantee and does not depend on which arm is wrapped. The ranking signal and the post-hoc calibrated intervals are the practically usable outputs, even where the point predictions are not competitive.

## S6.2 Identifiability and compensation

Lemma S1 asserts that the crystal/activity split is not recoverable from solubility data alone. Two diagnostics make this concrete. The first is a numerical Fisher audit of the SLE design, computed on synthetic data and independent of any particular activity closure. We evaluate the Fisher information matrix  $I$  of the observation model at a reference operating point and compute the conditioning  $\text{cond}(I)$  and the Cramér–Rao lower bound on the standard deviation of  $\Delta H_{\text{fus}}$  under three supervision regimes.

Every entry below is conditional on a design that must be stated, because a Cramér–Rao bound in J/mol scales directly with the assumed measurement noise. The operating point is  $\Delta H_{\text{fus}} = 25,000$  J/mol,  $T_m = 400$  K, and an activity term  $\ln \gamma_2^\infty(T) = a + b(T_{\text{ref}}/T - 1)$  with  $a = 1$ ,  $b = 200$  K and  $T_{\text{ref}} = 298.15$  K;  $x_2$  is not a free input but the SLE solution at each temperature, running from  $3 \times 10^{-8}$  to 0.15 across the window. The design is  $n = 9$  equally

Table S23: Fisher audit of the SLE design at  $\Delta T = 40$  K (synthetic, closure-independent).

| Supervision regime              | $\text{cond}(I)$  | CRLB $\text{sd}(\Delta H_{\text{fus}})$ [J/mol] |
|---------------------------------|-------------------|-------------------------------------------------|
| Solubility only                 | $2.3 \times 10^5$ | 19,895                                          |
| + $T_m$ label                   | $5.4 \times 10^3$ | 3,151                                           |
| + $T_m + \Delta H_{\text{fus}}$ | $1.6 \times 10^3$ | 1,689                                           |

spaced temperatures spanning  $\Delta T = 40$  K from 280 K, one measurement each. The noise levels are  $\sigma_{\ln x_2} = 0.3$  on each log-solubility observation,  $\sigma_{T_m} = 10$  K on a melting-point label and  $\sigma_{\Delta H_{\text{fus}}} = 2,000$  J/mol on an enthalpy label. Parameters are non-dimensionalised by the scales used in the model’s own configuration (5,000 J/mol, 50 K, 1, 300 K), so the condition number reports rank structure rather than unit choice. Table S23 reports the result at that window. With solubility measurements only, the design is severely ill-conditioned and the CRLB on  $\Delta H_{\text{fus}}$  is roughly 20,000 J/mol, comparable to the quantity being estimated. Adding a direct  $T_m$  label reduces the conditioning by nearly two orders of magnitude and lowers the bound to about 3,200 J/mol; supplying  $T_m$  and  $\Delta H_{\text{fus}}$  together lowers it to about 1,700 J/mol. The ill-conditioning is a property of the design rather than of noise level, and it is closed only by external single-component supervision, which the grounding streams supply.

Widening or narrowing the temperature window does not remove the ill-conditioning of the solubility-only design. Varying  $\Delta T$  over  $\{120, 80, 40, 20, 10\}$  K moves the solubility-only conditioning through  $1.7 \times 10^3$ ,  $5.6 \times 10^3$ ,  $2.3 \times 10^5$ ,  $6.5 \times 10^9$  and  $1.9 \times 10^{16}$ , and the CRLB on  $\Delta H_{\text{fus}}$  through 1,348, 2,249, 19,895,  $3.2 \times 10^6$  and  $8.0 \times 10^6$  J/mol. The point value quoted in the main text is the  $\Delta T = 40$  K entry of that sweep and is not a design-independent constant. Multi-temperature data alone therefore does not resolve the split; the two parameters remain confounded along the ideal-term direction. Restricting the activity slope rather than adding labels helps only partially: it reduces  $\text{sd}(\Delta H_{\text{fus}})$  to 8,251 J/mol, still well above the label-supervised regimes. A second, model-side diagnostic is not part of the primary argument. On the supervised test

set, with a group-contribution (Joback) crystal reference, the crystal and activity residuals are strongly anti-correlated ( $\text{corr}(\delta\Phi, \delta\ln\gamma_2) = -0.962$ ). This is an *accounting artifact*, not evidence of a learned split: the two residuals sum to the (negative)  $\ln x_2$  fit residual, so their correlation tends to  $-1$  as the fit improves, a permutation null that destroys any learned coupling still reproduces most of it, and the model’s own factors barely co-vary. The large apparent crystal offset is likewise not a usable mis-grounding metric — it is dominated by reference error, since the Joback reference is physically implausible on these drug-like polycyclics, and capping predicted  $T_m$  at a physical bound collapses the offset from  $\approx -7.1$  to  $\approx -1.5$ .<sup>1</sup> We therefore base the non-identifiability claim on the reference-free Fisher/CRLB audit above, and present the compensation figures in the SI (§S5) as a cautionary diagnostic rather than a decomposition-quality measurement.

### S6.3 Grounding the crystal factor: an arm underpowered by a pool the model already held

Whether the ceiling lies in the inputs rather than the closure can be tested by grounding a *different* factor—the crystal term—on abundant external data. We trained the identical model without and with an external single-component crystal-property pool of roughly 15,000 melting points (15,427 molecules), added through the grounding-stream mechanism after excluding any pool solute whose Bemis–Murcko scaffold appears in the validation or test split.

Two quantities follow and they are on different sets. On the held-out solutes, neither the property the pool supplies nor solubility improved: melting-point MAE  $45.0 \rightarrow 46.6$  K over the  $n = 2518$  test solutes carrying a  $T_m$  label, a paired difference of  $+1.59$  K with a solute-clustered 95% interval of  $[+0.31, +2.85]$ , and

<sup>1</sup>Permutation null (learned coupling destroyed)  $-0.90$  to  $-0.93$ ; model-factor  $\text{corr}(\hat{\Phi}, \ln\gamma_2) = -0.19$ ; the Joback reference predicts  $T_m > 1000$  K for a substantial fraction of these polycyclics.

$\ln x_2$  MAE  $1.81 \rightarrow 1.88$  over the 5608 labelled test rows. The melting-point half is therefore resolved against the grounded arm, and it is a shift rather than a change of regime: the middle half of the 2518 solutes move between  $-17$  and  $+21$  K, and the grounded arm is the worse of the two on 1329 of them. On the pool itself—a different set, and the one the next paragraph is about—the movement is resolved, on the pool as a whole and on its net-new part, and it runs the other way.

That held-out-solute comparison did not test external crystal grounding, because the pool was barely external. Of its 15,427 molecules, 15,082—97.76%—already carried a  $T_m$  label in the training corpus, 98.2% of them agreeing with the in-corpus value to within a millikelvin (mean  $|\Delta|$  0.07 K); the net-new contribution is 345 molecules, 2.24%. The two label sets are drawn from the same open melting-point tables, and nothing in the pool builder counts the overlap. Where the pool did carry something new, the crystal head moved. Scored on the pool itself,  $T_m$  MAE falls  $35.85 \rightarrow 35.21$  K (paired  $-0.636$  K, 95% CI  $[-1.09, -0.20]$ ), splitting into  $-0.371$  K  $[-0.81, +0.05]$  on the 15,082 duplicated molecules and  $-12.233$  K  $[-18.25, -6.27]$  on the 345 new ones—thirty times the movement per molecule on the fraction that supplied information.

Dose was not the constraint. The exported state is the one entering phase 3, after 79 epochs, by which point the stream had run 632 optimiser steps—2.62 passes over the pool, 93.0% of it seen at least once, the loader reshuffling each epoch—and 240 of those steps ran at loss weight 1.0, the phase-1 melting-point weight that an unset stream weight inherits, not at the phase-2 value 0.05. The downstream half of the held-out-solute comparison is in turn weaker than its row count suggests: the  $+0.065$   $\ln x_2$  MAE gap against the grounded arm is 5608 rows from 147 test solutes, and clustered by solute it is  $+0.088$  with a 95% interval of  $[-0.127, +0.284]$ , crossing zero. Nor is it the activity branch absorbing the crystal branch. Between the two arms the predicted  $\Phi$  differ by 1.25  $\ln$  units per row on average; regressing the predicted  $\ln\gamma_2$  on that difference gives a slope of

$-0.08$  where full compensation would be  $-1$ , so 91% of the crystal head’s disagreement passes into  $\ln x_2$ . The activity-path  $B_{\text{closure}}/B_{\text{insuff}}$  decomposition does not apply on this path in any case: the crystal “closure” is the ideal-solubility term  $\Phi = (\Delta H_{\text{fus}}/R)(1/T - 1/T_m)$ , a fixed function of two solute properties rather than a fitted map, so it has no free parameters to be misspecified in. Exact it is not, and on one part of this row set it is not even the right equation: 796 of these 5608 rows carry a solute of more than one component, for which the solid in equilibrium is not the one-component crystal  $\Phi$  is written for (§S3.2).

What the experiment licenses is narrower than a null on external crystal labels, and more useful: a grounding pool has to be measured by its net-new content before it can test anything. This one was scaffold-guarded correctly—none of the 2518 evaluation solutes appears in it—and was redundant all the same, because the guard protects the estimand and says nothing about what the model already holds. A real test declares the net-new count in advance, the builder reporting it and refusing to run below a floor of a few thousand molecules, which in turn needs a melting-point source disjoint from the one the training corpus itself drew on. Compute is not the obstacle: the grounded arm here was an eight-hour CPU run, so three seeds by two arms is on the order of 50 CPU-hours. Sourcing the labels is.

## S6.4 Chemistry-resolved structure of the asymptotic physics penalty

*Scope.* The two arms compared in this section differ from each other in capacity and in the per-arm optimisation settings of §2.4, and no hyperparameter-matched control was run, so nothing here is an accuracy claim about the thermodynamic bottleneck: Table S3 records that  $\Delta\text{MAE}$  as a difference between separately tuned pipelines, and what follows is a hypothesis-generating description of where two particular pipelines differ (§3.7). “Physics penalty” names that measured differ-

ence throughout, not an effect of the bottleneck.

Across the full test set the two arms are separated by little more than label noise. On the 3-seed scaffold split the DirectGNN control reaches an  $\ln x_2$  MAE of  $1.70 \pm 0.03$ , while the grounded physics arm (learned- $\sigma$  COSMO-SAC closure) reaches  $1.85 \pm 0.05$ , for a global  $\Delta\text{MAE} = \text{MAE}(\text{physics}) - \text{MAE}(\text{control}) = +0.14$  (positive meaning the closure adds cost).<sup>2</sup> At this granularity the difference between the two pipelines is small and near label noise. Once the error is resolved along chemical axes, that difference is an average over regimes that disagree in sign.

The solvent axis is the most pronounced. Binning test pairs by solvent class and recomputing  $\Delta\text{MAE}$  per seed (Table S24) shows the cost concentrated in solvents that hydrogen-bond strongly and directionally, precisely the regime in which the COSMO-SAC closure is least faithful and  $|\ln \gamma_2|$  is largest. Water carries the heaviest penalty, and the effect decreases monotonically as the donor strength of the class weakens. One class inverts the sign: amide solvents, which are aprotic dipolar hydrogen-bond acceptors and therefore closer to the closure’s well-specified regime, are the seed-robust stratum in which the penalty reverses sign ( $\Delta\text{MAE} = -0.22 \pm 0.11$  over seeds). This is exploratory rather than a confirmed improvement: across a family of nine simultaneous solvent-class comparisons the amide interval is an uncorrected per-class pair bootstrap, and the three-seed spread alone does not exclude zero; the halogenated class shows a nominal improvement of similar magnitude but is not seed-robust. The robust reading is the *direction* — the asymptotic physics penalty shrinks and can turn negative as solvent hydrogen-bond donation weakens toward the acceptor-dominated regime the closure was built for — not the significance of any single class. This solvent-resolved pattern is the chemistry-resolved counterpart of the synthetic closure-fidelity sweep (§S10).

<sup>2</sup>The per-seed  $\Delta\text{MAE}$  spans  $+0.05$  to  $+0.25$ , so the point estimate is an imprecise 3-seed mean; being computed on unrounded per-seed means, the rounded  $1.70/1.85$  differ by  $0.15$  rather than  $0.14$ .

Table S24: Per-solvent-class asymptotic physics penalty  $\Delta\text{MAE} = \text{MAE}(\text{physics}) - \text{MAE}(\text{control})$  on the scaffold split (mean  $\pm$  sd over 5 seeds; positive means the closure adds cost). Only the physics side is five-seed, from the leak-free re-run; the DirectGNN control stays at its published three.

| Solvent class   | $\Delta\text{MAE}$ (mean $\pm$ sd) |
|-----------------|------------------------------------|
| Water           | $+0.52 \pm 0.35$                   |
| Carboxylic acid | $+0.39 \pm 0.16$                   |
| Sulfoxide       | $+0.26 \pm 0.24$                   |
| Nitrile         | $+0.20 \pm 0.19$                   |
| Alcohol         | $+0.15 \pm 0.14$                   |
| Aromatic        | $+0.03 \pm 0.63$                   |
| Hydrocarbon     | $+0.02 \pm 0.50$                   |
| Halogenated     | $-0.29 \pm 0.36$                   |
| Amide           | $-0.22 \pm 0.11$                   |

Table S25: Per-solute-class and novelty-stratum asymptotic physics penalty  $\Delta\text{MAE} = \text{MAE}(\text{physics}) - \text{MAE}(\text{control})$  on the scaffold split (positive means the closure adds cost). Only the physics side is five-seed, from the leak-free re-run; the DirectGNN control stays at its published three. Entries carry the 5-seed spread where it was computed: the two Tanimoto strata and the charged/salt class. The other four solute classes are point estimates.

| Solute stratum              | $\Delta\text{MAE}$ |
|-----------------------------|--------------------|
| Oxygenated                  | $+0.32$            |
| Polyaromatic                | $+0.27$            |
| Heterocycle                 | $+0.17$            |
| Halogenated-aromatic        | $\approx 0$        |
| Charged / salt              | $-0.04 \pm 0.19$   |
| Tanimoto $\leq 0.3$ (novel) | $-0.04 \pm 0.18$   |
| Tanimoto $> 0.8$ (familiar) | $+0.51 \pm 0.09$   |

The solute axis shows a compatible but weaker pattern (Table S25). The closure costs most on ordinary neutral organics—oxygenated solutes, polyaromatics, and heterocycles—while the charged, high- $|\ln \gamma_2|$  stratum of salts is neutral. On this axis the penalty therefore does not track  $|\ln \gamma_2|$ : it falls on neutral organics, not on the ionic tail.

The novelty axis inverts the usual coverage expectation: the physics arm is neutral on the most novel chemistry (nearest-train Tanimoto  $\leq 0.3$ ) and hurts most on the most familiar ( $> 0.8$ ; Table S25). This is exploratory rather than a confirmed effect: the seed spread is not a within-bin sampling interval over which solutes fall in the stratum, and the Tanimoto binning is one of several strata families we examine (solvent class, solute class, similarity) without multiplicity control. Taken as a direction rather than a tested quantity, the inversion suggests that the ceiling on the physics path is not coverage or extrapolation but the closure’s systematic misfit. That misfit is masked on novel solutes by the large errors both arms already incur there, and exposed on well-covered solutes where the direct control is accurate enough for a structural bias in the closure to become the dominant residual, which is the same closure-misspecification reading as the rest of the paper. No multiplicity-controlled, within-bin-bootstrapped test of it was run, and the separate-tuning confound stands over both maps, so neither is an accuracy claim in either direction.

## S6.5 The data-efficiency sweep, and what survives its confounds

The strongest a-priori case for a physics prior is data scarcity: with few training molecules, an inductive bias toward the correct factorisation should outperform a data-hungry black box. We test it by subsampling the training set *by solute* at fractions 0.05–1.0 and training both arms on the identical subsample (Table S26, Fig. S7, seed 42). The physics arm is worse at every fraction, and against the hypothesis the gap is *smallest* at the least data ( $\Delta\text{MAE} = +0.08$  at 5%) — the one direction in this sweep that favours the prior rather than this paper’s reading of it.

Nothing in the sweep is attributable to the thermodynamic bottleneck, and no accuracy claim is made from it. The two arms differ in capacity and in the per-arm optimisation set-

Table S26: Data-efficiency sweep (train sub-sampled by solute, fixed scaffold test, seed 42).  $\Delta\text{MAE} = \text{MAE}(\text{physics}) - \text{MAE}(\text{direct})$ .

| train fraction<br>(by solute) | MAE     |        | $\Delta\text{MAE}$<br>(phys-dir) | $R^2$   |        |
|-------------------------------|---------|--------|----------------------------------|---------|--------|
|                               | physics | direct |                                  | physics | direct |
| 0.05                          | 2.28    | 2.20   | +0.08                            | -0.09   | +0.07  |
| 0.10                          | 2.44    | 2.07   | +0.37                            | -0.25   | +0.15  |
| 0.25                          | 2.93    | 1.81   | +1.12                            | -0.75   | +0.38  |
| 0.50                          | 3.12    | 1.98   | +1.14                            | -0.98   | +0.22  |
| 1.00                          | 2.98    | 1.97   | +1.01                            | -0.79   | +0.22  |

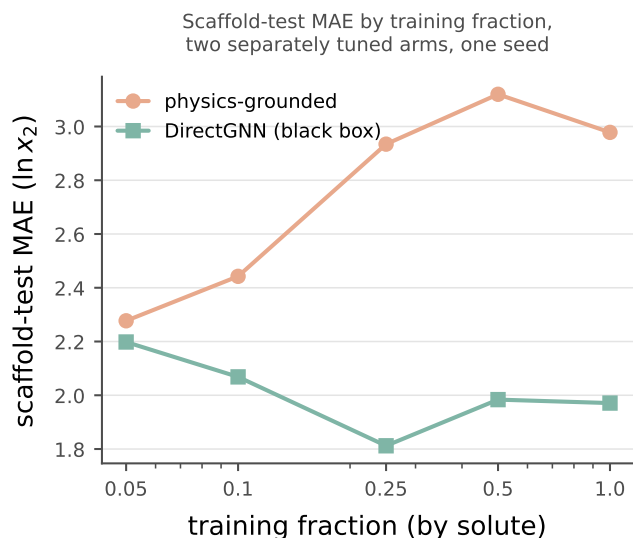


Figure S7: Data-efficiency curve: scaffold-test MAE of the physics-grounded arm (salmon) and of DirectGNN (teal) at five training-data fractions, one seed. The plotted values are the MAE columns of Table S26.

tings of §2.4, both are wider than the three-seed arms (hidden width 256 and 128 here, 64 there), the sweep is single-seed with non-monotone physics-arm MAEs, and its endpoint does not reproduce the main comparison: at fraction 1.0 it gives 2.98 against 1.97 where the three-seed runs of Table S4 give 1.7950 against 1.7022, a  $\Delta\text{MAE}$  an order of magnitude apart (+1.01 here, +0.09 there). Both arms are worse than their three-seed counterparts and the physics arm much the more so (+1.18 against +0.27 MAE), so the trend measures how a light per-fraction budget interacts with training-set size at least as much as how a physics prior does. The one inverting metric is top-1 solvent ranking (0.42 against 0.23 at 5%), which does not translate into lower error.

## S6.6 The 2002 $\rightarrow$ 2010 arms, the chemistry the closure error sits in, and the size of the profile-database exposure

The article’s §3.5.3 states two things—that the fidelity lever it can demonstrate is confined to the VT-2005-matched set’s chemistry, and that the closure error is chemically placed and placed differently on the two sets. The arithmetic behind both is in §S3.10: the input-fixed control and its three limits, and the three arms on both sets under both conventions with the block split’s third block (Table S9). This subsection holds what neither of them carries.

**The dispersion arm’s coverage, and the two facts behind the block split.** The coverage of the arm that supplies the dispersion term is the first of them. The 3% that term recovers on the broad IDAC set is a null whose interval is an order of magnitude wider than the effect, and its run on the smaller set is on  $n=57$ , three halogenated pairs having no tabulated dispersion value—a coverage limit of the model offered as the lever. No claim rests on the 0.765-versus-0.785 ordering, which is not an artifact of the differing  $n$ .

Two facts sit behind the block split, which is §3.5.3’s evidence that the reversal is chemical and neither the profile database nor the temperature range. The VT-2005-matched set is a low- $\gamma$ , hydrogen-bond-dominated sub-design, the regime the donor/acceptor-typed 2010 kernel was built for. And the map does not corroborate the reversal from the other side, which we do not ask it to: the typed kernel is not an improvement on the broad set, of whose squared error the glycol ethers carry 45%, and the classes that would explain a reversal—the halogenated solvents, the aryl ethers and water as the solute, largely one row set, all 13 aryl-ether rows and 18 of the 30 halogenated being water-as-solute rows from one publication—are the cells the rule excludes. That split is the whole of the evidence for the reversal being chemical.

The floor checks are same-database, same-



latent *and* same-convention. The *UD-native* check on the smaller set, the one commensurable with the broad set, is at zero under the residual-only rule and +0.69 under the full convention. The two aggregates under the 2010 kernel’s own typed latent, and the reading of them as loss of resolution rather than a demonstrated flip to the inputs, are in §3.5.3, and the resolution ratios behind them in §S3.10; a flip would need a *lower* bound on  $B_{\text{insuff}}$ . We know of no large-scale infinite-dilution benchmark of the 2002  $\rightarrow$  2010 step, and our search was not exhaustive, so the broad-set result is an absence of evidence rather than evidence of absence; it lies within the range of magnitudes the comprehensive assessment<sup>S11</sup> reports. That is why the lever this paper can demonstrate is confined to the smaller set’s chemistry, and why the broad set’s null—additionally a small-molecule and glycol-heavy one—leaves the lever on drug-like solids untested rather than absent.

**Which half of the donor reading carries to the published evaluation, and how large the profile exposure is.** The kernel-residual negative, its fold design and the leakage argument that makes only the grouped design admissible are §S3.10’s. What that negative supports is a statement about what a correction of *that shape* can absorb, and not about which component of the composite the error came from—the composite being the fixed kernel together with the one computed conformer it reads (§2.1.1), which nothing here separates.

§3.5.3 places the failure chemically—on the VT-2005-matched set 90% of the squared error on strong hydrogen-bond-donor solvents, near-exactness on acceptor-only ones—and states that half of that reading carries to the published evaluation of Table S27 and half does not. Which half is which is here. On that evaluation’s rows, 130 times as many, the acceptor-only solvents take 10% of the squared error at MSE 0.60, the lowest of its three donor classes, so the acceptor half carries. The donor half does not carry intact—strong donors take 56% there rather than 90%, and its inert solvents, on which the smaller set’s reading is silent, take 34% at the highest per-row error of the

three. Either way it places the error in a *chemistry* and not in one component of the composite: the reference profile for each of those solvents is one gas-phase conformer too, and on the glycol ethers—donor and acceptor on one flexible chain, the class this paper’s own cascade names conformer-dependent (§S3.5)—a displaced reference would load onto exactly the rows that carry the error. Neither set separates that from the kernel. On the broad set that reading does not reproduce, and neither does its replacement, an absent dispersion term: which of the two the broad set points at is decided in either direction by the same publication the map already declines to state, so we state neither, and the H-bond reading remains the VT-2005-matched set’s, neither corroborated nor replaced by the broad set. The residual’s rank ordering, its failure to gain over the free 2002  $\rightarrow$  2010 step, and the Staverman–Guggenheim counter-test in which the physically *more* complete combinatorial term *increases* error on size-asymmetric pairs are in §S3.10.

## S6.7 The deployed closure against a published evaluation of the same closure

Reproducing a reference implementation fixes the code, and it leaves open whether the error this paper decomposes is the error a published COSMO-SAC has or something peculiar to this data assembly. Table S27 answers that. De Souza et al.<sup>S17</sup> score the original Lin–Sandler COSMO-SAC on the infinite-dilution activity-coefficient database distributed with *The Properties of Gases and Liquids*, at  $\text{AAD} = \mathbb{E}|\ln \gamma_{\text{calc}}^{\infty} - \ln \gamma_{\text{exp}}^{\infty}|$  of 1.75 with HF-TZVP profiles and 0.68 with BP-TZVPD-FINE: one closure, two profile databases, a factor of 2.6 between them. An evaluation of COSMO-SAC is in that sense an evaluation of the profiles it was handed, which is why  $B_{\text{closure}}$  is defined above on the composite of map and profile calculation and not on the kernel. We ran our own differentiable layer over the same records (§S8), and on AAD it lands inside the bracket the pub-

lished closure spans across profile databases, under both conventions. On  $R^2$  it is inside in the residual-only convention and level with the bracket’s lower end in the full one, the two printing alike at 0.88. Our own two sets are inside on AAD alone, and in the residual-only convention only: their  $R^2$ , 0.61 and 0.40, is below the published range. Two limits stay open. The profile database is the one axis we cannot match, VT-2005 being what this paper deploys and the axis that publication shows dominates. And whether the publication’s 6977 points are the same rows as the 7889 scored here is unproven, the record selection not being stated file by file. Subject to those, the closure error decomposed in §3.5 is, in AAD, the error a published COSMO-SAC has. Which half of this paper’s chemical localisation carries across to that set, on 130 times the data and someone else’s compilation, and which half does not, is in §3.5.3. The same publication scores a later kernel on the same records, and its rows are in the table because they put the deployed one in its place: COSMO-SAC-HB2, which types segments as donors or acceptors, reaches AAD 0.46 to 0.48 at  $R^2$  0.96 to 0.97, about half the deployed closure’s 0.77 and a third of the 1.75 that anchors the profile-database bracket. We run the 2002 kernel because it is the one whose differentiable re-implementation can be checked against a public reference here, and the results below are substitutions inside a fixed map rather than claims that the map is the best available. §3.5.3 runs the 2002  $\rightarrow$  2010 step on our own sets and finds the improvement confined to the smaller set’s chemistry. Every ceiling this paper reports is accordingly this kernel’s ceiling and not COSMO-SAC’s.

One arithmetic sizes the exposure this closure’s low fidelity creates, and §3.5.3 states why that exposure is the contribution rather than a qualification of the result. Its working is here. The ground truth charging  $B_{\text{closure}}$  is quantum chemistry and not experiment: the VT-2005 profiles are computed for a single reference conformer and protonation state, and the successor Delaware database revises those conformations for about a third of its compounds;<sup>S11</sup> where the reference is displaced, even a perfectly spec-

ified  $g$  carries a term in  $B_{\text{closure}}$  through its curvature (§S8). Table S27 measures that axis directly—one closure, two profile databases, HF-TZVP against BP-TZVPD-FINE, a factor of 2.6 in AAD, 6.6 in squared error if the errors scale uniformly—while the glycol-ether stratum’s margin, read on the UD profiles the broad set is matched to and not on the VT-2005 profiles the solubility pipeline deploys, closes only once its squared error falls below twice its 0.108 bound, a more than ten-fold reduction (Table S18). A profile source better than UD by the whole of that published bracket would leave that sign standing. No side of that arithmetic is on the same database as another: the bracket enters as the size of a database-to-database effect, transported to the stratum, and not as a comparison run on it. The arithmetic holds the insufficiency bound fixed, which a new profile source would not, so it sizes the exposure and does not bound the share.

## S6.8 Conventions, admissible cells, latent magnitudes, probe scarcity and the untested modern closure: limitations (ii), (iii), (v), (vii)–(xi) and (xiii)

(ii), (iii) and (v): the convention rule, the other admissible cells, and the constant-offset bound. (ii) Under the residual-only convention no *aggregate* margin is resolved on either set and only the literature-standard convention resolves one, while the headline convention is the residual-only one; the aggregate falls for the stronger reason of §3.5.1 rather than for want of resolution. (iii) Of the map’s six admissible cells the glycol ethers are the only row set also positive and left standing by clause (c), and not the only cell whose deletions all run; the halogenated solvents and the mono-alcohols keep their sign under the one deletion each admits, so they are unverified rather than shown to fail. (v) The constant-offset margin is convention-specific and inverts under the residual-only default on both sets; there the separation rests on the binning bound and

Table S27: The deployed closure against a published evaluation of the same closure, and this paper’s own two sets through it. AAD is the mean absolute error in  $\ln \gamma^\infty$ ,  $n$  the number of scored records.

| COSMO-SAC-2002 on                                                                     | profiles      | $n$  | AAD  | $R^2$ |
|---------------------------------------------------------------------------------------|---------------|------|------|-------|
| <i>As published<sup>S17</sup></i>                                                     |               |      |      |       |
| PGL 6th ed. IDAC set                                                                  | HF-TZVP       | 6977 | 1.75 | 0.88  |
| PGL 6th ed. IDAC set                                                                  | BP-TZVPD-FINE | 6977 | 0.68 | 0.96  |
| <i>A later kernel on the same evaluation<sup>S17</sup>: COSMO-SAC-HB2<sup>b</sup></i> |               |      |      |       |
| PGL 6th ed. IDAC set                                                                  | HF-TZVP       | 6977 | 0.48 | 0.96  |
| PGL 6th ed. IDAC set                                                                  | BP-TZVP       | 6977 | 0.48 | 0.97  |
| PGL 6th ed. IDAC set                                                                  | BP-TZVPD-FINE | 6977 | 0.46 | 0.97  |
| <i>The same set, through this paper’s closure</i>                                     |               |      |      |       |
| full convention <sup>a</sup>                                                          | VT-2005       | 7889 | 0.98 | 0.88  |
| residual-only                                                                         | VT-2005       | 7889 | 0.77 | 0.92  |
| non-aqueous rows                                                                      | VT-2005       | 4347 | 0.47 | 0.68  |
| aqueous rows                                                                          | VT-2005       | 3542 | 1.13 | 0.83  |
| <i>This paper’s own sets, residual-only convention</i>                                |               |      |      |       |
| broad IDAC set                                                                        | UD            | 477  | 1.12 | 0.61  |
| VT-2005-matched set                                                                   | VT-2005       | 60   | 0.81 | 0.40  |

<sup>a</sup> The like-for-like row, a published COSMO-SAC carrying its Staverman–Guggenheim term. The residual row is the setting this paper’s  $B_{\text{closure}}$  is measured in.

<sup>b</sup> A donor/acceptor-typed hydrogen-bonding kernel, rather than the kernel this paper deploys. Printed because it is on the same 6977 records and beats the deployed closure by a factor of about two.

the more-physics-worse effect. That bound is not itself convention-independent—only the estimand  $B_{\text{insuff}}$  is, Lemma 1(d)—so the headline claim rests on it *computed under the residual-only convention*.

The article’s §3.7 carries limitations (i), (iv) and (vi), which scope what the two data sets and the estimator can deliver, (xii), which the leak-free re-run of §2.2 rewrites, and (xiv), which names two terms the deployed model sets to zero. The remaining nine are here, keeping those numerals. Each of the nine is stated in one line in that subsection’s list and again in the section whose claim it qualifies, so what is added here is the detail that qualifies it and not the qualification itself.

(vii) Absolute scaffold accuracy is encoder-bound but well *above* the inter-lab noise floor of §2.2, so label noise does not explain the ceiling. (viii) The latent-departure result is three-seed  $\text{mean} \pm \text{sd}$  ( $n=44$ ): what it establishes is the *sign* and a transferable shared direction, the magnitudes ( $51 \pm 2\%/72 \pm 1\%/4.1 \pm 0.5\times$ ) being reported but not stable across analysis configurations. Which null each of the two distinct top-two shares clears, and that none of

it is measured on the arms of Fig. 2, whose own latent departure is not measured, are in §3.3. (ix) The probe on the unconstrained control is bounded by the same reference scarcity: its crystal-term and temperature tests rest on 13 held-out solutes and cannot be enlarged, so the negative there is a bound and not a demonstration that no crystal information is present. How low that bound reaches is not computed: no power calculation was run on those 13 clusters, and each of the three measured  $R^2$  ( $-0.506/+0.111/+0.027$ ) sits at or below its own 1000-draw permutation null’s ceiling (95th percentiles 0.105/0.117/0.096, maxima 0.316/0.380/0.346), so nothing below that ceiling is excluded. The between-seed spread is not read against the null’s support: the null is two-sided and reaches below the lowest measured value, so its support is the wider of the two. The crystal-slope test is additionally limited by the corpus, whose empirical slope carries  $-\Delta H_{\text{fus}}/R$  at  $r = +0.24$ . The scaffold-organisation result needs no external reference, but its two partitions nearly coincide by construction, so a ratio near one is weak evidence of within-scaffold indistinguishability

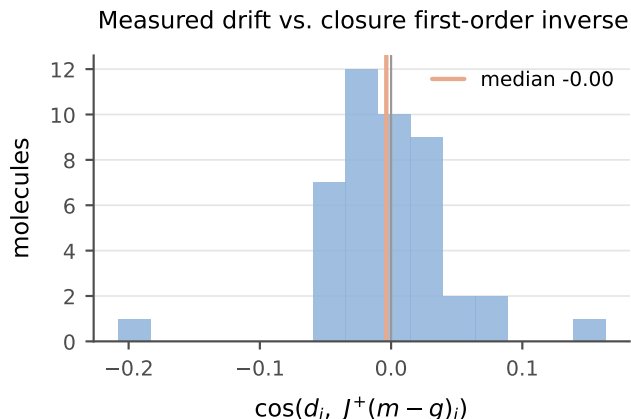


Figure S8: Per-molecule cosine between the measured drift  $\hat{\sigma} - \sigma$  and the first-order closure compensation  $J^+(m - g)$ ; the distribution is centred on zero (median  $-0.00$ ).

(§S9). (x) The closure-fidelity contrast compares *oracle-input* activity error across kernels, so  $\mathcal{G}$  itself is not measured under a modern closure on either axis, and whether the *solubility* paradox survives a 2010/dsp closure is untested; both need a typed  $\sigma$ -profile head trained end to end through the 2010 kernel, and the solubility test a  $\text{UD} \cap \text{BigSolDB}$  matched set. (xi) No difference in accuracy between the physics and control arms is attributable to the thermodynamic bottleneck; the confound and what survives it are stated above. (xiii) Several of the numbers the verdict rests on are checkable only by re-running the released code, whose input artifacts are larger than the repository carries; the two decomposition sets are therefore deposited as row-level tables (Data Sets S1 and S2), so the sets and their margins can be recomputed without code, which does not deliver the provenance of the closure evaluations in them or of the trained arms.

## S6.9 Diagnostics that do not work, and why

**The drift is not the closure’s signed first-order inverse—but the test is under-resolved.** A candidate account of the compensating drift is the closure’s linear-response inverse,  $\hat{\sigma} - \sigma \approx J^+(m - g(\sigma))$  (with  $J = \partial g / \partial \sigma$ ), predictable a priori from  $J$  before any

solubility training. On the matched set the per-molecule drift shows no signed alignment with  $J^+(m - g)$  (median cosine  $-0.00$ ; Fig. S8).<sup>3</sup> We read this as a signed negative only, and do not build on it, for two reasons intrinsic to the test. First, the matched set is sparse (median one pair per molecule), so each per-molecule Jacobian subspace is one-dimensional and the dimension-matched row-space controls are uninformative—a resolved direction common across molecules makes a molecule’s “own” and “other” subspaces coincide. Second, and more fundamentally, the single-pair  $J^+(m - g)$  is a poor *estimator* of the first-order compensation: the activity Fisher is near rank-one (PR  $\approx 1.4$  of 51, §S5), so the closure’s first-order inverse is ill-posed along all but one direction. A null here is therefore uninformative about whether the drift is first-order, not evidence that it is not. The load-bearing claim is the under-determination of  $\sigma$  by activity (§S5), not the geometry of the drift.

The instrument behind that under-determination—the rank of the activity Fisher  $J^\top J$  ( $J = \partial \ln \gamma_2^\infty / \partial \sigma$ ), read as a participation ratio  $\text{PR} = (\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$ —is computationally inexpensive and oracle-free, which is its appeal for probing any physics-grounded latent. It is also easy to misread in three ways that inflate a null or a rank in the favourable direction without any warning sign; we record them because the corrections are not self-evident.

**(i) Finite differences inflate the participation ratio.** PR is a *tail* statistic, governed by the small eigenvalues. A closure evaluated by an unrolled fixed point carries a small, step-*independent* gradient noise (from the convergence tolerance, not truncation), so central finite differences lift the exactly- and near-zero eigenvalues onto a floor and PR rises. On the differentiable COSMO-SAC-2002 layer, autograd and finite differences on *identical* profiles give PR = 1.00 and 1.25—a 25% inflation from

<sup>3</sup>The residual COSMO-SAC closure runs a segment-activity fixed point, so its autograd Jacobian is numerically unstable ( $\|\partial g / \partial \sigma_{\text{solvent}}\| \sim 10^{15}$  at infinite dilution);  $J$  is computed by finite differences on the converged closure.

noise alone, though the per-gradient cosine is 0.995 (the direction is right, the tail is not). It is worse on a typed ( $3 \times 51$ ) profile basis. A closure that ignores the typing (2002) has 102 exactly-redundant coordinates that finite differences decorrelate into a spurious floor, inflating its PR from 1.0 toward 2.0, while a closure that uses the typing (2010) has no such redundancy. The same step therefore inflates the two *asymmetrically* and can manufacture a rank ordering with no physical content. *Autograd is the appropriate instrument where the closure is differentiable; with only finite differences, PR is interpretable against a known-rank or permuted control, never absolutely.*

(ii) **“Own vs. other” controls are powerless when the resolved direction is common.** A dimension- and smoothness-matched control aligns a molecule’s drift against its own predicted compensation and against other molecules’. It has power only if the predictions differ. When the aggregate Fisher is rank-one (§3.3), however, the resolved direction is *common*, so “own” and “other” point almost the same way and the diagonal must sit inside the null— $p \approx 0.5$  regardless of the truth. On the matched set sparsity compounds this: a median of one pair per molecule makes each per-molecule Jacobian subspace one-dimensional, so “own” and “other” subspaces coincide exactly. *A control’s null is interpretable only once its power is established: if the compared predictions have median cosine near one, the test cannot reject.*

(iii) **Energy-in-subspace baselines need the effective rank and a cluster null.** A random zero-mean vector in  $D$  dimensions places  $k_{\text{eff}}/(D-1)$  of its energy in a subspace of *effective* rank  $k_{\text{eff}}$  (the participation ratio of its singular values), which for near-collinear Jacobian rows is far below the nominal row count. On the matched set the per-molecule subspace is effectively one-dimensional, so the baseline is  $\approx 2\%$  and the observed 5.5% is a two-to-threefold *enrichment*, not the “no concentration” a raw own-versus-other equality suggests: own equals other because both one-dimensional

subspaces are the same common direction, not because the drift avoids the resolved one. With that direction common,  $N$  enriched molecules are not  $N$  independent draws—the enrichment needs a cluster null (§3.5), without which its significance is over-stated by  $\sqrt{N/N_{\text{eff}}}$ . We therefore report the 5.5% as evidence in neither direction. *The baseline must be set by effective rank, and the null clustered by the shared direction.*

Each failure converts a spectral concentration statistic into a structural claim it does not support. The surviving statements of §3.3 avoid that conversion by carrying their null model and a power check inline.

## S6.10 The second domain in full: construction, bounds, and the reversal

The instrument generalises beyond COSMO-SAC, and a second domain runs it where the sign of grounding can be predicted before the measurement. The closure is a Hammett linear free-energy relationship  $g(z^*) = \text{p}K_{a,0} - \rho \sigma_{\text{H}}^*$ , with  $\text{p}K_{a,0}$  and  $\rho$  the tabulated Hansch–Leo–Taft constants of the reaction series, one literature pair per scaffold: (4.20, 1.00) benzoic acid, (9.99, 2.23) phenol, (4.60, 2.77) anilinium, (5.25, 5.90) pyridinium. Those pairs are literature values, unfitted on these data, so  $g$  is fixed in the sense Lemma 1 requires and the same  $g$  runs inside every split. It reads two things, so  $z^*$  is the pair  $(s, \sigma_{\text{H}}^*)$ : the series  $s$ , which selects that literature pair, and the substituent constant. The constant is the one learned quantity, the intermediate  $\hat{\sigma}_{\text{H}}$  being it predicted from structure. The external reference is the Hansch–Leo table<sup>S18</sup> and the data are the OPERA compilation<sup>S19</sup> under a uniform, closure-independent quality filter (§S7).

A rule that reads structure alone, never the residual, assigns the reference constant and thereby defines two poles.  $\sigma_{\text{H}}^*$  is the sum over substituted ring positions of that substituent’s tabulated constant at that position,  $\sigma_m$  meta and  $\sigma_p$  para, from a fixed list of nineteen; an ortho position, and a substituent outside the list,

enter as zero. The *clean* pole is the 158 records whose every substituted position is meta or para with a tabulated substituent. The *degraded* pole is the 846 where at least one is neither (571 ortho, 275 untabulated), and there  $\sigma_H^*$  is a partial sum over the positions the table can see. The split cuts across benzenoid and heteroaromatic alike, all four series occurring on both sides, 32 of the clean pole’s records being pyridinium.

The relation’s two classic failure modes fall onto the two channels of Lemma 1: *resonance saturation*, a curvature beyond a linear closure, lands in  $B_{\text{closure}}$ , and what the zeroed positions carry (ortho field and steric effects, and whatever an untabulated substituent contributes) is information the intermediate lacks and lands in  $B_{\text{insuff}}$ . On the clean pole, the chemistry the relation was built for, the closure predicts to RMSE 0.55 and the two available bounds on  $B_{\text{closure}}$  disagree. The binning-free floor  $\text{MSE} - B_{\text{insuff}}^{\text{up}}$  is  $-0.2$  (95% CI  $[-0.4, -0.0]$ ) and establishes nothing: the out-of-fold  $B_{\text{insuff}}^{\text{up}}$  is 0.500 against an MSE of 0.302, a difference negative whatever the closure, which says that a flexible regressor on the closure’s own argument predicts worse than the closure does. The Lemma 1(b) bound runs the other way and is informative: the per-scaffold systematic offsets reach 0.429 in  $\text{p}K_a$  on phenol, and the  $n$ -weighted mean of their squares bounds  $B_{\text{closure}}$  below at 0.083, 27% of the pole’s total. At least a quarter of the clean pole’s error is closure error, so its specification stays in doubt. On the degraded pole the closure reaches RMSE 2.26, of which the out-of-fold within-series regressor places at most 4.5 of the 5.1 total in input insufficiency and hence at least 0.6 in the closure, both one-sided and in the same direction as the solubility bounds.

Two properties of that estimator carry into how the poles may be read. It differs from the eight-bin cell of §2.5, and the functional differs too: these are *floors*  $\text{MSE} - B_{\text{insuff}}^{\text{up}}$  on  $B_{\text{closure}}$ , establishing  $B_{\text{closure}} > 0$  when positive, and not the separation margin  $\text{MSE} - 2B_{\text{insuff}}^{\text{up}}$  of Table S1, which would establish the stronger  $B_{\text{closure}} > B_{\text{insuff}}$ . And conditioning on  $s$  is forced, not a conservative choice: pool the

four series and the same estimator returns 8.3 against the degraded pole’s MSE of 5.1, under which  $g$  is unmeasurable, Lemma 1 withholds its licence, and the floor it leaves is vacuous anyway. The folds are free of the leakage §3.5.1 refuses a fitted estimator for on solubility: the regressor reads  $\sigma_H^*$  within a series and nothing besides, so a fold hands it only what the closure’s own argument already carries. The quality filter is closure-independent (net formal charge  $\neq 0$ , applied to both poles, blind to any residual) and drops five pre-ionized microstates, and it decides the clean pole’s level: it moves that pole by a factor of eleven in MSE, 3.33 to 0.30, and leaves the degraded pole where it was, so a large part of the poles’ separation in level is the filter’s. On the 163 rows before it the same floor is  $+0.55$  rather than  $-0.2$ , positive, establishing  $B_{\text{closure}} > 0$  by the other route as well.

The decisive test is the same oracle swap, measured *in-distribution* across  $K=20$  within-scaffold splits: the learned  $\hat{\sigma}_H$  replaced by the tabulated  $\sigma_H^*$  in the same closure, the series label read from structure in both arms so the swap moves one coordinate of  $z^*$ . It flips. On the clean pole the reference *improves* accuracy, MAE 0.48 learned against 0.315 with the reference, a gap of  $-0.165$  at a 90% CI of  $[-0.28, -0.04]$ , better in 19 of 20 splits and in three of the four series: an internal positive control on real data, and one on the swap’s *sign* and not on the closure’s specification, since the bound above puts at least a quarter of that pole’s error in the closure. On the degraded pole the same swap *degrades* it, 0.83 against 1.62, worse in 20 of 20 splits and in all four series, a gap of  $+0.79$   $[+0.57, +0.95]$ , the freely learned constant absorbing what the zeroed positions carry and the tabulated one lacks. The sign of grounding reverses inside one closure, in the direction the poles’ construction fixes in advance: which ring positions the Hansch–Leo table can see, complete on one pole and a partial sum on the other. Remark 1 supplies the vocabulary and not the prediction.

That the reversal runs through the insufficiency channel, and not through the closure-error cancellation of §3.3, is unsupported on

the clean side, where the bound above leaves at least 27% of the error in the closure. On the degraded side it rests on the flip itself and on how the pole is built; the Jensen bound adds only compatibility, leaving at most 87% of the error in insufficiency, and an upper bound is as consistent with none of it as with all. The out-of-fold regressor answers the same question on the same pole at 4.5 of the 5.1 and is an upper bound too, so the two agree while leaving the attribution to rest elsewhere. A competence sweep adds that the oracle side of the comparison holds fixed across the training budget while the learned side moves: the learned arm stays above the clean pole’s threshold ( $\approx 0.315$ ) at all four budgets swept, and beats the degraded pole’s (1.62) across the whole interpolation range, down to 12% of the data, grounding reverting to helping only under scaffold *extrapolation* where the learned constant breaks ( $2.6 \pm 2.0$ ) back above the threshold, in the mean over six splits, three of which still have grounding hurting (§S7).

**The flip is an in-distribution result and the scaffold protocol does not certify it.** Re-running the same  $K=20$  certification on a scaffold split rather than a stratified one, the reference helps on *both* poles: the clean pole’s gap is  $-0.61$  and the degraded pole’s  $-1.10$ , with a series-cluster interval of  $[-1.86, +0.37]$  and  $P = 0.28$  there, so the degraded pole’s “grounding hurts” half fails to certify, and so does the reversal. We measure every solubility result in this paper on a scaffold split, and this is the one place the protocol changes. What the scaffold run confirms is the mechanism and not the flip: the learned constant breaks under extrapolation and rises above the oracle threshold on both poles. The clustering unit is the Hammett reaction series, of which only  $G=4$  exist, so the bootstrap resolves series-level sign agreement and nothing finer, read as a sign test on four units:  $p=1/16$  on the degraded pole, where all four series agree that grounding hurts, and  $p \approx 0.31$  on the clean one, where three do. That test carries the degraded pole and not the positive control, which rests on its splits and its gap instead.

## S7 The pKa closure: full construction, measured sweep, and trained comparison

This section gives the full construction behind §3.6: a fixed Hammett linear-free-energy relationship (LFER)  $g(z^*) = pK_{a,0} - \rho \sigma_H^*$  over  $z^* = (s, \sigma_H^*)$ : the reaction series  $s$ , which fixes the per-scaffold Hansch–Leo–Taft constants  $(pK_{a,0}, \rho)$ , together with the Hammett substituent constant  $\sigma_H$ , whose tabulated Hansch–Leo values are the external reference for the coordinate the network predicts. That constant carries its subscript throughout, so that it cannot be confused with the  $\sigma$ -profile of §2.1. The analogy is close: as with the  $\sigma$ -profile through COSMO-SAC, a learnable physical descriptor is pushed through a fixed, approximate, differentiable map. The two classic LFER failure modes fall cleanly onto the two error channels of Lemma 1: *resonance saturation*—a curvature in  $\sigma_H$  that the linear closure structurally cannot represent—is a mis-map of a function of  $z^*$  and lands in  $B_{\text{closure}}$ ; *ortho/steric* field effects, absent from the tabulated constants, are information the intermediate lacks and land in  $B_{\text{insuff}}$ . The chemical scope is a fidelity sweep, over the four reaction series alike (benzoic acid, phenol, anilinium, pyridinium): records whose every substituted ring position is meta or para with a tabulated substituent are the near-exact ( $B_{\text{closure}} \approx 0$ ) regime, and records carrying an ortho substituent or one outside the tabulated list—both of which enter  $\sigma_H^*$  as zero—the low-fidelity one. The assignment rule is a function of structure alone and never reads the residual, and its mechanics are next.

**The rule that selects a reaction series and types a ring position.** The reference constant is assigned mechanically, and that rule is also what defines the two poles of §3.6. The four kinds are tried in the fixed order benzoic acid, phenol, anilinium, pyridinium, and a record enters under the first of them whose ionizing group matches exactly once in the molecule, at

a ring atom lying in exactly one ring and that ring six-membered and aromatic. An ionizing site the entering kind does not consume does not exclude the record: on the same ring it is typed as an ordinary substituent, tabulated or not. It enters only if its ring system is also unambiguous: no aromatic nitrogen anywhere in the molecule for the three carbon-ring series, exactly one and no *N*-oxide for pyridinium. The kind a record enters under names the reaction series and with it  $(\text{p}K_{a,0}, \rho)$ . Ring positions are typed by shortest ring-bond distance from that ring atom—the one bearing the ionizing group, which for pyridinium is the ring nitrogen itself—one bond *ortho*, two *meta*, three *para*; and  $\sigma_{\text{H}}^*$  is the sum over the substituted positions of the Hansch–Leo constant of that substituent *at that position*,  $\sigma_m$  meta and  $\sigma_p$  para, taken from the fixed list of nineteen substituents in Table S28; an unsubstituted parent has  $\sigma_{\text{H}}^* = 0$ . The zeroing of ortho and untabulated positions, and with it the definition of the clean and degraded poles and their counts, is §3.6’s.

**The estimator, its identity, and its fold design.** §3.6 scopes the two floors it prints: they are not measured with the eight-bin estimator cell of §2.5, and the functional is a floor on  $B_{\text{closure}}$  and not the separation margin of Table S1. The estimator’s identity and its fold design are here. The closure reads the reaction series too— $\text{p}K_{a,0}$  and  $\rho$  are per-series—so the argument Lemma 1 requires is the pair  $z^* = (s, \sigma_{\text{H}}^*)$ , series label and constant, and the estimand is  $\mathbb{E}[\text{Var}(\text{p}K_a \mid s, \sigma_{\text{H}}^*)]$ . Neither coordinate needs binning:  $s$  takes four values and  $\sigma_{\text{H}}^*$  is a scalar, so the conditional variance is estimated directly—a random forest of  $\text{p}K_a$  on  $\sigma_{\text{H}}^*$  alone, fitted *within* each series and pooled over series by  $n$ , five-fold, its *out-of-fold* residual mean square. That regressor  $\hat{r}$  reads  $(s, \sigma_{\text{H}}^*)$  and nothing else, and predicts each row from the folds that exclude it, which is what makes the split exact for it:  $\mathbb{E}[(m - \hat{r})^2] = \mathbb{E}[\text{Var}(m \mid s, \sigma_{\text{H}}^*)] + \mathbb{E}[(\mathbb{E}[m \mid s, \sigma_{\text{H}}^*] - \hat{r})^2]$ , so the estimate exceeds  $B_{\text{insuff}}$  by the regressor’s own approximation error—an upper figure, leaving a lower one. Conditioning on  $s$  is not a free choice and

it is not the conservative one: pool the four series and the same estimator returns 8.3 against the degraded pole’s MSE of 5.1, a conditioning under which  $g$  is not measurable, so Lemma 1 does not license it and the closure floor it leaves would in any case be vacuous. The floor of 0.6 therefore rests on  $s$  belonging in  $z^*$ , which is what the lemma demands of a closure that reads it. §3.5.1 declines fitted estimators on solubility because random folds let a held-out row borrow strength from a near-twin through the representation; here the regressor’s only input is  $\sigma_{\text{H}}^*$  within a series, so no fold can hand it what the closure’s argument does not already carry. What the identity needs of the fold design is only that a row be held out of the fit that predicts it.

The nineteen substituents that rule can see, and the constants it sums, are Table S28. They are standard Hammett  $\sigma_m/\sigma_p$  values<sup>S18</sup> to two decimals; they are a constant of the builder, not fitted or refit on these data, which is the sense in which  $g$  is fixed for Lemma 1. The order matters: a ring position takes the constant of the first pattern in the list that matches at it, and the list is ordered specific before generic. A substituent outside this list makes the position untabulated, and the record degraded.

This decomposition is directly visible before any training. Pushed through the fixed Hammett closure, *p*-nitrophenol reads  $\text{p}K_a \approx 8.3$  against an experimental  $\approx 7.1$  (and *p*-nitroaniline overshoots similarly): the linear closure is off by  $\sim 1$  unit exactly where through-conjugation ( $\sigma_{\text{H}}^-$ ) breaks it— $B_{\text{closure}}$  observed in practice.

A controlled semi-synthetic construction on real tabulated  $\sigma_{\text{H}}$  (known ground-truth channels) validates the estimator and reproduces the paper’s central distinction in this second closure (Fig. S9). Sweeping closure fidelity with a sufficient intermediate, the grounding gap  $\mathcal{G}$  tracks  $B_{\text{closure}}$  (the plug-in estimate recovers the truth to within a few percent, e.g. 0.20 vs 0.21, 0.81 vs 0.81), the case where Theorem S1’s proxy is valid. Sweeping the ortho/steric channel with a *well-specified* closure gives  $B_{\text{closure}} = 0$  at every knob while  $\mathcal{G}$  rises with the insufficiency fraction, from  $-0.035$  at zero insufficiency through



Table S28: The fixed substituent list behind  $\sigma_{\text{H}}^*$ : nineteen groups with the meta and para Hammett constants summed over the substituted positions (§3.6). Patterns are tried in the order shown—down the first block, then the second, then the third—and the first match at a ring position wins. An ortho position, and any group not in this list, enters the sum as zero.

| group                           | $\sigma_m$ | $\sigma_p$ | group                            | $\sigma_m$ | $\sigma_p$ | group                            | $\sigma_m$ | $\sigma_p$ |
|---------------------------------|------------|------------|----------------------------------|------------|------------|----------------------------------|------------|------------|
| NO <sub>2</sub>                 | 0.71       | 0.78       | OCH <sub>3</sub>                 | 0.12       | -0.27      | Br                               | 0.39       | 0.23       |
| CF <sub>3</sub>                 | 0.43       | 0.54       | OH                               | 0.12       | -0.37      | I                                | 0.35       | 0.18       |
| CN                              | 0.56       | 0.66       | N(CH <sub>3</sub> ) <sub>2</sub> | -0.16      | -0.83      | C(CH <sub>3</sub> ) <sub>3</sub> | -0.10      | -0.20      |
| COOH                            | 0.37       | 0.45       | NHC(O)CH <sub>3</sub>            | 0.21       | 0.00       | C <sub>2</sub> H <sub>5</sub>    | -0.07      | -0.15      |
| CO <sub>2</sub> CH <sub>3</sub> | 0.37       | 0.45       | NH <sub>2</sub>                  | -0.16      | -0.66      | CH <sub>3</sub>                  | -0.07      | -0.17      |
| COCH <sub>3</sub>               | 0.38       | 0.50       | F                                | 0.34       | 0.06       |                                  |            |            |
| CHO                             | 0.35       | 0.42       | Cl                               | 0.37       | 0.23       |                                  |            |            |

zero near a fraction of 0.25 to +0.34 at 0.7: at the three fractions where it is positive it is positive with  $B_{\text{closure}}$  identically zero, which is the pKa realization of the conflation that makes  $\mathcal{G}$  an indicator rather than a certificate. The two negative points are the same statement in the other direction—a well-specified closure with a nearly sufficient intermediate leaves the oracle marginally ahead. The oracle- $\sigma_{\text{H}}$  pipeline (scaffold-resolved substituent  $\rightarrow \sigma_{\text{H}}$  assignment) and this validation extend directly to real data. On the OPERA experimental pKa compilation<sup>S19</sup> (7,904 records under the same net-neutral filter), applying the same decomposition to the fixed Hammett closure yields a *measured* fidelity sweep. On the clean meta/para series ( $n=158$ ) the closure predicts experimental pKa to RMSE 0.55. The leakage-immune lower bound MSE  $-B_{\text{insuff}}^{\text{up}}$  is -0.2 (95% CI [-0.4, -0.0]), below zero, and it is uninformative here rather than reassuring: the out-of-fold  $B_{\text{insuff}}^{\text{up}}$  is 0.5003 against an MSE of 0.3023, so that difference is negative for every closure whatever, and what the 158 rows across four series establish is that the within-scaffold regressor predicts worse than the LFER does. The second bound is informative and points the other way. Lemma 1(b) bounds  $B_{\text{closure}}$  below by the *square* of the per-scaffold systematic offset, and the deposit’s per-scaffold Jensen field is already that square: its entries 0.0918/0.0011/0.1837/0.0392 are the squares of the offsets +0.303 anilinium, +0.034 benzoic acid, +0.429 phenol, -0.198 pyridinium. The offsets reach 0.429 in pKa, and the  $n$ -weighted

mean of the squares is 0.083 against a total of 0.303, or 27%; on phenol alone it is 0.184 against that series’ own MSE of 0.371, or half of it. At least a quarter of this pole’s error is closure error, so the pole is not shown to be well specified, and neither bound available on it says that it is. The degraded pole ( $n=846$ ) degrades to RMSE 2.26, dominated by input insufficiency. Here  $z^*$  is the pair  $(s, \sigma_{\text{H}}^*)$ —the reaction series, which fixes  $(\text{p}K_{a,0}, \rho)$ , and the constant—and neither coordinate needs binning, four labels and a scalar, so—unlike the solubility regime— $B_{\text{insuff}} = \mathbb{E}[\text{Var}(\text{p}K_a \mid s, \sigma_{\text{H}}^*)]$  is estimated *directly* rather than through a coarsening of the closure’s argument (§2.1.1): a within-series out-of-fold regression estimator gives  $B_{\text{insuff}} = 4.5$  (95% CI [4.0, 5.1];  $\approx 89\%$  of the error, on the unrounded 4.528 against 5.094— $\sigma_{\text{H}}^*$  carries nothing from the zeroed positions, exactly the a-priori channel assignment, which the measurement is consistent with rather than establishes), leaving a closure misspecification  $B_{\text{closure}} = 0.6$  ([0.3, 0.8], a stratified within-scaffold molecule bootstrap; both intervals are percentile intervals of 4000 resamples of molecules within each series, at their sizes). Directly estimated is not point-identified: the estimator is an out-of-fold residual mean square, which is an upper estimate of  $\mathbb{E}[\text{Var}(\text{p}K_a \mid s, \sigma_{\text{H}}^*)]$ , so 4.5 is an upper figure and the 0.6 it leaves is a lower one—one-sided in the same direction as everywhere else in this paper, and narrower only because a four-valued label and a scalar need no binning (§2.1.1, §3.6). The two poles differ by a factor of 17 in MSE,

and that is the transition, measured rather than asserted. The clean pole is the milder regime and not the well-specified one. Run the same Jensen bound on both and the closure floor is 0.083 there against 0.654 on the degraded pole, 27% of the first pole’s error and 13% of the second’s, so what grows between the poles is the total and the insufficiency inside it—at most 73% on the clean pole and at most 87% on the degraded—while a closure error of at least a quarter is present on the pole the relation was built for.

**Data quality control (uniform, closure-independent).** The trained runs use the canonical full OPERA set (7904 records). A small number of entries are supplied as pre-ionized microstates—an  $[\text{NH}^+]$ -protonated aminopyridine at  $\text{p}K_a = -7.78$ , a pyrylium  $[\text{O}^+]$ , among others—inconsistent with a neutral-scaffold Hammett assignment, and they degrade the clean pole (pyridinium RMSE  $0.63 \rightarrow 3.83$ ). We drop them with one rule, *net formal charge*  $\neq 0$ , applied uniformly to both strata. It is closure-independent (it never references the oracle residual, so it cannot be circular) and keeps net-neutral substituents such as nitro  $[\text{N}^+][\text{O}^-]=\text{O}$ . This removes 5 of 163 clean-pole rows (restoring that pole to RMSE  $\approx 0.55$ ) and 352 of 1198 degraded-pole rows (leaving 846; that pole’s RMSE is essentially unchanged, so the filter does not manufacture the sign), for 1004 Hammett-amenable rows. The clean pole’s floor does turn on it:  $\text{MSE} - B_{\text{insuff}}^{\text{up}}$  is +0.55 on the 163 unfiltered rows against the  $-0.2$  reported on the 158 filtered. That sensitivity decides which of the two bounds carries the pole and not whether the closure is well specified there, since the Jensen bound puts  $B_{\text{closure}} \geq 0.083$  on the filtered rows regardless.

**Evidence for the flip.** We train the physics arm (encoder  $\rightarrow \hat{\sigma}_{\text{H}} \rightarrow$  fixed LFER) on  $K=20$  stratified 80/20-within-scaffold splits—a Bemis–Murcko split collapses the meta/para pole to two ring systems and is unstable—and record the paired per-molecule error against the deterministic oracle, per stratum. Clean pole:

learned MAE  $0.48 \pm 0.09$  vs oracle  $0.315 \pm 0.07$ , gap  $-0.165$ , oracle better in 19/20 splits, 90% CI  $[-0.28, -0.04]$ . Degraded pole: learned  $0.83 \pm 0.07$  vs oracle  $1.62 \pm 0.08$ , gap  $+0.79$ , oracle worse in 20/20 splits, 90% CI  $[+0.57, +0.95]$  (both intervals the 5th and 95th percentiles of 4000 draws of the scaffold-cluster bootstrap, one cluster per reaction series). Both halves certify, and the flip with them.

**The same certification on a scaffold split, where it fails.** The run above is in-distribution, and the companion run at the same  $K=20$  on a Bemis–Murcko scaffold split is deposited beside it. There the reference helps on both poles and nothing flips. Clean pole: learned  $0.97 \pm 0.66$  vs oracle  $0.37 \pm 0.14$ , gap  $-0.61$ , oracle better in 20/20 splits, cluster CI  $[-1.13, -0.24]$ —the “reference helps” half certifies. Degraded pole: learned  $2.73 \pm 1.72$  vs oracle  $1.63 \pm 0.32$ , gap  $-1.10$ , oracle better in 12 of 20 splits, cluster CI  $[-1.86, +0.37]$  at  $P = 0.28$ —the “reference hurts” half does not certify, and neither does the reversal. Both arms degrade under the harder protocol and the learned one degrades further, which is the crossover mechanism of the paragraph below acting on both poles at once; the flip itself is a within-distribution phenomenon and is reported as one. Every solubility result in this paper is measured on a scaffold split, so this is the one place the protocol differs from the rest, and the reason it does is that a scaffold split collapses the meta/para pole to two ring systems.

**What the two bootstraps do and do not resolve.** We report the sign statements above rather than the bootstrap  $P$  values, because neither bootstrap here carries the resolution a two-decimal  $P$  implies. The cluster unit is the Hammett reaction series, and only  $G=4$  exist (benzoic acid, phenol, anilinium, pyridinium): the resampling law then has  $\binom{7}{4}=35$  distinct outcomes and a finest atom  $(1/4)^4 \approx 0.004$ , so the cluster-bootstrap  $P(\text{oracle worse})=1.00$  on the degraded pole is exactly the statement that all four series means favour the learned arm, and  $P=0.004$  on the clean pole that exactly

one of them does, the other three favouring the oracle. Read as a sign test on four units those are  $p=1/16$  and  $p\approx 0.31$ ; the interval endpoints quoted above are percentiles of the same 35-outcome law and are no finer. This is the same small- $G$  caution applied on the solubility side (§S3.11), where the  $P$  values are likewise read as a coarse band rather than as two-decimal quantities. The split bootstrap adds nothing independent: the 20 splits resample train/test assignments over the same molecules (each molecule lands in  $\approx 4$  test sets), so it prices split-to-split noise rather than molecule sampling, and with all 20 degraded-pole gaps positive it returns 1.000 by construction—a restatement of the 20/20, not a second confirmation. What the flip rests on is therefore the sign consistency across splits and series and the size of the degraded pole’s gap (+0.79 against an oracle MAE of 1.62,  $\approx 49\%$ ), not on a  $P$  value. A leave-one-series-out refit, the estimate that would settle the series-level question at  $G=4$ , was not run.

#### The crossover and its scope boundary.

The sign of the grounding gap is by construction the sign of the learned arm’s error against the oracle’s. What a competence sweep (train fraction  $\in \{1.0, 0.5, 0.25, 0.12\}$  and, as the low-competence extreme, scaffold extrapolation; six seeds each) adds is that the oracle side of that comparison is fixed by the fidelity set and not by the budget: its threshold is the same at each of the four budgets, while the learned side moves. On the clean pole the learned arm rises from 0.46 to 0.98 as data shrinks but never reaches the 0.31 oracle threshold, so grounding helps throughout. On the degraded pole the learned arm stays below the 1.62 threshold across the whole interpolation range (0.85 at full data, 1.31 at 12%), so grounding hurts throughout; only scaffold extrapolation, where  $\hat{\sigma}_H$  breaks to  $2.6 \pm 2.0$ , pushes it back above the threshold and reverts the sign. That reversion is a property of the mean over the six splits and not of every split: three of the six still have grounding hurting, and the negative mean gap ( $-1.04$ ) is carried by two splits at  $-3.2$  and  $-3.8$ . The flip is therefore a within-

distribution, fidelity-driven crossover, and the extrapolation reversion follows the same comparison. It is the sharp, two-sided form of the penalty sign rule (Remark 1): the half saying that a fitted intermediate can beat the true one is known—plug-in efficiency under correct specification,<sup>S8</sup> pseudo-true fitting under misspecification<sup>S20</sup>—and what is added here is the fidelity-set threshold at which the two halves change places. A trained physics-vs-DirectGNN comparison agrees in direction overall but has weak absolute errors and an untuned control; the load-bearing second-domain results are the decomposition above and the sign-consistent flip.

## S8 Data provenance and reproducibility

**The broad IDAC set ( $n=477$ ), and the UD profile database.** The UD  $\sigma$ -profile database is the 2261-compound COSMO set reported by Fingerhut et al.<sup>S11</sup> and redistributed with the benchmark implementation of Ref.<sup>S1</sup> Of it, 1941 compounds come from the University of Delaware collection assembled in Sandler’s group, which extends the freely available VT set with revised molecular conformations for about a third of its compounds. A further 320 were generated by the authors of Ref.<sup>S11</sup> It ships both the Mullins-averaged single profile the 2002 kernel prescribes and the typed Hsieh three-profile set the 2010 kernel prescribes, with the cavity volume and, for most compounds, a tabulated dispersion parameter. The broad IDAC set is the intersection of that database with our expanded ThermoML infinite-dilution pull, at every temperature rather than only 298 K.

*How the 14,900 raw rows are extracted.* The pull walks the NIST ThermoML archive JSON records for four journals. A row is kept only where five conditions hold. The dataset has exactly two components. The property is an activity coefficient at infinite dilution, either declared so by a dataset constraint or carried at a solute mole fraction of zero. Exactly one component is the non-solute. A temper-

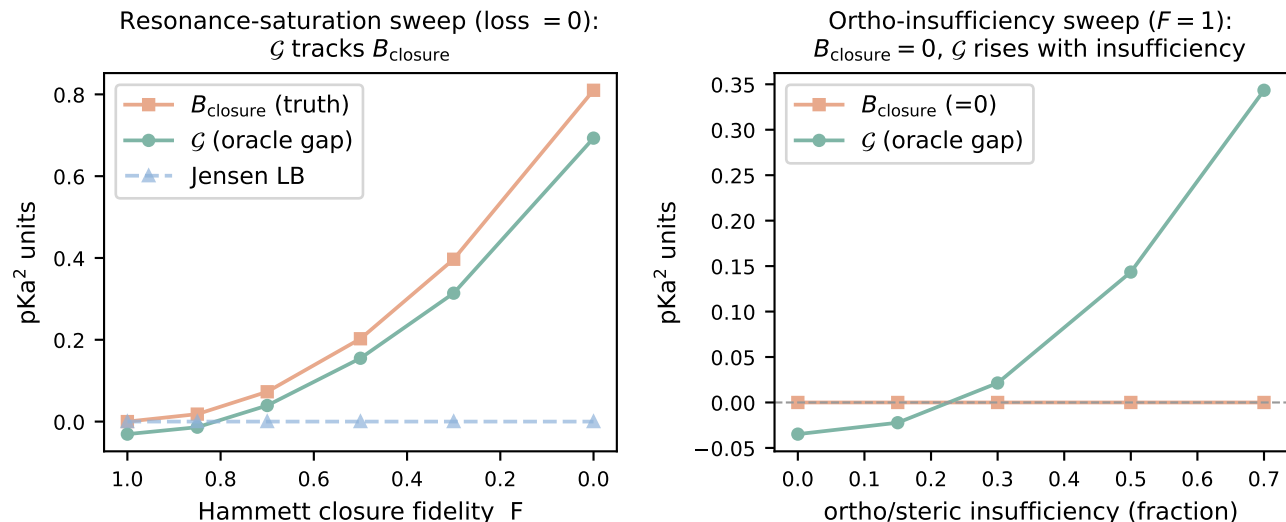


Figure S9: pKa through a Hammett LFER on a construction with known ground-truth channels. *Left*: closure fidelity varied with a sufficient intermediate; grounding gap  $\mathcal{G}$ , closure bias  $B_{\text{closure}}$ , and the assumption-light Jensen bound. *Right*: an ortho/steric insufficiency channel varied with a *well-specified* closure;  $B_{\text{closure}} = 0$  at each knob while  $\mathcal{G}$  rises with the insufficiency fraction, crossing zero near 0.25.

Table S29: Run families: which runs produced which numbers, and at how many seeds. The deposit names the analysis script and the artifact behind each row.

| Run family                                                            | Seeds                                                                                        | What it produced                                                                                                                         |
|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| $\sigma$ -grounding arms, GPU, full corpus                            | 42, 43, 44, 45, 46, mean $\pm$ sd on the $n=5608$ labelled test rows                         | Figs. 2 and 3, Table S4, the +0.41 and -0.11 of §3.1                                                                                     |
| Co-adaptation and channel-swap oracle controls; gated output residual | 42 only: a sign, not an effect size                                                          | the +0.18 and +0.27 of §S3, which the article does not claim; §S5                                                                        |
| Decomposition, VT-2005-matched                                        | none: the closure has no learnable parameters                                                | §3.5.2; the estimator grids                                                                                                              |
| Decomposition, broad IDAC, with its design and provenance audit       | none                                                                                         | §3.5.1: the +0.51, the +0.95, the deletion curve                                                                                         |
| The stratified map                                                    | none                                                                                         | the 59 strata, Tables S12–S18                                                                                                            |
| Closure-fidelity kernels                                              | none                                                                                         | §3.5.3, Table S9                                                                                                                         |
| Kernel residual                                                       | five fit seeds; $\pm$ is their standard error                                                | the $46 \pm 5\%$ and $-5 \pm 16\%$ of §3.5.3                                                                                             |
| Latent against its reference                                          | 0, 1, 2; mean $\pm$ sd                                                                       | §3.3, Fig. S5                                                                                                                            |
| Closure validation                                                    | none                                                                                         | the 2002 NIST reproduction; Table S27. The 2010 reproduction comes from a validator that wrote no artifact and is regenerated by neither |
| External baselines                                                    | one run each, on a <i>by-solute</i> split and, for two of the rows, a <i>random-pair</i> one | Tables S30 and S20                                                                                                                       |
| pKa second domain: decomposition, Hammett probe and bootstrap         | $K=20$ within-scaffold splits; $G=4$ series as the cluster unit                              | §3.6                                                                                                                                     |
| Noise floor and encoder probe                                         | none                                                                                         | §2.2, §2.1.2                                                                                                                             |

The  $\sigma$ -grounding row’s seed-44 oracle prediction file was deposited truncated and has been recovered complete, so every row of every arm at every seed is re-analisable from the deposit.

ature is resolvable for the value. And  $\gamma^\infty$  is finite and positive. Components are mapped to SMILES through the record’s own InChI. Duplicate (solute, solvent, temperature) records are then aggregated, with the record and DOI counts retained per row, which takes 14,910 ex-

tracted values to 14,900 rows. No solubility-side, method-side or quality-side filter is applied at this stage.

*Construction ladder.* The pull starts at 14,900 raw IDAC rows, and 14,900 of them carry a finite  $\ln \gamma_2^\infty$  and a temperature. Of those, 490

have solute *and* solvent resolving into the UD compound list by InChIKey, first-block fallback allowed, and the same 490 have both the Mullins and the typed profile files present and parsing at the expected 51 and 153 bins. The last step takes 477: those whose components both carry a tabulated dispersion  $\epsilon$ . The 2010, *dispersion on arm* needs it, and we impose it on every arm so that the three kernels are scored on identical rows. That leaves 78 solutes, 36 solvents and 35 distinct temperatures from 273.2 to 403.1 K, with  $\text{Var}(m) = 4.85$  and  $\overline{m} = 3.42$ . It is a harder and far less ideal regime than the VT-2005-matched set’s  $\overline{\ln \gamma_2^\infty} = 0.75$ . It is not disjoint from the VT-2005-matched set: 57 of the 60 VT-2005-matched pairs appear in it at 298 K, evaluated there on UD rather than VT-2005 profiles. The two are therefore the same chemistry measured against two profile databases, not independent samples, and we never combine them. The 3 VT-2005-matched pairs it excludes are exactly the 3 that carry no tabulated dispersion  $\epsilon$ : bromobenzene and iodobenzene in 1,2-ethanediol, and 2-bromopropane in pyridine. The “dispersion-defined 57” of §3.5.3 and the “57 shared with the broad IDAC set” are therefore one set and not two.

*Why neither set can observe  $B_{\text{insuff}}$  directly.*  $B_{\text{insuff}} = \mathbb{E}[\text{Var}(m \mid z^*, T)]$  would be point-identified only by two or more independent determinations at a common closure argument, and neither design supplies one. VT-2005 assigns exactly one profile per compound, and the 44 matched compounds have distinct profiles: the closest pair sits at  $L_2=3.4$  against a median profile norm of 44. No pair carries a replicate either, so on the VT-2005-matched set no two rows sit at a common  $z^*$ . The broad IDAC set comes closer without reaching it: 359 of its 477 rows share a molecule pair with another row, but at a different temperature and hence at a different  $(z^*, T)$ . On these two sets  $B_{\text{insuff}}$  is accordingly reached only as an upper bound, by coarsening in the binning-bound cells and out of fold in the fifteen fitted rows of Table S10, and  $B_{\text{closure}}$  only as a lower one. Where the latent is a scalar the headline estimate is the out-of-fold residual, one-sided in the same direction

(§2.1.1, §S7).

*The matching rule, and its asymmetry with the VT-2005-matched set.* The VT-2005-matched set is matched by canonical SMILES and this set by InChIKey with a 14-character first-block fallback, which is looser. It is also looser in the anti-conservative direction: a  $z^*$  imported across a tautomer, protonation or stereochemical variant inflates MSE and hence  $B_{\text{closure}}$ . We therefore count it. Of the 90 distinct molecules, 86 resolve by full InChIKey; four resolve only through the first block (C=CC1CC=CCC1, CCC(C)N, CC(C1)CC1, CCC(C)C(C)=O), touching 12 of the 477 rows. In all four the discrepancy is the stereochemical layer and not a tautomer or a protonation state: each query key ends -UHFFFAOYSA-N, an unspecified stereocentre, against a stereo-labelled UD entry with the same connectivity. On the 465 exactly matched rows (178 distinct pairs) the decomposition is unchanged. It is read at the headline cell of the grid: eight equal-count bins, unbiased within-bin variance, edge values to the upper bin. That cell recomputes these two lines from the deposit: residual-only convention MSE 1.930,  $B_{\text{insuff}} \lesssim 0.699$ , margin +0.53 at  $P \approx 0.9$ ; full convention 1.926, 0.495, +0.94 at  $P > 0.95$ .

*Composition, and where the squared error sits.* The set’s chemistry is small in both channels: *n*-alkanes, aromatic hydrocarbons and ketones supply 243 of the 477 solute rows, and water and the four ethylene-glycol oligomers 293 of the solvent rows. Its overlap with this paper’s solubility corpus is correspondingly asymmetric. Only 0.46% of corpus rows carry a solute from this set, while 34 of its 36 solvents cover 59% of corpus rows. That asymmetry is what the main text means by the set covering the solvent side of deployment and not the solute side. The error concentrates in the same places. Under the residual-only convention 45% of the squared error sits in the four ethylene-glycol solvents and 32% in the 37 rows carrying water as solute; under the full convention 49% sits in those same four solvents. The fidelity null of §3.5.3 is therefore a small-molecule and glycol-heavy null, and the same concentration is what makes the aggregate margin a

composition statistic (Table S12). Broken out by the classification of that table, three solvent classes holding 47% of the rows carry 74% of the squared error, while the acceptor-only aprotics carry 0.6% of it on 12% of the rows. By publication, the three largest contributors carry 76% between them, at 32%, 24% and 20%.

*Design and stratification.* The 477 rows are 185 distinct solute-solvent pairs: 118 pairs carry one row, 38 carry five or more, and the largest carries 15. Counted per row the design ratio is  $\dim(z^*, T)/n = 103/477 \approx 0.22$ ; counted per distinct pair, 0.56, against 1.7 on the VT-2005-matched set. Molecule-pair identity coarsens  $(z^*, T)$ , so  $\mathbb{E}[\text{Var}(m \mid \text{pair})]$  is a valid  $B_{\text{insuff}}$  upper bound over the 359 rows that support it. It is 0.046: under 1% of  $\text{Var}(m)$ , and thirteen times tighter than the binning bound on those same rows (0.61 residual-only, 0.39 full). We report it and do not promote it, because all 67 multi-row pairs come from a *single* publication each. Those rows are one laboratory’s temperature series, so the bound omits the between-determination component of the superpopulation (§2.1.1). The binning bound, whose eight bins mix all 15 sources, does not omit it. It bounds a sub-component, so the reading that holds is that the binning bound is slack rather than that  $B_{\text{insuff}}$  is near zero. The temperature axis carries under 1% of  $\text{Var}(m)$  at a fixed profile pair, and that same fact makes the choice between conditioning on  $z^*$  and on  $(z^*, T)$  numerically immaterial here. Only the latter is licensed by Lemma 1, and it is the one we use. Collapsing to one row per pair (each the row nearest 298 K) gives  $\text{MSE} = 1.679$ ,  $B_{\text{insuff}} \lesssim 0.495$  and a margin of +0.69 at  $P > 0.95$  under the full convention (1.540, 0.635, +0.27,  $P \approx 0.8$  residual-only). By method, 334 rows are chromatographic, 37 chemical or heterogeneous equilibration and 106 carry no method field; by source, 15 publications in four journals (*J. Chem. Eng. Data* 203, *J. Chem. Thermodyn.* 114, *Fluid Phase Equilib.* 108, *Thermochim. Acta* 52), the largest single publication contributing 105 rows. On the 334 chromatographic rows alone, re-binned within the stratum, the activity error is  $\text{MSE} = 1.783$  against  $B_{\text{insuff}} \lesssim 0.36$  under the full convention. That

is a margin of +1.06 ( $P > 0.95$ ), wider than the full set’s +0.95. Under the residual-only convention it is 1.474 against  $B_{\text{insuff}} \lesssim 0.58$ , a margin of +0.31 ( $P \approx 0.9$ ) and narrower than the full set’s +0.51. The cut therefore does not move the sign under either convention, and it does not uniformly widen the margin either. A DOI-clustered bootstrap of the full-set margin gives  $P > 0.95$  with 90% [+0.50, +1.57] (full) and  $P \approx 0.9$  with [−0.07, +1.74] (residual-only), and a pair-clustered one  $P > 0.95$  with [+0.71, +1.27] and [+0.23, +1.16] respectively. This is the stratification the 60-pair VT-2005-matched set cannot support, and it is why the broad IDAC set carries the map. The quality cut that *inverts* the enlarged 298 K set (§3.7, limitation (i)) leaves the sign alone here under either convention. It does not buy an aggregate verdict, for the reason Table S12 and §3.5.1 give: the chromatographic cut and the pair collapse each leave the sign standing, and the chemistry cut compounded with the pair collapse does not.

**Scoring the deployed closure against the published evaluation.** Our differentiable COSMO-SAC layer, with its own constants and damped fixed point, was run over the same three record files De Souza et al.<sup>S17</sup> score. Components were resolved through the repository’s CAS-to-InChIKey map and matched by exact InChIKey to this paper’s VT-2005 artifact, which gives 7889 of 8138 usable records (96.9%) over 2279 distinct pairs, each at its own temperature. Those are the rows of Table S27’s middle block, and the same 7889 rows are the out-of-sample set of §3.5.1.

Within that block the error is not evenly spread, and the worst cell is water carried as the dilute solute in an organic solvent: AAD 2.02 with a +1.92 bias on 468 rows. At 298 K and non-aqueous, the regime the  $n=60$  VT-2005-matched set occupies, AAD is 0.42 on 1033 rows against that set’s own 0.81. This paper’s set is if anything the harder one.

**Sources.** Solubility is BigSolDB 2.0,<sup>S21</sup> partitioned by Bemis–Murcko scaffold: molecules grouped by ring system plus linker, side chains

Table S30: Solubility accuracy in two blocks that may not be read across: this work on the solute-scaffold split, external models re-run here on an easier by-solute split. Two sample sizes appear where a row’s two metric columns are scored on different rows, one where they coincide. The grounded, ungrounded and  $\sigma$ -oracle arms of the first block come from the five-seed leak-free re-run.

| Model                                                                                                                                 | $n$<br>( $\ln x_2 / \log_{10} S$ ) | MAE<br>$\ln x_2$ | RMSE<br>$\log_{10} S$ | $R^2$<br>$\ln x_2$ |
|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|------------------|-----------------------|--------------------|
| <i>This work, solute-scaffold split; mean <math>\pm</math> s.d. over seeds: five for the two COSMO-SAC rows, three for the others</i> |                                    |                  |                       |                    |
| DirectGNN, same encoder, no closure ( <i>control</i> )                                                                                | 5608 / 5440                        | 1.70 $\pm$ 0.03  | 1.00 $\pm$ 0.02       | 0.42 $\pm$ 0.03    |
| TGNN, NRTL closure (physics)                                                                                                          | 5608 / 5440                        | 1.79 $\pm$ 0.07  | 1.54 $\pm$ 0.07       | 0.34 $\pm$ 0.03    |
| TGNN, COSMO-SAC closure, supervised $\hat{\sigma}$ (physics)                                                                          | 5608 / 5440                        | 1.93 $\pm$ 0.09  | 1.44 $\pm$ 0.11       | 0.29 $\pm$ 0.07    |
| TGNN, COSMO-SAC, reference $\sigma^*$ substituted at evaluation                                                                       | 5608 / 5440                        | 2.34 $\pm$ 0.11  | 1.64 $\pm$ 0.09       | -0.07 $\pm$ 0.08   |
| <i>Scale reference on the same solute-scaffold rows</i>                                                                               |                                    |                  |                       |                    |
| predict the test-set mean of $\ln x_2$                                                                                                | 5608 / 5440                        | 2.32             | 1.33                  | 0.00               |
| <i>External models re-run on this corpus on a by-solute split, one run each—not a ranking against the block above</i>                 |                                    |                  |                       |                    |
| FastSolv, retrained on this corpus, re-scored                                                                                         | 10287                              | 1.44             | 0.84                  | 0.55               |
| SolProp cycle, released model + 3-parameter recalibration                                                                             | 10287                              | 2.34             | 1.34                  | -0.18              |
| SolProp cycle, released model, zero-shot                                                                                              | 10287                              | 2.36             | 1.87                  | -0.47              |
| SolProp encoder retrained on $\ln x_2$ , cycle discarded                                                                              | 10570 / 10287                      | 1.62             | 1.07                  | 0.39               |

stripped. Crystal  $T_m / \Delta H_{\text{fus}}$  come from an open melting-point pool plus a group-contribution prior. True  $\sigma$ -profiles come from the VT-2005 database, the Virginia Tech 2005 compilation of quantum-chemically computed screening-charge profiles for  $\sim 1,400$  compounds,<sup>S22</sup> ingested to 1432 compounds with the COSMO cavity volume  $V_{\text{cosmo}}$  for the combinatorial term. These are quantum-chemically computed for a single reference conformer and protonation state. Mismatch to the experimental measurement conditions is noise in  $z^*$  that inflates  $B_{\text{insuff}}$ . But under a Berkson-type displacement it also loads onto  $B_{\text{closure}}$  through the curvature of  $g$ , so we treat it as an unsigned limitation and *not* as conservative for the closure-dominance verdict (§3.5). Concretely, if the true latent is  $z^*$  displaced by a mean-zero (Berkson-type)  $\varepsilon$ , then even a perfectly specified  $g$  has  $B_{\text{closure}} = \mathbb{E}[(\mathbb{E}_\varepsilon[g(z^* + \varepsilon)] - g(z^*))^2]$ , a pure curvature effect that vanishes only for affine  $g$ —and under classical rather than Berkson error, not even then. We do not bound that curvature share.  $B_{\text{closure}}$  is accordingly named for the closure and measured on the composite of closure and profile calculation, which is how §2.1.1 defines it and how every attribution in this work reads it; §3.5.3 sizes the exposure against the two-database bracket of the published evaluation and leaves the share un-

bounded. Infinite-dilution activity coefficients: ThermoML, the NIST/IUPAC archive of experimental thermophysical data.<sup>S23</sup> Enlarging the reference  $\sigma$ -profile set would not materially grow the matched activity set: of the 298 K ThermoML IDAC pairs, only two solutes lack a VT-2005 profile while their solvent has one—the binding constraint is joint IDAC $\times$ profile coverage, not profile availability. What does enlarge the VT-2005-matched set is a broader extraction on the IDAC side: the expanded ThermoML pull released with the code matches 115 pairs at 298 K. We do not adopt it as a replacement set because its verdict is quality-cut sensitive: it widens the margin from 0.24 to 0.35 but *inverts* it to -0.13 on the 100 pairs labelled gas chromatography, and that sensitivity is unresolved at 298 K. It is not a verdict on the extraction, which matters because the broad IDAC set is built from the same pull: there the same cut leaves the sign alone under either convention (+0.51  $\rightarrow$  +0.31 residual-only, +0.95  $\rightarrow$  +1.06 full). What differs is resolution, not provenance—at 298 K a 15-row cut inverts a margin already inside the design’s resolution, whereas on 477 rows a 143-row cut moves one without inverting it—so we use the enlarged pull where it is stratifiable and keep the strict  $n=60$  set where the paradox’s own oracle is defined. The choice cannot be a choice

of answer, because no answer is taken from this set: its aggregate is reported as not standing, no stratum of it is stated as a finding, and both enlargements available to it point the same way as everything else about it. The expanded pull inverts the margin on its chromatographic subset; the looser InChIKey rule takes it to +0.004 on 42% more pairs. A set kept for a verdict would not be one whose verdict is withheld. The deployed VT-2005-matched set is also narrowly sourced: the curated IDAC compilation yields 74 VT-2005-matchable pairs under canonical SMILES (60 at 298 K plus 14 off-temperature), all by gas chromatography and all from three publications, 33 pairs from one, so neither a method-stratified nor a source-clustered re-estimate is available on it. The VT-2005-matched set is matched to VT-2005 by RDKit canonical SMILES on solute and solvent, the rule the deployed  $\sigma$ -oracle itself uses; matching by InChIKey instead admits 85 pairs over 46 solutes and 19 solvents at 298 K in place of 60/41/17 (both counts are recorded in the deposited decomposition artifact). We keep the stricter rule so that the  $z^*$  charged against each label is the profile the deployed pipeline would supply, rather than one imported across a tautomer or protonation variant—the same mismatch that makes the reference imperfect above. We have re-estimated the decomposition under the looser rule. At the cell of §2.5 the 85 pairs give  $\text{MSE} = 1.243$  against  $B_{\text{insuff}} \lesssim 0.620$ , a margin of +0.004 where the reported 60 give +0.045, and +0.52 against +0.58 under the full convention; the recomputation reproduces the reported set’s own margin exactly before it is pointed at the larger one. The sign survives under both conventions and the residual-only margin does not: it goes to zero. That is the same verdict this set already carries — its aggregate does not stand, and no stratum of it is stated as a finding — reached on 42% more pairs, and nothing in the paper rests on it either way. The broad IDAC set uses the looser InChIKey rule, so the two sets are matched under different standards; that asymmetry is quantified above (four molecules, 12 rows, verdict unchanged on the exactly matched subset) rather than left implicit. Second-domain

pKa: the OPERA experimental compilation<sup>S19</sup> (7904 QSAR-ready records, public-domain US EPA; cited, not redistributed); the uniform net-formal-charge filter reduces this to the 1004 Hammett-amenable rows (158 meta/para, 846 ortho/heteroaromatic) analysed in §S7. The Bemis–Murcko scaffold split follows the main text. Because the seeded split is not stable across pipeline versions (§2.2), the numbers in this paper are computed on a *single, pinned* split, whose per-file SHA-256 each run’s manifest records and whose files are deposited with the checkpoints rather than versioned—with the one run whose split provenance was not logged, recorded in the footnote.<sup>4</sup> Melting-point tables are auto-detected as °C or K to avoid a +273 K double-conversion. One limit of that apparatus is worth stating beside what it catches: value-conservation checks do not detect loss of *role*, because a numeral goes on printing elsewhere in the submission while the statement that gave it meaning is deleted. Three such losses in the editing of this manuscript—two null *p*-values and the comparator of a two-MSE check—were found by a clause-level audit and not by the numeric gate, and are restored in §S5.

**Grounding-stream provenance, and one build that violates the guard.** The  $\sigma$ -stream builder drops any pool molecule whose Bemis–Murcko scaffold appears in the validation or test split and aborts rather than proceed when a split file is missing; a correct build retains 1319 of the 1427 VT-2005 molecules that parse into the pool. The stream file was subsequently rebuilt with the exclusion pointed at the splits of a small debug dataset: it carries 1425 rows, of which 106 (99 train, 7 validation) have a scaffold in the real test  $\cup$  validation union of 2702 scaffolds. The overlap is not confined to scaffold relatives: 91 of those 106

<sup>4</sup>The three-seed surrogate isolation (§3.3) was produced on a separate run whose per-run split provenance was not logged; its matched set is the same  $n=44$  molecules, derivable from the committed split independently of the model, so it is consistent with the reported split. We record the split hash in the runner so future runs are auditable directly.



rows are themselves held-out solutes by canonical SMILES — 65 rows carrying 64 distinct test-split molecules and 26 carrying 26 validation ones — and the correctly guarded build leaves none of them.

Those are counts over the split *files*, which carry the auxiliary melting-point rows (auxiliary-only, no solubility label) alongside the labelled ones and so span 2634 and 2467 distinct solutes. Every test metric in this paper is computed on the 5608 labelled rows, which span 147 solutes, and that is the set a leak can act on. Of those 147, 7 have their own reference profile in the defective stream and 8—those 7 among them—have a scaffold relative in it, so the second count is the union and not an addition, on 297 and 351 of the 5608 rows, 5.3% and 6.3%. For a scaffold split solute coverage is the load-bearing statistic rather than the fraction of stream rows, so these are the figures the disclosure of §2.2 carries; the correctly guarded build leaves zero of each. The remaining 57 leaked molecules appear in the test file only as melting-point rows and enter no reported metric, and the split holds nothing out on the solvent side, so the stream’s coverage of test solvents violates no guarantee this split makes. The crystal stream’s build on disk records the correct 2702-scaffold exclusion, so among the streams feeding the runs reported here the defect is confined to the  $\sigma$  one. Per-run stream provenance was not logged, so the artifacts cannot certify which build fed which arm.

What the file times establish is narrower than their ordering first suggests. The local stream path was created on 2026-06-21 and last written on 2026-07-06, and the content that write left is the defective one. Every prediction file of §3.1 postdates the creation, so the ordering separates the *last write* from the runs rather than the defective build from them: the prediction files behind Fig. 2 and Table S4 were created before 2026-07-06 and those of the two single-seed oracle-application controls (§S3.2) and the output-residual arms (§S5) after it. Creation, not modification: the seed-44  $\sigma$ -oracle copy carries a later write, from the recovery described below. A guarded build overwritten on 6 July and a build defective from 21 June are equally

consistent with that. These are in any case modification times on local copies: the runs read an uploaded data bundle, the one bundle retained on disk was assembled on 2026-07-13, after all of these runs, and a local write does not date a read. The ordering therefore licenses no build provenance for any arm, and in particular does not establish that the two single-seed controls differ in build from their seed-42 comparator. The +0.18 and +0.27 of §S3 are accordingly single-seed penalties of uncertified stream provenance, and “confound-free” names the test-time input swap they remove and nothing about the stream.

Three facts bound what the leak can do, and each bounds one contrast. It is common-mode across the *paradox* contrast, the learned- $\sigma$  and  $\sigma$ -oracle rows being one checkpoint evaluated two ways, so it cannot generate the gap between them; DirectGNN uses no grounding stream, so in the *physics-versus-control* comparison a leak can only favour the physics arm that loses to it; and on the  $n=44$  matched set of §3.3 a correct guard would have removed one molecule (piperidine). A fourth contrast none of the three reaches is the *grounded-versus-ungrounded* supervision contrast. The ungrounded arm sets its warm-up and stream budget to zero and so carries no  $\sigma$  stream either, which leaves the grounded arm as the only arm a leak can reach, in the direction the guard exists to prevent. The 0.20 supervision gain is therefore the one reported contrast that a leak could inflate rather than one it is bounded on, and it stays uncertified against the scaffold split until the leak-free re-run. The defect does compromise auditability, and with it the reach of the split guarantee. An arm trained on that build would have had reference  $\sigma$ -profiles for the 7 evaluated test solutes above, though no solubility labels for them. The guarantee stated in §2 is therefore a property of the released builder rather than of every stream file we trained on. Closing this requires hashing each stream build beside the split hash, which was not done for these runs.

**What the leak-free re-run reports, and what it decides. This specification is**

**frozen as it was written, before the run existed, and the run has since been made.** The second branch below is the one that fired: the per-seed supervision gains straddle zero, so supervision is reported with no established sign, and every display this paragraph enumerates has been regenerated at five seeds. Nothing below is edited to match the outcome, because a protocol rewritten after its result is not one; read the future tense as the tense it was fixed in. The re-run is specified here rather than described afterwards, so that its protocol is fixed before its numbers exist. It trains the ungrounded and the grounded arm at five seeds (42 through 46) under the COSMO-SAC configuration of the published arms, unchanged. The  $\sigma$  stream is rebuilt with the scaffold guard pointed at the validation and test files of *this* split, and is certified inside each training run rather than at the builder. The run re-reads the stream it is about to consume and the split files it will be scored against, then asserts that they share no molecule by canonical SMILES and none by Bemis–Murcko scaffold. The stream’s SHA-256 and row count are written into that run’s manifest beside the SHA-256 of each pinned split file, so the artifact states which build fed which arm. The ungrounded arm’s manifest records that it carried no stream at all, the absence being the thing asserted there. Accuracy is read on the same 5608 labelled test rows as every other number in this paper, under the same cross-arm intersection rule (§2.4), and the five per-seed MAE values are reported for each arm rather than a mean and a standard deviation alone. Two further readings are taken on those same checkpoints: the evaluation-only substitution of §3.1, which is within-checkpoint and therefore a consistency check rather than a test, and the  $n=44$  profile comparison of §3.3 with the one molecule the leak touches removed. Quoting the magnitude of the co-adaptation control as something other than one seed would need a separate pass at the same five seeds, which this protocol does not commit to running; the article reports no magnitude for it at all.

The decision the re-run settles is stated in advance and in one direction only, since it is the

supervision contrast alone that hangs on it. If the five per-seed gains are of one sign, the two senses of grounding keep their opposite signs, §3.1 and Table S3 print the re-run’s per-seed values in place of the three uncertified ones, and the abstract’s scope clause moves from three seeds to five, with the refusal to order the two operations by magnitude unchanged. If the per-seed gains straddle zero, supervision has no established sign at five seeds. The paper then reports one operation with a sign, substituting the reference at inference hurting, and records the other as unresolved. The abstract, the heading of §3.1, the opening sentence of the Conclusions (§4) and the supervision row of Table S3 are rewritten to say so. If the gains are of one sign and it is the opposite one, that is what is reported. In either of the last two cases the graphic for the table of contents, which prints the two senses of grounding and their opposite signs, is redrawn to what is then reported, and the freeze contrast of §3.3 is restated as well, because the warm-up state whose quality it turns on is the supervised one. The substitution result is unaffected in all three cases: it is one checkpoint evaluated two ways, and no  $\sigma$ -stream build can reach it.

Which displays the five-seed values replace is fixed here as well, and in all three branches alike, because a five-seed table printed beside a three-seed figure would be a contradiction, and which displays move is not a choice to be taken once the numbers are known. Three arms carry those values: the ungrounded arm, the grounded arm, and the  $\sigma$ -oracle arm, which is the grounded checkpoint re-evaluated with  $z^*$  substituted and therefore moves with it. Each display named next carries at least one of the three and is regenerated from the five-seed arms by the script that produced it:

- the abstract’s seed count—its *over three seeds*, which is these arms, and not its *separate three-seed runs*, which is the  $n=44$  surrogate family at seeds 0, 1 and 2 and stays at three—its leak-free clause, which goes in all three branches alike, and the three  $\ln x_2$  MAE means it prints—the ungrounded 2.043, the grounded 1.846, the  $\sigma$ -oracle 2.252—which every branch replaces

with five-seed means at the same three decimals, the word ceiling forbidding per-seed values in this display alone: branch 1 also moves that first seed count from three to five, leaving the refusal to order the two operations by size standing over them; branch 2 also rewrites the sentence carrying the first two, supervision then having no established sign; branch 3 prints that pair in the direction it then runs; and the 2.252 moves with the grounded checkpoint throughout, the contrast it belongs to being unaffected in all three;

- §3.1 entire, together with the two sections that carry the rest of its prose—§3.2, which carries the regression slopes read off Fig. 3, the ranking contrast and the solute-blind comparison, and §3.4, which carries its channel paragraph—the per-seed gains and the seed-42 values included wherever they now print;
  - the five rows of Table S3 under *the two senses of grounding*, and the +0.14 of the accuracy row of Table S3, the ledger’s third part;
  - the  $\sigma$ -grounding row of Table S29 entire—its seed list, the mean $\pm$ sd cell that carries it, and the +0.41 and  $-0.20$  its third column prints, so that no cell of that row states a seed policy the cell beside it contradicts;
  - the ungrounded, grounded and oracle bars of Fig. 2 with their  $R^2$ ;
  - panels (b), (c) and (d) of Fig. 3;
  - the block-1 grounded and  $\sigma$ -oracle rows, which print in Tables S30 and S20 off one source of record, so a regenerated value lands in both, with the clip sweep in the note of the second, the fuller sweep of §S4 and the readings §3.7 takes from both;
  - Table S4 and the unrounded means, per-seed values and arm orderings §S3 reads off it;
  - Table S22 with the ranking and per-group recalibration readings §S6.1 sets around it, whose two columns are the grounded checkpoint and its own  $\sigma$ -oracle re-evaluation and nothing else;
  - §S6.4 with Tables S24 and S25, which resolve that same +0.14 along chemical axes and re-
- state the two arm means it is taken from;
- the row-class split of the substitution in §S1, whose  $-0.000$ ,  $+0.431$  and  $+0.006$  put the  $+0.41$  on the solvent channel;
  - the label-noise-floor paragraph of §2.2, whose scaffold-split range 1.70 to 1.85 takes its upper endpoint, and the factor of five it reads against the floor, from the grounded arm;
  - the deposit-recovery paragraph below, whose before-and-after cells are that same  $\sigma$ -oracle row and move with it, the repair standing as a fact about the published deposit;
  - the freeze contrast of §3.3, whose closing clause sets its 65% against *the +0.41 arms of §3.1* by name and so prints an enrolled value outside the section that owns it, the runs it contrasts them with staying as published;
  - the three printed seed statements the abstract’s enrolment does not reach—§2.4’s row-set paragraph, which says the controlled comparisons are run at *three* seeds and that the lock’s finite-prediction clause removes no row from any arm *at any seed*, and, in §S8, the compute ledger’s (*three seeds*) and the seeds-and-determinism paragraph’s {42,43,44}, which is a second printed seed list for the same three arms, and in §S1 the substitution-perturbation paragraph’s *over the three seeds*, whose three maxima are taken across the same arms;
  - the opening two paragraphs of §S3.2, which are bookkeeping over these arms’ deposits and move on two axes at once—the 2495-row remainder decomposes *identically at all three seeds*, and the finiteness audit runs over *twenty-two* per-row prediction files, the five intersecting arms and the  $\sigma$ -oracle *at each of the three seeds* plus the four seed-42 control files, so the count, the seed statements and the parenthetical *twenty-one as first deposited* are rewritten together, and the same count where the deposit-recovery paragraph prints it as *all twenty-two pass* moves with them, the two copies being one number and not two;
  - the Conclusions’ second sentence (§4), whose first half asserts that training the intermedi-

ate toward its reference improved the model and whose second half asserts that supplying it at prediction time hurt *at every seed*—a per-seed universal over the grounded and  $\sigma$ -oracle arms, two of whose five seeds do not exist yet, which the consistency of the substitution across builds fixes in sign but does not verify seed by seed;

- the same universal where §1 states it—*here it degrades prediction at every seed*—which is the identical assertion over the identical two arms and is rewritten with it;
- and the graphic for the table of contents, whose caveat that training on the reference is not certified leak-free is removed in all three branches alike, the re-run certifying the stream inside each training run whichever way its per-seed gains fall.

Seven of those floats—Figs. 2 and 3, Tables S30, S20, S4, S3 and S3—then carry re-run arms beside arms that were not re-run. Each therefore states in its caption which of its arms came from the re-run. The block-1 header of the two base-lines floats already carries the form of words for it, and Fig. 3 remains one seed for all four of its panels, the new seed 42 in three of them and the published one in the control. The two ledger parts are on that list by the criterion and not by exception. The first prints three re-run rows beside the two seed-42 co-adaptation rows, whose +0.18 and 0.59 the sentence below keeps published with the five-seed value beside them; that is a caption statement by disposition. The third prints the enrolled +0.14 beside the four mechanism rows, measured on the  $\{0, 1, 2\}$  surrogate runs and named in the same cell as not the arms of Fig. 2. Table S3, the ledger’s second part, carries no re-run arm at all—its three blocks are the two activity-axis closures and the  $pK_a$  domain—and stays out. Table S29 meets the criterion too, its  $\sigma$ -grounding row sitting among eleven families that are not re-run, but discharges it in its seed column row by row, which is what that column is for, so its caption takes no addition. Table S22 needs no such statement, both its arms being re-run; Tables S24 and S25 print no arm at all, every cell being a difference against the control, so each

says in its caption that only the physics side is five-seed. Three numbers instead keep their published value with the five-seed one printed beside them and the caption saying so. The reason is the same in each case: the re-run supplies one side of the comparison and not the other. They are the Staverman–Guggenheim contrast  $1.8457 \rightarrow 1.8788$ , whose combinatorial arm is not retrained; the channel swap’s +0.27, a seed-42 checkpoint that is not retrained either; and the co-adaptation control’s +0.18 together with the 0.59 share, both of §S3 and neither claimed in the article, which move only if the separate five-seed pass above is run. Four passages carry no arm value but state that the 0.20 is uncertified: the disclosure of §2.2, limitation (xii) of §3.7, the grounding-stream provenance paragraph above, and the Conclusions’ second sentence (§4), whose *on a contrast this work does not certify leak-free* is that caveat in that section’s register. The in-run certification answers all four, so each is rewritten to what the re-run certifies, and limitation (xii)’s 0.20 moves with §3.1’s. Nothing else moves, and the boundary is worth stating because two neighbouring numbers sit just outside it: neither the DirectGNN control nor the NRTL arm is retrained, so their rows in Tables S30, S20 and S4, and the endpoint comparison of §S6.5—which is the NRTL arm against that control—stand as printed; and §3.5 uses no trained model at all.

**The two decomposition sets, deposited row by row.** Because the closure/insufficiency verdict rests on a newly constructed set, both sets are deposited with the Supporting Information as row-level tables that need no code to inspect. The results that otherwise require re-running the code are, specifically, the broad IDAC set with its design and provenance audit, the 2010 kernel’s own-latent floors, the fitted kernel residual and the estimator grid. **Data Set S1** carries one row per measurement: pair identifier, solute and solvent SMILES, the UD compound key matched to each and which rule matched it (full InChIKey or first-block fallback), temperature, the experimental  $\ln \gamma_2^\infty$ , the *four* closure outputs—2002

and 2010-dispersion-off, each in the full and the residual-only combinatorial convention, so that the residual-only-convention kernel contrast of §3.5.3 is recomputable from the file—the ThermoML method field, the source DOI and journal, and a flag for the 57 rows shared with the **Data Set S2** (printed in full as Table S31) carries the same fields for the 60 VT-2005-matched pairs, with three differences that follow from how the VT-2005-matched set is defined. Its closure outputs are the VT-2005 ones the deployed  $\sigma$ -oracle uses, in both combinatorial conventions, alongside the four UD-profile outputs that make it commensurable with Data Set S1. Its matching rule is canonical SMILES throughout, so it carries neither the per-row match-rule columns nor the UD compound keys. It also names each compound in words as well as in SMILES. It additionally carries a per-row record count, 1 throughout. All 60 rows sit at 298 K, all are chromatographic, and their three source DOIs divide the set 33/20/7. The counts, margins, stratified re-estimates and clustered bootstraps reported for either set are recomputable from these two files with the estimator definitions of §2.5; what they do *not* restore is the provenance of the closure evaluations themselves, which still requires the code.

**Compute ledger.** The headline  $\sigma$ -grounding comparison (Fig. 2, Table S4) is a full-corpus GPU run, at five seeds for the arms the leak-free re-run trains and three for the rest; the closure/insufficiency decomposition (§3.5), the Fisher audit (§S6.2), the noise floor, the encoder probe, and the temperature and mechanism diagnostics of §S5 are CPU-reproducible from deposited artifacts. The prediction CSVs, the model checkpoints and the larger analysis artifacts will be deposited separately, in a Zenodo archive [**whose DOI is registered at submission**], rather than in the repository. The broad IDAC set’s design and provenance audit regenerates in one CPU run of the deposited script. It covers distinct-pair counts, within-pair variance across temperature, pair-blocked out-of-fold estimators, method and source-DOI composition, and the

method-stratified and cluster-resampled margins. It also covers the kernel contrast split three ways, into the shared 57, the complementary 420 and the 80 complementary rows that are also at 298 K, together with the VT-2005-matched floor recomputed on UD profiles and both sets’ floors under the 2010 kernel’s own typed latent. A second CPU run, the convention audit, produces the convention-rule material: both sets’ margins and clustered bootstraps under each combinatorial convention, the per-cell broad-set grid of Table S6, the InChIKey match-rule audit and its exactly matched re-estimate, the pair-conditional bound and its single-source diagnosis, the typed latent’s dimensions, and the two deposited row-level tables. Those files are regenerated by the released code rather than deposited, which is the exposure limitation (xiii) records; the two deposited tables are the part of this material that is checkable without running anything.

**Seeds and determinism.** The controlled comparisons of §3.1 use seeds {42, 43, 44, 45, 46}, the leak-free re-run’s, and are reported on the labelled test rows (5608 of 8103; §S3.2); the surrogate isolation (§3.3) uses seeds {0, 1, 2}; the training-time and channel-swap oracle controls, the output-residual arms (§S5), the data-efficiency curve (§S6.5) and the loss-simplification ablation (§S1) are seed 42 only. The decomposition is seed-independent (its closure has no learnable parameters).

**One deposited file was partial, and has been repaired.** The released per-row prediction file for the seed-44 oracle solubility arm carried 295 of the run’s 8103 rows. The cause was not the run: the file the run wrote is intact on the compute volume, and the transfer that produced the deposited copy stopped mid-record—the deposited file is an exact byte-prefix of the complete one, 198,190 bytes of 5,213,388, ending part-way through the 295th line. The complete file has been recovered from that volume and now stands in the deposit, and it is the file the run scored: every field of that run’s own summary— $n_{\text{rows}} = 8103$ ,  $n_{\text{labelled}} = 5608$ , MAE 2.24199, RMSE 2.96058,  $R^2 = 0.01768$ ,

Table S31: **Data Set S2** in full: the VT-2005-matched set ( $n=60$ ), row by row, deposited so the set can be inspected without running code. Read in two blocks of 30, left then right.

| #  | solute <sup>a</sup> | solvent          | $m$   | $g_{\text{full}}$ | $g_{\text{res}}$ | broad | #  | solute         | solvent      | $m$   | $g_{\text{full}}$ | $g_{\text{res}}$ | broad |
|----|---------------------|------------------|-------|-------------------|------------------|-------|----|----------------|--------------|-------|-------------------|------------------|-------|
| 1  | Brc1ccccc1          | OCCO             | 3.81  | 2.33              | 2.40             | n     | 31 | CCc1ccccc1     | c1ccncc1     | 0.36  | 0.35              | 0.39             | y     |
| 2  | C#CCCCC             | c1ccncc1         | 0.44  | -0.42             | -0.34            | y     | 32 | CO             | C1CCNCC1     | -0.84 | -2.38             | -2.21            | y     |
| 3  | C1=CCCCC1           | c1ccncc1         | 0.91  | 0.75              | 0.75             | y     | 33 | CO             | CC(C)(C)N    | -1.24 | -3.15             | -2.99            | y     |
| 4  | C1CCOC1             | c1ccncc1         | -0.07 | 0.03              | 0.03             | y     | 34 | CO             | CCC(C)N      | -0.97 | -3.20             | -3.02            | y     |
| 5  | C=CCCCC             | c1ccncc1         | 1.27  | 0.93              | 1.03             | y     | 35 | CO             | CCCCCN       | -0.97 | -3.23             | -2.87            | y     |
| 6  | CC#N                | c1ccncc1         | 0.38  | 0.25              | 0.30             | y     | 36 | CO             | CCCCN        | -1.05 | -3.00             | -2.79            | y     |
| 7  | CC(C)(C)N           | CO               | -1.39 | -2.07             | -1.85            | y     | 37 | CO             | CCCNCCC      | -0.37 | -2.33             | -1.97            | y     |
| 8  | CC(C)(C)N           | c1ccccc1         | 0.28  | 0.39              | 0.38             | y     | 38 | CO             | CCNCC        | -0.67 | -2.35             | -2.15            | y     |
| 9  | CC(C)=O             | c1ccncc1         | 0.17  | -0.13             | -0.12            | y     | 39 | CO             | Cc1cccc(C)N1 | -0.49 | -0.78             | -0.49            | y     |
| 10 | CC(C)Br             | c1ccncc1         | 0.30  | 0.22              | 0.21             | n     | 40 | CO             | Cc1cccn1     | -0.24 | -1.07             | -0.86            | y     |
| 11 | CC(Cl)CCl           | c1ccncc1         | -0.07 | -0.05             | -0.05            | y     | 41 | CO             | Cc1cccn1     | -0.06 | -1.06             | -0.85            | y     |
| 12 | CCC(C)N             | CO               | -0.29 | -1.99             | -1.71            | y     | 42 | CO             | Cc1ccncc1    | -0.39 | -1.11             | -0.90            | y     |
| 13 | CCC(C)N             | c1ccccc1         | 0.14  | 0.42              | 0.42             | y     | 43 | CO             | NC1CCCCC1    | -1.11 | -3.21             | -2.98            | y     |
| 14 | CCCCCCC             | c1ccncc1         | 1.91  | 1.60              | 1.71             | y     | 44 | Cc1ccc(C)cc1   | c1ccncc1     | 0.44  | 0.39              | 0.45             | y     |
| 15 | CCCCCCCC            | c1ccncc1         | 2.07  | 1.74              | 1.91             | y     | 45 | Cc1cccc(C)c1   | c1ccncc1     | 0.43  | 0.37              | 0.43             | y     |
| 16 | CCCCCCCC            | c1ccncc1         | 2.17  | 1.86              | 2.12             | y     | 46 | Cc1cccc(C)N1   | CO           | -0.06 | 0.14              | 0.64             | y     |
| 17 | CCCCCCCCC           | c1ccncc1         | 2.26  | 1.97              | 2.32             | y     | 47 | Cc1ccccc1      | OCCO         | 4.53  | 2.61              | 2.67             | y     |
| 18 | CCCCCCCCC           | c1ccncc1         | 2.32  | 2.08              | 2.52             | y     | 48 | Cc1ccccc1C     | c1ccncc1     | 0.34  | 0.33              | 0.37             | y     |
| 19 | CCCCCCN             | CO               | 2.62  | -2.35             | -1.66            | y     | 49 | Cc1cccn1       | CO           | -0.29 | -0.21             | 0.12             | y     |
| 20 | CCCCCCN             | c1ccccc1         | 2.79  | 0.51              | 0.59             | y     | 50 | Cc1cccn1       | CO           | -0.20 | -0.36             | -0.03            | y     |
| 21 | CCCCCOC(C)=O        | c1ccncc1         | 0.11  | 0.09              | 0.26             | y     | 51 | Cc1cccn1       | c1ccccc1     | -0.04 | -0.01             | -0.01            | y     |
| 22 | CCCCCl              | c1ccncc1         | 0.43  | 0.37              | 0.38             | y     | 52 | Cc1ccncc1      | CO           | 0.00  | -0.41             | -0.08            | y     |
| 23 | CCCCN               | CO               | -0.78 | -2.18             | -1.85            | y     | 53 | C1c1ccccc1     | OCCO         | 3.69  | 2.29              | 2.35             | y     |
| 24 | CCCCOC(C)=O         | c1ccncc1         | 0.14  | 0.04              | 0.14             | y     | 54 | Fc1ccccc1      | OCCO         | 3.47  | 2.03              | 2.06             | y     |
| 25 | CCCN(CCC)CCC        | CCCCCCCCCCCCCCCC | 0.01  | -0.03             | 0.07             | y     | 55 | Ic1ccccc1      | OCCO         | 3.74  | 2.45              | 2.54             | n     |
| 26 | CCCN(CCC)CCC        | CO               | 2.17  | 1.96              | 3.15             | y     | 56 | NC1CCCCC1      | CO           | 1.57  | -2.17             | -1.81            | y     |
| 27 | CCCN(CCC)CCC        | c1ccccc1         | 1.03  | 0.54              | 0.79             | y     | 57 | NC1CCCCC1      | c1ccccc1     | 1.86  | 0.45              | 0.46             | y     |
| 28 | CCCNCCC             | CO               | -0.09 | -0.31             | 0.40             | y     | 58 | c1ccc2ccccc2c1 | OCCO         | 5.47  | 2.71              | 2.85             | y     |
| 29 | CCCNCCC             | c1ccccc1         | 0.39  | 0.28              | 0.35             | y     | 59 | c1ccccc1       | OCCO         | 3.47  | 2.11              | 2.12             | y     |
| 30 | CCNCC               | CO               | -1.11 | -0.64             | -0.30            | y     | 60 | c1ccncc1       | CO           | 0.18  | -0.28             | -0.09            | y     |

<sup>a</sup> Columns:  $m$ , the experimental  $\ln \gamma_2^\infty$  at 298 K;  $g_{\text{full}}$  and  $g_{\text{res}}$ , the VT-2005-input COSMO-SAC-2002 output under the full and residual-only conventions; *broad*, the 57 pairs that also appear in the broad IDAC set, the three that do not being the bromine- and iodine-containing pairs of §3.5.3. The 477-row broad IDAC set is the companion machine-readable Data Set S1.

bias 1.00945, and the matched-subset block’s  $n = 5571$  with MAE 2.24373—recomputes from it to the last printed digit, at zero absolute difference in float64. Its 8103 rows carry no non-finite  $\ln x_2$  prediction, its labelled subset is the same 5608 rows in the same order as all five other seed-44 arms, and on the 37 labelled rows where the oracle matched neither molecule its  $\ln \gamma_2$  agrees with the grounded arm’s to  $5 \times 10^{-6}$ , the two being the same checkpoint forwarded twice. Those checks are not a claim to be taken on trust: a deposited verification pass runs all of them over the deposit in one CPU pass—row count, labelled count, whether the last record is complete, non-finite predictions, cross-arm row order, and each file against its own summary—and exits non-zero if any file fails. All thirty-seven pass.

Nothing computed at run time moves: Fig. 2, Table S4 and the +0.41 were aggregated from

the in-memory predictions before any file was written, and they are unchanged. What the re-pair changes is what could be recomputed afterwards. The  $\sigma$ -oracle row of Table S30 had been the mean over the two intact deposits and is now a three-seed mean like the rest of its block, moving from  $2.26 \pm 0.02$  to  $2.25 \pm 0.02$  in MAE, from  $1.57 \pm 0.17$  to  $1.60 \pm 0.14$  in  $\log_{10} S$  RMSE, and from  $-0.05 \pm 0.02$  to  $-0.03 \pm 0.04$  in  $R^2$ —the last of which is the same  $-0.03 \pm 0.04$  Table S4 reads, the truncation having been the whole of the disagreement between the two tables. The clip sweep of §S4 likewise carries three seeds for that arm.

The learned- $\sigma$  and DirectGNN solubility files and the activity-level ( $n=60$ ,  $n=44$ ) artifacts were complete for every seed throughout.

## S9 Black-box probe programme: full tables

All three studies use the same current-split DirectGNN checkpoints (seeds 42/43/44, test MAE 1.749/1.674/1.684) whose manifests pin the train/val/test SHA-256 to the split on disk. The probed object throughout is the control’s 788-dimensional pair representation. Every number here is CPU-computed; none required a GPU.

### Crystal-term decodability (Table S32).

Probe fitted on 152 solutes / 14,348 rows, scored on 13 held-out solutes / 527 rows, solute overlap zero.  $\Phi^*$  is 92.2% between-solute variance, so the effective  $n$  is 13 clusters.

Controls: positive—model’s own prediction  $R^2 = 0.980$ , temperature  $R^2 = 0.9997$ ; negative—random target  $-0.0006$ , molecule-blind constant profile  $-0.015$ , raw RDKit descriptors  $-0.514$  (set A) and  $-0.928$  (set B). Null  $p_{95} = +0.105/+0.117/+0.096$  and max =  $+0.316/+0.380/+0.346$  over 1000 draws.

Two features of Table S32 bear on how such probes should be read. First, the between-seed spread (0.616 range, sd 0.334) exceeds every individual value: a single-seed run would have been unreadable, and seed 43 alone would have read as a near-miss. Second, a decision rule of the common form  $R^2(\text{model}) - R^2(\text{raw}) \geq 0.10$  would have declared success here on all three seeds under descriptor set B—including seed 42, whose probe is worse than a molecule-blind constant—and on two of three under set A. The subtrahend is an unregistered analyst choice worth more than the effect; we report  $R^2(\text{raw})$  as a level beside  $R^2(\text{model})$  and never as a difference.

### Temperature structure (Table S33).

Slopes are fitted at fixed (solute, solvent); a slope pooled across solvents would confound activity differences with temperature. Ceiling, measured on the data before any model:  $r = +0.239$  on train (1353 pairs / 151 solutes) and  $-0.125$  held out (66 pairs / 13 solutes, permutation  $p = 0.581$ ).

**Organisation of the representation (Table S34).** Excess over a permutation null matched to the factor’s level, on the full test set.

As a share of the solute-identity ceiling: scaffold 100.3/99.9/99.8%, log  $P$  13.6/8.2/4.7%, TPSA 9.0/10.3/7.1%, MolWt 5.3/2.3/3.1%. That 99.8–100.3% scaffold range is the organisation measure §3.7 points here for. The scaffold share is pinned near one by the split rather than by the representation: of the 124 scaffolds spanned by the 147 test solutes only 11 hold more than one solute, so only 34 solutes share a Bemis–Murcko scaffold with another, carrying 492 of the 5608 rows (8.8%), and the two partitions coincide on 91% of the data. The raw variance shares run the other way (seed 42: scaffold 0.636 against solute 0.639), so the *raw* ratio falls short of one by 0.4/0.7/0.9% across the three seeds, and it reaches one only through the differing size-matched permutation nulls (0.022 against 0.026); no interval is attached to that 0.4–0.9% gap. The row therefore bounds what the representation adds beyond scaffold and does not measure within-scaffold discriminability (§3.7). A direct measure on the 34 solutes that do share a scaffold sizes what it bounds: two same-scaffold solutes in a common solvent sit at 0.21–0.40 of the between-scaffold representation distance across the three seeds (112 pairs over 11 scaffolds; 90% scaffold-cluster intervals inside [0.14, 0.57]). Molecules sharing a scaffold are compressed rather than unresolved.

Temperature is decodable from this vector at  $R^2 = 0.9997$  yet occupies only 3–4% of its variance (rows above): decodability of a quantity is not evidence that a representation is organised around it, which is why the crystal-term probe is run as a held-out transfer test rather than an in-sample decode.

On the 331 rows carrying a measured  $\Phi^*$ ,  $\Phi^*$  accounts for 18.3/42.5/18.0% of that subset’s own solute-identity ceiling. This does not contradict the held-out null above.  $\Phi^*$  is a solute-level quantity and the representation is organised by solute identity, so any solute-level variable shares variance with it in sample; the decodability probe above asks the different and sharper question of whether the mapping trans-

Table S32: Crystal-term decodability per seed, with the permutation null, the drop-one-solute jackknife and the leverage carried by the worst solutes.

| seed | $R^2(\text{model})$ | perm. $p$ | jackknife range    | sign flips | top-3 leverage |
|------|---------------------|-----------|--------------------|------------|----------------|
| 42   | -0.506              | 0.894     | $[-1.151, -0.228]$ | 0/13       | 56.2%          |
| 43   | +0.111              | 0.055     | $[-0.286, +0.333]$ | 3/13       | 59.8%          |
| 44   | +0.027              | 0.177     | $[-0.253, +0.237]$ | 6/13       | 57.2%          |

Table S33: The model’s temperature response against the empirical slope and against  $-\Delta H_{\text{fus}}/R$ , with the curvature of both in  $1/T$ .

|                                               | seed 42           | seed 43           | seed 44           |
|-----------------------------------------------|-------------------|-------------------|-------------------|
| corr(model, empirical slope)                  | +0.294            | +0.406            | +0.195            |
| corr(model, $-\Delta H/R$ )                   | +0.066            | -0.283            | -0.028            |
| median  model - empirical  slope              | 1329              | 1088              | 1007              |
| curvature in $1/T$ (data: $6.8 \times 10^5$ ) | $2.0 \times 10^6$ | $2.1 \times 10^6$ | $2.3 \times 10^6$ |

Table S34: Share of representation variance per organising factor, as an excess over a permutation null matched to the level at which that factor varies.

| factor                                    | seed 42 | seed 43 | seed 44 | null level |
|-------------------------------------------|---------|---------|---------|------------|
| solute identity (147 levels)              | +0.613  | +0.587  | +0.597  | row        |
| solute scaffold (124 levels)              | +0.614  | +0.586  | +0.595  | row        |
| scaffold beyond arbitrary solute grouping | +0.088  | +0.082  | +0.082  | solute     |
| solvent identity (70 levels)              | +0.278  | +0.301  | +0.271  | row        |
| temperature                               | +0.040  | +0.029  | +0.037  | row        |
| log $P$ (solute)                          | +0.083  | +0.048  | +0.028  | solute     |
| TPSA (solute)                             | +0.055  | +0.061  | +0.042  | solute     |
| MolWt (solute)                            | +0.032  | +0.013  | +0.018  | solute     |

fers to solutes the probe never saw. In sample the organisation is partly  $\Phi^*$ -aligned; out of sample that alignment does not carry.

## S10 Closure fidelity sets the reference-input penalty: a controlled synthetic family

To isolate the mechanism we construct a controlled synthetic experiment in which only the closure’s fidelity varies (input quality, sample size, and estimator held fixed). A teacher draws a latent  $z^*$  and a target through a known map  $T$ ; a fixed candidate closure  $g_F$  reconstructs it under a tunable misspecification of scalar fidelity  $F$  ( $F = 1$  recovers  $T$  exactly, smaller  $F$  pushes  $g_F$  away). The reference-input

penalty  $\Delta R^2 = R_{\text{oracle}}^2 - R_{\text{head}}^2$  is a measure of  $B_{\text{closure}}$  in this construction—the free head recovers  $\mathbb{E}[m \mid z^*]$  while  $g_F$  is the misspecified map. Because  $\Delta R^2$  equals  $B_{\text{closure}}$  by design here, the sweep verifies that the instrument responds to fidelity as constructed; it is a check on the construction, not an independent test of the hypothesis. As  $F$  falls the penalty grows monotonically: averaged over six misspecification forms (linear, quadratic, cubic, absolute-value, sinusoidal, cross-term)  $F=1.00$  gives  $\Delta R^2 \approx 0$ ,  $F=0.38$  gives  $\approx -0.33$ , and an over-rotation stress test ( $F=-0.50$ ) gives  $\approx -2.0$  (Fig. S10); all 24 family  $\times$  form sweeps are individually monotone, and the ordering is identical across four unrelated teacher families (linear, monotone-nonlinear, kinetics-exponential, PDE-field), whose curves are visually coincident. That coincidence says the teacher factor carries no information *in this*



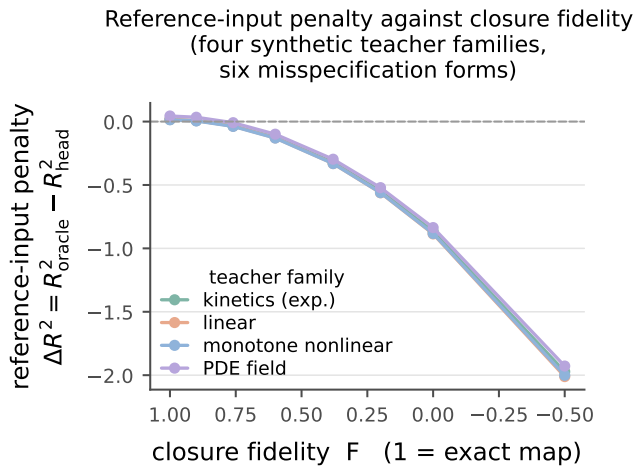


Figure S10: Closure-fidelity sweep on synthetic, non-chemical teacher functions. As the closure is made more misspecified (fidelity  $F$  falling from 1; axis reversed), the reference-input penalty  $\Delta R^2 = R^2_{\text{oracle}} - R^2_{\text{head}}$  falls monotonically below zero. Each curve is one of four teacher families, averaged over six misspecification forms, which the panel does not resolve; the four curves are visually coincident. No point is marked for the real closure.

*construction*; it is not evidence that the effect is domain-general, and the panel does not resolve the six misspecification forms, which do spread (Fig. S10). Placing the real system on the sweep only illustratively (the real-closure-to- $F$  map is a synthetic construction, not a calibrated measurement), the COSMO-SAC closure over predicted  $\sigma$ -profiles falls at  $F \lesssim 0.4$ : its solubility  $\Delta R^2 \approx -0.36$  is what a closure of this fidelity predicts, consistent with  $B_{\text{closure}}$  dominating  $B_{\text{insuff}}$ . The two  $\Delta R^2$  values are measured on different targets—synthetic  $m$  on the sweep,  $\ln x_2$  through the SLE solver and the crystal head in the real system—which is the other reason the placement carries no calibration.

The same construction shows why the grounding gap is an indicator, not a certificate (§2.1.2). Closure fidelity and input sufficiency are varied separately: when  $z^*$  is sufficient,  $\mathcal{G}$  tracks  $B_{\text{closure}}$  as fidelity varies (Theorem S1’s proxy is exact); when  $z^*$  is lossy but the closure is *well specified*,  $B_{\text{closure}} = 0$  throughout yet  $\mathcal{G} > 0$  grows with the discarded variance.

A positive grounding gap can thus be pure input insufficiency, not closure error—so what closure dominance is claimed on rests on the binning bound of the decomposition, computed under the residual-only convention (the bound is convention-specific even though the estimand it bounds is not; §2.1.1). That claim is made stratum by stratum and on one row set, the glycol-ether solvents, and through a one-sided bound; the aggregate does not stand (§3.5.1).  $\mathcal{G}$  only corroborates it.

## S11 Positioning against adjacent literatures

§1 places the instrument against the model-discrepancy problem of Kennedy and O’Hagan<sup>S24</sup> and its calibration critique,<sup>S25</sup> which is the frame the paper works in. Four further literatures bear on it without being that frame. Separating whether predictions are right on average from whether they are sharp is Murphy and DeGroot’s reliability–resolution decomposition:<sup>S26,S27</sup> the closure/insufficiency split is an object of the same kind, an identity that partitions a squared error, but on a *fixed* operator with an external reference for the intermediate rather than on a forecast–outcome pair. A misspecified model still converges to a pseudo-true limit under White’s quasi-maximum-likelihood account,<sup>S20</sup> which is what  $\Delta_\infty$  of Remark 1 names for a composed predictor. That a decomposition may be only weakly constrained by the data is the subject of parameter sloppiness<sup>S28,S29</sup> and of structural versus practical identifiability<sup>S30,S31</sup>—the setting of Lemma S1. That apparent skill may reflect pipeline choices rather than physics is the concern of underspecification,<sup>S32</sup> leakage<sup>S33</sup> and compositional risk,<sup>S34</sup> which is why every arm difference in this paper is reported as a difference between two separately tuned pipelines (§3.7). Finally, that a constrained prior lowers variance at the cost of a bias confined to the directions the data leave undetermined is the null-space reading of regularisation<sup>S35,S36</sup> and of null-space priors in inverse problems,<sup>S37</sup> which is the frame in which Remark 1 is stated.

### S11.1 The nearest published construction, and the account from fluid dynamics

**The nearest published construction, which reports the opposite sign.** TeNNet-SAC,<sup>S38</sup> a  $\sigma$ -profile predictor feeding a COSMO-SAC-shaped segment-activity model, reports accuracy consistently better than COSMO-SAC across its benchmarks. The two works score different operations. Its gain is a training-time gain from supervising and then fine-tuning a learned profile, and over five leak-free seeds that operation has no established sign here (§3.1). Its profile predictor, supervised externally on quantum-chemical  $\sigma$ -profiles for 39,745 molecules, then stays frozen while only the segment-activity output head is fine-tuned, so the closure’s gradient never reaches the intermediate. Our negative result is about substituting a tabulated profile at inference, which TeNNet-SAC leaves unreported, on an intermediate trained through the closure with neither safeguard. Its segment-activity kernel is trainable besides, where this one is fixed at the 2002 parameterisation. If the sign of the substitution turns on whether the kernel can move, our finding is about a fixed 2002 kernel and not about routing a prediction through a physical intermediate. §3.4 already records that no grounding gap has been measured under any modern closure. Retraining the two arms against the 2010/dispersion kernel end to end is limitation (x)’s named run and remains to be made.

**Whether the closure amplifies the substitution, or the two profiles are simply different objects.** An account of this degradation is available from outside chemistry, and it is not the one above. Wu et al.<sup>S39</sup> substitute near-exact Reynolds stresses into the RANS equations and recover large velocity errors, which they trace to ill-conditioning: the fixed model amplifies a small perturbation of its input. By that metric this closure is ill-conditioned: the local relative condition number  $\|J\| \|\sigma\| / |\ln \gamma^\infty|$  at the learned profile is 32 [15, 52] at eight segment iterations and 72

[55, 124] at thirty. It is nonetheless not the account of what follows, because  $\kappa$  is a worst-case, first-order number and the substitution is neither infinitesimal nor worst-case. Ill-conditioning requires the step to be small. On the 354 test rows whose solute and solvent both carry a VT-2005 profile (64 solutes, 18 solvents, 4.4% of the split), it is not. The reference profile stands 1.53 [1.39, 1.80] times the learned profile’s own norm away from it, so the two are different objects rather than an estimate and its target. Neither does the closure amplify selectively along that direction. A random direction of the same norm produces a *larger* relative change in  $\ln \gamma^\infty$  on 71% [62, 76] of those rows at eight segment iterations and on 81% [70, 86] at thirty. The amplification ratio itself is not stable across that pair (median 0.60 [0.37, 1.14] against 1.29 [0.64, 2.20]), which is the segment count behaving as the free hyperparameter §2.5 says it is. The step size is a property of the two profiles and does not move with it. Both readings are of the retired head of this section and price the quantity rather than the arms of Fig. 2.

## References

- (S1) Bell, I. H.; Mickoleit, E.; Hsieh, C.-M.; Lin, S.-T.; Vrabec, J.; Breitenkopf, C.; Jäger, A. A Benchmark Open-Source Implementation of COSMO-SAC. *Journal of Chemical Theory and Computation* **2020**, *16*, 2635–2646.
- (S2) Hsieh, C.-M.; Sandler, S. I.; Lin, S.-T. Improvements of COSMO-SAC for Vapor–Liquid and Liquid–Liquid Equilibrium Predictions. *Fluid Phase Equilibria* **2010**, *297*, 90–97.
- (S3) Hsieh, C.-M.; Lin, S.-T.; Vrabec, J. Considering the Dispersive Interactions in the COSMO-SAC Model for More Accurate Predictions of Fluid Phase Behavior. *Fluid Phase Equilibria* **2014**, *367*, 109–116.
- (S4) Renon, H.; Prausnitz, J. M. Local Compositions in Thermodynamic Excess Functions for Liquid Mixtures. *AIChE Journal* **1968**, *14*, 135–144.
- (S5) Fredenslund, A.; Jones, R. L.; Prausnitz, J. M. Group-Contribution Estimation of Activity Coefficients in Nonideal Liquid Mixtures. *AIChE Journal* **1975**, *21*, 1086–1099.
- (S6) Rampásek, L.; Galkin, M.; Dwivedi, V. P.; Luu, A. T.; Wolf, G.; Beaini, D. Recipe for a General, Powerful, Scalable Graph Transformer. Advances in Neural Information Processing Systems (NeurIPS). 2022.
- (S7) van der Vaart, A. W. *Asymptotic Statistics*; Cambridge Series in Statistical and Probabilistic Mathematics; Cambridge University Press, 1998; Chapter 25 (semiparametric efficiency: efficient score and information).
- (S8) Hirano, K.; Imbens, G. W.; Ridder, G. Efficient Estimation of Average Treatment Effects Using the Estimated Propensity Score. *Econometrica* **2003**, *71*, 1161–1189.
- (S9) Robins, J. M.; Ritov, Y. Toward a Curse of Dimensionality Appropriate (CODA) Asymptotic Theory for Semi-Parametric Models. *Statistics in Medicine* **1997**, *16*, 285–319.
- (S10) Chernozhukov, V.; Chetverikov, D.; Demirer, M.; Duflo, E.; Hansen, C.; Newey, W.; Robins, J. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal* **2018**, *21*, C1–C68.
- (S11) Fingerhut, R.; Chen, W.-L.; Schedemann, A.; Cordes, W.; Rarey, J.; Hsieh, C.-M.; Vrabec, J.; Lin, S.-T. Comprehensive Assessment of COSMO-SAC Models for Predictions of Fluid-Phase Equilibria. *Industrial & Engineering Chemistry Research* **2017**, *56*, 9868–9884.
- (S12) Lin, S.-T.; Sandler, S. I. A Priori Phase Equilibrium Prediction from a Segment Contribution Solvation Model. *Industrial & Engineering Chemistry Research* **2002**, *41*, 899–913.
- (S13) Attia, L.; Burns, J. W.; Doyle, P. S.; Green, W. H. Data-driven organic solubility prediction at the limit of aleatoric uncertainty. *Nature Communications* **2025**, *16*, 7497, FASTSOLV model; earlier ChemRxiv 2024 preprint doi:10.26434/chemrxiv-2024-93qp3.
- (S14) Vermeire, F. H.; Chung, Y.; Green, W. H. Predicting Solubility Limits of Organic Solutes for a Wide Range of Solvents and Temperatures. *Journal of the American Chemical Society* **2022**, *144*, 10785–10797.
- (S15) Boobier, S.; Hose, D. R. J.; Blacker, A. J.; Nguyen, B. N. Machine learning with physicochemical relationships: solubility prediction in organic solvents and water. *Nature Communications* **2020**, *11*, 5753.

- (S16) Sanchez Medina, E. I.; Linke, S.; Stoll, M.; Sundmacher, K. Gibbs–Helmholtz graph neural network: capturing the temperature dependency of activity coefficients at infinite dilution. *Digital Discovery* **2023**, *2*, 781–798.
- (S17) de Souza, E. T., Jr.; Alcantara, M. L.; Staudt, P. B.; Coutinho, J. a. A. P.; Soares, R. d. P. Development of a COSMO-SAC Parametrization with Advanced QM Method TZVPD-FINE. *Industrial & Engineering Chemistry Research* **2025**, *64*, 14700–14711.
- (S18) Hansch, C.; Leo, A.; Taft, R. W. A Survey of Hammett Substituent Constants and Resonance and Field Parameters. *Chemical Reviews* **1991**, *91*, 165–195.
- (S19) Mansouri, K.; Cariello, N. F.; Korotcov, A.; Tkachenko, V.; Grulke, C. M.; Sprankle, C. S.; Allen, D.; Casey, W. M.; Kleinstreuer, N. C.; Williams, A. J. Open-source QSAR models for pKa prediction using multiple machine learning approaches. *Journal of Cheminformatics* **2019**, *11*, 60.
- (S20) White, H. Maximum Likelihood Estimation of Misspecified Models. *Econometrica* **1982**, *50*, 1–25.
- (S21) Krasnov, L.; Malikov, D.; Kiseleva, M.; Tatarin, S.; Sosnin, S.; Bezzubov, S. Big-SolDB 2.0, dataset of solubility values for organic compounds in different solvents at various temperatures. *Scientific Data* **2025**, *12*, 1236.
- (S22) Mullins, E.; Oldland, R.; Liu, Y. A.; Wang, S.; Sandler, S. I.; Chen, C.-C.; Zwolak, M.; Seavey, K. C. Sigma-Profile Database for Using COSMO-Based Thermodynamic Methods. *Industrial & Engineering Chemistry Research* **2006**, *45*, 4389–4415.
- (S23) Frenkel, M.; Chirico, R. D.; Diky, V. V.; Yan, X.; Dong, Q.; Muzny, C. ThermoML: An XML-Based Approach for Storage and Exchange of Experimental and Critically Evaluated Thermophysical and Thermochemical Property Data. *Journal of Chemical Information and Computer Sciences* **2003**, *43*, 1344–1359.
- (S24) Kennedy, M. C.; O’Hagan, A. Bayesian Calibration of Computer Models. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* **2001**, *63*, 425–464.
- (S25) Brynjarsdóttir, J.; O’Hagan, A. Learning about physical parameters: the importance of model discrepancy. *Inverse Problems* **2014**, *30*, 114007.
- (S26) Murphy, A. H. A New Vector Partition of the Probability Score. *Journal of Applied Meteorology* **1973**, *12*, 595–600.
- (S27) DeGroot, M. H.; Fienberg, S. E. The Comparison and Evaluation of Forecasters. *Journal of the Royal Statistical Society: Series D (The Statistician)* **1983**, *32*, 12–22.
- (S28) Transtrum, M. K.; Machta, B. B.; Brown, K. S.; Daniels, B. C.; Myers, C. R.; Sethna, J. P. Perspective: Sloppiness and emergent theories in physics, biology, and beyond. *The Journal of Chemical Physics* **2015**, *143*, 010901.
- (S29) Gutenkunst, R. N.; Waterfall, J. J.; Casey, F. P.; Brown, K. S.; Myers, C. R.; Sethna, J. P. Universally Sloppy Parameter Sensitivities in Systems Biology Models. *PLoS Computational Biology* **2007**, *3*, 1871–1878.
- (S30) Raue, A.; Kreutz, C.; Maiwald, T.; Bachmann, J.; Schilling, M.; Klingmüller, U.; Timmer, J. Structural and practical identifiability analysis of partially observed dynamical models by exploiting the profile likelihood. *Bioinformatics* **2009**, *25*, 1923–1929.

- (S31) Villaverde, A. F.; Barreiro, A.; Papachristodoulou, A. Structural Identifiability of Dynamic Systems Biology Models. *PLoS Computational Biology* **2016**, *12*, e1005153.
- (S32) D’Amour, A.; Heller, K.; Moldovan, D.; Adlam, B.; Alipanahi, B.; Beutel, A.; Chen, C.; Deaton, J.; Eisenstein, J.; Hoffman, M. D.; Hormozdiari, F.; Houlisby, N.; Hou, S.; Jerfel, G.; Karthikesalingam, A.; Lucic, M.; Ma, Y.; McLean, C.; Mincu, D.; Mitani, A.; Montanari, A.; Nado, Z.; Natarajan, V.; Nielson, C.; Osborne, T. F.; Raman, R.; Ramasamy, K.; Sayres, R.; Schrouff, J.; Seneviratne, M.; Sequeira, S.; Suresh, H.; Veitch, V.; Vladymyrov, M.; Wang, X.; Webster, K.; Yadlowsky, S.; Yun, T.; Zhai, X.; Sculley, D. Underspecification Presents Challenges for Credibility in Modern Machine Learning. *Journal of Machine Learning Research* **2022**, *23*, 1–61.
- (S33) Kapoor, S.; Narayanan, A. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns* **2023**, *4*, 100804.
- (S34) Mahajan, D.; Pezeshki, M.; Arnal, C.; Mitliagkas, I.; Ahuja, K.; Vincent, P. Compositional Risk Minimization. Proceedings of the 42nd International Conference on Machine Learning. 2025; pp 42503–42553.
- (S35) Engl, H. W.; Hanke, M.; Neubauer, A. *Regularization of Inverse Problems; Mathematics and Its Applications*; Kluwer Academic Publishers, 1996; Vol. 375.
- (S36) Hansen, P. C. *Rank-Deficient and Discrete Ill-Posed Problems: Numerical Aspects of Linear Inversion*; SIAM Monographs on Mathematical Modeling and Computation; SIAM, 1998.
- (S37) Schwab, J.; Antholzer, S.; Haltmeier, M. Deep Null Space Learning for Inverse Problems: Convergence Analysis and Rates. *Inverse Problems* **2019**, *35*, 025008.
- (S38) Yang, Y.; Lin, S.-T. Physics-Embedded Machine Learning Model for Phase Equilibrium Prediction in Multicomponent Systems. *Journal of Chemical Information and Modeling* **2025**, *65*, 10180–10193.
- (S39) Wu, J.; Xiao, H.; Sun, R.; Wang, Q. Reynolds-averaged Navier–Stokes equations with explicit data-driven Reynolds stress closure can be ill-conditioned. *Journal of Fluid Mechanics* **2019**, *869*, 553–586.